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ON RINGS OF OPERATORS. IV

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INTRODUCTION

§1. The object of this paper is the investigation of the isomorphism properties of those operator rings which are factors and which are the substratum of several earlier publications of the authors. (Cf. [5, 6, 8]. For the detailed definitions and also for the cases of factors—of which there will be more to say below—cf. [5], p. 138, Definition 3.1.2, p. 172, Theorem VIII.) The discrete cases—i.e. the cases (I)—have been exhaustively dealt with before. (Cf. [5], p. 173, Lemma 8.6.1, p. 139, Definitions 3.2.1 and 3.2.2 and Lemmas 3.2.1–3.2.3.) The purely infinite case—i.e. the case (III)—is the most refractory of all and we have, at least for the time being, scarcely any tools to investigate it. ([8] deals mainly with these factors.) Thus we are left with the continuous cases—i.e. the cases (II)—and they are our main objective in this paper.

An added justification of this program may be found in the fact that among all factors those of the finite continuous case (II_1) have the strongest immediate interest. (Cf. [5], part V, [6], Chapter IV and Appendix.)

It will be seen however that the discrete cases can be included in our discussion with scarcely any extra effort. So we shall deal with them, i.e. with all cases but the purely infinite one.

For the discrete and continuous cases, the finite ones—i.e. the cases (I_n) ($n = 1, 2, \ldots$) and (II_1) respectively—are basic because the infinite ones—i.e. the cases (I_∞) and (II_∞) respectively—can be subsequently described with their help. (Cf. Theorem IX and Lemma 3.1.6.) Therefore we shall direct our main effort on the finite cases. And since the discrete ones—(the cases (I_n), ($n = 1, 2, \ldots$))—are just the finite order matrix rings, this means essentially the above-mentioned continuous finite case (II_1).

§2. Let us now state the main problems of isomorphism more precisely.

Consider an operator ring $\mathbf{M}$ in a Hilbert space $\mathfrak{H}$ which contains 1. (We do not restrict $\mathbf{M}$ in any other way yet. For the significant definitions, cf. [2], pp. 388–389, Definitions 1–3.) Then there exist two kinds of notions in $\mathbf{M}$: First, those which can be expressed in terms of the entity 1 (the unit operator) and the operations αA (α any complex number), A^*, $A + B$, $A \cdot B$ alone and referring only to the operators belonging to $\mathbf{M}$; Second those which need other things as well, e.g. operators outside of $\mathbf{M}$, elements of $\mathfrak{H}$, etc. The former notions are purely algebraical, the latter ones are not; we shall also call them spatial.

These notions were already investigated by the second author in [9].

It seems worth while to formulate this distinction in terms of isomorphisms. Let, for each $i = 1, 2$, a Hilbert space $\mathfrak{H}_i$ and an operator ring $\mathbf{M}_i$ in $\mathfrak{H}_i$ be given. We now define

A) Spatial isomorphism of $\mathbf{M}_1$ and $\mathbf{M}_2$. This is a one-to-one isomorphic mapping of $\mathfrak{H}_1$ on $\mathfrak{H}_2$ (i.e. one which is linear and isometric—unitary) which carries $\mathbf{M}_1$ into $\mathbf{M}_2$.

B) Algebraical ring isomorphism of $\mathbf{M}_1$ and $\mathbf{M}_2$. This is a one-to-one mapping of $\mathbf{M}_1$ onto $\mathbf{M}_2$ which leaves the entity 1 and the operations αA (α any complex number), A^*, $A + B$, AB (when passing from $\mathbf{M}_1$ to $\mathbf{M}_2$) invariant.

(Cf. also [5], p. 145, §5.2, in particular Definition 5.2.1. We have added the requirement that 1 be invariant.)

Any spatial property is invariant under the spatial isomorphisms of A), while the purely algebraical properties are characterized by their invariance under the algebraical isomorphisms of B).

Clearly A) implies B) while the converse need not be true.

An important ring theoretical notion which is only spatial is $\mathbf{M}'$. (Cf. [2], p. 388, Def. 3. For the spatial character, cf. the detailed discussion of §3.3.) Nevertheless the factor property, i.e. $\mathbf{M} \cap \mathbf{M}' = (\alpha \cdot 1)$ (cf. §3.3, loc. cit., and also the beginning of §1) is purely algebraical since it states that the operators $\alpha \cdot 1$ exhaust the center of $\mathbf{M}$ (cf. [5], p. 138, Def. 3.1.2).

Now our program is subdivided as follows.

Question I: When does B) hold?

Question II: Under what additional conditions does B) imply A)?

Question II was already investigated in a special case in [6]. It was shown there that under certain conditions A) and B) are equivalent. (Cf. [6], p. 244, Theorem XI.) We shall obtain a complete answer to Question II in Theorem X.

Question I is more difficult. It coincides with Problem 6 in [5], p. 172, and the present paper contains what progress the authors have been able to make in that direction. The main results are these: An extensive class of factors of case (II_1) which are all isomorphic to each other in the sense B) will be determined. These factors are called "approximately finite". (Cf. Theorem XII, Theorem XIV, which are based on Definition 4.1.1, Definition 4.3.1, Definition 4.5.2 and Definition 4.6.1 below.) The isomorphisms announced in [5], p. 229, (v) are contained in this class. On the other hand, certain factors of case (II_1) which are not isomorphic to the approximately finite ones will be constructed. (Cf. Theorem XVI, Theorem XVI′.)

§3. There are indications to the effect that the approximately finite factors are the simplest among those of case (II_1) but the evidence is not quite conclusive. It is true that every factor in case (II_1) has an approximately finite sub-ring. (Cf. Theorem XIII.) However, this "imbedding theorem" does not settle the matter, since the analogue of the Cantor–Bernstein "equivalence theorem" is not true: Two factors in the case (II_1) might be such that each is isomorphic to a sub-ring of the other, but the factors themselves may not be isomorphic. (An example of this is given in the appendix.) Hence the possibility exists that any factor in the case (II_1) is isomorphic to a sub-ring of any other such factor.

§4. The best we can do at present concerning the isomorphism problem in the case (II_1) is this: The limits of the approximately finite case are extended rather far in §5.2 and §5.6. The existence of non-approximately-finite factors in this case is established in Theorem XVI, Theorem XVI′. Certain algebraical invariants of factors in the case (II_1) are formed. ((1), (2), in §4.8, and the property Γ in Definition 6.1.1) of which the first two are probably of greater general significance, but the last one has so far been put to greater practical use. (Cf. the remark at the beginning of §6.1.)

The isomorphism questions of the case (II_∞) are reduced to those of the case (II_1). (This follows from Theorem IX.) Those of the discrete cases—the cases (I)—have been settled before (cf. above at the beginning of §1; also α) in Theorem IX).

This enumeration exhausts our present program. It answers Problems 6, 7 in [5], p. 172, and [5] eod., p. 229, as far as possible at this moment.

§5. We add that §5.3 contains a new technique of constructing factors in the case (II_1). This seems to be considerably simpler than our previous procedures, but it is closely related to them. (Cf. §5.4, §5.5.) It also throws some light on the meaning of this work from the point of view of the unitary representation theory of groups (cf. the remarks after Lemma 5.3.4). Further generalizations are probably possible and important. (Cf. the remark at the end of §5.6.)

The result that not all factors in the case (II_1) are isomorphic to each other, expressed in Theorem XVI′, deserves some further comment. From the point of view of the systematic build-up of the paper, it seemed best to put it at the end. It is, however, intelligible and of interest in itself, and it can be derived independently of most of the paper. The reader who is primarily interested in this particular result need only read §5.3, Definition 6.1.1, Lemma 6.1.1, Lemma 6.2.1, Lemma 6.2.2. The entire remainder of this paper is unnecessary for this purpose, in particular the extensive theories of algebraical and spatial types, of genera, and of approximate finiteness and its various equivalent forms. But we think, nevertheless, that knowledge of the whole paper will help to see this result too in the proper perspective.

§6. The notations are the same as in the papers referred to in the bibliography, particularly [5] and [6].

We use the Kronecker–Weierstrass symbol in the general sense:

$$\delta_{a,b} = \begin{cases} 1 & \text{for } a = b, \\ 0 & \text{otherwise,} \end{cases}$$

for any objects a, b.

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CONTENTS

CHAPTER I. THE METRIC OF M. PURELY ALGEBRAICAL CHARACTER OF VARIOUS NOTIONS

§1.1 Consider a Hilbert space $\mathfrak{H}$ and an operator ring $\mathbf{M}$ in $\mathfrak{H}$. We know from §2 in the introduction as well as from [9] that the purely algebraical character of certain notions in $\mathbf{M}$ is neither obvious nor certain.

It was demonstrated in [9], however, that the following notions in $\mathbf{M}$ are purely algebraical.

Definiteness of an $A \in \mathbf{M}$; the numerical value $|||A|||$ of the bound of an $A \in \mathbf{M}$; convergence in the sense of the strong and of the weak operator topologies. (Convergence in the sense of the strongest topology is the same as in the strong one. Cf. [4], p. 112, end of §2.)

We did not establish the same thing for the operator topologies themselves,

i.e. for the strongest, the strong and the weak. (For the definitions, cf. [4, p. 111, §2], and [2, pp. 381–382].) The uniform topology (cf. [2, pp. 384–386]) causes no difficulty, since $|||A|||$ is purely algebraical, but it will play no role in our discussions. This is very unsatisfactory since the notion of a subring of $\mathbf{M}$ must be based on one of these topologies. (Cf. [4, p. 112, beginning of §3]; [2, p. 396, end of §3; p. 388, Definition 1]. Best consulted in the reverse order.) They cannot be replaced by convergence notions, nor by the uniform topology. (Cf. in particular [2, pp. 382–383 and p. 384].) And yet the notion of a subring ought to be purely algebraical.

For our present purposes it will be adequate to settle these questions for a restricted class of rings $\mathbf{M}$ and this we now do.

§1.2 Observe first that the following notions are purely algebraical.

The factor character of $\mathbf{M}$ and the case to which $\mathbf{M}$ belongs, the numerical value of the relative dimension function $D_{\mathbf{M}}(E)$ (E any projection $\in \mathbf{M}$) apart from its normalization and even the standard normalization in the cases where one is defined; the numerical value of the relative trace $\mathrm{Tr}_{\mathbf{M}}(A)$, ($A \in \mathbf{M}$) in the finite cases.

These assertions were established in [5, p. 145, Lemma 5.2.1; p. 173, Theorem IX]; [6, p. 219, Property IV] (since that characterization is purely algebraical).

We assume now that $\mathbf{M}$ is a factor in a finite case.[1] We shall use the relative dimension function $D_{\mathbf{M}}(E)$ and it will be advantageous not to normalize it. We shall also use the relative trace $\mathrm{Tr}_{\mathbf{M}}(A)$ with its usual properties. We define for every $A \in \mathbf{M}$

$$[\![A]\!] = \sqrt{\mathrm{Tr}_{\mathbf{M}}(A^*A)} = \sqrt{\mathrm{Tr}_{\mathbf{M}}(AA^*)}. \tag{1.2.α}$$

(Cf. [6, p. 241, Lemma 4.3.2]. The considerations of [8, p. 102, Definition 1.3.1 and Theorem II] are somewhat more general but of the same type.) Owing to our above remarks, the numerical value of $[\![A]\!]$ is purely algebraical too.

By [6, p. 235, Theorem III], there exists a fixed finite system $g_1, \ldots, g_m \in \mathfrak{H}$ (depending on $\mathbf{M}$ only), such that for all $A \in \mathbf{M}$

$$\mathrm{Tr}_{\mathbf{M}}(A) = \sum_{i=1}^{m} (Ag_i, g_i). \tag{1.2.β}$$

Combining (1.2.α) with (1.2.β), and remembering that $(A^*Ag, g) = \|Ag\|^2$, $(AA^*g, g) = \|A^*g\|^2$, we obtain

$$[\![A]\!] = \sqrt{\sum_{i=1}^{m} \|Ag_i\|^2} = \sqrt{\sum_{i=1}^{m} \|A^*g_i\|^2}. \tag{1.2.γ}$$

This formula is useful for many purposes, although it obscures the purely algebraical character of $[\![A]\!]$.

$[\![A]\!]$ has all the properties of a *norm* in a linear space, and accordingly $[\![A - B]\!]$ has all those of a *distance*. (Cf. for these and for some further properties [6, p. 242, Property II°].) We need these for $\mathbf{M}$ only, and not, as loc. cit., for its extension $Q(\mathbf{M})$. All these properties can be read off the representation (1.2.γ)—the decisive

[1] A finite case is either a (I_n), $n = 1, 2, \ldots$, or a (II_1). The inclusion of the former is unnecessary, since our real interest is in case (II_1). But these considerations have the totality of all finite cases as their natural scope.

fact being, of course, that we have them both.) Thus $\mathbf{M}$ has become a topological space in a new way, with a purely algebraical metric.[2]

We now define

Definition 1.2.1. *A sequence $A_1, A_2, \ldots \in \mathbf{M}$ is metrically convergent to $A \in \mathbf{M}$ if $\lim_{\nu\to\infty} [\![A_\nu - A]\!] = 0$. It is restrictedly metrically convergent to $A \in \mathbf{M}$ if $\lim_{\nu\to\infty} [\![A_\nu - A]\!] = 0$ and the (numerical) sequence $|||A_1|||, |||A_2|||, \ldots$ is bounded.*

The notions of closure for subsets $\mathbf{S}$ of $\mathbf{M}$ which are induced by these two notions of convergence, are the "metrical closure" and the "restricted metrical closure" of $\mathbf{S}$.[3]

We shall also need:

Definition 1.2.2. *A subset $\mathbf{S}$ of $\mathbf{M}$ is linear if $A, B \in \mathbf{S}$ imply $\alpha A, A + B \in \mathbf{S}$. It is an algebra if $A, B \in \mathbf{S}$ imply $\alpha A, A^*, A + B, AB \in \mathbf{S}$.*

Clearly both these definitions define purely algebraical notions.

Now a subring $\mathbf{S}$ of $\mathbf{M}$ is a closed algebra, where closure may be understood in any one of the three operator topologies: the strongest, the strong, or the weak (cf. the end of §1.1). We shall establish the purely algebraical character of the notion of a subring by proving the equivalence of the notion of closure in all these topologies to the purely algebraical closures of Definition 1.2.1 for a set $\mathbf{S}$ which is an algebra.[4]

§1.3 Observe first:

Lemma 1.3.1. *For the $g_1, \ldots, g_m$ of (1.2.β) and (1.2.γ) the $A'g_i$, ($A' \in \mathbf{M}'$, $i = 1, \ldots, m$) span the closed linear set $\mathfrak{H}$.*

Proof. For a fixed i $(= 1, \ldots, m)$ the $A'g_i$, $A' \in \mathbf{M}'$, span the closed linear set $\mathfrak{M}^{\mathbf{M}'}_{g_i}$ (cf. [5, p. 143, Definition 5.1.1]). Put $\mathfrak{N} = [\mathfrak{M}^{\mathbf{M}'}_{g_i},\ i = 1, \ldots, m]$. We must prove

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[2] The uniform topology, already mentioned in §1.1, is actually of the same character: $|||A|||$, $|||A - B|||$ are a norm and a metric, just as $[\![A]\!]$, $[\![A - B]\!]$. But only the latter is useful in the theory of operator rings. This is due to its connection with the strong topology, which will appear in §1.4 and §1.5.

[3] The metrical closure is the conventional topological notion, based on the specified metric. The restricted metrical closure is not of that nature. Thus it is not evident that the set of all limits of all restrictedly metrically convergent sequences from a given set $\mathbf{S}$ is necessarily closed in that sense. Actually this is not true for all sets $\mathbf{S}$ although when $\mathbf{S}$ is an algebra it does follow from the results of §1.5 (cf. there).

[4] Observe that restrictions on both the ring $\mathbf{M}$ and its subset $\mathbf{S}$ are necessary in order to secure this result. $\mathbf{M}$ must be a factor in a finite case, $\mathbf{S}$, an algebra.

$\mathfrak{N} = \mathfrak{H}$. Now $\mathfrak{N}\ \eta\ \mathbf{M}'$ (cf. loc. cit., Lemma 5.1.1). Hence $\mathfrak{N}\ \eta\ \mathbf{M}$ and $E = P_{\mathfrak{N}} \in \mathbf{M}$. Also $g_i \in \mathfrak{M}_{g_i}^{\mathbf{M}'} \subset \mathfrak{N}$, so

$$Eg_i = g_i. \tag{1.3.α}$$

Now (1.2.γ) shows that if $A \in \mathbf{M}$ and $Ag_i = 0$ for $i = 1, \ldots, m$, then $A = 0$. Apply this to $A = 1 - E$. Then (1.3.α) yields $1 - E = 0$, $P_{\mathfrak{N}} = E = 1$, as desired.

Lemma 1.3.2. *The sequence $A_1, A_2, \ldots \in \mathbf{M}$ is strongly convergent to $A \in \mathbf{M}$ if and only if it is restrictedly metrically convergent to this A.*

Proof. Necessity: Assume that $A_1, A_2, \ldots$ is strongly convergent to A. Then the (numerical) sequence $|||A_1|||, |||A_2|||, \ldots$ is bounded by [2, p. 382, footnote 35], and $\lim_{\nu\to\infty} [\![A_\nu - A]\!] = 0$ by (1.2.γ). Hence we have restricted metric convergence.

Sufficiency: Assume that $A_1, A_2, \ldots$ are restrictedly metrically convergent to A. We must show that they converge strongly too, i.e. that the set $\mathfrak{M}$ of all f with $\lim_{\nu\to\infty}\|(A_\nu - A)f\| = 0$ is $\mathfrak{H}$. $\mathfrak{M}$ is obviously a linear set. Since the (numerical) sequence $|||A_1|||, |||A_2|||, \ldots$ is bounded, $\mathfrak{M}$ is also closed. Hence it suffices to show that $\mathfrak{M}$ spans the closed linear set $\mathfrak{H}$.

Now by (1.2.γ), $\|(A_\nu - A)g_i\| \leq [\![A_\nu - A]\!]$. Hence $\lim_{\nu\to\infty}\|(A_\nu - A)g_i\| = 0$. Therefore for every $A' \in \mathbf{M}'$, $\lim_{\nu\to\infty}\|A'(A_\nu - A)g_i\| = 0$. But $A'(A_\nu - A) = (A_\nu - A)A'$ since $A' \in \mathbf{M}'$ and $A_\nu, A \in \mathbf{M}$. Consequently $\lim_{\nu\to\infty}\|(A_\nu - A)A'g_i\| = 0$ and all $A'g_i \in \mathfrak{M}$. Hence $\mathfrak{M}$ spans the closed linear set $\mathfrak{H}$ by Lemma 1.3.1, and the proof is completed.

Thus closure with respect to strong convergence is equivalent to restricted metric closure for every subset $\mathbf{S}$ of $\mathbf{M}$. This however is not enough to establish the purely algebraical character of the notion of a subring, since that involves topological closure which cannot be replaced by convergence closure (cf. the end of §1.1). In fact, from that point of view we have made no progress at all, since we knew all along from [9] that strong convergence is a purely algebraical notion for all rings $\mathbf{M}$. Besides our present restrictions on $\mathbf{M}$, we shall also have to restrict $\mathbf{S}$ to algebras before we can get any further (cf. the end of §1.2).

§1.4 Consider first the following two simple results, valid for all subsets $\mathbf{S}$ of $\mathbf{M}$.

Lemma 1.4.1. *Strong closure of $\mathbf{S}$ implies the restricted metric closure.*

Proof. Strong closure of $\mathbf{S}$ implies closure with respect to strong convergence, and by Lemma 1.3.2 the latter is equivalent to restricted metric closure.

Lemma 1.4.2. *Metric closure of $\mathbf{S}$ implies its strong closure.*

Proof. Let $\mathbf{S}$ be metrically closed and consider a strong condensation point A of $\mathbf{S}$.

Since $\mathbf{S} \subset \mathbf{M}$ and $\mathbf{M}$ is strongly closed, this implies $A \in \mathbf{M}$.

For every $\nu = 1, 2, \dots$ choose an $A_\nu \in \mathbf{S}$ with $\|(A_\nu - A)g_i\| \leq 1/\nu$ for $i = 1, \dots, m$. This can be done since, by the definition of a limit point, we must have at least one $A_\nu \in \mathbf{M}$ in the strong neighborhood $U_3(A, g_1, \dots, g_m; 1/\nu)$ of A (cf. [2, §2, p. 381]). Then (1.2.γ) gives $[\![A_\nu - A]\!] \leq \sqrt{m}/\nu$ and hence $\lim_{\nu\to\infty} [\![A_\nu - A]\!] = 0$. Thus $A_1, A_2, \dots$ converges metrically to A and since $\mathbf{S}$ is metrically closed, $A \in \mathbf{S}$. This completes the proof.

Now if we can show that the restricted metric closure implies metric closure, the above lemmas will yield the equivalence of the strong, the metric and the restricted metric notions of closure. As pointed out at the end of §1.3, this is what we need. The assumption that $\mathbf{S}$ is an algebra will be needed to secure the above implication.

§1.5 In what follows we shall make repeated use of the notion and properties of the functions of bounded Hermitian and of unitary operators,[5] and it will be convenient to omit specific references to that theory.

Lemma 1.5.1. *(i) If A is bounded Hermitian, then $(A - i1)^{-1}$ can be formed and*

$$\frac{A + i1}{A - i1} = (A - i1)^{-1}(A + i1) = (A + i1)(A - i1)^{-1}$$

is unitary.

(ii) If $A, B \in \mathbf{M}$ and Hermitian, then $\frac{A + i1}{A - i1}, \frac{B + i1}{B - i1} \in \mathbf{M}$ too, and

$$\left[\!\!\left[\frac{A + i1}{A - i1} - \frac{B + i1}{B - i1}\right]\!\!\right] \leq 2\,[\![A - B]\!]\,.$$

Proof. Ad (i). Consider the two functions

$$f(\lambda) = \frac{1}{\lambda - i}, \qquad g(\lambda) = \frac{\lambda + i}{\lambda - i}.$$

Since they are continuous for all real λ we can form $f(A)$ and $g(A)$. We also have

$$(\lambda - i)f(\lambda) = f(\lambda)(\lambda - i) = 1, \qquad (\lambda + i)f(\lambda) = f(\lambda)(\lambda + i) = g(\lambda).$$

[5]Cf., e.g., [3, pp. 205 and 220]. The real function-theory methods of [3] are actually much more than what is needed for the limited objectives of §1.5, as only continuous functions (sometimes even polynomials) of operators occur. ((i) in Lemma 1.5.1 can even be proven by elementary evaluations.) But it would take up too much space to go into such methodological questions systematically.

Therefore $f(A) = (A - i1)^{-1}$ and $g(A) = (A + i1)f(A) = f(A)(A + i1)$. Thus $g(A) = (A + i1)(A - i1)^{-1} = (A - i1)^{-1}(A + i1)$ or $g(A) = (A + i1)/(A - i1)$ in the sense indicated above.

Since $\overline{g(\lambda)}g(\lambda) = 1 = g(\lambda)\overline{g(\lambda)}$ for all real λ, $g(A)^*g(A) = 1 = g(A)g(A)^*$ and hence $g(A)$ is unitary. Thus the proof of (i) is completed. We note also that inasmuch as $|f(\lambda)| \leq 1$ for all real λ, $\left|\left|\left| f(A) \right|\right|\right| \leq 1$ or

$$\left|\left|\left|(A - i1)^{-1}\right|\right|\right| \leq 1. \tag{1.5.α}$$

Ad (ii). $A, B \in \mathbf{M}$ imply $g(A), g(B) \in \mathbf{M}$ or $(A+i1)/(A-i1), (B+i1)/(B-i1) \in \mathbf{M}$. We have

$$\begin{aligned}
\frac{A + i1}{A - i1} - \frac{B + i1}{B - i1} &= (A - i1)^{-1}(A + i1) - (B + i1)(B - i1)^{-1} \\
&= (A - i1)^{-1}\{(A + i1)(B - i1) - (A - i1)(B + i1)\}(B - i1)^{-1} \\
&= -2i(A - i1)^{-1}(A - B)(B - i1)^{-1}.
\end{aligned}$$

Consequently

$$\left[\!\left[\frac{A + i1}{A - i1} - \frac{B + i1}{B - i1}\right]\!\right] \leq 2\left|\left|\left|(A - i1)^{-1}\right|\right|\right| [\![A - B]\!] \left|\left|\left|(B - i1)^{-1}\right|\right|\right|,$$

and by (1.5.α)

$$\left[\!\left[\frac{A + i1}{A - i1} - \frac{B + i1}{B - i1}\right]\!\right] \leq 2\,[\![A - B]\!].$$

Lemma 1.5.2. *Let a function $\varphi(\xi)$ be given, defined and continuous for all ξ with $|\xi| = 1$. Then there exists a function $\omega_1(\epsilon)$ (given φ, ω_1 is determined) defined and > 0 for all $\epsilon > 0$, with the following property: If U, V are unitary and $\in \mathbf{M}$ then $\varphi(U), \varphi(V) \in \mathbf{M}$ and $[\![U - V]\!] < \omega_1(\epsilon)$ implies $\left[\!\left[\varphi(U) - \varphi(V)\right]\!\right] < \epsilon$.*

Proof. That U, V unitary and $\in \mathbf{M}$ imply $\varphi(U), \varphi(V) \in \mathbf{M}$ is obvious. We shall now show that it suffices to prove our assertion for all trigonometrical polynomials

$$\varphi(\xi) = \sum_{\nu=-n}^{n} a_\nu \xi^\nu.$$[6]

Assume accordingly that it is true for these and consider a general $\varphi(\xi)$.

[6]Since $|\xi| = 1$ we may put $\xi = e^{i\alpha}$, α real, modulo 2π. Now $\varphi(\xi) = \sum_{\nu=-n}^{n} a_\nu \xi^\nu = \sum_{\nu=-n}^{n} a_\nu e^{i\nu\alpha}$. Thus these $\varphi(\xi)$ are the trigonometrical polynomials with their familiar approximation properties. Cf. Zygmund, Antoni, "Trigonometrical Series," Warsaw (1935), §3.23(iii), p. 47.

Consider an $\epsilon' > 0$. Choose a trigonometrical polynomial

$$\varphi^{\epsilon'}(\xi) = \sum_{\nu=-n^{\epsilon'}}^{n^{\epsilon'}} a_\nu^{\epsilon'} \xi^\nu$$

such that $|\varphi(\xi) - \varphi^{\epsilon'}(\xi)| \leq \frac{1}{3}\epsilon'$ for all ξ with $|\xi| = 1$. Then the unitarity of U, V implies $\left|\left|\left|\varphi(U) - \varphi^{\epsilon'}(U)\right|\right|\right| \leq \frac{1}{3}\epsilon'$ and $\left|\left|\left|\varphi(V) - \varphi^{\epsilon'}(V)\right|\right|\right| \leq \frac{1}{3}\epsilon'$. Hence, *a fortiori*,

$$\left[\left[\varphi(U) - \varphi^{\epsilon'}(U)\right]\right] \leq \tfrac{1}{3}\epsilon', \qquad \left[\left[\varphi(V) - \varphi^{\epsilon'}(V)\right]\right] \leq \tfrac{1}{3}\epsilon'.$$

Denote the $\omega_1(\epsilon)$ of $\varphi^{\epsilon'}(\xi)$ by $\omega_1^{\epsilon'}(\epsilon)$. Then $[[U - V]] < \omega_1^{\epsilon'}(\frac{1}{3}\epsilon')$ implies

$$\left[\left[\varphi^{\epsilon'}(U) - \varphi^{\epsilon'}(V)\right]\right] < \tfrac{1}{3}\epsilon'$$

and so, owing to the above,

$$\left[\left[\varphi(U) - \varphi(V)\right]\right] < \epsilon'.$$

Thus $\omega_1(\epsilon') = \omega_1^{\epsilon'}(\frac{1}{3}\epsilon')$ meets our requirements for the original $\varphi(\xi)$.

It only remains therefore to prove our assertion for the trigonometrical polynomials

$$\varphi(\xi) = \sum_{\nu=-n}^{n} a_\nu \xi^\nu.$$

Obviously we may even restrict ourselves to the monomials

$$\varphi(\xi) = \xi^\nu, \qquad (\nu = 0, \pm 1, \pm 2, \ldots).$$

For $\nu = 0$ this is trivial, and for $\nu < 0$ the unitarity of U, V implies

$$[[U^\nu - V^\nu]] = [[(U^\nu - V^\nu)^*]] = [[U^{*\nu} - V^{*\nu}]] = [[U^{-\nu} - V^{-\nu}]].$$

Therefore we may even assume $\nu > 0$. In this case, however,

$$\begin{aligned} U^\nu - V^\nu &= \sum_{\mu=1}^{\nu} \left(U^\mu V^{\nu-\mu} - U^{\mu-1} V^{\nu-\mu+1}\right) \\ &= \sum_{\mu=1}^{\nu} U^{\mu-1}(U - V)V^{\nu-\mu}, \end{aligned}$$

$$[[U^\nu - V^\nu]] = \left[\left[\sum_{\mu=1}^{\nu} U^{\mu-1}(U - V)V^{\nu-\mu}\right]\right]$$

$$\leq \sum_{\mu=1}^{\nu} |||U^{\mu-1}||| \, [\![U - V]\!] \, |||V^{\nu-\mu}|||$$

$$= \sum_{\mu=1}^{\nu} [\![U - V]\!] = \nu \, [\![U - V]\!] \, .$$

Thus $\omega_1(\epsilon) = (1/\nu)\epsilon$ meets our requirements for these $\varphi(\xi)$.

Lemma 1.5.3. *Let a function $\psi(\lambda)$ be given, defined and continuous for all real λ and for which $\lim_{\lambda\to\pm\infty}\psi(\lambda)$ exists.*[7] *Then there exists a function $\omega_2(\epsilon)$ (given ψ, ω_2 is determined) defined and > 0 for all $\epsilon > 0$ with the following property: If A, B are Hermitian and $\in \mathbf{M}$ then $\psi(A), \psi(B) \in \mathbf{M}$ and $[\![A - B]\!] < \omega_2(\epsilon)$ implies $\left[\!\left[\psi(A) - \psi(B)\right]\!\right] < \epsilon$.*

Proof. That A, B Hermitian and $\in \mathbf{M}$ imply $\psi(A), \psi(B) \in \mathbf{M}$ is obvious. The correspondence

$$\xi = \frac{\lambda + i}{\lambda - i}, \qquad \text{or equivalently} \qquad \lambda = i\frac{\xi + 1}{\xi - 1}$$

is a one-to-one and bicontinuous mapping of ξ with $|\xi| = 1$ on all real λ and $\lambda = \pm\infty$ where $\xi = 1$ corresponds to $\lambda = \pm\infty$. Hence

$$\varphi(\xi) = \begin{cases} \psi\left(i\dfrac{\xi+1}{\xi-1}\right), & |\xi| = 1, \ \xi \neq 1, \\ \lim\limits_{\lambda\to\pm\infty} \psi(\lambda), & \xi = 1, \end{cases}$$

is defined and continuous for all ξ with $|\xi| = 1$. Clearly

$$\psi(\lambda) = \varphi\left(\frac{\lambda + i}{\lambda - i}\right).$$

We recall the $g(\lambda) = (\lambda + i)/(\lambda - i)$ used in the proof of Lemma 1.5.1 and let

$$U = g(A) = \frac{A + i1}{A - i1}, \qquad V = g(B) = \frac{B + i1}{B - i1}.$$

Then the above relations imply

$$\psi(A) = \varphi(U), \qquad \psi(B) = \varphi(V).$$

[7] We require $\lim_{\lambda\to\infty}\psi(\lambda) = \lim_{\lambda\to-\infty}\psi(\lambda)$. This condition could be removed, but it simplifies the proof and suffices for our present purposes.

By Lemma 1.5.1, U and V are unitary and $\in \mathbf{M}$ and

$$[\![U - V]\!] \leq 2\,[\![A - B]\!]\,.$$

Now Lemma 1.5.2 applies to $\varphi(\xi)$. Consider the resulting $\omega_1(\epsilon)$. Then $[\![A - B]\!] < \frac{1}{2}\omega_1(\epsilon)$ implies $[\![U - V]\!] < \omega_1(\epsilon)$ and hence $\left[\!\left[\varphi(U) - \varphi(V)\right]\!\right] < \epsilon$ or $\left[\!\left[\psi(A) - \psi(B)\right]\!\right] < \epsilon$. Therefore $\omega_2(\epsilon) = \frac{1}{2}\omega_1(\epsilon)$ meets our requirements.

Lemma 1.5.4. *Let an algebra $\mathbf{S}$ in $\mathbf{M}$ be given and a sequence $A_1, A_2, \ldots \in \mathbf{S}$ which converges metrically to an $A \in \mathbf{M}$. Then there exists another sequence $A'_1, A'_2, \ldots \in \mathbf{S}$ which converges restrictedly metrically to A.*

Proof. $A_1, A_2, \ldots$ converges metrically to A; hence $A_1^*, A_2^*, \ldots$ converges metrically to A^*.[8] Consequently $\frac{1}{2}(A_1 + A_1^*), \frac{1}{2}(A_2 + A_2^*), \ldots$ converges metrically to $\frac{1}{2}(A + A^*)$, and $\frac{1}{2i}(A_1 - A_1^*), \frac{1}{2i}(A_2 - A_2^*), \ldots$ converges metrically to $\frac{1}{2i}(A - A^*)$. The operators occurring in the two last assertions are all Hermitian and $\in \mathbf{S}$. Also $A = \frac{1}{2}(A + A^*) + i\frac{1}{2i}(A - A^*)$.

Now suppose our result has been established in the special case in which all the operators $A_1, A_2, \ldots$ are Hermitian. Then to treat the general case, we should apply the special result to the above-mentioned Hermitian "real and imaginary parts" and by taking the linear combination we should obtain the desired result since the notion of "restricted metrical convergence" holds for linear combinations of series if it holds for the series themselves. Thus we may assume in what follows that $A, A_1, A_2, \ldots$ are Hermitian operators.

Now form the function

$$\psi(\lambda) = \begin{cases} \lambda, & |\lambda| \leq |||A|||\,, \\ \dfrac{|||A|||^2}{\lambda}, & |\lambda| \geq |||A|||\,. \end{cases}$$

Lemma 1.5.3 applies to $\psi(\lambda)$ yielding $\omega_2(\epsilon)$.

Consider a $\nu = 1, 2, \ldots$. Choose a $\nu' = 1, 2, \ldots$ (depending on ν as well as on $A, A_1, A_2, \ldots$) with

$$[\![A_{\nu'} - A]\!] < \omega_2\left(\frac{1}{\nu}\right).$$

Then Lemma 1.5.3 gives $\left[\!\left[\psi(A_{\nu'}) - \psi(A)\right]\!\right] < 1/\nu$. Since $\psi(\lambda) = \lambda$ for all λ with $|\lambda| \leq |||A|||$, $\psi(A) = A$.[9] Thus we have

$$\left[\!\left[\psi(A_{\nu'}) - A\right]\!\right] < \frac{1}{\nu}. \tag{1.5.β}$$

[8]Since $[\![A]\!] = [\![A^*]\!]$, $A \mapsto A^*$ is an isometric mapping.

[9]The set $|\lambda| \leq |||A|||$ contains the entire spectrum of A.

Form next the function

$$\theta(\lambda)=\begin{cases}1, & |\lambda|\le |||A|||\,,\\ \dfrac{|||A|||^2}{\lambda^2}, & |\lambda|\ge |||A|||\,.\end{cases}$$

It is clear that $\psi(\lambda)=\lambda\theta(\lambda)$. Choose a polynomial $p_\nu(\lambda)$ with

$$\theta(\lambda)\ge p_\nu(\lambda)\ge \mathrm{Max}\left(\theta(\lambda)-\frac{1}{\nu|\lambda|},0\right)\quad \text{for } |\lambda|\le |||A_{\nu'}|||\,.$$ [10]

Then the polynomial $q_\nu(\lambda)=\lambda p_\nu(\lambda)$ has no constant term and we have

$$|q_\nu(\lambda)-\psi(\lambda)|\le\frac{1}{\nu}\quad\text{for } |\lambda|\le |||A_{\nu'}|||\,.$$

Furthermore $0\le q_\nu(\lambda)\le\psi(\lambda)\le |||A|||$ and thus

$$|q_\nu(\lambda)|\le |||A|||\quad\text{for } |\lambda|\le |||A_{\nu'}|||\,.$$

These two inequalities imply

$$\Big|\Big|\Big|q_\nu(A_{\nu'})-\psi(A_{\nu'})\Big|\Big|\Big|\le\frac{1}{\nu},$$ [11]

and hence, *a fortiori*,

$$\Big[\Big[q_\nu(A_{\nu'})-\psi(A_{\nu'})\Big]\Big]\le\frac{1}{\nu},\tag{1.5.γ}$$

and

$$\Big|\Big|\Big|q_\nu(A_{\nu'})\Big|\Big|\Big|\le |||A|||\,.\tag{1.5.δ}$$ [11]

Combining (1.5.β) and (1.5.γ) gives

$$\Big[\Big[q_\nu(A_{\nu'})-A\Big]\Big]<\frac{2}{\nu}.\tag{1.5.ϵ}$$

Now put $A'_\nu=q_\nu(A_{\nu'})$. Then $A_{\nu'}\in\mathbf{S}$ implies $q_\nu(A_{\nu'})\in\mathbf{S}$ since $q_\nu(\lambda)$ has no constant term. Hence the sequence $A'_1,A'_2,\ldots$ consists of elements of $\mathbf{S}$, and it is restrictedly metrically convergent to A by (1.5.δ) and (1.5.ϵ).

Observe that in the above proof we even had $|||A'_\nu|||\le |||A|||$ when A is Hermitian. This could be extended to all A but we shall not pursue this matter further.

§1.6 The desired results are now immediate.

[10] This is an easy variant of the Weierstrass (polynomial) approximation theorem.

[11] The set $|\lambda|\le |||A_{\nu'}|||$ contains the entire spectrum of $A_{\nu'}$.

Lemma 1.6.1. *Restricted metric closure of an algebra* **S** *implies its metric closure.*

Proof. Immediate by Lemma 1.5.4.

Theorem I. *For an algebra* **S** *within a finite* **M** *the following eight notions of closure are equivalent to each other:*
α) *Restricted metric closure*
β) *Metric closure*
γ) *Weak (topological) closure*
δ) *Weak convergence closure*
ϵ) *Strong (topological) closure*
θ) *Strong convergence closure*
ζ) *Strongest (topological) closure*
η) *Strongest convergence closure.*

Proof. The implications $\epsilon) \Rightarrow \alpha), \beta) \Rightarrow \epsilon), \alpha) \Rightarrow \beta)$ were established in Lemma 1.4.1, Lemma 1.4.2, Lemma 1.6.1, respectively. Consequently $\alpha) \Leftrightarrow \beta) \Leftrightarrow \epsilon)$. Also Lemma 1.3.2 yields $\alpha) \Leftrightarrow \theta)$. From [2, p. 396, end of §3], we have $\gamma) \Leftrightarrow \epsilon)$. [4, p. 112, beginning of §3] yields $\gamma) \Leftrightarrow \zeta)$. Also [4, p. 112, end of §2] implies $\theta) \Leftrightarrow \eta)$. These equivalences give together

$$\alpha) \Leftrightarrow \beta) \Leftrightarrow \gamma) \Leftrightarrow \epsilon) \Leftrightarrow \theta) \Leftrightarrow \zeta) \Leftrightarrow \eta).$$

And obviously

$$\gamma) \Rightarrow \delta) \Rightarrow \theta).$$

Thus δ) also is equivalent to the others.

Thus all those topologizations of the space of all operators, which may be reasonably used in this connection, turn out to be equivalent to each other for the algebras **S** in a factor **M** which is in a finite case.

We also state explicitly

Theorem II. *The notion of a subring* **S** *of* **M**, *for a finite* **M**, *is purely algebraical.*

Proof. The equivalences of Theorem I now permit us to apply the argument given at the end of §1.2.

Thus the desideratum expressed at the end of §1.1 is fully realized.

CHAPTER II. THE ALGEBRAICAL TYPE AND ITS OPERATIONS. THE FUNDAMENTAL GROUP

§2.1 We shall call the abstraction of a ring $\mathbf{M}$ with respect to algebraical isomorphism, its algebraical type, i.e. two rings $\mathbf{M}_1$ and $\mathbf{M}_2$ will be said to have the same algebraical type if and only if they are algebraically isomorphic. We denote the algebraic type of the ring $\mathbf{M}$ by $\overline{\mathbf{M}}$. Notions in $\mathbf{M}$ which are purely algebraical, i.e. invariant under algebraical isomorphisms, will be said to be notions concerning the algebraical type $\overline{\mathbf{M}}$ (and not $\mathbf{M}$ itself).[12]

Thus the properties enumerated in §1.1 (or equally in [9]) are properties of $\overline{\mathbf{M}}$ for every ring $\mathbf{M}$. The notion of a subring of $\mathbf{M}$ concerns $\overline{\mathbf{M}}$ when $\mathbf{M}$ is a factor in a finite class (cf. Theorem II). This is also true of $D_{\mathbf{M}}(E)$, $\mathrm{Tr}_{\mathbf{M}}(A)$, $[\![A]\!]$ (cf. the beginning of §1.2), and the various notions of closure, as enumerated in Theorem I, when they concern an algebra $\mathbf{S}$. Thus when $\mathbf{M}$ is in a finite class, the properties concerning $\overline{\mathbf{M}}$ are particularly extensive.

§2.2 As in the Introduction, §2, let for each $i = 1, 2$ a Hilbert space $\mathfrak{H}_i$ and an operator ring $\mathbf{M}_i$ in $\mathfrak{H}_i$ be given. We generalize B), loc. cit., as follows:

B*) General (algebraic ring) isomorphism of $\mathbf{M}_1$ and $\mathbf{M}_2$. This is a one-to-one mapping of $\mathbf{M}_1$ on $\mathbf{M}_2$ which leaves the entity 1 and the operations A^*, $A + B$ invariant; while it carries the entity αA (α any complex number) into either αA always (case a') or into $\overline{\alpha} A$ always (case b') and AB into AB always (case a'') or into BA always (case b''). More specifically, we talk according to which combination of the above cases occurs, of an $[a', a'']$ (or "proper"), $[b', a'']$ (or "conjugate"), $[a', b'']$ (or "dual") or $[b', b'']$ (or "conjugate dual") isomorphism.

The existence of a general isomorphism, of any one of the above four kinds, between $\mathbf{M}_1$ and $\mathbf{M}_2$ is clearly a property of the types $\overline{\mathbf{M}}_1$ and $\overline{\mathbf{M}}_2$. If a (proper) isomorphism holds, we have $\overline{\mathbf{M}}_1 = \overline{\mathbf{M}}_2$. If a conjugate or a dual isomorphism holds, then $\overline{\mathbf{M}}_1$ determines $\overline{\mathbf{M}}_2$ uniquely, assuming its existence. Hence we may express these relationships by $\overline{\mathbf{M}}_1^c = \overline{\mathbf{M}}_2$ or $\overline{\mathbf{M}}_1' = \overline{\mathbf{M}}_2$, respectively.

We must now deduce the existential and other properties of the operations $\overline{\mathbf{M}}^c$ and $\overline{\mathbf{M}}'$.

§2.3 Let a Hilbert space $\mathfrak{H}$ be given. Modify the fundamental operations αf, $f + g$, (f, g) in $\mathfrak{H}$ by replacing them by $\overline{\alpha} f$, $f + g$, $\overline{(f, g)}$. Denote the set $\mathfrak{H}$ with the new definition of its fundamental operations by $\mathfrak{H}_c$. Clearly $\mathfrak{H}_c$ is also a Hilbert space; every (linear bounded) operator A in $\mathfrak{H}$ is also one in $\mathfrak{H}_c$, and every ring of operators $\mathbf{M}$ in $\mathfrak{H}$ is also one in $\mathfrak{H}_c$. But we shall denote the $A, \mathbf{M}$ of $\mathfrak{H}$, when considered in $\mathfrak{H}_c$, by $A_c, \mathbf{M}_c$.

Now consider an operator ring $\mathbf{M}$ in $\mathfrak{H}$. The identical mapping $\mathfrak{I}^\circ$ then maps $\mathbf{M}$

[12] This definition should be compared and contrasted with that of spatial type in §3.3.

in $\mathfrak{H}$ on $\mathbf{M}_c$ in $\mathfrak{H}_c$ and it is in this aspect a conjugate isomorphism of $\mathbf{M}$ and $\mathbf{M}_c$.

Consider next the mapping $\mathfrak{I}^*$, $A \mapsto A^*$. This maps $\mathbf{M}$ in $\mathfrak{H}$ on $\mathbf{M}$ in $\mathfrak{H}$ and it is, in this aspect, a conjugate dual isomorphism of $\mathbf{M}$ and $\mathbf{M}$. Consequently $\mathfrak{I}^*\mathfrak{I}^\circ$ (i.e. $\mathfrak{I}^*$ in a new aspect) is a dual isomorphism of $\mathbf{M}$ and $\mathbf{M}_c$.

Thus we have

Theorem III. $\overline{\mathbf{M}}^c$ *and* $\overline{\mathbf{M}}'$ *always exist and are single-valued. They are both equal to the above* $\overline{\mathbf{M}}_c$. *We shall denote them by* $\overline{\mathbf{M}}^c$.

Clearly

$$\overline{\mathbf{M}}^{cc} = \overline{\mathbf{M}} \tag{2.3.α}$$

since $\mathfrak{H}_{c,c} = \mathfrak{H}$, $A_{c,c} = A$, $\mathbf{M}_{c,c} = \mathbf{M}$.

Observe that the definitions of a factor and of the case to which it belongs are unaffected by conjugate isomorphisms. For these definitions make use of αA only in defining the set of all $\alpha 1$ but never for a specific numerical value of α. To see this, reconsider [5, p. 138, Definition 3.1.2; p. 173, Theorem IX]. Hence the operation $\overline{\mathbf{M}}^c$ (viewed as $\overline{\mathbf{M}}'$) does not affect the factor character of $\mathbf{M}$ nor the case to which it belongs.

We now are able to prove:

§2.4 In what follows we shall consider matrices of operators. Given a $p = 1, 2, \ldots, \infty$, a p-order matrix $\langle A_{t,s}\rangle$ is a scheme or array of operators $A_{t,s}$ in which the indices s and t range from 1 to p for a finite p and over $1, 2, \ldots$ if $p = \infty$. (This last will be abbreviated $s, t \leq p$ in both cases.) An $\langle A_{t,s}\rangle$ is finite if there exists a finite $q = 1, 2, \ldots$ such that $A_{t,s} = 0$ if either $t > q$ or $s > q$. For a finite p this is automatically fulfilled, but it is a real restriction for $p = \infty$.

Now as in the Introduction, §2, let, for each $i = 1, 2$, a Hilbert space $\mathfrak{H}_i$ and an operator ring $\mathbf{M}_i$ in $\mathfrak{H}_i$ be given. We generalize B), loc. cit., as follows:
B_p) p**-matrix (algebraic ring) isomorphism of $\mathbf{M}_1$ over $\mathbf{M}_2$.** This is a one-to-one mapping of $\mathbf{M}_1$ on a set $\mathfrak{I}$ of p-order matrices $\langle A_{t,s}\rangle$ over $\mathbf{M}_2$ (i.e. $A_{t,s} \in \mathbf{M}_2$) which is algebraically ring-isomorphic in the matrix sense, i.e. it carries 1, αA°, $A^{\circ *}$, $A^\circ + B^\circ$, $A^\circ B^\circ$ in $\mathbf{M}_1$ into

$$\langle \delta_{t,s} 1\rangle, \quad \langle \alpha A_{t,s}\rangle, \quad \langle A^*_{s,t}\rangle, \quad \langle A_{t,s} + B_{t,s}\rangle, \quad \left\langle \sum_{r=1}^{p} A_{r,s} B_{t,r} \right\rangle$$

in $\mathfrak{I}$. The $\sum_{r=1}^{\infty}$ in the above expression, when $p = \infty$, must converge in the sense of the strong operator topology, irrespective of the order of summation.

The set $\mathfrak{I}$ must, at any rate, contain all finite matrices $\langle A_{t,s}\rangle$ over $\mathbf{M}_2$. (Cf. [5, pp. 136–137, Lemma 2.4.3], where the same notions are used, the same multiplication convention obtains, as well as the same notion of convergence for $\sum_{r=1}^{\infty}$.)

For a given $p, \mathbf{M}_1, \mathbf{M}_2$ the existence of such a p-matrix isomorphism of $\mathbf{M}_1$ over $\mathbf{M}_2$ is a property of types $\overline{\mathbf{M}}_1, \overline{\mathbf{M}}_2$. This is obvious for $p = 1, 2, \ldots$ while for $p = \infty$ it is only necessary to remember the purely algebraical character of the notion of strong convergence (cf. the beginning of §1.1, or [9, Theorem II]). Whether $\overline{\mathbf{M}}_1$ (assuming its existence) is uniquely determined by $\overline{\mathbf{M}}_2$ or not, for a given $p = 1, 2, \ldots, \infty$, depends obviously on whether the set $\mathfrak{J}$ in B_p) is uniquely determined or not. For $p = 1, 2, \ldots$ (p finite), the last part of B_p) implies that the set $\mathfrak{J}$ must be the set of all $\langle A_{t,s} \rangle$ with $A_{t,s} \in \mathbf{M}_2$ since they are all finite. Hence in this case $\overline{\mathbf{M}}_1$ is uniquely determined. For $p = \infty$ the question of uniqueness is not so immediately settled. We shall prove that $\mathfrak{J}$ and with it $\overline{\mathbf{M}}_1$ are again unique, but we delay this discussion to the next section. Nevertheless, for $p = 1, 2, \ldots, \infty$ we express the existence of a p-order matrix isomorphism of $\mathbf{M}_1$ over $\mathbf{M}_2$ as a relationship between $\overline{\mathbf{M}}_1$ and $\overline{\mathbf{M}}_2$ by $\overline{\mathbf{M}}_1 = \overline{\mathbf{M}}_2^p$. But then, quite apart from the question of existence, we may assume that $\overline{\mathbf{M}}^p$ is single-valued only for a finite p. At present we must consider the possibility that for $p = \infty$, $\overline{\mathbf{M}}^p$ is many valued.

The general existence of $\overline{\mathbf{M}}^p$ belongs in abstract algebra, but we shall need $\overline{\mathbf{M}}^p$ as an operator ring. For this reason, as well as for the sake of general orientation, we shall construct it explicitly within the framework of [5, pp. 135–137, §2.4].

Let a Hilbert space $\mathfrak{H}$ and a ring $\mathbf{M}$ in it be given, and a $p = 1, 2, \ldots, \infty$. We perform the constructions of [5, loc. cit.], the $\mathfrak{H}_2$ there being our $\mathfrak{H}$ and the $\mathfrak{H}_1$ there, any p-dimensional unitary space. (This is not the $\mathfrak{H}_1$ of B_p) above. We conform to the notations of [5, loc. cit.].) Form $\mathfrak{H}_1 \otimes \mathfrak{H}_2$ and the ring $\mathbf{B}^{\otimes}$ of all its bounded operators. Then all elements of $\mathbf{B}^{\otimes}$ can be represented as p-order matrices $\langle A_{t,s} \rangle$ ($t, s \leq p$, $A_{t,s}$ in $\mathfrak{H}_2 = \mathfrak{H}$). (Cf. [5, p. 136, Definition 2.4.2 and Lemma 2.4.2].) For the given ring $\mathbf{M}$, form the set $\mathbf{M}_1$ consisting of all operators in $\mathfrak{H}_1 \otimes \mathfrak{H}_2$ which have a matrix $\langle A_{t,s} \rangle$ with all $A_{t,s} \in \mathbf{M}$. $\mathfrak{J}$ is the set of all those $\langle A_{t,s} \rangle$ with $A_{t,s} \in \mathbf{M}$ which correspond to some operator in $\mathfrak{H}_1 \otimes \mathfrak{H}_2$. From this it follows immediately that $\mathfrak{J}$ contains all finite matrices $\langle A_{t,s} \rangle$ with $A_{t,s} \in \mathbf{M}$.[13] Now $\mathbf{M}_1$ in $\mathfrak{H}_1 \otimes \mathfrak{H}_2$ and $\mathbf{M}$ in $\mathfrak{H} = \mathfrak{H}_2$ together with the interpretation of the matrix scheme $\langle A_{t,s} \rangle$ which we are now using, clearly fulfill all the requirements of B_p). (We have our $\mathfrak{H}_1 \otimes \mathfrak{H}_2$, $\mathfrak{H} = \mathfrak{H}_2$, $\mathbf{M}_1$, $\mathbf{M}$ in place of the $\mathfrak{H}_1$, $\mathfrak{H}_2$, $\mathbf{M}_1$, $\mathbf{M}_2$ of B_p).)

Summing up the results obtained so far, we have

[13]For a finite matrix $\langle A_{t,s} \rangle$ an operator A° in $\mathfrak{H}_1 \otimes \mathfrak{H}_2$ with $A^\circ \sim \langle A_{t,s} \rangle$ can be constructed in the sense of [5, p. 136, Definition 2.4.2], without any convergence difficulties.

Theorem IV. *$\overline{\mathbf{M}}^p$ exists for all $p = 1, 2, \ldots, \infty$ and it is certainly one-valued for $p = 1, 2, \ldots$ (p finite).*

Our definitions in B_p) and that of §2.3 give together

$$(\overline{\mathbf{M}}^c)^p = (\overline{\mathbf{M}}^p)^c \qquad (p = 1, 2, \ldots), \tag{2.4.α}$$

and B_p) gives

$$(\overline{\mathbf{M}}^p)^q = \overline{\mathbf{M}}^{pq} \qquad (p, q = 1, 2, \ldots). \tag{2.4.β}$$

(The exponent ∞ will be considered in the next section.)

§2.5 In this section we shall make use of the operations $A_{(\mathfrak{M})}$, $\mathbf{M}_{(\mathfrak{M})}$ defined in [1, p. 78], and [5, p. 186, Definition 11.3.1]. If A is an operator in $\mathfrak{H}$ which is reduced by the closed linear set $\mathfrak{M}$ then $A_{(\mathfrak{M})}$ is its part in $\mathfrak{M}$. And $\mathbf{M}_{(\mathfrak{M})}$ is the set of all $A_{(\mathfrak{M})}$ for all $A \in \mathbf{M}$ which are reduced by $\mathfrak{M}$. $A_{(\mathfrak{M})}, \mathbf{M}_{(\mathfrak{M})}$ are in $\mathfrak{M}$ while $A, \mathbf{M}$ were in $\mathfrak{H}$.

Lemma 2.5.1. *Assume $E = P_{\mathfrak{M}} \in \mathbf{M}$. Let $\mathbf{M}^E$ be the set of all $A \in \mathbf{M}$ for which $EA = AE = A$. Then we have*

(i) $A_{(\mathfrak{M})}$ can be formed for all $A \in \mathbf{M}^E$.

(ii) $\mathbf{M}_{(\mathfrak{M})}$ coincides with $(\mathbf{M}^E)_{(\mathfrak{M})}$.

(iii) $A \mapsto A_{(\mathfrak{M})}$ is a one-to-one mapping of $\mathbf{M}^E$ on $\mathbf{M}_{(\mathfrak{M})}$ which carries E (in $\mathbf{M}^E$, i.e. in $\mathfrak{H}$) into 1 (in $\mathbf{M}_{(\mathfrak{M})}$, i.e. in $\mathfrak{M}$) and leaves the operations $\alpha A, A^*, A + B, AB$ invariant.

Proof. Ad (i). Every $A \in \mathbf{M}^E$ commutes with $E = P_{\mathfrak{M}}$, i.e. it is reduced by $\mathfrak{M}$.

Ad (ii). $A \in \mathbf{M}$ implies $EAE \in \mathbf{M}$ and thence clearly $EAE \in \mathbf{M}^E$. Now if $\mathfrak{M}$ reduces A, then both A and EAE have the same part in $\mathfrak{M}$, i.e. $A_{(\mathfrak{M})} = (EAE)_{(\mathfrak{M})}$. Thus $(\mathbf{M}^E)_{(\mathfrak{M})} \supseteq \mathbf{M}_{(\mathfrak{M})}$. But $\mathbf{M}^E \subseteq \mathbf{M}$ hence also $(\mathbf{M}^E)_{(\mathfrak{M})} \subseteq \mathbf{M}_{(\mathfrak{M})}$. So $(\mathbf{M}^E)_{(\mathfrak{M})} = \mathbf{M}_{(\mathfrak{M})}$.

Ad (iii). Every $A \in \mathbf{M}^E$ is reduced by $\mathfrak{M}$ (cf. (i) above) and its part in $\mathfrak{M}^\perp$ ($= \mathfrak{H} \ominus \mathfrak{M}$) is clearly 0. So we may view it as an operator in $\mathfrak{M}$ as well as in $\mathfrak{H}$, and in its former aspect it is $A_{(\mathfrak{M})}$. All our assertions are now immediate.

Consider now $\mathbf{M}_1, \mathbf{M}_2$ and $\mathfrak{H}_1, \mathfrak{H}_2$ and an ∞-matrix isomorphism of $\mathbf{M}_1$ over $\mathbf{M}_2$ as in B_∞) in the last section. For any $q = 1, 2, \ldots$ form the set $\mathfrak{I}^q$ of all matrices $\langle A_{t,s} \rangle$ for which all $A_{t,s} \in \mathbf{M}_2$ and where $A_{t,s} = 0$ when $t > q$ or $s > q$. Form also the matrix $\langle e^q_{t,s} \cdot 1 \rangle$ where $e^q_{t,s} = 1$ if $t = s \leq q$ and $e^q_{t,s} = 0$ otherwise. Then it is easy to establish that

Lemma 2.5.2. *(i) $\mathfrak{I}^q \subseteq \mathfrak{I}$.*

(ii) $\langle e^q_{t,s} \cdot 1 \rangle \in \mathfrak{I}^q$ corresponds to a projection $E^q \in \mathbf{M}_1$. Put $E^q = P_{\mathfrak{M}^q}$.

(iii) *$E^1 \le E^2 \le \cdots$ and $\lim_{q\to\infty} E^q = 1$ in the strong sense.*

(iv) *Every matrix $\langle A_{t,s}\rangle \in \mathfrak{J}^q$ can be viewed both as a qth order matrix and as an ∞-order matrix. It corresponds in its former aspect to an element of a ring $\mathbf{M}_2^q$ with $\overline{\mathbf{M}_2^q} = \overline{\mathbf{M}}_2^q$ and in its latter aspect to an element of $(\mathbf{M}_1)^{E^q} \subseteq \mathbf{M}_1$. The latter correspondence can be continued by Lemma 2.5.1 to one with an element of $(\mathbf{M}_1)_{(\mathfrak{M}^q)}$. These elements exhaust the sets $\mathbf{M}_2^q$, $(\mathbf{M}_1)^{E^q}$ and $(\mathbf{M}_1)_{(\mathfrak{M}^q)}$, respectively.*

(v) *The correspondence of (iv) is an algebraic ring isomorphism of $\mathbf{M}_2^q$ and $(\mathbf{M}_1)_{(\mathfrak{M}^q)}$. Hence $\overline{(\mathbf{M}_1)_{(\mathfrak{M}^q)}} = \overline{\mathbf{M}_2^q} = \overline{\mathbf{M}}_2^q$.*

Proof. Ad (i). Immediate by the final clause of B_∞).

Ad (ii). Clearly $\langle e^q_{t,s} \cdot 1\rangle \in \mathfrak{J}^q$ hence it is $\in \mathfrak{J}$ and thus corresponds to an element of $\mathbf{M}_1$. The rules of matrix computation show that this element is a projection.

Ad (iii). The rules of matrix computation show that $E^1 \le E^2 \le \cdots$. Now if $f = \langle f_1, f_2, \ldots\rangle \in \mathfrak{H}_1 \otimes \mathfrak{H}_2$ then $E^q f = \langle f_1, \ldots, f_q, 0, \ldots\rangle \to f$ as $q \to \infty$. Hence $\lim_{q\to\infty} E^q$ is 1.

Ad (iv). The $\langle A_{t,s}\rangle \in \mathfrak{J}^q$ are clearly characterized by

$$\langle e^q_{t,s} \cdot 1\rangle\langle A_{t,s}\rangle = \langle A_{t,s}\rangle\langle e^q_{t,s} \cdot 1\rangle = \langle A_{t,s}\rangle$$

and thus we may add by (i), $\langle A_{t,s}\rangle \in \mathfrak{J}$. Hence the second correspondence mentioned maps the $\langle A_{t,s}\rangle \in \mathfrak{J}^q$ precisely on the $A^\circ \in \mathbf{M}_1$ with $E^q A^\circ = A^\circ E^q = A^\circ$, i.e. on the $A^\circ \in (\mathbf{M}_1)^{E^q}$. All other assertions are obvious, remembering in particular Lemma 2.5.1.

Ad (v). The corresponding assertions for the relationship between $\mathbf{M}_2^q$ and $(\mathbf{M}_1)^{E^q}$ follow immediately from the rules of matrix computation. They are transferred to $\mathbf{M}_2^q$ and $(\mathbf{M}_1)_{(\mathfrak{M}^q)}$ by using Lemma 2.5.1.

We now go further:

Lemma 2.5.3. *A matrix $\langle A_{t,s}\rangle \in \mathfrak{J}^q$ corresponds to three operators in the sense of (iv) in Lemma 2.5.2, i.e. to an operator of $\mathbf{M}_2^q$, to one in $(\mathbf{M}_1)^{E^q}$ and to one in $(\mathbf{M}_1)_{(\mathfrak{M}^q)}$. All three operators have the same bound, and we denote this common bound by $|||\langle A_{t,s}\rangle|||$.*

Proof. The operator in $(\mathbf{M}_1)^{E^q}$ and that one in $(\mathbf{M}_1)_{(\mathfrak{M}^q)}$ have the same bound, since we are considering the same operator, once in $\mathfrak{H}_1$ and once in $\mathfrak{M}^q$.[14] The operator in $\mathbf{M}_2^q$ and that one in $(\mathbf{M}_1)_{(\mathfrak{M}^q)}$ have the same bound since they obtain from one another by an algebraical ring isomorphism (cf. (v) in Lemma 2.5.2) and since the numerical value of the bound is a purely algebraical notion (cf. §1.1). This completes the proof.

[14]The part of the first-mentioned operator in $(\mathfrak{M}^q)^\perp$ is 0.

Lemma 2.5.4. *Consider a matrix $\langle A_{t,s}\rangle$, all $A_{t,s} \in \mathbf{M}_2$. Define the matrices $\langle A^q_{t,s}\rangle \in \mathfrak{J}^q$ by the equations $A^q_{t,s} = A_{t,s}$ if $t \le q$ and $s \le q$, and $A^q_{t,s} = 0$ otherwise. Then $\langle A_{t,s}\rangle \in \mathfrak{J}$ if and only if the numerical sequence $\left|\left|\left|\langle A^1_{t,s}\rangle\right|\right|\right|, \left|\left|\left|\langle A^2_{t,s}\rangle\right|\right|\right|, \ldots$ (cf. Lemma 2.5.3 above) is bounded.*

Proof. Necessity: Assume $\langle A_{t,s}\rangle \in \mathfrak{J}$ and let it correspond to $A^\circ \in \mathbf{M}_1$. Then, clearly,

$$\langle A^q_{t,s}\rangle = \langle e^q_{t,s} \cdot 1\rangle\langle A_{t,s}\rangle\langle e^q_{t,s} \cdot 1\rangle,$$

i.e. $\langle A^q_{t,s}\rangle$ corresponds to the operator $E^q A^\circ E^q \in \mathbf{M}_1$. Clearly $E^q A^\circ E^q \in (\mathbf{M}_1)^{E^q}$. Thus the second definition of Lemma 2.5.3 gives $\left|\left|\left|\langle A^q_{t,s}\rangle\right|\right|\right| = |||E^q A^\circ E^q|||$. Now we have in $\mathbf{M}_1$ that

$$|||E^q A^\circ E^q||| \le |||E^q|||^2\, |||A^\circ||| = |||A^\circ|||\,.$$

So $\left|\left|\left|\langle A^q_{t,s}\rangle\right|\right|\right| \le |||A^\circ|||$. This holds for $q = 1, 2, \ldots$ and hence the sequence $\left|\left|\left|\langle A^1_{t,s}\rangle\right|\right|\right|, \left|\left|\left|\langle A^2_{t,s}\rangle\right|\right|\right|, \ldots$ is bounded.

Sufficiency: Assume that the numerical sequence $\left|\left|\left|\langle A^1_{t,s}\rangle\right|\right|\right|, \left|\left|\left|\langle A^2_{t,s}\rangle\right|\right|\right|, \ldots$ is bounded. $\langle A^q_{t,s}\rangle \in \mathfrak{J}^q$, so it corresponds to an $A^q \in (\mathbf{M}_1)^{E^q} \subseteq \mathbf{M}_1$. Owing to the definition of Lemma 2.5.3, $\left|\left|\left|\langle A^q_{t,s}\rangle\right|\right|\right| = |||A^q|||$, hence

$$\text{The (numerical) sequence } \left|\left|\left|A^1\right|\right|\right|, \left|\left|\left|A^2\right|\right|\right|, \ldots \text{ is bounded.} \tag{2.5.α}$$

For $q \ge r$ clearly

$$\langle e^r_{t,s} \cdot 1\rangle\langle A^q_{t,s}\rangle\langle e^r_{t,s} \cdot 1\rangle = \langle A^r_{t,s}\rangle,$$

hence $E^r A^q E^r = A^r$. Now consider an $f \in \mathfrak{M}^r$.

For $q \ge r$, let $f_q = A^q f$. Since $\|f_q\|^2 \le |||A^q|||^2\, \|f\|^2$, these f_q's are bounded. Also $f \in \mathfrak{M}^r \subseteq \mathfrak{M}^q$, so $E^q f = f$. Hence for $q' \ge q \ge r$, $E^q f_{q'} = E^q A^{q'} f = E^q A^{q'} E^q f = A^q f = f_q$. So $f_{q'} - f_q$ and f_q are orthogonal, $\|f_{q'} - f_q\|^2 + \|f_q\|^2 = \|f_{q'}\|^2$, i.e.

$$\|f_{q'} - f_q\|^2 = \|f_{q'}\|^2 - \|f_q\|^2 \qquad (q' \ge q \ge r). \tag{2.5.β}$$

Thus the $\|f_r\|^2, \|f_{r+1}\|^2, \ldots$ form a monotone bounded sequence, and hence the $\lim_{q\to\infty}\|f_q\|^2$ exists. Therefore the right-hand side of (2.5.β) converges to zero for $q, q' \to \infty$. Hence (2.5.β) now implies that the strong limit $A^q f$ exists for $q \to \infty$. The restriction $q \ge r$ may now be dropped.

The set of all f for which $A^q f$ is strongly convergent is closed, since the $|||A^q|||$ are uniformly bounded. This set is obviously linear and by the above it contains

$\mathfrak{M}^1, \mathfrak{M}^2, \ldots$. Thus we see

$$\begin{cases}\text{The sequence } A^1, A^2, \ldots \text{ converges for all } f \text{ in the closed linear} \\ \text{set determined by } \mathfrak{M}^1, \mathfrak{M}^2, \ldots.\end{cases} \tag{2.5.γ}$$

Since the $\mathfrak{M}^1, \mathfrak{M}^2, \ldots$ are together dense, the set of (2.5.γ) is necessarily $\mathfrak{H}$ itself. Denote the limit of $A^1 f, A^2 f, \ldots$ by Af. Then A belongs to $\mathbf{M}_1$ along with $A^1, A^2, \ldots$. Thus we have shown

$$\begin{cases}\text{The sequence } A^1, A^2, \ldots \text{ converges for all } f \in \infty \otimes \mathfrak{H} \text{ to an } A \in \mathbf{M}_1 \\ \text{in the strong sense.}\end{cases} \tag{2.5.δ}$$

Let A correspond to the matrix $\langle B_{t,s} \rangle$ in B_∞). Of course $\langle B_{t,s} \rangle \in \mathfrak{J}$. Since for $q \geq r$, $E^r A^q E^r = A^r$, (2.5.δ) implies that $E^r A E^r = A^r$. Hence

$$\langle e^r_{t,s} \cdot 1 \rangle \langle B_{t,s} \rangle \langle e^r_{t,s} \cdot 1 \rangle = \langle A^r_{t,s} \rangle.$$

Thus the rules of matrix computation yield $B_{t,s} = A_{t,s}$ if $t, s \leq r$. Now let t, s be given and choose $r = \mathrm{Max}(t, s)$. Then the above formula gives $B_{t,s} = A_{t,s}$. Since t, s were arbitrary, this means $\langle B_{t,s} \rangle = \langle A_{t,s} \rangle$ and consequently $\langle A_{t,s} \rangle \in \mathfrak{J}$ too.

Thus the proof is completed.

Lemma 2.5.5. *The set $\mathfrak{J}$ can be uniquely and purely algebraically characterized in terms of $\mathbf{M}_2$ alone.*

Proof. A unique characterization of $\mathfrak{J}$ was given in Lemma 2.5.4. By using the first definition of Lemma 2.5.3 for $\left|\left|\left|\langle A^q_{t,s} \rangle\right|\right|\right|$, that one in terms of $\mathbf{M}_2^q$, it becomes clear that this characterization is purely algebraical in terms of $\mathbf{M}_2$ and of the $\mathbf{M}_2^q$, $q = 1, 2, \ldots$. But we have already reduced the $\mathbf{M}_2^q$, i.e. the $\overline{\mathbf{M}}_2^q$ for q finite, to $\mathbf{M}_2$ by Theorem IV.

The proof is thus completed.

The remarks made in §2.4, preceding Theorem IV, in conjunction with the above Lemma 2.5.5, permit us now to assert

Theorem IV′. $\overline{\mathbf{M}}^\infty$ *too is one-valued.*

Our definitions in B_∞) and that of §2.3 give together

$$(\overline{\mathbf{M}}^c)^\infty = (\overline{\mathbf{M}}^\infty)^c. \tag{2.5.ϵ}$$

(This corresponds to (2.4.α). We could extend (2.4.β) too, but we shall not need it here. Furthermore this extension is an easy consequence of Lemma 3.1.6.)

§2.6 We undertake now to find the inverse operation to $\overline{\mathbf{N}}^p$. At first we restrict neither the ring $\mathbf{N}$ nor the exponent $p = 1, 2, \ldots, \infty$.

Definition 2.6.1. *A system of p^2 operators $W_{\mu,\nu}$ $(\mu, \nu \leq p)$ is a system of p-matrix units if it possesses the following properties:*

(i) $W^*_{\mu,\nu} = W_{\nu,\mu}$,

(ii)

$$W_{\mu,\nu}W_{\sigma,\omega} = \begin{cases} W_{\mu,\omega}, & \nu = \sigma, \\ 0, & \nu \neq \sigma. \end{cases}$$

Some immediate properties of these systems are given in

Lemma 2.6.1. *Every system of p-matrix units $W_{\mu,\nu}$ $(\mu, \nu \leq p)$ possesses the following properties:*

(i) The $W_{\mu,\mu}$ $(\mu \leq p)$ are pairwise orthogonal projections.

(ii) Every $W_{\mu,\nu}$ is partially isometric with the initial projection $W_{\nu,\nu}$ and the final projection $W_{\mu,\mu}$ (cf. [5, §4.3]).

(iii) Form $E_0 = \sum_{\mu=1}^{p} W_{\mu,\mu}$. This $\sum_{\mu=1}^{p}$ is either a finite sum or when $p = \infty$ converges in the sense of the strong operator topology, irrespective of the order in which the $\mu = 1, 2, \ldots$ are gone through. This E_0 is a projection and it is the unit of the system $W_{\mu,\nu}$, i.e. always

$$E_0 W_{\mu,\nu} = W_{\mu,\nu} E_0 = W_{\mu,\nu}.$$

Proof. Ad (i), (iii): The convergence in (iii) is a consequence of (i), as is seen from familiar considerations concerning pairwise orthogonal projections (cf. e.g. [5, pp. 76–78], or the proof of (2.5.δ) in the proof of Lemma 2.5.4 above). All other assertions are immediately verified by using (i), (ii) in Definition 2.6.1.

Ad (ii). Definition 2.6.1 gives directly

$$W^*_{\mu,\nu}W_{\mu,\nu} = W_{\nu,\nu}, \qquad W_{\mu,\nu}W^*_{\mu,\nu} = W_{\mu,\mu}.$$

Since $W_{\nu,\nu}$, $W_{\mu,\mu}$ are projections by (i) above, these formulae imply our assertions by [5, p. 142, Lemma 4.3.2 (the last two criteria), and p. 141, Lemma 4.3.1].

We prove now

Theorem V. $\overline{\mathbf{M}} = \overline{\mathbf{N}}^p$ *is equivalent to this: There exists a system of p-matrix units $W_{\mu,\nu} \in \mathbf{M}$ $(\mu, \nu \leq p)$ with the unit 1 (cf. (iii) in Lemma 2.6.1) and with the following further property:*

$W_{1,1}$ is a projection. Let $\mathfrak{M}$ be the closed linear set with $P_{\mathfrak{M}} = W_{1,1}$. Then $\overline{\mathbf{M}_{(\mathfrak{M})}}$ must be algebraically isomorphic to $\overline{\mathbf{N}}$. [15]

(Concerning the omission of this extra condition, cf. the corollary below.)

Proof. Sufficiency: Assume the existence of $W_{\mu,\nu}$ ($\mu, \nu \leq p$) as described. $W_{1,1} \in \mathbf{M}$ is a projection. Let $\mathfrak{M}$ be such that $P_{\mathfrak{M}} = E = W_{1,1}$.

For every $A^0 \in \mathbf{M}$ define

$$A'_{t,s} = W_{1,s}A^0W_{t,1} \qquad (t, s \leq p). \tag{2.6.α}$$

Clearly $A'_{t,s} \in \mathbf{M}$ and one verifies immediately that $EA'_{t,s} = A'_{t,s}E = A'_{t,s}$ ($E = W_{1,1}$). Hence $A'_{t,s} \in \mathbf{M}^E$. Now apply Lemma 2.5.1 and form

$$A_{t,s} = (A'_{t,s})_{(\mathfrak{M})}. \tag{2.6.β}$$

Accordingly $A_{t,s} \in \mathbf{M}_{(\mathfrak{M})}$.

Thus to every $A^0 \in \mathbf{M}$ there corresponds a p-order matrix $\langle A_{t,s}\rangle$ ($t, s \leq p$) with all $A_{t,s} \in \mathbf{M}_{(\mathfrak{M})}$. Let $\mathfrak{J}$ be the set of these $\langle A_{t,s}\rangle$ which correspond to an $A^0 \in \mathbf{M}$. We shall now show that this correspondence establishes a p-matrix automorphism of $\mathbf{M}$ over $\mathbf{M}_{(\mathfrak{M})}$ in the sense of B_p) in §2.4. That means $\overline{\mathbf{M}} = \overline{\mathbf{M}_{(\mathfrak{M})}}^p$ and as $\overline{\mathbf{M}_{(\mathfrak{M})}} = \overline{\mathbf{N}}$, so $\overline{\mathbf{M}} = \overline{\mathbf{N}}^p$. This will complete the proof of the sufficiency. We prove the above assertions in three successive steps.

The correspondence is one-to-one—i.e. if $A^0, B^0 \in \mathbf{M}$ have $\langle A_{t,s}\rangle = \langle B_{t,s}\rangle$ then $A^0 = B^0$. Now $\langle A_{t,s}\rangle = \langle B_{t,s}\rangle$ means $A_{t,s} = B_{t,s}$ for $t, s \leq p$ and hence $(A'_{t,s})_{(\mathfrak{M})} = (B'_{t,s})_{(\mathfrak{M})}$ and $A'_{t,s} = B'_{t,s}$ since both are $\in \mathbf{M}^E$. Thus $W_{1,s}A^0W_{t,1} = W_{1,s}B^0W_{t,1}$. Multiplication by $W_{s,1}$ on the left and by $W_{1,t}$ on the right gives $W_{s,s}A^0W_{t,t} = W_{s,s}B^0W_{t,t}$. If we sum over s and t we obtain $A^0 = B^0$ as desired, since by hypothesis $E_0 = 1 = \sum_{s=1}^p W_{s,s}$.

The correspondence is a matrix isomorphism. This can be verified by the use of Definition 2.6.1, $E_0 = \sum_{\mu=1}^p W_{\mu,\mu} = 1$ and the equations (2.6.α) and (2.6.β). For these

[15]Observe that in this presentation $\overline{\mathbf{N}}$ is determined by $\overline{\mathbf{M}}$ and $W_{1,1}$ together. Whether $\overline{\mathbf{M}}$ alone suffices to determine $\overline{\mathbf{N}}$ is a different question. If, however, p is finite and $\mathbf{M}$ is a factor, we do have that $\overline{\mathbf{M}}$ and p determine $\overline{\mathbf{N}}$ uniquely if there is such an $\overline{\mathbf{N}}$. For if $\mathfrak{M}^{(1)}$ is the range of $W_{1,1}$ for one choice of the p-matrix units and $\mathfrak{M}^{(2)}$ that for another, then $D_{\mathbf{M}}(\mathfrak{M}^{(1)}) = (1/p)D_{\mathbf{M}}(\mathfrak{H}) = D_{\mathbf{M}}(\mathfrak{M}^{(2)})$. Hence Lemma 2.8.1 (of our present text) implies $\overline{\mathbf{M}_{(\mathfrak{M}^{(1)})}} = \overline{\mathbf{M}_{(\mathfrak{M}^{(2)})}}$. Hence $\overline{\mathbf{M}_{(\mathfrak{M})}}$ depends only on $\overline{\mathbf{M}}$ and p.

If however p is ∞ then $\mathbf{M}$ itself is in an infinite case by (iii) of Lemma 2.7.1. Under these circumstances, $\overline{\mathbf{M}}$ fails to determine $\overline{\mathbf{N}}$ as may be seen from δ) in Lemma 3.1.5 (write $\overline{\mathbf{M}}$ for its $\overline{\mathbf{M}}^\infty$).

show that by the use of Equation (2.6.α) we correspond to $1, \alpha A^0, A^{0*}, A^0+B^0, A^0B^0$ (in $\mathbf{M}$) matrices

$$\langle \delta_{t,s}E\rangle, \quad \langle \alpha A'_{t,s}\rangle, \quad \langle A'^{*}_{s,t}\rangle, \quad \langle A'_{t,s}+B'_{t,s}\rangle, \quad \left\langle \sum_{r=1}^{p} A'_{r,s}B'_{t,r}\right\rangle$$

of elements of $\mathbf{M}^E$ and then by the use of Equation (2.6.β) we correspond these to

$$\langle \delta_{t,s}1\rangle, \quad \langle \alpha A_{t,s}\rangle, \quad \langle A^{*}_{s,t}\rangle, \quad \langle A_{t,s}+B_{t,s}\rangle, \quad \left\langle \sum_{r=1}^{p} A_{r,s}B_{t,r}\right\rangle,$$

whose elements are in $\mathbf{M}_{(\mathfrak{M})}$.

$\mathfrak{J}$ contains all finite matrices. Consider a finite matrix $\langle A_{t,s}\rangle$ where $A_{t,s} \in \mathbf{M}_{(\mathfrak{M})}$. There is a $q < \infty$ such that $A_{t,s} = 0$ when $t > q$ or when $s > q$. Let $A'_{t,s} \in \mathbf{M}^E$ satisfy (2.6.β) and put

$$A^0 = \sum_{t=1}^{q}\sum_{s=1}^{q} W_{s,1}A'_{t,s}W_{1,t}.$$

Then $A^0 \in \mathbf{M}$ and it follows from the rules of Definition 2.6.1 that (2.6.α) holds. Thus $\langle A_{t,s}\rangle$ corresponds to this A^0 and therefore it belongs to $\mathfrak{J}$.

Necessity: Assume $\overline{\mathbf{M}} = \overline{\mathbf{N}}^p$ in the sense of B_p) in §2.4. For every pair $\mu, \nu \le p$ form the matrix $\langle \delta_{t,\mu}\delta_{s,\nu}1\rangle$. This matrix is clearly finite and hence it belongs to $\mathfrak{J}$. Thus it corresponds to an element of $\mathbf{M}$ which we denote by $W_{\mu,\nu}$. Then the matrix computation rules of B_p) give immediately the rules of Definition 2.6.1. Thus the results of Lemma 2.6.1 are now available. Now in the notation of (ii) in Lemma 2.5.2, $E^q = \sum_{\mu=1}^{q} W_{\mu,\mu}$ $(q = 1, 2, \ldots)$. Hence Lemma 2.5.2 (iii) and Lemma 2.6.1 (iii) imply $E_0 = \sum_{\mu=1}^{p} W_{\mu,\mu} = 1$.

Put $P_{\mathfrak{M}} = E = W_{1,1}$. $A^0 \in \mathbf{M}^E$ means $EA^0 = A^0E = A^0$ or, by the above-mentioned rules, that the matrix of A^0 has the form $\langle \delta_{t,1}\delta_{s,1}A\rangle$, $A \in \mathbf{N}$. This correspondence between the $A^0 \in \mathbf{M}^E$ and the $A \in \mathbf{N}$ is clearly one-to-one, it carries $E \in \mathbf{M}^E$ (which has the matrix $\langle \delta_{t,1}\delta_{s,1}1\rangle$) into $1 \in \mathbf{N}$ and it leaves the operations $\alpha A, A^*, A+B, AB$ invariant. These results, together with Lemma 2.5.1, show therefore that $\mathbf{M}_{(\mathfrak{M})}$ is algebraically isomorphic to $\mathbf{N}$.

Thus all our requirements are fulfilled.

Corollary. *Given $\overline{\mathbf{M}}$ and p, the equation $\overline{\mathbf{M}} = \overline{\mathbf{N}}^p$ possesses a solution $\overline{\mathbf{N}}$ if and only if there exists a system of p-matrix units, $W_{\mu,\nu} \in \mathbf{M}$ $(\mu, \nu \le p)$, with the unit* 1.

Proof. This becomes clear when we omit the last condition in the above theorem, since the only effect of that condition is to determine $\overline{\mathbf{N}}$ in terms of the $W_{\mu,\nu}$.

Under certain conditions we may even go further.

Lemma 2.6.2. *If* $\mathbf{M}$ *is a factor not in case* (I_n)*, and* $p = 2, 3, \ldots$ *is finite, then there is a system of p-matrix units* $W_{\mu,\nu} \in \mathbf{M}$ $(\mu, \nu \leq p)$ *with unit* $E_0 = 1$.

Proof. We first obtain a set $E_1, \ldots, E_p$ of projections in $\mathbf{M}$ such that $D_{\mathbf{M}}(E_i) = D_{\mathbf{M}}(E_1)$ and $\sum_{n=1}^{p} E_n = 1$. Suppose first that $\mathbf{M}$ is in a case (II_1). We may assume that $D_{\mathbf{M}}(1) = 1$. The range of $D_{\mathbf{M}}$ contains $1/p$ (cf. [5, p. 172, Theorem VIII]), and hence there is an $E_1 \in \mathbf{M}$ with $D_{\mathbf{M}}(E_1) = 1/p$. Since $D_{\mathbf{M}}(E_1) = 1/p \leq 1 - 1/p = D_{\mathbf{M}}(1 - E_1)$, there exists a projection $E_2 \leq 1 - E_1$ with $D_{\mathbf{M}}(E_2) = 1/p = D_{\mathbf{M}}(E_1)$ (cf. [5, p. 167, Lemma 8.3.1]). Furthermore $D_{\mathbf{M}}(1 - E_1 - E_2) = 1 - 2/p$. If $p > 2$ we can find an $E_3 \leq 1 - E_1 - E_2$ such that $D_{\mathbf{M}}(E_3) = 1/p$. Thus we may continue until we obtain a set of p pairwise orthogonal projections $E_1, \ldots, E_p \in \mathbf{M}$ with $\sum_{n=1}^{p} E_n = 1$, $D_{\mathbf{M}}(E_n) = 1/p$.

If $\mathbf{M}$ is in an infinite case, we can easily modify the proof of [5, p. 157, Lemma 7.2.3] to yield that there are p pairwise orthogonal, dimensionally equivalent, closed, linear sets $\mathfrak{M}_1, \ldots, \mathfrak{M}_p$ which together span $\mathfrak{H}$. In that proof it is only necessary to separate the $\mathfrak{N}_i$'s into p sets rather than 2. Then if E_i is the projection on $\mathfrak{M}_i$, $\sum_{n=1}^{p} E_n = 1$ and $D_{\mathbf{M}}(E_n) = \infty = D_{\mathbf{M}}(E_1)$.

Since $D_{\mathbf{M}}(E_t) = D_{\mathbf{M}}(E_1)$ there is a partially isometric $W_{t,1} \in \mathbf{M}$ with initial set $\mathfrak{M}_1$ and final set $\mathfrak{M}_t$ by the definition of the dimensionality function. If we now define $W_{t,s}$ by the equation $W_{t,s} = W_{t,1} W_{s,1}^{*}$, the properties of the partially isometric $W_{t,1}$ (cf. [5, §4.3]) readily yield the equations of Definition 2.6.1 above. Hence the $W_{t,s}$ constitute a set of p-matrix units with $E_0 = 1$, $W_{t,s} \in \mathbf{M}$.

Lemma 2.6.3. *If* $\mathbf{M}$ *is in case* (I_n) *then* $\mathbf{M}$ *possesses a set of p-matrix units with* $E_0 = 1$ *if and only if p divides n.*

Proof. Necessity: If $\mathbf{M}$ has a set of p-matrix units, with $E_0 = 1$, then by Lemma 2.6.1, the $W_{1,1}, \ldots, W_{p,p}$ are p pairwise orthogonal projections each with same relative dimension, and whose sum is 1. It follows that $D_{\mathbf{M}}(W_{t,t}) = (1/p) D_{\mathbf{M}}(\mathfrak{H})$. If we take $D_{\mathbf{M}}(\mathfrak{H}) = n$, we know that $D_{\mathbf{M}}(W_{t,t}) = (1/p) D_{\mathbf{M}}(\mathfrak{H})$ is a whole number and hence p divides n (cf. [5, p. 172, Theorem VIII]).

Sufficiency: If p divides n, we can find p dimensionally equivalent pairwise orthogonal projections $E_1, \ldots, E_p \in \mathbf{M}$ and with $\sum_{n=1}^{p} E_n = 1$. For if $D_{\mathbf{M}}(\mathfrak{H}) = n$, we can find a projection $E_1 \in \mathbf{M}$ with $D_{\mathbf{M}}(E_1) = n/p$ and we proceed as in the proof of Lemma 2.6.2 to find $E_2, \ldots, E_p$.

The rest of that proof then applies to the present situation.

Theorem V yields

Lemma 2.6.4. *(i) If* $\mathbf{M}$ *is a factor, not in case* (I_n) *and* p *is a finite integer, then there is an* $\overline{\mathbf{N}}$ *such that* $\overline{\mathbf{N}}^p = \overline{\mathbf{M}}$.

(ii) If $\mathbf{M}$ *is in case* (I_n) *there is an* $\overline{\mathbf{N}}$ *such that* $\overline{\mathbf{N}}^p = \overline{\mathbf{M}}$ *if and only if* p *divides* n.

(iii) If there is an $\overline{\mathbf{N}}$ *such that* $\overline{\mathbf{N}}^p = \overline{\mathbf{M}}$ *for* $p < \infty$, $p = 2, 3, \ldots$, *then there is a manifold* $\mathfrak{M}$ *such that* $\overline{\mathbf{M}_{(\mathfrak{M})}} = \overline{\mathbf{N}}$ *and* $D_{\mathbf{M}}(\mathfrak{M}) = (1/p)D_{\mathbf{M}}(\mathfrak{H})$.

As we remarked in footnote 15, we may add

Lemma 2.6.5. *If* p *is finite,* $\mathbf{M}$ *is a factor and if* $\overline{\mathbf{N}}$ *is such that* $\overline{\mathbf{N}}^p = \overline{\mathbf{M}}$ *then* $\overline{\mathbf{N}}$ *is uniquely determined.*

§2.7 It remains to consider the nature of $\overline{\mathbf{N}}$ and also what happens if $p = \infty$.

Lemma 2.7.1. *We have for any fixed* $p = 1, 2, \ldots, \infty$,

(i) $\overline{\mathbf{N}}$ *is a factor if and only if* $\overline{\mathbf{N}}^p$ *is one.*

(ii) If $\overline{\mathbf{N}}, \overline{\mathbf{N}}^p$ *are factors, then they belong to the same one of the Cases (I), (II), (III).*

(iii) If $\overline{\mathbf{N}}, \overline{\mathbf{N}}^p$ *are factors, then* $\overline{\mathbf{N}}^p$ *is in a finite case if and only if* $\overline{\mathbf{N}}$ *is in a finite case and* p *is finite.*

Proof. Choose an $\mathbf{M}$ with $\overline{\mathbf{M}} = \overline{\mathbf{N}}^p$.

Ad (i). An $A^0 \in \mathbf{M}$ belongs to the center of $\mathbf{M}$ if and only if its matrix $\langle A_{t,s}\rangle$ (all $A_{t,s} \in \mathbf{N}$) commutes with all matrices $\langle B_{t,s}\rangle \in \mathfrak{J}$. For this it is necessary that it commute with all finite matrices $\langle B_{t,s}\rangle$ (all $B_{t,s} \in \mathbf{N}$, cf. B_p)). This is immediately seen to imply that $A_{t,s} = \delta_{t,s}A$ $(t, s \le p)$ for an A belonging to the center of $\mathbf{N}$. This condition is also clearly sufficient. Hence the A^0 of the center of $\mathbf{M}$ are precisely the $\alpha 1$ corresponding to the matrices $\langle \alpha\delta_{t,s}1\rangle$ if and only if the A of the center of $\mathbf{N}$ are precisely the $\alpha 1$, i.e. $\mathbf{M}$ is a factor if and only if $\mathbf{N}$ is one.

This completes the proof.

Ad (ii). By Theorem V, $\mathbf{N}$ is isomorphic to an $\mathbf{M}_{(\mathfrak{M})}$. Hence our assertion follows from [5, p. 189, Lemma 11.4.3].

Ad (iii). The $W_{\mu,\mu}$ $(\mu \le p)$ are pairwise orthogonal projections $\in \mathbf{M}$ all with the same relative dimension in $\mathbf{M}$ owing to (i), (ii) in Lemma 2.6.1.

Hence

$$D_{\mathbf{M}}(1) = \sum_{\mu=1}^{p} D_{\mathbf{M}}(W_{\mu,\mu}) = pD_{\mathbf{M}}(W_{1,1}).^{16} \qquad (2.7.\alpha)$$

Now $W_{1,1}$ is not zero, since if it were we should have $W_{\mu,\mu} = 0$ and $\sum_{\mu=1}^{p} W_{\mu,\mu} = 0$ for $p = 1, 2, \ldots, \infty$. Hence $D_{\mathbf{M}}(W_{1,1}) \neq 0$. Now $\mathbf{M}$ is in a finite case if and only if $D_{\mathbf{M}}(1)$ is finite. By the above, this is equivalent to the statement that both p and $D_{\mathbf{M}}(W_{1,1})$ are finite. And the finiteness of $D_{\mathbf{M}}(W_{1,1}) = D_{\mathbf{M}}(\mathfrak{M})$, $(P_{\mathfrak{M}} = W_{1,1})$ is, according to [5, p. 189, Lemma 11.4.3], equivalent to $\mathbf{M}_{(\mathfrak{M})}$ being in a finite case, i.e. by Theorem V, to $\mathbf{N}$ being in a finite case.

This completes the proof.

§2.8 Let $\mathbf{M}$ be a factor in a Hilbert space $\mathfrak{H}$. As usual we denote by $D_{\mathbf{M}}(E)$ and by $D_{\mathbf{M}}(\mathfrak{M})$, $(E = P_{\mathfrak{M}} \in \mathbf{M})$ a fixed relative dimension function of $\mathbf{M}$. We consider again the rings $\mathbf{M}^E$ and $\mathbf{M}_{(\mathfrak{M})}$ as described in the beginning of §2.5.

Lemma 2.8.1. *If $\mathfrak{M}_1, \mathfrak{M}_2$ are two closed linear sets η $\mathbf{M}$ with $D_{\mathbf{M}}(\mathfrak{M}_1) = D_{\mathbf{M}}(\mathfrak{M}_2)$ then $\mathbf{M}_{(\mathfrak{M}_1)}$ and $\mathbf{M}_{(\mathfrak{M}_2)}$ (in $\mathfrak{M}_1$ and $\mathfrak{M}_2$, respectively) are spatially isomorphic.*

Proof. Since $D_{\mathbf{M}}(\mathfrak{M}_1) = D_{\mathbf{M}}(\mathfrak{M}_2)$ there exists a partially isometric $U \in \mathbf{M}$ with the initial set $\mathfrak{M}_1$ and final set $\mathfrak{M}_2$. This is a one-to-one isomorphic mapping of $\mathfrak{M}_1$ on $\mathfrak{M}_2$ and it obviously carries $\mathbf{M}_{(\mathfrak{M}_1)}$ into $\mathbf{M}_{(\mathfrak{M}_2)}$. Hence it establishes the desired spatial isomorphism.

For the remainder of this chapter, we assume that the factor $\mathbf{M}$ is in a finite case.

Consider a real number α and a projection $E \in \mathbf{M}$ or equivalently a closed linear set $\mathfrak{M}\,\eta\,\mathbf{M}$ with $E = P_{\mathfrak{M}}$. We assume $E \neq 0$, i.e. $\mathfrak{M} \neq \{0\}$ and

$$\alpha = \frac{D_{\mathbf{M}}(E)}{D_{\mathbf{M}}(1)}. \qquad (2.8.\alpha)$$

Then the above Lemma 2.8.1 shows that $\overline{\mathbf{M}_{(\mathfrak{M})}}$ depends only on $\overline{\mathbf{M}}$ and α but not on the $E, \mathfrak{M}$ themselves. (A spatial isomorphism implies an algebraic isomorphism.)

Consider now two algebraically isomorphic $\mathbf{M}_1$ and $\mathbf{M}_2$. Choose $E_1 \in \mathbf{M}_1$ with $\alpha = D_{\mathbf{M}_1}(E_1)/D_{\mathbf{M}_1}(1)$. Then the isomorphism carries E_1 into an $E_2 \in \mathbf{M}_2$ with $\alpha = D_{\mathbf{M}_2}(E_2)/D_{\mathbf{M}_2}(1)$. Let $\mathfrak{M}_1, \mathfrak{M}_2$ be η $\mathbf{M}_1, \eta$ $\mathbf{M}_2$ respectively, and such that $P_{\mathfrak{M}_1} = E_1$, $P_{\mathfrak{M}_2} = E_2$. The isomorphism carries $\mathbf{M}_1^{E_1}$ into $\mathbf{M}_2^{E_2}$, in particular E_1 into E_2, and it leaves the operations $\alpha A, A^*, A + B, AB$ invariant. Hence it generates, by Lemma 2.5.1, an algebraic isomorphism of $(\mathbf{M}_1)_{(\mathfrak{M}_1)}$ and $(\mathbf{M}_2)_{(\mathfrak{M}_2)}$.

[16] Here as in some subsequent instances (proof of Lemma 5.3.4; III in §5.5) it is convenient to use the convention $\infty \cdot 0 = 0$.

Thus $\overline{\mathbf{M}_{(\mathfrak{M})}}$ (assuming its existence) is uniquely determined by $\overline{\mathbf{M}}$ and α. We denote this $\overline{\mathbf{M}_{(\mathfrak{M})}}$ by $\overline{\mathbf{M}}^{\alpha}$.

This notation could conflict with that of Theorem IV when $p = \alpha = 1$. But in that case clearly $\overline{\mathbf{M}}^{1}$ exists and is equal to $\overline{\mathbf{M}}$ under both definitions.

The α, for which $\overline{\mathbf{M}}^{\alpha}$ can be formed, comprise precisely the range of $D_{\mathbf{M}}(E)/D_{\mathbf{M}}(1)$ for all $E \in \mathbf{M}$, $E \neq 0$. This is the set $(1/n, 2/n, \ldots, 1)$ if $\mathbf{M}$ is in a case (I_n) and the set $0 < \alpha \leq 1$ if $\mathbf{M}$ is in a case (II_1).

Summing up,

Theorem VI. $\overline{\mathbf{M}}^{\alpha}$ *exists if and only if* α *belongs to the following set:* $(1/n, 2/n, \ldots, 1)$ *if* $\mathbf{M}$ *is in a case* (I_n), *(*$n = 1, 2, \ldots$*)*, $(0 < \alpha \leq 1)$ *if* $\mathbf{M}$ *is in the case* (II_1).

There is no conflict between this notation and that of Theorem IV.

Since the relative dimension $D_{\mathbf{M}}(E)$ can be characterized in a way which is invariant under conjugate isomorphisms (cf. the remark at the end of §2.3), our present definition and that of §2.3 yield together

$$(\overline{\mathbf{M}}^{c})^{\alpha} = (\overline{\mathbf{M}}^{\alpha})^{c} \qquad (\alpha \text{ in the range of Theorem VI}). \tag{2.8.β}$$

Consider next two projections $E, F \in \mathbf{M}$, $E, F \neq 0$ and $E \leq F$, or equivalently the closed linear sets $\mathfrak{M}, \mathfrak{N} \,\eta\, \mathbf{M}$ with $P_{\mathfrak{M}} = E$, $P_{\mathfrak{N}} = F$. Hence $\mathfrak{M} \neq (0)$, $\mathfrak{N} \neq (0)$ and $\mathfrak{M} \subseteq \mathfrak{N}$. To avoid misunderstanding, let 1 be the unit operator in $\mathfrak{H}$ and $1' = F_{(\mathfrak{N})}$ be the unit in $\mathfrak{N}$. Put

$$\alpha = \frac{D_{\mathbf{M}}(F)}{D_{\mathbf{M}}(1)}, \qquad \beta = \frac{D_{\mathbf{M}_{(\mathfrak{N})}}(E_{(\mathfrak{N})})}{D_{\mathbf{M}_{(\mathfrak{N})}}(1')} = \frac{D_{\mathbf{M}}(E)}{D_{\mathbf{M}}(F)}.$$

Clearly $\alpha\beta = D_{\mathbf{M}}(E)/D_{\mathbf{M}}(1)$ and $\mathbf{M}_{(\mathfrak{M})} = (\mathbf{M}_{(\mathfrak{N})})_{(\mathfrak{M})}$. Our definitions now yield immediately

$$(\overline{\mathbf{M}}^{\alpha})^{\beta} = \overline{\mathbf{M}}^{\alpha\beta} \qquad (\alpha, \beta \text{ in the range of Theorem VI}). \tag{2.8.γ}$$

§2.9 We continue the discussion of the preceding section.

Lemma 2.9.1. *If* p *is finite and* $\mathbf{M}$ *and* $\mathbf{N}$ *are factors in a finite case,* $\overline{\mathbf{N}} = \overline{\mathbf{M}}^{1/p}$ *is equivalent to the statement* $\overline{\mathbf{M}} = \overline{\mathbf{N}}^{p}$. *(The statement "*$\overline{\mathbf{M}}^{1/p} = \overline{\mathbf{N}}$*" includes "*$\overline{\mathbf{M}}^{1/p}$ *exists.")*

Proof. Assume $\overline{\mathbf{M}} = \overline{\mathbf{N}}^{p}$. It follows from Theorem V that there is an E with $D_{\mathbf{M}}(E) = (1/p)D_{\mathbf{M}}(1)$ such that $\overline{\mathbf{N}} = \overline{\mathbf{M}_{(\mathfrak{M})}}$ where $\mathfrak{M}$ is such that $E = P_{\mathfrak{M}}$. Hence, by the above definition, $\overline{\mathbf{M}}^{1/p} = \overline{\mathbf{N}}$.

Conversely, suppose $\overline{\mathbf{M}}^{1/p} = \overline{\mathbf{N}}$. It follows that $1/p$ is in the set of Theorem VI and hence, if $\mathbf{M}$ is in a case (I_n), p divides n. Thus Lemma 2.6.2 and Lemma 2.6.3 imply that there is an $\overline{\mathbf{N}}_0$ such that $\overline{\mathbf{N}}_0^p = \overline{\mathbf{M}}$. Hence the above result (with $\overline{\mathbf{N}}_0$ in place of $\overline{\mathbf{N}}$) gives $\overline{\mathbf{N}}_0 = \overline{\mathbf{M}}^{1/p}$, i.e. $\overline{\mathbf{N}}_0 = \overline{\mathbf{N}}$. Consequently $\overline{\mathbf{N}}^p = \overline{\mathbf{N}}_0^p = \overline{\mathbf{M}}$.

This completes the proof.

Lemma 2.9.2. *If $\overline{\mathbf{M}}^\alpha$ exists (in the sense of Theorem VI) and if $p\alpha = q\beta$, $(p, q = 1, 2, \ldots)$ then $(\overline{\mathbf{M}}^\alpha)^p = (\overline{\mathbf{M}}^q)^\beta$. (This is to include the statement that $(\overline{\mathbf{M}}^q)^\beta$ exists.)*

Proof. $p\alpha = q\beta$. Hence $\alpha/q = \beta/p$, $\overline{\mathbf{M}}^\alpha$ exists. From this we shall draw inferences so that the existence of each expression that appears will be established when we write it down. In this way (2.8.γ) and Lemma 2.9.1 give

$$\overline{\mathbf{M}}^\alpha = \left((\overline{\mathbf{M}}^q)^{1/q}\right)^\alpha = (\overline{\mathbf{M}}^q)^{\alpha/q} = (\overline{\mathbf{M}}^q)^{\beta/p}.$$

Hence by Lemma 2.9.1,

$$(\overline{\mathbf{M}}^\alpha)^p = \left((\overline{\mathbf{M}}^q)^{\beta/p}\right)^p = \left(\left((\overline{\mathbf{M}}^q)^\beta\right)^{1/p}\right)^p = (\overline{\mathbf{M}}^q)^\beta.$$

Lemma 2.9.3. *For a given $\overline{\mathbf{M}}$ consider all α of the set in Theorem VI, and all $p = 1, 2, \ldots$. Then we have*

(i) The range of $\theta = p\alpha$ is the set $(1/n, 2/n, \ldots)$ if $\mathbf{M}$ is in a case (I_n) $(n = 1, 2, \ldots)$, and the set $0 < \theta < \infty$ if $\mathbf{M}$ is in the case (II_1).

(ii) $(\overline{\mathbf{M}}^\alpha)^p$ depends only on $\theta = p\alpha$ and not on the individual values of α and p.

Proof. Ad (i). Immediate by Theorem V.

Ad (ii). Lemma 2.9.2 yields, when $p\alpha = q\beta$, $(\overline{\mathbf{M}}^\alpha)^p = (\overline{\mathbf{M}}^q)^\beta$ and $(\overline{\mathbf{M}}^\beta)^q = (\overline{\mathbf{M}}^p)^\alpha$. Hence $(\overline{\mathbf{M}}^\alpha)^p = (\overline{\mathbf{M}}^\beta)^q$.

We denote the above $(\overline{\mathbf{M}}^\alpha)^p$, $\theta = p\alpha$, by $\overline{\mathbf{M}}^\theta$.

This notation does not conflict with those of Theorem IV and Theorem VI. This can be seen by putting $\alpha = 1$ or $p = 1$, respectively.

Summing up,—

Theorem VII. *$\overline{\mathbf{M}}^\theta$ exists if and only if θ belongs to the following set: $(1/n, 2/n, \ldots)$ if $\mathbf{M}$ is in a case (I_n), $(n = 1, 2, \ldots)$, $(0 < \theta < \infty)$ if $\mathbf{M}$ is in the case (II_1).*

There is no conflict between this notation and that of Theorem IV and Theorem VI.

Combining (2.4.α) and (2.8.β) gives

$$(\overline{\mathbf{M}}^c)^\theta = (\overline{\mathbf{M}}^\theta)^c \qquad (\theta \text{ in the set of Theorem VII}). \tag{2.9.α}$$

Also, by the use of (2.4.β), (2.8.γ) and Lemma 2.9.2, we obtain

$$\left(((\overline{\mathbf{M}}^{\alpha})^{p})^{\beta}\right)^{q} = \left(((\overline{\mathbf{M}}^{\alpha})^{\beta})^{p}\right)^{q} = (\overline{\mathbf{M}}^{\alpha\beta})^{pq}.$$

Hence if we put $\theta = p\alpha$, $\xi = q\beta$, then

$$(\overline{\mathbf{M}}^{\theta})^{\xi} = (\overline{\mathbf{M}}^{\xi})^{\theta} = \overline{\mathbf{M}}^{\xi\theta} \qquad (\theta \text{ and } \xi \text{ in the set of Theorem VII}). \tag{2.9.β}$$

(Concerning the exponent ∞, consider the remark at the end of §2.5.)

§2.10 Denote the set of all θ for which $\overline{\mathbf{M}}^{\theta}$ exists and equals $\overline{\mathbf{M}}$ by

$$\mathfrak{G} = \mathfrak{G}(\overline{\mathbf{M}}). \tag{2.10.α}$$

Obviously $1 \in \mathfrak{G}$ and $\theta, \xi \in \mathfrak{G}$ imply $\theta\xi \in \mathfrak{G}$. Since $(\overline{\mathbf{M}}^{\theta})^{1/\theta}$ exists and equals $\overline{\mathbf{M}}$, $\overline{\mathbf{M}}^{\theta} = \overline{\mathbf{M}}$ implies $\overline{\mathbf{M}}^{1/\theta} = \overline{\mathbf{M}}$, i.e. $\theta \in \mathfrak{G}$ implies $1/\theta \in \mathfrak{G}$. (All this is due to (2.9.β) in §2.9.)

Theorem VIII. *The set $\mathfrak{G}$ of (2.10.α) above is a subgroup of P, the multiplication group of the real numbers θ, $0 < \theta < \infty$. We call $\mathfrak{G}$ the fundamental group of $\overline{\mathbf{M}}$.*

Some properties of $\mathfrak{G}$.

Lemma 2.10.1. $\overline{\mathbf{M}}^{\theta} = \overline{\mathbf{M}}^{\xi}$ *(both sides are assumed to exist) is equivalent to $\theta/\xi \in \mathfrak{G}$.*

Proof. Sufficiency: $\theta/\xi \in \mathfrak{G}$ implies $\overline{\mathbf{M}}^{\theta/\xi} = \overline{\mathbf{M}}$. Hence $\overline{\mathbf{M}}^{\theta} = (\overline{\mathbf{M}}^{\theta/\xi})^{\xi} = \overline{\mathbf{M}}^{\xi}$.

Necessity: Assume $\overline{\mathbf{M}}^{\theta} = \overline{\mathbf{M}}^{\xi}$. $(\overline{\mathbf{M}}^{\xi})^{1/\xi}$ exists and equals $\overline{\mathbf{M}}$. Hence $(\overline{\mathbf{M}}^{\theta})^{1/\xi} = \overline{\mathbf{M}}^{\theta/\xi}$ exists and equals $\overline{\mathbf{M}}$. Thus $\theta/\xi \in \mathfrak{G}$.

Lemma 2.10.2. $\overline{\mathbf{M}}$, $\overline{\mathbf{M}}^{c}$ *and all* $\overline{\mathbf{M}}^{\theta}$ *have the same fundamental group.*

Proof. Ad $\overline{\mathbf{M}}^{c}$. Immediate by (2.9.$\alpha$).

Ad $\overline{\mathbf{M}}^{\theta}$. Immediate by Lemma 2.10.1.

Before we conclude this chapter, we determine the behavior of the discrete finite cases, i.e. the (I_n), $n = 1, 2, \ldots$.

Consider the ring $\mathbf{\Theta}$ of all operators $\alpha \cdot 1$. This is obviously the prototype of all factors of the case (I_1), its type $\overline{\mathbf{\Theta}}$ is that of the ring of all complex numbers.

Now we have

Lemma 2.10.3. *If* $\mathbf{M}$ *is in case (I_n), $(n = 1, 2, \ldots)$, then* $\overline{\mathbf{M}}^{\alpha}$ *(α finite) exists if and only if $n\alpha = 1, 2, \ldots$, and then* $\overline{\mathbf{M}}^{\alpha} = \overline{\mathbf{\Theta}}^{n\alpha}$.

Proof. The range asserted for α is the same as that of Theorem VII. Put $n\alpha = k$. Then

$$\overline{\mathbf{M}}^{\alpha} = \overline{\mathbf{M}}^{k/n} = \left(\overline{\mathbf{M}}^{1/n}\right)^{k}.$$

Now consider $\overline{\mathbf{M}}^{1/n}$. Apply the definition of §2.8, preceding Theorem VII. For the E in question, $D_{\mathbf{M}}(E)/D_{\mathbf{M}}(1)$ assumes its minimal value, $1/n$. Hence, this E is minimal (cf. [5, pp. 143–144, Definition 5.1.2]). Consequently $\mathbf{M}^{E}$ is the set of all αE (cf. [5, p. 144, the last part of the proof of Lemma 5.1.3]—or it may be proved directly). Hence $\mathbf{M}_{(\mathfrak{M})}$ is the set of all $\alpha \cdot 1$ in $\mathfrak{M}$, i.e.

$$\overline{\mathbf{M}}^{1/n} = \overline{\mathbf{M}_{(\mathfrak{M})}} = \overline{\mathbf{\Theta}}.$$

Consequently

$$\overline{\mathbf{M}}^{\alpha} = \overline{\mathbf{\Theta}}^{k}, \qquad k = n\alpha,$$

as desired.

Corollary. *If* $\mathbf{M}$ *is in a case* (I_n) $(n = 1, 2, \ldots)$ *then* $\overline{\mathbf{M}} = \overline{\mathbf{\Theta}}^{n}$.

Proof. Put $\alpha = 1$ in the above lemma.

Lemma 2.10.4. *If* $\mathbf{M}$ *is in a case* (I_n) $(n = 1, 2, \ldots)$ *then* $\overline{\mathbf{M}}^{c} = \overline{\mathbf{M}}$ *and* $\mathfrak{G} = \mathfrak{G}(\overline{\mathbf{M}}) = (1)$.

Proof. [17] Ad $\overline{\mathbf{M}}^{c} = \overline{\mathbf{M}}$. Clearly $\overline{\mathbf{\Theta}}^{c} = \overline{\mathbf{\Theta}}$. Hence the corollary to Lemma 2.10.3, and (2.4.α) or (2.9.α) imply $\overline{\mathbf{M}}^{c} = \overline{\mathbf{M}}$.

Ad $\mathfrak{G} = (1)$. By Lemma 2.10.3 and its corollary, $\overline{\mathbf{M}}^{\alpha}$ is in case (I_k), $k = n\alpha$. Hence $\overline{\mathbf{M}}^{\alpha} = \overline{\mathbf{M}}$ implies $n\alpha = n$ or $\alpha = 1$. So $\mathfrak{G} = (1)$.

The validity or invalidity of the equation $\overline{\mathbf{M}}^{c} = \overline{\mathbf{M}}$ and the fundamental group $\mathfrak{G} = \mathfrak{G}(\overline{\mathbf{M}})$ are algebraical isomorphism invariants of $\mathbf{M}$. Lemma 2.10.4 showed us how they behave when $\mathbf{M}$ is in a discrete finite case, i.e. in a case (I_n), $n = 1, 2, \ldots$. The really interesting factors are the remaining finite ones, those in the continuous finite case (II_1). We shall see that they are not all isomorphic to each other, but we shall achieve this differentiation with the help of other invariants. (Cf. the end of §5.6 and the beginning of §6.1.)

For all $\mathbf{M}$ in case (II_1) for which we have succeeded in settling this question, $\overline{\mathbf{M}}^{c} = \overline{\mathbf{M}}$ and $\overline{\mathbf{M}}^{\theta} = \overline{\mathbf{M}}$ (θ in the set of Theorem VII), i.e. $\mathfrak{G} = P$, was found. (Cf. Theorem XV and the end of §5.6. Observe Lemma 2.10.4 by comparison.) There seems to be, however, no reason to believe the general validity of these relations. The general behavior of the above invariants remains therefore an open question.

[17] Both assertions could be proved directly by matrix considerations.

CHAPTER III. CHARACTERIZATION OF ALL SPATIAL TYPES IN TERMS OF ALGEBRAICAL TYPES

§3.1 We have classified all operator rings $\mathbf{M}$ according to their types $\overline{\mathbf{M}}$, i.e. with respect to algebraical isomorphism. Now we shall introduce a broader classification, to be called the *genus*, each genus consisting of one or more types. It will be necessary, however, to restrict this new classification to factors $\mathbf{M}$.

We need some auxiliary lemmas which lead up to the desired definition. Throughout this section $\mathbf{M}$ will be a factor in a Hilbert space $\mathfrak{H}$. As before, we denote by $D_{\mathbf{M}}(E)$ and $D_{\mathbf{M}}(\mathfrak{M})$ a fixed relative dimension function of $\mathbf{M}$ applicable to the $E \in \mathbf{M}$ or the $\mathfrak{M}\ \eta\ \mathbf{M}$.

Lemma 3.1.1. *If $\mathfrak{M}$ is a closed linear set, $\mathfrak{M}\ \eta\ \mathbf{M}$, and such that $\mathfrak{M} \neq (0)$ and $D_{\mathbf{M}}(\mathfrak{H} \ominus \mathfrak{M}) = \infty$, then*

$$\overline{\mathbf{M}} = \left(\overline{\mathbf{M}_{(\mathfrak{M})}}\right)^{\infty}.$$

Proof. Under our hypothesis, $\mathfrak{M}$ "divides" $\mathfrak{H} \ominus \mathfrak{M}$ an infinite number of times, i.e. there are an infinite number of pairwise orthogonal closed linear sets $\mathfrak{M}_2, \mathfrak{M}_3, \ldots$ such that $D_{\mathbf{M}}(\mathfrak{M}_i) = D_{\mathbf{M}}(\mathfrak{M})$ and $\mathfrak{M}_i \subset \mathfrak{H} \ominus \mathfrak{M}$ for $i = 2, 3, \ldots$. If $\mathfrak{M}$ is relatively finite-dimensional, this is obvious. If $\mathfrak{M}$ is infinite-dimensional, we apply a variant of [5, p. 157, Lemma 7.2.3] to $\mathfrak{H} \ominus \mathfrak{M}$, which shows that $\mathfrak{H} \ominus \mathfrak{M}$ is determined by a denumerably infinite number of pairwise orthogonal closed linear sets $\mathfrak{M}'_2, \mathfrak{M}'_3, \ldots$, each of which is relatively infinite-dimensional. (To prove the variant, it is only necessary, in the proof of [5, Lemma 7.2.3], to divide the $\mathfrak{P}_1, \mathfrak{P}_2, \ldots$ into an infinite number of sets, each having an infinite number of $\mathfrak{P}_i$'s, rather than into two sets.)

If we now apply [5, p. 155, Lemma 7.1.2] to $\mathfrak{M}$ and $\mathfrak{H} \ominus \mathfrak{M}$, we see that $\mathfrak{H} \ominus \mathfrak{M}$ is determined by a denumerably infinite number of sets $\mathfrak{M}_2, \mathfrak{M}_3, \ldots$, which are pairwise orthogonal and for which $D_{\mathbf{M}}(\mathfrak{M}_i) = D_{\mathbf{M}}(\mathfrak{M})$, for $i = 2, 3, \ldots$. Let $\mathfrak{M} = \mathfrak{M}_1$.

The second paragraph of the proof of Lemma 2.6.2 can now be used to show the existence in $\mathbf{M}$ of a set of ∞-matrix units $W_{\mu,\nu}$, with $W_{\mu,\mu} = P_{\mathfrak{M}_\mu}$, $W_{1,1} = P_{\mathfrak{M}_1} = P_{\mathfrak{M}}$. For the hypothesis $p < \infty$ is not used here. Furthermore,

$$\begin{aligned} E_0 &= \sum_{\mu=1}^{\infty} W_{\mu,\mu} = W_{1,1} + \sum_{\mu=2}^{\infty} W_{\mu,\mu} \\ &= P_{\mathfrak{M}_1} + \sum_{\mu=2}^{\infty} P_{\mathfrak{M}_\mu} = P_{\mathfrak{M}} + P_{\mathfrak{H} \ominus \mathfrak{M}} = 1. \end{aligned}$$

Thus Theorem V in §2.6 above yields the desired result.

Lemma 3.1.2. *If $\mathfrak{M}$ is a closed linear set, $\mathfrak{M}\ \eta\ \mathbf{M}$ and $\mathfrak{M} \neq (0)$, and if $\mathbf{M}$ is in an infinite case, then*

$$\overline{\mathbf{M}} = \left(\overline{\mathbf{M}_{(\mathfrak{M})}}\right)^{\infty}.$$

Proof. If $D_{\mathbf{M}}(\mathfrak{H} \ominus \mathfrak{M}) = \infty$, the result is implied by Lemma 3.1.1. Hence we may assume that $D_{\mathbf{M}}(\mathfrak{H} \ominus \mathfrak{M})$ is finite. Since $\mathbf{M}$ is in an infinite case, $D_{\mathbf{M}}(\mathfrak{H})$ is infinite and hence

$$D_{\mathbf{M}}(\mathfrak{M}) = D_{\mathbf{M}}(\mathfrak{H}) - D_{\mathbf{M}}(\mathfrak{H} \ominus \mathfrak{M})$$

is infinite also.

Since $D_{\mathbf{M}}(\mathfrak{H})$ is infinite, there exists by [5, p. 157, Lemma 7.2.3] a closed linear set $\mathfrak{N}$ such that $D_{\mathbf{M}}(\mathfrak{N})$ and $D_{\mathbf{M}}(\mathfrak{H} \ominus \mathfrak{N})$ are also infinite. Hence $\overline{\mathbf{M}_{(\mathfrak{N})}} = \overline{\mathbf{M}_{(\mathfrak{M})}}$ by Lemma 2.8.1 (we need only the algebraical, not the spatial, isomorphism), and $\overline{\mathbf{M}} = \left(\overline{\mathbf{M}_{(\mathfrak{N})}}\right)^{\infty}$ by Lemma 3.1.1. These give together $\overline{\mathbf{M}} = \left(\overline{\mathbf{M}_{(\mathfrak{M})}}\right)^{\infty}$.

Lemma 3.1.3. *A factor $\mathbf{M}$ is in an infinite case if and only if*

$$\overline{\mathbf{M}} = \overline{\mathbf{M}}^{\infty}.$$

Proof. Sufficiency: Immediate by (iii) in Lemma 2.7.1.

Necessity: Put $\mathfrak{M} = \mathfrak{H}$ in Lemma 3.1.2.

Definition 3.1.1. *If, for two factors, $\mathbf{M}, \mathbf{N}$,*

$$\overline{\mathbf{N}} = \overline{\mathbf{M}_{(\mathfrak{M})}}$$

for a suitable closed set $\mathfrak{M} \neq (0)$ and $\mathfrak{M}\ \eta\ \mathbf{M}$ in the Hilbert space $\mathfrak{H}$ of $\mathbf{M}$, then we say that $\overline{\mathbf{M}}$ is a multiple *of $\overline{\mathbf{N}}$, or that $\overline{\mathbf{N}}$ is a* divisor *of $\overline{\mathbf{M}}$.*

It is clear that this relationship concerns $\overline{\mathbf{M}}$ and $\overline{\mathbf{N}}$. Cf. the argument immediately preceding Theorem VI in §2.8. It is also obviously transitive.

Lemma 3.1.4. *$\overline{\mathbf{N}}$ is a divisor of $\overline{\mathbf{M}}$ if and only if*

α) $\overline{\mathbf{M}} = \overline{\mathbf{N}}$, when $\overline{\mathbf{N}}$ is in an infinite case;

β) $\overline{\mathbf{M}} = \overline{\mathbf{N}}^{\theta}$, with a $\theta \geq 1$ in the range of Theorem VII or of Theorem IV′ (i.e. $\theta = \infty$), when $\overline{\mathbf{N}}$ is in a finite case.

Proof. Ad α). Sufficiency: Obvious.

Necessity: Since $\overline{\mathbf{N}} = \overline{\mathbf{M}_{(\mathfrak{M})}}$ is in an infinite case, $D_{\mathbf{M}}(\mathfrak{M})$ is infinite by [5, p. 189, Lemma 11.4.3]. Hence $\overline{\mathbf{M}}$ is in an infinite case and thus Lemma 3.1.2 gives

$$\overline{\mathbf{M}} = \left(\overline{\mathbf{M}_{(\mathfrak{M})}}\right)^{\infty} = \overline{\mathbf{N}}^{\infty}.$$

On the other hand, $\overline{\mathbf{N}} = \overline{\mathbf{N}}^{\infty}$ by Lemma 3.1.3. Thus $\overline{\mathbf{M}} = \overline{\mathbf{N}}$.

Ad β). Sufficiency: θ finite: Then $\overline{\mathbf{N}} = \overline{\mathbf{M}}^{\alpha}$, with $\alpha = 1/\theta \leq 1$. Hence the assertion follows by our definition of $\overline{\mathbf{M}}^{\alpha}$ for $\alpha \leq 1$ in §2.8. θ infinite: Immediate by Theorem V.

Necessity: $\overline{\mathbf{M}}$ in a finite case: Clearly $\overline{\mathbf{N}} = \overline{\mathbf{M}_{(\mathfrak{M})}} = \overline{\mathbf{M}}^{\alpha}$, with $\alpha = D_{\mathbf{M}}(E)/D_{\mathbf{M}}(1) \leq 1$, $E = P_{\mathfrak{M}}$. Hence $\overline{\mathbf{M}} = \overline{\mathbf{N}}^{\theta}$, with $\theta = 1/\alpha \geq 1$. $\overline{\mathbf{M}}$ in an infinite case: $\overline{\mathbf{M}} = \left(\overline{\mathbf{M}_{(\mathfrak{M})}}\right)^{\infty} = \overline{\mathbf{N}}^{\infty}$ by Lemma 3.1.2.

Lemma 3.1.5. *The four following statements are equivalent to each other:*

α) $\overline{\mathbf{M}}$ is either a multiple or divisor of $\overline{\mathbf{N}}$;
β) $\overline{\mathbf{M}}$ and $\overline{\mathbf{N}}$ possess a common multiple;
γ) $\overline{\mathbf{M}}$ and $\overline{\mathbf{N}}$ possess a common divisor;
δ) $\overline{\mathbf{M}}^{\infty} = \overline{\mathbf{N}}^{\infty}$.

Proof. The implications

$$\alpha) \longrightarrow \beta), \qquad \alpha) \longrightarrow \gamma)$$

are obvious. Furthermore,

$$\delta) \longrightarrow \beta)$$

follows from Theorem V, since this theorem shows that $\overline{\mathbf{M}}^{\infty} = \overline{\mathbf{N}}^{\infty}$ is a multiple of $\overline{\mathbf{M}}$ and of $\overline{\mathbf{N}}$.

We prove next

$$\beta) \longrightarrow \alpha).$$

Let $\overline{\mathbf{L}}$ be the common multiple of both $\overline{\mathbf{M}}$ and $\overline{\mathbf{N}}$. Let closed linear sets $\mathfrak{M}$ and $\mathfrak{N}$ be chosen in the $\mathfrak{H}$ associated with $\mathbf{L}$, so that $\overline{\mathbf{M}} = \overline{\mathbf{L}_{(\mathfrak{M})}}$ and $\overline{\mathbf{N}} = \overline{\mathbf{L}_{(\mathfrak{N})}}$. Since $\mathbf{L}$ is a factor, either $\mathfrak{M}$ has the same relative dimension (in $\mathbf{L}$) as a $\mathfrak{N}' \subseteq \mathfrak{N}$, or $\mathfrak{N}$ has the same relative dimension as a $\mathfrak{M}' \subseteq \mathfrak{M}$. Cf. [5, p. 153, Lemma 6.2.3, or p. 154, Theorem VI]. By symmetry we may assume the former. By Lemma 2.8.1 we may now replace $\mathfrak{M}$ by $\mathfrak{N}'$, i.e. we can assume that $\mathfrak{M} \subseteq \mathfrak{N}$.

Thus

$$\overline{\left(\mathbf{L}_{(\mathfrak{N})}\right)_{(\mathfrak{M})}} = \overline{\mathbf{L}_{(\mathfrak{M})}},$$

so that $\overline{\mathbf{L}_{(\mathfrak{N})}}$ is a multiple of $\overline{\mathbf{L}_{(\mathfrak{M})}}$. Hence $\overline{\mathbf{N}}$ is a multiple of $\overline{\mathbf{M}}$.

Finally,

$$\gamma) \longrightarrow \delta).$$

Let $\overline{\mathbf{K}}$ be a common divisor of $\overline{\mathbf{M}}$ and $\overline{\mathbf{N}}$. Since $\overline{\mathbf{M}}$ is a divisor of $\overline{\mathbf{M}}^{\infty}$ by Theorem V, so $\overline{\mathbf{K}}$ is a divisor of $\overline{\mathbf{M}}^{\infty}$ too. Now $\overline{\mathbf{M}}^{\infty}$ is in an infinite case by (iii) in Lemma 2.7.1. Hence $\overline{\mathbf{K}}^{\infty} = \overline{\mathbf{M}}^{\infty}$ by Lemma 3.1.2. Similarly $\overline{\mathbf{K}}^{\infty} = \overline{\mathbf{N}}^{\infty}$. Consequently $\overline{\mathbf{M}}^{\infty} = \overline{\mathbf{N}}^{\infty}$.

All these equivalences give together

$$\alpha) \longrightarrow \gamma) \longrightarrow \delta) \longrightarrow \beta) \longrightarrow \alpha),$$

thus completing the proof.

Definition 3.1.2. *If the equivalent conditions of Lemma 3.1.5 are satisfied, then we say that $\overline{\mathbf{M}}, \overline{\mathbf{N}}$ are* commensurable.

Condition δ) of Lemma 3.1.5 implies that the notion of commensurability is reflexive, symmetric and transitive.

Lemma 3.1.6. *$\overline{\mathbf{M}}, \overline{\mathbf{N}}$ are commensurable if and only if*

α) $\overline{\mathbf{M}} = \overline{\mathbf{N}}$, when $\overline{\mathbf{M}}, \overline{\mathbf{N}}$ are both in infinite cases;

β) $\overline{\mathbf{M}} = \overline{\mathbf{N}}^{\infty}$, when $\overline{\mathbf{M}}$ is in an infinite case and $\overline{\mathbf{N}}$ in a finite case;

γ) $\overline{\mathbf{M}}^{\infty} = \overline{\mathbf{N}}$, when $\overline{\mathbf{M}}$ is in a finite case and $\overline{\mathbf{N}}$ in an infinite case;

δ) $\overline{\mathbf{M}} = \overline{\mathbf{N}}^{\theta}$, with a θ in the range of Theorem VII, when $\overline{\mathbf{N}}$ and $\overline{\mathbf{M}}$ are both in finite cases.

Proof. Ad α)–γ). Condition δ) of Lemma 3.1.5 for commensurability is $\overline{\mathbf{M}}^{\infty} = \overline{\mathbf{N}}^{\infty}$. Lemma 3.1.3 shows that when $\overline{\mathbf{M}}$ is in an infinite case we may substitute $\overline{\mathbf{M}}$ for $\overline{\mathbf{M}}^{\infty}$ in this condition, and correspondingly for $\overline{\mathbf{N}}$. When $\overline{\mathbf{M}}$ and $\overline{\mathbf{N}}$ are both infinite, this yields α), while if only one is infinite we obtain β) or γ).

Ad δ). By Lemma 3.1.4, β), $\overline{\mathbf{M}} = \overline{\mathbf{N}}^{\theta}$, for $\theta \geq 1$ in the range of Theorem VII, is equivalent to $\overline{\mathbf{N}}$ a divisor of $\overline{\mathbf{M}}$. Since $\overline{\mathbf{M}}$ is finite, θ is finite by Lemma 2.7.1 (iii).

On the other hand, $\overline{\mathbf{M}} = \overline{\mathbf{N}}^{\theta}$, for a $\theta \leq 1$ in the range of Theorem VII (for $\overline{\mathbf{N}}$), is equivalent to $\overline{\mathbf{M}}^{1/\theta} = \overline{\mathbf{N}}$, with $1/\theta$ in the range of Theorem VII (for $\overline{\mathbf{M}}$). (Cf. (2.9.β). That θ is in the correct sets in the discrete cases is readily shown by reference to Lemma 2.10.3 and its corollary.) Lemma 3.1.4, with $\overline{\mathbf{M}}$ and $\overline{\mathbf{N}}$ interchanged, shows that this is equivalent to $\overline{\mathbf{N}}$ is a multiple of $\overline{\mathbf{M}}$.

Thus $\overline{\mathbf{M}} = \overline{\mathbf{N}}^{\theta}$, for a θ in the range of Theorem VII, is equivalent to $\overline{\mathbf{N}}$ is a divisor of $\overline{\mathbf{M}}$ or $\overline{\mathbf{N}}$ is a multiple of $\overline{\mathbf{M}}$, i.e. to the condition of Lemma 3.1.5 for commensurability.

We call the abstraction of type $\overline{\mathbf{M}}$ (for factors) with respect to commensurability its *genus*, i.e. two types $\overline{\mathbf{M}}_1$ and $\overline{\mathbf{M}}_2$ will be said to have the same genus if and only if they are commensurable. We denote the genus of the type $\overline{\mathbf{M}}$ by $\widetilde{\overline{\mathbf{M}}}$. Notions invariant under commensurability will be said to be notions concerning the genus $\widetilde{\overline{\mathbf{M}}}$ (and not $\overline{\mathbf{M}}$ or $\mathbf{M}$ itself).

Thus the cases I, II, III are notions concerning the genus. This follows from (ii) in Lemma 2.7.1 (with $p = \infty$) and the characterization δ) in Lemma 3.1.5.

Similarly $\overline{\mathbf{M}}^c$ is an operation concerning the genus. This follows from (2.5.ϵ) and the above-mentioned characterization. Therefore we can define

$$\widetilde{\overline{\mathbf{M}}}^c = \widetilde{\overline{\mathbf{M}}^c}.$$

Observe finally that $\overline{\mathbf{M}_{(\mathfrak{M})}}$, $\overline{\mathbf{M}}^p$, $p = 1, 2, \ldots, \infty$, and (when it is defined, i.e. for $\overline{\mathbf{M}}$ in a finite case) $\overline{\mathbf{M}}^\theta$ (θ in the range of Theorem VII) have the same genus as $\overline{\mathbf{M}}$ (for $\overline{\mathbf{M}_{(\mathfrak{M})}}$ this follows from α) in Lemma 3.1.5; for $\overline{\mathbf{M}}^p$ this is implied by Theorem V and Lemma 3.1.5, α); and for $\overline{\mathbf{M}}^\theta$ this follows from Lemma 3.1.6, δ).)

It is clear then that being in a finite or an infinite case does not concern the genus (cf. Theorem IX for details).

§3.2 Since the cases (I), (II), (III) are notions concerning the genus, each one of these cases is the sum of one or more genera. And of course each genus is the sum of one or more types. These are the details.

Theorem IX. *α) Case (I) consists of exactly one genus. This genus contains exactly one type in an infinite case:* (I_∞)*; and for every positive integer n exactly one type* (I_n)*, these being all its types in finite cases.*

β) Case (II) consists of one or more genera. Each such genus contains exactly one type in an infinite case (II_∞) *and one or more types in finite cases* (II_1).

γ) Case (III) consists of one or more genera. Each such genus contains exactly one type which is, of course, in an infinite case (III_∞).

Proof. Ad α). Each one of the cases (I_1), (I_2), $\ldots$, (I_∞) contains exactly one type by [5, p. 173, Lemma 8.6.1]. They can obviously all be obtained from the last one by the operations $\mathbf{M}_{(\mathfrak{M})}$. Hence they are all of the same genus as this last by α) of Lemma 3.1.5 and Definition 3.1.1.

Ad β). Consider a genus $\widetilde{\overline{\mathbf{M}}}$ of case (II). Then there exists an $\overline{\mathbf{M}_{(\mathfrak{M})}}$ in a finite case, for any $\mathfrak{M}$ with $D_{\mathbf{M}}(\mathfrak{M}) < \infty$ will do. Thus $\overline{\mathbf{M}_{(\mathfrak{M})}}$ is finite and it belongs to the given genus of $\overline{\mathbf{M}}$ by condition α) of Lemma 3.1.5. $\overline{\mathbf{M}}^\infty$ belongs also to this genus for the same reason, since $\overline{\mathbf{M}}$ is a divisor of $\overline{\mathbf{M}}^\infty$ by Theorem V of §2.6. The infinite case in a particular genus is unique by α) of Lemma 3.1.6.

Ad γ). Consider a genus $\widetilde{\overline{\mathbf{M}}}$ of case (III_∞). Then $\overline{\mathbf{M}}$ must be in an infinite case, and consequently it is unique by α) of Lemma 3.1.6.

Observe that none of the three cases (I), (II) or (III) is empty. Cf. [8, p. 94, §1, in the introduction].

Lemma 3.2.1. *Consider a genus* $\widetilde{\overline{\mathbf{M}}}$ *which contains types* $\overline{\mathbf{M}}$ *in finite cases, i.e. one in the cases (I), (II) (cf. Theorem IX above). Then the fundamental group* $\mathfrak{G}(\overline{\mathbf{M}})$ *is a notion concerning the genus* $\widetilde{\overline{\mathbf{M}}}$*, i.e. it is the same for all the types* $\overline{\mathbf{M}}$ *of the genus which are in a finite case.*

Proof. Immediate by δ) in Lemma 3.1.6 and Lemma 2.10.2.

We shall therefore denote the fundamental group from now on by $\mathfrak{G}(\widetilde{\overline{\mathbf{M}}})$. Some simple properties of $\mathfrak{G}(\widetilde{\overline{\mathbf{M}}})$ follow.

Lemma 3.2.2. *Consider a genus* $\widetilde{\overline{\mathbf{M}}}$ *as above. Form the ring* $\mathbf{M}$ *and two closed linear sets* $\mathfrak{M}, \mathfrak{N}\ \eta\ \mathbf{M}$*, which are both* $\neq (0)$ *and of finite relative dimension. Then we have*

(i)

$$\overline{\mathbf{M}_{(\mathfrak{M})}} = \overline{\mathbf{M}_{(\mathfrak{N})}}^{D_{\mathbf{M}}(\mathfrak{M})/D_{\mathbf{M}}(\mathfrak{N})};$$

(ii)

$$\overline{\mathbf{M}_{(\mathfrak{M})}} = \overline{\mathbf{M}_{(\mathfrak{N})}} \quad \textit{if and only if} \quad \frac{D_{\mathbf{M}}(\mathfrak{M})}{D_{\mathbf{M}}(\mathfrak{N})} \in \mathfrak{G}(\widetilde{\overline{\mathbf{M}}}).$$

Proof. Ad (i). Suppose first that $D_{\mathbf{M}}(\mathfrak{M}) \leq D_{\mathbf{M}}(\mathfrak{N})$. Here $D_{\mathbf{M}}(\mathfrak{M}) = D_{\mathbf{M}}(\mathfrak{N}')$ for a $\mathfrak{N}' \subseteq \mathfrak{N}$, and since by Lemma 2.8.1 $\overline{\mathbf{M}_{(\mathfrak{N}')}} = \overline{\mathbf{M}_{(\mathfrak{M})}}$, we may suppose $\mathfrak{M} \subseteq \mathfrak{N}$. If $\mathfrak{P}$ is a set η $\mathbf{M}$ and $\subseteq \mathfrak{N}$, $D_{\mathbf{M}}(\mathfrak{P})$ is a dimension function for $\mathbf{M}_{(\mathfrak{N})}$. Since $\mathfrak{M} \subseteq \mathfrak{N}$, by definition

$$\overline{\mathbf{M}_{(\mathfrak{M})}} = \left(\overline{\mathbf{M}_{(\mathfrak{N})}}\right)^{\alpha}, \qquad \alpha = \frac{D_{\mathbf{M}}(\mathfrak{M})}{D_{\mathbf{M}}(\mathfrak{N})}.$$

(Cf. §2.8; $\mathfrak{H}, \mathbf{M}$ are replaced by $\mathfrak{N}, \mathbf{M}_{(\mathfrak{N})}$.)

On the other hand, if $D_{\mathbf{M}}(\mathfrak{M}) > D_{\mathbf{M}}(\mathfrak{N})$, the above argument yields

$$\left(\overline{\mathbf{M}_{(\mathfrak{M})}}\right)^{1/\alpha} = \overline{\mathbf{M}_{(\mathfrak{N})}}.$$

Equation (2.9.β) can now be used to show that $\overline{\mathbf{M}_{(\mathfrak{M})}} = \left(\overline{\mathbf{M}_{(\mathfrak{N})}}\right)^{\alpha}$, suitable use of Lemma 2.10.3 and its corollary being made in the discrete case.

Ad (ii). By the definition of $\mathfrak{G}(\overline{\mathbf{M}_{(\mathfrak{N})}})$, (i) implies $\overline{\mathbf{M}_{(\mathfrak{M})}} = \overline{\mathbf{M}_{(\mathfrak{N})}}$ if and only if $\alpha \in \mathfrak{G}(\overline{\mathbf{M}_{(\mathfrak{N})}})$. But $\mathfrak{G}(\overline{\mathbf{M}_{(\mathfrak{N})}}) = \mathfrak{G}(\widetilde{\overline{\mathbf{M}}})$ by Lemma 3.2.1.

Lemma 3.2.3. *Consider a genus* $\widetilde{\overline{\mathbf{M}}}$ *as above. Then it contains precisely one type in a finite case if and only if* $\mathfrak{G}(\widetilde{\overline{\mathbf{M}}}) = P$ *(cf. Theorem VIII).*

Proof. Sufficiency: Immediate by δ) in Lemma 3.1.6.

Necessity: Assume $\mathfrak{G}(\widetilde{\overline{\mathbf{M}}}) \neq P$. Consider an $\overline{\mathbf{M}}$ of this genus in a finite case. If we find a $\theta \notin \mathfrak{G}(\widetilde{\overline{\mathbf{M}}})$ which is in the range of Theorem VII, then $\overline{\mathbf{M}}^{\theta}$ has the same genus, is also in a finite case and yet $\neq \overline{\mathbf{M}}$, thus completing the proof.

Now in case (I), $\mathfrak{G}(\widetilde{\overline{\mathbf{M}}}) = (1)$ by Lemma 2.10.4. Hence any $\theta = 2, 3, \ldots$ will do. And in case (II), P is the range of Theorem VII, so $\mathfrak{G}(\widetilde{\overline{\mathbf{M}}}) \neq P$ guarantees the existence of the desired θ.

§3.3 The moment has now come when it is opportune to introduce the notion of a spatial type (cf. footnote 12). We shall call the abstraction of $\mathbf{M}$ with respect to spatial isomorphism its spatial type. Thus two rings $\mathbf{M}_1$ and $\mathbf{M}_2$, in two Hilbert spaces $\mathfrak{H}_1$ and $\mathfrak{H}_2$, respectively, will be said to have the same spatial type if and only if they are spatially isomorphic. We denote the spatial type of the ring $\mathbf{M}$ by $\widetilde{\mathbf{M}}$.

We shall now answer Question II in §2 in the introduction, for the cases (I) and (II)[18]; i.e. determine when an algebraic isomorphism determines a spatial one. Thus we shall establish the relationship of $\widetilde{\mathbf{M}}$ to $\overline{\mathbf{M}}$. For the cases (I) and (II) we shall determine all spatial types $\widetilde{\mathbf{M}}$ in terms of the algebraical types $\overline{\mathbf{M}}$.

The notions $\overline{\mathbf{M}}^{\theta}$, $\mathfrak{G}(\widetilde{\overline{\mathbf{M}}})$, as well as the genus $\widetilde{\overline{\mathbf{M}}}$, will be basic in these considerations.

A spatial isomorphism of $\mathbf{M}_1$ in $\mathfrak{H}_1$ and $\mathbf{M}_2$ in $\mathfrak{H}_2$ (cf. Definition A in §2 in the Introduction) will also carry $\mathbf{M}_1'$ into $\mathbf{M}_2'$. So the spatial type $\widetilde{\mathbf{M}}$ of a ring $\mathbf{M}$ determines not only the algebraical type $\overline{\mathbf{M}}$, but also the algebraical type $\overline{\mathbf{M}'}$. Now $\overline{\mathbf{M}}$ does not determine $\overline{\mathbf{M}'}$ uniquely, not even for a factor $\mathbf{M}$[19]. Consequently a discussion of the spatial type $\widetilde{\mathbf{M}}$ of a factor $\mathbf{M}$ must begin with an investigation of the relation between $\overline{\mathbf{M}}$ and $\overline{\mathbf{M}'}$. At this point the notion of genus comes in.

Lemma 3.3.1. *For every factor* $\mathbf{M}$ *in the cases (I) and (II),*

$$\overline{\mathbf{M}'} = \overline{\mathbf{M}}^{c}.$$

Proof. Due to our assumptions concerning $\mathbf{M}$, there exists a closed linear set $\mathfrak{M} \,\eta\, \mathbf{M}$ which is not $= (0)$ and has a finite $D_{\mathbf{M}}(\mathfrak{M})$. Choose an $f \in \mathfrak{M}$, with $f \neq 0$, and form $\mathfrak{M}_f^{\mathbf{M}'}$, $\mathfrak{M}_f^{\mathbf{M}}$ by [5, p. 143, Def. 5.1.1]. These are closed linear sets, clearly η $\mathbf{M}$ and

[18] In the cases (I) of course the answer is known. Nevertheless we include them for the reason given in footnote 1.

[19] The direct factors described in [5, p. 139, or p. 173, Lemma 8.6.1] give elementary examples of this. The exhaustive results in this respect are contained in the last part of our Theorem X.

η $\mathbf{M}'$, respectively, both $\neq (0)$, and $\mathfrak{M}_f^{\mathbf{M}'} \subseteq \mathfrak{M}$, since $f \in \mathfrak{M} \ \eta\ \mathbf{M}$. Hence $D_{\mathbf{M}}(\mathfrak{M}_f^{\mathbf{M}'})$ is finite.

We now normalize $D_{\mathbf{M}}$ so as to make $D_{\mathbf{M}}(\mathfrak{M}_f^{\mathbf{M}'}) = 1$, and then $D_{\mathbf{M}'}$ so as to make $C = 1$ for the C of [5, p. 182, Theorem X]. Hence

$$D_{\mathbf{M}}(\mathfrak{M}_f^{\mathbf{M}'}) = D_{\mathbf{M}'}(\mathfrak{M}_f^{\mathbf{M}}) = 1. \tag{3.3.α}$$

For the sake of brevity we write

$$\mathfrak{M}_1 = \mathfrak{M}_f^{\mathbf{M}'}, \qquad \mathfrak{M}_1' = \mathfrak{M}_f^{\mathbf{M}}. \tag{3.3.β}$$

Let us now use the considerations of [5, pp. 188–190, §11.4]. We form, in accord with them, $\mathbf{M}_{(\mathfrak{M}_1 \cdot \mathfrak{M}_1')}$, $\mathbf{M}'_{(\mathfrak{M}_1 \cdot \mathfrak{M}_1')}$. These are coupled factors in $\mathfrak{M}_1 \cdot \mathfrak{M}_1' \neq (0)$, just as $\mathbf{M}, \mathbf{M}'$ are in $\mathfrak{H}$. (Cf. the remarks, loc. cit., at the beginning of §11.4, and immediately preceding Lemma 11.4.3.) We can use $D_{\mathbf{M}}$ and $D_{\mathbf{M}'}$ to obtain relative dimension functions in $\mathbf{M}_{(\mathfrak{M}_1 \cdot \mathfrak{M}_1')}$, $\mathbf{M}'_{(\mathfrak{M}_1 \cdot \mathfrak{M}_1')}$, respectively, as described in Lemma 11.4.2, loc. cit. Our equations (3.3.α) and (3.3.β) are now seen to imply:

First, the relative dimension of $\mathfrak{M}_1 \cdot \mathfrak{M}_1'$ is 1 for $\mathbf{M}_{(\mathfrak{M}_1 \cdot \mathfrak{M}_1')}$ and for $\mathbf{M}'_{(\mathfrak{M}_1 \cdot \mathfrak{M}_1')}$. Hence both factors are in a finite case. Besides $C = 1$ (cf. above). Hence they are either both in a case (I_n), $n = 1, 2, \ldots$, with $1/n$ times the standard normalization, or both in case (II_1), with the standard normalization.[20]

Second, they fulfill the requirements of [6, p. 235, beginning of §4.1], on which the Theorem VI on p. 239 eod. is based.[21] Now this theorem states that the coupled factors to which it applies are dual isomorphic.[22] Again, by [5, p. 188, Lemma 11.4.1], $\mathbf{M}_{(\mathfrak{M}_1)}$, $\mathbf{M}_{(\mathfrak{M}_1 \cdot \mathfrak{M}_1')}$ are algebraically ring isomorphic and, similarly, $\mathbf{M}'_{(\mathfrak{M}_1')}$, $\mathbf{M}'_{(\mathfrak{M}_1 \cdot \mathfrak{M}_1')}$. Thus $\mathbf{M}_{(\mathfrak{M}_1)}$, $\mathbf{M}'_{(\mathfrak{M}_1')}$ are dual isomorphic, i.e.

$$\overline{\mathbf{M}'_{(\mathfrak{M}_1')}} = \overline{\mathbf{M}_{(\mathfrak{M}_1)}}^{\,c}. \tag{3.3.γ}$$

Consequently,

$$\overline{\mathbf{M}'} = \overline{\mathbf{M}'_{(\mathfrak{M}_1')}} = \overline{\mathbf{M}_{(\mathfrak{M}_1)}}^{\,c} = \overline{\mathbf{M}}^{\,c},$$

as desired.

[20] This discrepancy is of course due to the different ways in which we defined the standard normalization in the cases (I) and (II) (cf. [5, p. 172, Theorem VIII]).

[21] The theorem referred to is stated for the case (II_1) only. It holds however for the cases (I_n), $n = 1, 2, \ldots$, too, and with precisely the same proof. Our remark in footnote 1 could have been applied to [6] also.

[22] Called anti-isomorphic, loc. cit.

This lemma determines the direction of further analysis of our present topic. It is easy to obtain now more detailed information.

For this purpose we define

Definition 3.3.1. *For every factor* $\mathbf{M}$ *in the cases (I) and (II) we form the number*

$$\theta = \frac{1}{C}\frac{D_{\mathbf{M}'}(\mathfrak{H})}{D_{\mathbf{M}}(\mathfrak{H})} \tag{3.3.δ}$$

with any normalization of $D_{\mathbf{M}}, D_{\mathbf{M}'}$*, and the corresponding* C *of [5, p. 182, Theorem X].*

The expression for θ is clearly independent of the normalization of $D_{\mathbf{M}}, D_{\mathbf{M}'}$. Since $0 < C < \infty$, $0 < D_{\mathbf{M}}(\mathfrak{H}), D_{\mathbf{M}'}(\mathfrak{H}) \leq \infty$, (3.3.$\delta$) characterizes θ as a well-defined number with

$$0 \leq \theta \leq \infty, \tag{3.3.ϵ}$$

except when $D_{\mathbf{M}}(\mathfrak{H}) = D_{\mathbf{M}'}(\mathfrak{H}) = \infty$, i.e. when $\mathbf{M}, \mathbf{M}'$ both belong to infinite cases. If this happens, we construe (3.3.δ) to mean

$$\theta = \frac{\infty}{\infty}, \tag{3.3.ζ}$$

the symbol ∞/∞ being considered as an entity different from all numbers of (3.3.ϵ).

Lemma 3.3.2. *Let* $\mathbf{M}$ *and* θ *be as above. Then we have*

α) If $\mathbf{M}, \mathbf{M}'$ *are both in infinite cases, then*

$$\overline{\mathbf{M}'} = \overline{\mathbf{M}}^{c} \quad \textit{and} \quad \theta = \frac{\infty}{\infty}.$$

β) If $\mathbf{M}$ *is in an infinite case, and* $\mathbf{M}'$ *is in a finite case, then*

$$\overline{\mathbf{M}} = \left(\overline{\mathbf{M}'}^{c}\right)^{\infty} \quad \textit{and} \quad \theta = 0, \quad \textit{so } 1/\theta = \infty.$$

γ) If $\mathbf{M}$ *is in a finite case and* $\mathbf{M}'$ *is in an infinite case, then*

$$\overline{\mathbf{M}'} = \left(\overline{\mathbf{M}}^{c}\right)^{\infty} \quad \textit{and} \quad \theta = \infty.$$

δ) If $\mathbf{M}, \mathbf{M}'$ *are both in finite cases, then*

$$\overline{\mathbf{M}'} = \left(\overline{\mathbf{M}}^{c}\right)^{\theta}, \quad \textit{or equivalently} \quad \overline{\mathbf{M}} = \left(\overline{\mathbf{M}'}^{c}\right)^{1/\theta}, \qquad 0 < \theta < \infty.$$

Proof. All assertions concerning θ are immediate by Definition 3.3.1. Let us now consider the other statements.

Ad α), β), γ). Lemma 3.3.1 states that $\overline{\mathbf{M}'}$ and $\overline{\mathbf{M}}^{c}$ are commensurable. Hence our α), β), γ) follow from α), β), γ) in Lemma 3.1.6, respectively.

Ad δ). The equivalence of these two equations is obvious. We consider the first one.

We choose the normalization of $D_{\mathbf{M}}, D_{\mathbf{M}'}$ exactly as in the proof of Lemma 3.3.1. Now (3.3.α), (3.3.β) in that proof give

$$\overline{\mathbf{M}_{(\mathfrak{M}_1)}} = \overline{\mathbf{M}}^{1/D_{\mathbf{M}}(\mathfrak{H})}, \qquad \overline{\mathbf{M}'_{(\mathfrak{M}'_1)}} = \overline{\mathbf{M}'}^{1/D_{\mathbf{M}'}(\mathfrak{H})}.$$

Then (3.3.γ) and (2.9.α) yield

$$\overline{\mathbf{M}'}^{1/D_{\mathbf{M}'}(\mathfrak{H})} = \left(\overline{\mathbf{M}}^{1/D_{\mathbf{M}}(\mathfrak{H})}\right)^{c} = \left(\overline{\mathbf{M}}^{c}\right)^{1/D_{\mathbf{M}}(\mathfrak{H})},$$

or

$$\overline{\mathbf{M}'} = \left(\overline{\mathbf{M}}^{c}\right)^{D_{\mathbf{M}'}(\mathfrak{H})/D_{\mathbf{M}}(\mathfrak{H})} = \left(\overline{\mathbf{M}}^{c}\right)^{\theta},$$

as desired.

Remark. The first formula of δ) holds clearly for γ) as well, while it becomes meaningless for α), β). The second formula of δ) holds clearly for β) as well, while it becomes meaningless for α), γ).

It is furthermore apparent from our proof that for δ) the exponents $\theta, 1/\theta$ are in the range of Theorem VII.

Lemma 3.3.3. *Let* $\mathbf{M}$ *and* θ *be as above. Then the spatial type* $\widetilde{\mathbf{M}}$ *is uniquely determined by the algebraical types* $\overline{\mathbf{M}}, \overline{\mathbf{M}'}$ *and the number* θ.

Proof. As in the introduction, §2, for each $i = 1, 2$ a Hilbert space $\mathfrak{H}_i$ and an operator ring $\mathbf{M}_i$ in $\mathfrak{H}_i$ must be given. We assume that both $\mathbf{M}_i$ are factors in case (I) or (II), and form for each $\mathbf{M}_i$ its θ_i in the sense of Definition 3.3.1. We assume further that

$$\overline{\mathbf{M}_1} = \overline{\mathbf{M}_2}, \qquad \overline{\mathbf{M}'_1} = \overline{\mathbf{M}'_2}, \qquad \theta_1 = \theta_2 = \theta.$$

Then our task is to establish $\widetilde{\mathbf{M}}_1 = \widetilde{\mathbf{M}}_2$, i.e. to find an isomorphic mapping of $\mathfrak{H}_1$ on $\mathfrak{H}_2$ (cf. A) in the Introduction, §2) which carries $\mathbf{M}_1$ into $\mathbf{M}_2$.

In proving this we must distinguish several alternatives corresponding to various values of θ (the joint value of θ_i).

Observe first that the isomorphic mapping of $\mathfrak{H}_1$ on $\mathfrak{H}_2$ which carries $\mathbf{M}_1$ into $\mathbf{M}_2$ also carries $\mathbf{M}'_1$ into $\mathbf{M}'_2$. Thus we can always replace each $\mathbf{M}_i$ by its $\mathbf{M}'_i$. This has the consequence that $\theta_1 = \theta_2 = \theta$ is replaced by $1/\theta_1 = 1/\theta_2 = 1/\theta$.

Let us now consider the alternatives for θ.

First alternative: $0 < \theta < \infty$. Since we may replace θ by $1/\theta$, we can even assume that $0 < \theta \leq 1$. For both $i = 1, 2$, the factors $\mathbf{M}_i, \mathbf{M}'_i$ are in finite cases. Choose the normalizations of both $D_{\mathbf{M}_i}, D_{\mathbf{M}'_i}$ so that $D_{\mathbf{M}'_i}(\mathfrak{H}_i) = 1$, $C_i = 1$ (we use the notation of Definition 3.3.1 for both $i = 1, 2$). So $D_{\mathbf{M}_i}(\mathfrak{H}_i) = 1/\theta_i = 1/\theta \geq 1$. Hence 1 belongs to the range of $D_{\mathbf{M}_i}$ (cf. [5, p. 182, the discussion of the ranges of $\Delta, \Delta_0, \Delta', \Delta'_0$ in Theorem X and preceding it]). Choose accordingly a $\mathfrak{K}_i \ \eta \ \mathbf{M}_i$, with $D_{\mathbf{M}_i}(\mathfrak{K}_i) = 1$. There exists furthermore an ϵ in the range of $D_{\mathbf{M}_1}$ for which there is a $p = 1, 2, \ldots$, with $0 < \epsilon \leq 1$, $p\epsilon = 1/\theta$.[23] Choose accordingly a $\mathfrak{L}_1 \ \eta \ \mathbf{M}_1$, with $D_{\mathbf{M}_1}(\mathfrak{L}_1) = \epsilon$, $\mathfrak{L}_1 \subseteq \mathfrak{K}_1$.

Let us now use the considerations of [5, pp. 188–190, §11.4]. We form, in accord with them, $\mathbf{M}_{i(\mathfrak{K}_i)}, \mathbf{M}'_{i(\mathfrak{K}_i)}$. (We choose $\mathfrak{H}_i, \mathfrak{K}_i, \mathfrak{K}_i$ and $\mathbf{M}_i, \mathbf{M}'_i$ for the $\mathfrak{H}, \mathfrak{M}, \mathfrak{M}'$ and $\mathbf{M}, \mathbf{M}'$, loc. cit.) These are coupled factors in $\mathfrak{K}_i \neq (0)$, just as $\mathbf{M}_i, \mathbf{M}'_i$ are in $\mathfrak{H}_i$. We have

$$\overline{\mathbf{M}'_{i(\mathfrak{K}_i)}} = \overline{\mathbf{M}'_i}.$$

Hence $\overline{\mathbf{M}'_1} = \overline{\mathbf{M}'_2}$ gives

$$\overline{\mathbf{M}'_{1(\mathfrak{K}_1)}} = \overline{\mathbf{M}'_{2(\mathfrak{K}_2)}}.$$

Besides,

$$D_{\mathbf{M}_{i(\mathfrak{K}_i)}}(\mathfrak{K}_i) = D_{\mathbf{M}_i}(\mathfrak{K}_i) = 1,$$
$$D_{\mathbf{M}'_{i(\mathfrak{K}_i)}}(\mathfrak{K}_i) = D_{\mathbf{M}'_i}(\mathfrak{H}_i) = 1,$$

and $C_i = 1$, as before. Hence [6, p. 244, Theorem XI] applies with our $\mathfrak{K}_i, \mathbf{M}_{i(\mathfrak{K}_i)}, \mathbf{M}'_{i(\mathfrak{K}_i)}$ in place of its $\mathfrak{H}_i, \mathbf{M}_i, \mathbf{M}'_i$. There exists an isomorphic mapping $\mathfrak{I}$ of $\mathfrak{K}_1$ on $\mathfrak{K}_2$ which carries $\mathbf{M}_{1(\mathfrak{K}_1)}$ into $\mathbf{M}_{2(\mathfrak{K}_2)}$, and $\mathbf{M}'_{1(\mathfrak{K}_1)}$ into $\mathbf{M}'_{2(\mathfrak{K}_2)}$. $\mathfrak{L}_1 \ \eta \ \mathbf{M}_1$, $\mathfrak{L}_1 \subseteq \mathfrak{K}_1$ imply $\mathfrak{L}_1 \ \eta \ \mathbf{M}_{1(\mathfrak{K}_1)}$. So $\mathfrak{I}$ carries $\mathfrak{L}_1$ into an $\mathfrak{L}_2 \ \eta \ \mathbf{M}_{2(\mathfrak{K}_2)}$. It follows that $\mathfrak{L}_2 \ \eta \ \mathbf{M}_2$ and $\mathfrak{L}_2 \subseteq \mathfrak{K}_2$. Now

$$D_{\mathbf{M}_2}(\mathfrak{L}_2) = D_{\mathbf{M}_{2(\mathfrak{K}_2)}}(\mathfrak{L}_2) = D_{\mathbf{M}_{1(\mathfrak{K}_1)}}(\mathfrak{L}_1) = D_{\mathbf{M}_1}(\mathfrak{L}_1) = \epsilon.$$

$\mathfrak{I}$ carries $\mathbf{M}_{1(\mathfrak{K}_1)(\mathfrak{L}_1)} = \mathbf{M}_{1(\mathfrak{L}_1)}$ into $\mathbf{M}_{2(\mathfrak{K}_2)(\mathfrak{L}_2)} = \mathbf{M}_{2(\mathfrak{L}_2)}$. Restrict $\mathfrak{I}$ from $\mathfrak{K}_1$ to $\mathfrak{L}_1$, and denote this restricted mapping by $\mathfrak{F}$. Then $\mathfrak{F}$ is an isomorphic mapping of $\mathfrak{L}_1$ on $\mathfrak{L}_2$, which carries $\mathbf{M}_{1(\mathfrak{L}_1)}$ into $\mathbf{M}_{2(\mathfrak{L}_2)}$.

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[23] The discussion of [5, p. 182] shows that the ranges of $D_{\mathbf{M}_1}$ and $D_{\mathbf{M}'_1}$ agree for the area $\leq$ both their maxima, i.e. $1, 1/\theta$. Hence we can argue as follows:

Case (I_n), $n = 1, 2, \ldots$: The range of $D_{\mathbf{M}_1}$ is $(0, 1/n, \ldots, 1)$; now $1/\theta \geq 1/n$, and $1/n$ is the smallest positive element of the range of $D_{\mathbf{M}'_1}$. Hence that range contains only integer multiples of $1/n$. So $1/\theta = p/n$, $p = 1, 2, \ldots$. Thus $\epsilon = 1/n$ meets our requirements.

Case (II_1). Choose $p = 1, 2, \ldots$ so that $1/(p\theta) \leq 1$. Then $\epsilon = 1/(p\theta)$ fulfills $p\epsilon = 1/\theta$, and besides $\epsilon \leq 1, 1/\theta$. Since ϵ belongs to the range of $D_{\mathbf{M}_1}$, it belongs to the range of $D_{\mathbf{M}'_1}$ also.

We have $p\epsilon = 1/\theta$, and therefore for both $i = 1, 2$,

$$pD_{\mathbf{M}_i}(\mathfrak{L}_i) = D_{\mathbf{M}_i}(\mathfrak{H}_i),$$

or, if we introduce the projection $E_i = P_{\mathfrak{L}_i}$, then

$$D_{\mathbf{M}_i}(E_i) = \frac{1}{p} D_{\mathbf{M}_i}(1).$$

Consequently we can proceed as in the proof of Lemma 2.6.2 or Lemma 2.6.3 to obtain a set of p-order matrix units $W_{i;u,v}$, $u, v = 1, \ldots, p$, with $W_{i;1,1} = E_i$, $\sum_{u=1}^{p} W_{i;u,u} = 1$. Let the projection $F_{i;u} = W_{i;u,u}$ have range $\mathfrak{N}_{i;u}$. We note that $\mathfrak{N}_{i;1} = \mathfrak{L}_i$, and that $W_{i;u,v}$ is partially isometric with initial set $\mathfrak{N}_{i;v}$ and final set $\mathfrak{N}_{i;u}$, i.e. it is an isomorphic mapping of $\mathfrak{N}_{i;v}$ on $\mathfrak{N}_{i;u}$.

Thus $W_{2;u,1}\mathfrak{F}W_{1;1,u} = \mathfrak{F}_u$ is an isomorphic mapping of $\mathfrak{N}_{1;u}$ on $\mathfrak{N}_{2;u}$, for $u = 1, 2, \ldots, p$. For $u = 1$, $W_{i;1,1} = F_{i;1} = E_i = P_{\mathfrak{L}_i}$, $\mathfrak{N}_{i;1} = \mathfrak{L}_i$, and hence $\mathfrak{F}_1$ coincides with $\mathfrak{F}$. Since the $\mathfrak{N}_{i;u}$, $u = 1, \ldots, p$, are pairwise orthogonal and span together the closed linear set $\mathfrak{H}_i$, we can combine the $\mathfrak{F}_u$, $u = 1, \ldots, p$, to one isomorphic mapping $\mathfrak{F}^*$ of $\mathfrak{H}_1$ on $\mathfrak{H}_2$. On $\mathfrak{L}_1$, this $\mathfrak{F}^*$ coincides with $\mathfrak{F}_1 = \mathfrak{F}$. Hence

$$\mathfrak{F}^* \text{ carries } \mathfrak{L}_1 \text{ into } \mathfrak{L}_2 \text{ and in it } \mathbf{M}_{1(\mathfrak{L}_1)} \text{ into } \mathbf{M}_{2(\mathfrak{L}_2)}. \tag{3.3.η}$$

By virtue of the properties of $W_{i;u,v}$ (cf. Definition 2.6.1), and owing to the definition of $\mathfrak{F}^*$, we have further

$$\mathfrak{F}^* \text{ carries } W_{1;u,v} \text{ into } W_{2;u,v}, \qquad u, v = 1, \ldots, p. \tag{3.3.ϑ}$$

Now $\mathbf{M}_i$ can be characterized in terms of $\mathbf{M}_{i(\mathfrak{L}_i)}$, $\mathfrak{L}_i$, and the $W_{i;u,v}$, $u, v = 1, \ldots, p$. Indeed: $A_i \in \mathbf{M}_i$ is equivalent to

$$\left(W_{i;1,v} A_i W_{i;u,1}\right)_{(\mathfrak{L}_i)} \in \mathbf{M}_{i(\mathfrak{L}_i)} \quad \text{for all } u, v = 1, \ldots, p.$$ [24]

Consequently (3.3.η) and (3.3.ϑ) imply that $\mathfrak{F}^*$ carries $\mathbf{M}_1$ into $\mathbf{M}_2$.

This completes the proof of the first alternative.

Second alternative: $\theta = 0$ or ∞. Since we may replace θ by $1/\theta$, it is clear that we need only consider the case $\theta = 0$. For both $i = 1, 2$, the factor $\mathbf{M}_i$ is in an

[24] Put $A_{i;u,v} = W_{i;1,v} A_i W_{i;u,1}$. The forward implication is obvious; the reverse follows from the formula

$$A_i = \sum_{u,v=1}^{p} W_{i;u,1} A_{i;u,v} W_{i;1,v},$$

which is easily verified.

infinite case and the factor $\mathbf{M}'_i$ is in a finite case. Choose the normalization of $D_{\mathbf{M}_i}, D_{\mathbf{M}'_i}$ so that

$$D_{\mathbf{M}'_i}(\mathfrak{H}_i) = 1, \qquad C_i = 1.$$

We may parallel our discussion of the first alternative, however with $\epsilon = 1$, $p = \infty$. We form the $\mathfrak{K}_i$ accordingly. Since $\epsilon = 1$, $\mathfrak{L}_1 = \mathfrak{K}_1$. $\mathfrak{I}$ obtains as loc. cit.; $\mathfrak{L}_1 = \mathfrak{K}_1$ yields $\mathfrak{L}_2 = \mathfrak{K}_2$ and $\mathfrak{F} = \mathfrak{I}$.

Let $E_i = P_{\mathfrak{L}_i}$. We proceed to obtain an infinite number of pairwise orthogonal manifolds $\mathfrak{N}_{i;1}, \mathfrak{N}_{i;2}, \ldots$, with $\mathfrak{N}_{i;1} = \mathfrak{L}_i$, $\mathfrak{N}_{i;u}\ \eta\ \mathbf{M}_i$, $D_{\mathbf{M}_i}(\mathfrak{N}_{i;u}) = 1$, for $u = 1, 2, \ldots$, and finally $\mathfrak{H}_i = [\mathfrak{N}_{i;1}, \mathfrak{N}_{i;2}, \ldots]$ (cf. [5, p. 155, Lemma 7.1.2]). We obtain the $W_{i;u,v}$ as in the second part of the proof of Lemma 2.6.2.

As in the discussion of the preceding alternative, we now obtain $\mathfrak{F}_u$, $u = 1, 2, \ldots$, and finally $\mathfrak{F}^*$; and by a literal repetition of that argument, the isomorphic mapping $\mathfrak{F}^*$ of $\mathfrak{H}_1$ on $\mathfrak{H}_2$ is seen to carry $\mathbf{M}_1$ into $\mathbf{M}_2$.[25]

Thus the second alternative too is settled.

Third alternative: $\theta = \infty/\infty$. For both $i = 1, 2$, the factors $\mathbf{M}_i, \mathbf{M}'_i$ are in infinite cases. Choose a finite $\mathfrak{K}_1\ \eta\ \mathbf{M}_1$, $\mathfrak{K}_1 \neq (0)$, and put $P_{\mathfrak{K}_1} = E_1 \in \mathbf{M}_1$. Under the (algebraic ring) isomorphism of $\mathbf{M}_1, \mathbf{M}_2$ (due to $\overline{\mathbf{M}_1} = \overline{\mathbf{M}_2}$), this projection $E_1 \in \mathbf{M}_1$ corresponds to a projection $E_2 \in \mathbf{M}_2$. Let $\mathfrak{K}_2$ be the range of E_2. Then $\mathfrak{K}_2\ \eta\ \mathbf{M}_2$, $\mathfrak{K}_2 \neq (0)$. Then the isomorphism of $\mathbf{M}_1, \mathbf{M}_2$ also induces one of $\mathbf{M}_{1(\mathfrak{K}_1)}, \mathbf{M}_{2(\mathfrak{K}_2)}$, owing to [5, p. 187, Lemma 11.3.3(i)]. Thus

$$\overline{\mathbf{M}_{1(\mathfrak{K}_1)}} = \overline{\mathbf{M}_{2(\mathfrak{K}_2)}}. \tag{3.3.ι}$$

Also by (ii) eod., $\overline{\mathbf{M}'_{i(\mathfrak{K}_i)}} = \overline{\mathbf{M}'_i}$; hence $\overline{\mathbf{M}'_1} = \overline{\mathbf{M}'_2}$ gives

$$\overline{\mathbf{M}'_{1(\mathfrak{K}_1)}} = \overline{\mathbf{M}'_{2(\mathfrak{K}_2)}}. \tag{3.3.κ}$$

Now choose the normalization of $D_{\mathbf{M}_i}, D_{\mathbf{M}'_i}$ so that $D_{\mathbf{M}_i}(\mathfrak{K}_i) = 1$, $C_i = 1$. Automatically $D_{\mathbf{M}'_i}(\mathfrak{H}_i) = \infty$. We may then write

$$D_{\mathbf{M}_{i(\mathfrak{K}_i)}}(\mathfrak{K}_i) = D_{\mathbf{M}_i}(\mathfrak{K}_i) = 1,$$
$$D_{\mathbf{M}'_{i(\mathfrak{K}_i)}}(\mathfrak{K}_i) = D_{\mathbf{M}'_i}(\mathfrak{H}_i) = \infty,$$

and $C_i = 1$. Apply Definition 3.3.1 to $\mathfrak{K}_i, \mathbf{M}_{i(\mathfrak{K}_i)}, \mathbf{M}'_{i(\mathfrak{K}_i)}$ in place of $\mathfrak{H}, \mathbf{M}, \mathbf{M}'$, and denote the resulting number by θ_i^0. The above equations give

$$\theta_1^0 = \theta_2^0 = \infty. \tag{3.3.λ}$$

[25] Footnote 24 applies to this case too. The convergence of $\sum_{u,v=1}^{\infty}$ in footnote 24 is readily seen to be a consequence of the convergence of the sum $\sum_{u=1}^{\infty} W_{i;u,u}$. And in connection with the latter convergence, the remarks made in the proof of (iii) in Lemma 2.6.1 apply again.

Equations (3.3.ι), (3.3.κ), and (3.3.λ) permit us to apply the second alternative, which we have already disposed of, to $\mathfrak{K}_i, \mathbf{M}_{i(\mathfrak{K}_i)}, \mathbf{M}'_{i(\mathfrak{K}_i)}$. There exists an isomorphic mapping $\mathfrak{I}$ of $\mathfrak{K}_1$ on $\mathfrak{K}_2$, which carries $\mathbf{M}_{1(\mathfrak{K}_1)}$ into $\mathbf{M}_{2(\mathfrak{K}_2)}$, and $\mathbf{M}'_{1(\mathfrak{K}_1)}$ into $\mathbf{M}'_{2(\mathfrak{K}_2)}$.

We can continue from here on exactly as in the discussion of the second alternative, i.e. paralleling the considerations of the first alternative. We have the $\mathfrak{K}_i$ already. Again we put $\epsilon = 1$, $p = \infty$. Since $\epsilon = 1$, $\mathfrak{L}_1 = \mathfrak{K}_1$. We already have $\mathfrak{I}$. $\mathfrak{L}_1 = \mathfrak{K}_1$ gives $\mathfrak{L}_2 = \mathfrak{K}_2$ and $\mathfrak{F} = \mathfrak{I}$. We have $E_i = P_{\mathfrak{L}_i} = P_{\mathfrak{K}_i}$. The $F_{i;u}$, $u = 1, 2, \ldots$, and $W_{i;u,v}$, $u, v = 1, 2, \ldots$, obtain as loc. cit. So do the $\mathfrak{F}_u$, $u = 1, 2, \ldots$, and finally $\mathfrak{F}^*$; and exactly as loc. cit., the isomorphic mapping $\mathfrak{F}^*$ of $\mathfrak{H}_1$ on $\mathfrak{H}_2$ is seen to carry $\mathbf{M}_1$ into $\mathbf{M}_2$.

Thus the third alternative too is settled and the entire proof is completed.

We settle now the question of existence.

Lemma 3.3.4. *A factor* $\mathbf{M}$ *in the cases (I) or (II), with given algebraical types* $\overline{\mathbf{M}}, \overline{\mathbf{M}'}$ *and given number* θ *(cf. Definition 3.3.1), exists if and only if these fulfill the alternative conditions* α)–δ) *in Lemma 3.3.2.*

Proof. Necessity: This is merely the assertion of Lemma 3.3.2.

Sufficiency: We shall continue to denote the two prescribed algebraical types by $\overline{\mathbf{M}}, \overline{\mathbf{M}'}$, although we have not yet identified them as belonging in this way to any coupled factors $\mathbf{M}, \mathbf{M}'$. (In fact, this is what we propose to prove.) The relations α)–δ) in Lemma 3.3.2, which we assumed, imply for the genera

$$\widetilde{\overline{\mathbf{M}'}} = \widetilde{\overline{\mathbf{M}}}^{c}. \tag{3.3.μ}$$

(This cannot be deduced from Lemma 3.3.1, since $\mathbf{M}, \mathbf{M}'$ are fictitious, cf. above.) The genus $\widetilde{\overline{\mathbf{M}}}$ contains a (unique) infinite type $\overline{\mathbf{R}}$ (cf. Theorem IX)—$\mathbf{R}$ being an actual factor in a Hilbert space $\mathfrak{R}$.

Consider now the factor $\mathbf{R}'$ in $\mathfrak{R}$. We repeat the construction in §2.4, immediately preceding Theorem IV′, with our $\mathfrak{R}, \mathbf{R}'$ in place of its $\mathfrak{H}, \mathbf{M}$, and with its $p = \infty$. Thus, using the notations of §2.4, loc. cit., our $\mathbf{R}'$ is a factor in $\mathfrak{R} = \mathfrak{H} = \mathfrak{H}_2$, and with the ($p = \infty$-dimensional) Hilbert space $\mathfrak{H}_1$, we obtain the Hilbert space $\mathfrak{H}_1 \otimes \mathfrak{H}_2$, and in it the ring $\mathbf{M}_1$ of all operators $\langle A'_{t,s} \rangle$, with all $A'_{t,s} \in \mathbf{R}'$ (cf. the similar argument before Theorem IV in §2.4). Put $\mathbf{R}_1 = \mathbf{M}'_1$, i.e. $\mathbf{M}_1 = \mathbf{R}'_1$. It is obvious, by the usual matrix computations, that $\mathbf{R}_1 = \mathbf{M}'_1$ is the set of all operators $\langle A\delta_{t,s} \rangle$, $A \in \mathbf{R}$. Hence $\overline{\mathbf{R}}_1 = \overline{\mathbf{R}}$. On the other hand, the discussions of §2.4 show that $\overline{\mathbf{R}'_1} = \overline{\mathbf{M}}_1 = \overline{\mathbf{R}'}^{\infty}$. Thus $\overline{\mathbf{R}'_1}$ is in an infinite case (by Lemma 2.7.1).

Now $\overline{\mathbf{R}}_1 = \overline{\mathbf{R}}$ permits us to replace $\mathfrak{R}, \mathbf{R}$ by $\mathfrak{H}_1 \otimes \mathfrak{H}_2, \mathbf{R}_1$. So we have shown: It is permissible to assume that $\overline{\mathbf{R}'}$ is in an infinite case. We do this, but we continue using the original notations $\mathfrak{R}, \mathbf{R}$.

By Lemma 3.3.1, $\widetilde{\overline{\mathbf{R}'}} = \widetilde{\overline{\mathbf{R}}}^{c}$. Now $\widetilde{\overline{\mathbf{R}}} = \widetilde{\overline{\mathbf{M}}}$; hence $\widetilde{\overline{\mathbf{R}'}} = \widetilde{\overline{\mathbf{M}}}^{c}$. By (3.3.$\mu$), this means $\widetilde{\overline{\mathbf{R}'}} = \widetilde{\overline{\mathbf{M}'}}$. Since $\overline{\mathbf{R}}, \overline{\mathbf{R}'}$ are both infinite, we have (cf. Theorem IX)

$$\overline{\mathbf{R}}, \overline{\mathbf{R}'} \text{ are the (unique) infinite types of the genera } \widetilde{\overline{\mathbf{M}}}, \widetilde{\overline{\mathbf{M}'}}, \text{ respectively.} \tag{3.3.ν}$$

Choose an $f_0 \neq 0$ in $\mathfrak{R}$, and form the closed linear sets $\mathfrak{M}^{\mathbf{R}'}_{f_0} \ \eta\ \mathbf{R}$, $\mathfrak{M}^{\mathbf{R}}_{f_0} \ \eta\ \mathbf{R}'$ (cf. [5, p. 143, Def. 5.1.1]). We may assume that $\mathfrak{M}^{\mathbf{R}'}_{f_0}$ is finite dimensional. (Choose a finite dimensional $\mathfrak{M} \ \eta\ \mathbf{R}$, $\mathfrak{M} \neq (0)$, and then $f_0 \in \mathfrak{M}$. Clearly $\mathfrak{M}^{\mathbf{R}'}_{f_0} \subseteq \mathfrak{M}$.) In this case, $\mathfrak{M}^{\mathbf{R}}_{f_0}$ is automatically finite dimensional. (Use [5, p. 182, Theorem X], for this and for the next step.) Hence both $D_{\mathbf{R}}(\mathfrak{M}^{\mathbf{R}'}_{f_0})$, $D_{\mathbf{R}'}(\mathfrak{M}^{\mathbf{R}}_{f_0})$ are $> 0, < \infty$, and

$$C = \frac{D_{\mathbf{R}'}(\mathfrak{M}^{\mathbf{R}}_{f_0})}{D_{\mathbf{R}}(\mathfrak{M}^{\mathbf{R}'}_{f_0})}.$$

Now normalize $D_{\mathbf{R}}, D_{\mathbf{R}'}$ so that $D_{\mathbf{R}}(\mathfrak{M}^{\mathbf{R}'}_{f_0}) = D_{\mathbf{R}'}(\mathfrak{M}^{\mathbf{R}}_{f_0})$. The above formula for C gives

$$C = 1. \tag{3.3.o}$$

Write $\mathfrak{M}_1 = \mathfrak{M}^{\mathbf{R}'}_{f_0}$, $\mathfrak{M}'_1 = \mathfrak{M}^{\mathbf{R}}_{f_0}$; then

$$D_{\mathbf{R}}(\mathfrak{M}_1) = D_{\mathbf{R}'}(\mathfrak{M}'_1). \tag{3.3.ξ}$$

We shall now again make use of the considerations of [5, pp. 188–190, §11.4]. In accord with them we form $\mathbf{R}_{(\mathfrak{M}_1 \cdot \mathfrak{M}'_1)}$, $\mathbf{R}'_{(\mathfrak{M}_1 \cdot \mathfrak{M}'_1)}$. These are coupled factors in $\mathfrak{M}_1 \cdot \mathfrak{M}'_1 \neq (0)$, just as $\mathbf{R}, \mathbf{R}'$ are in $\mathfrak{H}$, and (3.3.ξ) gives

$$D_{\mathbf{R}_{(\mathfrak{M}_1 \cdot \mathfrak{M}'_1)}}(\mathfrak{M}_1 \cdot \mathfrak{M}'_1) = D_{\mathbf{R}'_{(\mathfrak{M}_1 \cdot \mathfrak{M}'_1)}}(\mathfrak{M}_1 \cdot \mathfrak{M}'_1).$$

This, together with (3.3.o) and a normalization, shows that the requirements of [6, p. 239, Theorem VI] are fulfilled.[21] Hence $\mathbf{R}_{(\mathfrak{M}_1 \cdot \mathfrak{M}'_1)}$, $\mathbf{R}'_{(\mathfrak{M}_1 \cdot \mathfrak{M}'_1)}$ are dual isomorphic[22], i.e.

$$\overline{\mathbf{R}'_{(\mathfrak{M}_1 \cdot \mathfrak{M}'_1)}} = \overline{\mathbf{R}_{(\mathfrak{M}_1 \cdot \mathfrak{M}'_1)}}^{c}.$$

Now $\overline{\mathbf{R}_{(\mathfrak{M}_1)}} = \overline{\mathbf{R}_{(\mathfrak{M}_1 \cdot \mathfrak{M}'_1)}}$, $\overline{\mathbf{R}'_{(\mathfrak{M}'_1)}} = \overline{\mathbf{R}'_{(\mathfrak{M}_1 \cdot \mathfrak{M}'_1)}}$. Consequently

$$\overline{\mathbf{R}'_{(\mathfrak{M}'_1)}} = \overline{\mathbf{R}_{(\mathfrak{M}_1)}}^{c}. \tag{3.3.π}$$

Since $\overline{\widetilde{\widetilde{\mathbf{R}}}} = \overline{\widetilde{\widetilde{\mathbf{M}}}}$, $\overline{\mathbf{R}}, \overline{\mathbf{M}}$ are commensurable. $\overline{\mathbf{R}}$ is in an infinite case. Hence α), β) in Lemma 3.1.6 and Lemma 3.1.4 give together that $\overline{\mathbf{M}}$ is a divisor of $\overline{\mathbf{R}}$. Hence by Definition 3.1.1,

$$\overline{\mathbf{M}} = \overline{\mathbf{R}_{(\mathfrak{L})}} \quad \text{for a suitable } \mathfrak{L} \ \eta \ \mathbf{R}. \tag{3.3.ρ}$$

If $\mathfrak{L}$ is of finite relative dimension, we may assume by [5, Theorem X] that $\mathfrak{L} = \mathfrak{M}^{\mathbf{R}'}_{f_0}$ for some f_0. Similarly,

$$\overline{\mathbf{M}'} = \overline{\mathbf{R}'_{(\mathfrak{L}')}} \quad \text{for a suitable } \mathfrak{L}' \ \eta \ \mathbf{R}', \quad \mathfrak{L}' = \mathfrak{M}^{\mathbf{R}}_{g_0} \text{ when finite.} \tag{3.3.σ}$$

From here on, it is again convenient to distinguish several alternatives, corresponding to various values of θ.

First alternative: $0 < \theta < \infty$. We have the case δ) in Lemma 3.3.2. Hence $\overline{\mathbf{M}'} = (\overline{\mathbf{M}}^c)^\theta$, or by (3.3.$\rho$) and (3.3.$\sigma$),

$$\overline{\mathbf{R}'_{(\mathfrak{L}')}} = \left(\overline{\mathbf{R}_{(\mathfrak{L})}}^c\right)^\theta .$$

Now (3.3.π) and (i) in Lemma 3.2.2 give

$$\overline{\mathbf{R}'_{(\mathfrak{L}')}} = \left(\overline{\mathbf{R}_{(\mathfrak{L})}}^c\right)^\beta ,$$

where $\beta = D_{\mathbf{R}'}(\mathfrak{L}')/D_{\mathbf{R}}(\mathfrak{L})$. Hence (ii) in Lemma 3.2.2 gives

$$\alpha = \frac{\beta}{\theta} = \frac{1}{\theta}\frac{D_{\mathbf{R}'}(\mathfrak{L}')}{D_{\mathbf{R}}(\mathfrak{L})} \in \mathfrak{G}(\overline{\widetilde{\mathbf{R}}}) = \mathfrak{G}(\overline{\mathbf{R}_{(\mathfrak{L})}}).$$

The interchange of $\overline{\mathbf{M}}$ and $\overline{\mathbf{M}'}$, and with them of θ and $1/\theta$, as well as $\mathbf{R}, \mathfrak{L}$ and $\mathbf{R}', \mathfrak{L}'$, replaces α by $1/\alpha$. So we may assume $\alpha \leq 1$. Hence $\alpha \in \mathfrak{G}(\overline{\mathbf{R}_{(\mathfrak{L})}})$ secures the constructability of $\overline{\mathbf{R}_{(\mathfrak{L})}}^\alpha$ in the original sense of §2.8. Thus there exists an $\mathfrak{L}'' \subseteq \mathfrak{L}$, with $\mathfrak{L}'' \ \eta \ \mathbf{R}_{(\mathfrak{L})}$, and

$$\alpha = \frac{D_{\mathbf{R}_{(\mathfrak{L})}}(\mathfrak{L}'')}{D_{\mathbf{R}_{(\mathfrak{L})}}(\mathfrak{L})} = \frac{D_{\mathbf{R}}(\mathfrak{L}'')}{D_{\mathbf{R}}(\mathfrak{L})}.$$

($\mathfrak{L}'' \ \eta \ \mathbf{R}_{(\mathfrak{L})}$ implies $\mathfrak{L}'' \ \eta \ \mathbf{R}$.) The membership $\alpha \in \mathfrak{G}(\overline{\mathbf{R}_{(\mathfrak{L})}})$ gives

$$\overline{\mathbf{R}_{(\mathfrak{L}'')}} = \overline{\mathbf{R}_{(\mathfrak{L})}}^\alpha = \overline{\mathbf{R}_{(\mathfrak{L})}} = \overline{\mathbf{M}}.$$

Thus we may replace in all our considerations $\mathfrak{L}$ by $\mathfrak{L}''$. This carries $\alpha = 1$. Writing again $\mathfrak{L}$ for $\mathfrak{L}''$, we see: We may assume that $\alpha = 1$, i.e. that

$$\theta = \frac{D_{\mathbf{R}'}(\mathfrak{L}')}{D_{\mathbf{R}}(\mathfrak{L})}.$$

Now consider the coupled factors $\mathbf{R}_{(\mathfrak{L}\cdot\mathfrak{L}')}$, $\mathbf{R}'_{(\mathfrak{L}\cdot\mathfrak{L}')}$ in $\mathfrak{L}\cdot\mathfrak{L}' \neq (0)$. Combining our results, we see that we have

$$\begin{aligned}\overline{\mathbf{R}_{(\mathfrak{L}\cdot\mathfrak{L}')}} &= \overline{\mathbf{R}_{(\mathfrak{L})}} = \overline{\mathbf{M}},\\ \overline{\mathbf{R}'_{(\mathfrak{L}\cdot\mathfrak{L}')}} &= \overline{\mathbf{R}'_{(\mathfrak{L}')}} = \overline{\mathbf{M}'}.\end{aligned}$$

Thus the spatial type $\widetilde{\mathbf{R}}_{(\mathfrak{L}\cdot\mathfrak{L}')}$ meets all our requirements.

Second alternative: $\theta = 0$ or $\theta = \infty$. Since we may interchange $\overline{\mathbf{M}}$ and $\overline{\mathbf{M}'}$, and with them θ and $1/\theta$, we can even assume that $\theta = 0$. Then we have the case β) in Lemma 3.3.2.

Thus $\overline{\mathbf{M}}$ is in an infinite case. Hence (3.3.ν) gives $\overline{\mathbf{R}} = \overline{\mathbf{M}}$. Now consider the coupled factors $\mathbf{R}_{(\mathfrak{L}')}$, $\mathbf{R}'_{(\mathfrak{L}')}$ in $\mathfrak{L}' \neq (0)$. Combining our results, we see that we have

$$\begin{aligned}\overline{\mathbf{R}_{(\mathfrak{L}')}} &= \overline{\mathbf{R}} = \overline{\mathbf{M}},\\ \overline{\mathbf{R}'_{(\mathfrak{L}')}} &= \overline{\mathbf{M}'},\\ D_{\mathbf{R}_{(\mathfrak{L}')}}(\mathfrak{L}') &= D_{\mathbf{R}}(\mathfrak{H}) = \infty,\\ D_{\mathbf{R}'_{(\mathfrak{L}')}}(\mathfrak{L}') &= D_{\mathbf{R}'}(\mathfrak{L}') < \infty.\end{aligned}$$

Thus the spatial type $\widetilde{\mathbf{R}}_{(\mathfrak{L}')}$ meets all our requirements.

Third alternative: $\theta = \infty/\infty$. We have the case α) in Lemma 3.3.2. Hence $\overline{\mathbf{M}}, \overline{\mathbf{M}'}$ are both in infinite cases, and so (3.3.ν) gives

$$\overline{\mathbf{R}} = \overline{\mathbf{M}}, \qquad \overline{\mathbf{R}'} = \overline{\mathbf{M}'}.$$

Also

$$D_{\mathbf{R}}(\mathfrak{H}) = D_{\mathbf{R}'}(\mathfrak{H}) = \infty.$$

Thus the spatial type $\widetilde{\mathbf{R}}$ meets all our requirements.

This completes the proof.

Summing up the results of Lemma 3.3.3 and Lemma 3.3.4,

Theorem X. *Consider factors* $\mathbf{M}$ *in the cases (I) and (II). Then the spatial type* $\widetilde{\mathbf{M}}$ *is uniquely determined by the algebraical types* $\overline{\mathbf{M}}, \overline{\mathbf{M}'}$, *and the number* θ *(cf. Definition 3.3.1).*

$\widetilde{\mathbf{M}}$ *exists if and only if* $\overline{\mathbf{M}}, \overline{\mathbf{M}'}$ *and* θ *fulfill the alternative conditions* α)–δ) *in Lemma 3.3.2.*

Remark. In some cases, two of $\overline{\mathbf{M}}, \overline{\mathbf{M}'}$, and θ suffice to determine the third one and thus $\widetilde{\mathbf{M}}$. Specifically,

α) If $\overline{\mathbf{M}}$ is in a finite case, then $\overline{\mathbf{M}}$ and θ determine $\overline{\mathbf{M}'}$ by γ), δ) in Lemma 3.3.2.
β) If $\overline{\mathbf{M}'}$ is in a finite case, then $\overline{\mathbf{M}'}$ and θ determine $\overline{\mathbf{M}}$ by β), δ) in Lemma 3.3.2.
γ) If either $\overline{\mathbf{M}}$ or $\overline{\mathbf{M}'}$ is in an infinite case, then $\overline{\mathbf{M}}, \overline{\mathbf{M}'}$ determine θ by α), β), or γ) in Lemma 3.3.2.
δ) If $\overline{\mathbf{M}}, \overline{\mathbf{M}'}$ are both in finite cases, then $\overline{\mathbf{M}}, \overline{\mathbf{M}'}$ determine θ precisely up to a factor belonging to the group $\mathfrak{G}(\widetilde{\overline{\mathbf{M}}})$, owing to δ) in Lemma 3.3.2, and to (2.10.α) and Lemma 3.2.1. Consequently they determine θ outright if and only if $\mathfrak{G}(\widetilde{\overline{\mathbf{M}}}) = (1)$.[26]

Thus the only remaining objects for our inquiries in the cases (I) and (II) are the algebraical types $\overline{\mathbf{M}}$. Theorem IX and Theorem X (or Lemma 3.2.1 and Lemma 3.2.2 in place of the latter) show that the knowledge of these is equivalent to the knowledge of their genera $\widetilde{\overline{\mathbf{M}}}$. The case (I) is completely clear; hence the above concepts must be studied in the case (II). And we can analyze a genus $\widetilde{\overline{\mathbf{M}}}$ in case (II) by analyzing any one of its types $\overline{\mathbf{M}}$ in a finite case. Consequently we need only be concerned with (algebraical) types $\overline{\mathbf{M}}$ in case (II_1).

CHAPTER IV. APPROXIMATE FINITENESS

§4.1 In accordance with the above we begin the systematic investigation of the algebraical types $\overline{\mathbf{M}}$ in case (II_1). Throughout this chapter $\mathbf{M}$ will be a factor in case (II_1) in Hilbert space $\mathfrak{H}$. We use $D_{\mathbf{M}}(E)$, (but now in the standard normalization) $\mathrm{Tr}_{\mathbf{M}}(A)$, $[\![A]\!]$ as in §1.2. We shall also consider other factors in $\mathfrak{H}$, but they will be denoted by different letters. All our considerations however will take place in $\mathfrak{H}$.

We begin by stating some familiar facts about factors in the case (I_n), $n = 1, 2, \ldots$.

Lemma 4.1.1. *Let $\mathbf{N}$ be a factor in a case (I_p), $p = 1, 2, \ldots$. Then there exists a one-to-one correspondence between the two following sets of entities,*

(α) *The (algebraical ring) isomorphisms $\mathfrak{R}$ between $\mathbf{N}$ and the system of all complex p-order matrices*

$$a = \{a_{t,s}\}, \qquad (t, s = 1, \ldots, p).^{27}$$

(β) *The matrix bases of $\mathbf{N}$, i.e. the systems $W_{u,v}$, ($u, v = 1, \ldots, p$), of p-order matrix*

[26] Hence θ is never needed in the cases (I), by Lemma 2.10.4. The situation is different in the cases (II). Cf. Theorem XV and the remark at the end of §5.6.

[27] The (complex numerical) matrices which we consider now should be viewed as endowed with the same operations and defined in the same way as the (operator) matrices of $\mathbf{B}_p$ in §2.4.

units, that are contained in **N** *and have*

$$E_0 = \sum_{u=1}^{p} W_{u,u} = 1. \tag{4.1.α}$$

The correspondence is this:

(i) $\mathfrak{K}$ *is given:*

$$W_{u,v} = \mathfrak{K}^{-1}(\{\delta_{t,v}\delta_{s,u}\})\,.$$

(ii) *The* $W_{u,v}$ *are given:*

$$\mathfrak{K}A = a = \{a_{t,s}\}$$

is equivalent to

$$A = \sum_{u=1}^{p}\sum_{v=1}^{p} a_{v,u}W_{u,v}.$$

Proof. [28] (i) leads from (α) to (β). The conditions of Definition 2.6.1 as well as (4.1.α) are verified by direct matrix computation.

(ii) leads from (β) to (α): The rules of matrix computation are verified from Definition 2.6.1 and (4.1.α).

(i) implies (ii): Immediate by matrix computation.

(ii) implies (i). Put $a_{u,v} = \delta_{t,v}\delta_{s,u}$ in (ii).

Remark 1. It is clear by (ii) that the ring **N** is generated by the $W_{u,v}$, $u, v = 1, \ldots, p$,

$$\mathbf{N} = \mathbf{R}(W_{u,v};\ u, v = 1, \ldots, p).$$

Remark 2. The two sets (α), (β) are of course not empty. This follows in numerous ways from past results; the equivalent was even stated explicitly in the Corollary to Lemma 2.10.3.

Lemma 4.1.2. *Let* **N**, **O** *be two factors in the cases* (I_p), (I_q) *respectively,* $p, q = 1, 2, \ldots$. *Assume that*

$$\mathbf{N} \subseteq \mathbf{O}.$$

Then we have

(i) p *is a divisor of* q: $q = pr$, $r = 1, 2, \ldots$.

(ii) *To every matrix basis* $W_{u,v}$, $u, v = 1, \ldots, p$, *of* **N** *there exists a matrix base* $X_{o,w}$, $o, w = 1, \ldots, q$, *of* **O** *such that*

$$W_{u,v} = \sum_{i=1}^{r} X_{(u-1)r+i,(v-1)r+i}.$$

[28] Given only for the sake of completeness.

(*iii*) *If we pass from the matrix bases* $W_{u,v}$ *and* $X_{o,w}$ *in (ii) to the isomorphisms* $\mathfrak{K}$ *and* $\mathfrak{L}$ *which correspond to them by Lemma 4.1.1, then the correlation of (ii) is equivalent to this:*
For $A \in \mathbf{N} \subseteq \mathbf{O}$, *both* $\mathfrak{K}A = a = \{a_{t,s}\}$, $t, s = 1, \ldots, p$, *and* $\mathfrak{L}A = \bar{a} = \{\bar{a}_{k,l}\}$, $k, l = 1, \ldots, q$, *are defined and*

$$\bar{a}_{k,l} = a_{t,s}\delta_{i,j} \quad \textit{for} \quad k = (t-1)r + i, \qquad l = (s-1)r + j.$$

Proof. Ad (i). $D_{\mathbf{O}}(E)$ will do as $D_{\mathbf{N}}(E)$ for the projections $E \in \mathbf{N} \subseteq \mathbf{O}$ in the sense of [5, p. 165, Definition 8.2.1][29] and standard normalization. And since the dimension is uniquely characterized by those properties, therefore $D_{\mathbf{O}}(E) = D_{\mathbf{N}}(E)$ for these E.

Hence the total range of $D_{\mathbf{O}}(E)$ contains the total range of $D_{\mathbf{N}}(E)$, i.e. the set $(0, 1/q, \ldots, 1)$ contains the set $(0, 1/p, \ldots, 1)$. Consequently, p is a divisor of q.

Ad (ii). Consider a matrix base $W_{u,v}$ of $\mathbf{N}$ and an arbitrary matrix base $X'_{o,w}$ of $\mathbf{O}$. Put

$$W'_{u,v} = \sum_{i=1}^{r} X'_{(u-1)r+i,(v-1)r+i}. \tag{4.1.β}$$

Then one verifies by explicit computation (using Definition 2.6.1) that the $W'_{u,v}$ also form a system of p-order matrix units. Also

$$\sum_{u=1}^{p} W'_{u,u} = \sum_{o=1}^{q} X'_{o,o} = 1.$$

Since $D_{\mathbf{O}}(W'_{u,u}) = D_{\mathbf{O}}(W'_{1,1})$, we have $D_{\mathbf{O}}(W'_{1,1}) = 1/p$, and similarly $D_{\mathbf{O}}(W_{1,1}) = 1/p$. Therefore there exists a partially isometric U in $\mathbf{O}$ with the initial projection $W'_{1,1}$ and the final projection $W_{1,1}$. Now put

$$V = \sum_{u=1}^{p} W_{u,1} U W'_{1,u}.$$

Clearly $V \in \mathbf{O}$, and one verifies by direct computation

$$\begin{aligned} &(1) \quad V^*V = VV^* = 1, \quad \text{i.e. } V \text{ is unitary,} \\ &(2) \quad VW'_{u,v} = W_{u,v}V, \quad \text{i.e. } W_{u,v} = VW'_{u,v}V^{-1}. \end{aligned}$$

Hence replacement of $X'_{u,v}$ by $VX'_{u,v}V^{-1} = X_{u,v}$ will leave all its properties intact, but replace $W'_{u,v}$ by $W_{u,v} = VW'_{u,v}V^{-1}$. And now (4.1.$\beta$) yields the desired equation of (ii).

[29] Consider also [5, p. 170, Lemma 8.3.5]. We are using the fact that $\mathbf{O}$ belongs to a finite case.

Ad (iii). Apply (ii) in Lemma 4.1.1 to $\mathfrak{K}$ (with $W_{u,v}$) and $\mathfrak{L}$ (with $X_{o,w}$). Then the equation of (ii) goes over into the desired relation of (iii).

We now define

Definition 4.1.1. **M** *is approximately finite of type* $[p_1, p_2, \ldots]$ *if this is true. There exists a sequence of factors* $\mathbf{N}_1, \mathbf{N}_2, \ldots$ *with the following properties:*

(i) $\mathbf{N}_n$ *is in case* (I_{p_n}).

(ii) $\mathbf{N}_1 \subseteq \mathbf{N}_2 \subseteq \cdots$.

(iii) *The ring* **M** *is generated by the* $\mathbf{N}_1, \mathbf{N}_2, \ldots$,

$$\mathbf{M} = \mathbf{R}(\mathbf{N}_n;\ n = 1, 2, \ldots).$$

We have immediately:

Lemma 4.1.3. *Under the conditions of Definition 4.1.1, we have necessarily*

(i) p_n *is a divisor of* p_{n+1}, $p_{n+1} = p_n r_n$, $(r_n = 1, 2, \ldots)$.

(ii) $p_n \to \infty$ *for* $n \to \infty$, *i.e. infinitely many* $r_n > 1$.

Proof. Ad (i). As $\mathbf{N}_n \subseteq \mathbf{N}_{n+1}$, this follows from (i) in Lemma 4.1.2.

Ad (ii). Owing to (i), $p_1 \leq p_2 \leq \cdots$, so failure of (ii) implies $p_m = p_{m+1} = p_{m+2} = \cdots$ for some m. Hence equally $\mathbf{N}_m = \mathbf{N}_{m+1} = \mathbf{N}_{m+2} = \cdots$[30] and so (iii) in Definition 4.1.1 gives

$$\mathbf{M} = \mathbf{R}(\mathbf{N}_n;\ n = 1, 2, \ldots) = \mathbf{N}_m.$$

But this is impossible since **M** is in the case (II_1) and $\mathbf{N}_m$ in the case (I_{p_m}).

For every sequence $[p_1, p_2, \ldots]$ which meets these requirements there exists an **M** which is approximately finite of that type. But the concept introduced by Definition 4.1.1 is a preliminary one and we prefer to prove this existence only in conjunction with the final reformulation of the concept of approximate finiteness in Theorem XII and Theorem XIV. We now proceed in a different direction.

Lemma 4.1.4. *For any sequence of rings* $\mathbf{N}_1, \mathbf{N}_2, \ldots$ *which fulfills conditions (ii), (iii) in Definition 4.1.1, denote by* **S** *the set theoretical sum of that sequence. Then we have*

(i) **S** *is an algebra.*

(ii) **M** *is the set of elements of* **B** *which are limits of a convergent sequence from* **S**.

Proof. Ad (i): Every $\mathbf{N}_n$ is a ring; hence an algebra. Also $\mathbf{N}_1 \subseteq \mathbf{N}_2 \subseteq \cdots$. Hence the sum **S** is an algebra also.

[30] For $l \geq m$, $\mathbf{N}_m$ and $\mathbf{N}_l$ contain the same number of linearly independent elements: $p_l^2 = p_m^2$. As $\mathbf{N}_m \subseteq \mathbf{N}_l$, this implies $\mathbf{N}_m = \mathbf{N}_l$.

Ad (ii): Denote the set of limits of all metrically convergent sequences from $\mathbf{S}$ by $\mathbf{S}_1$. We must prove that $\mathbf{S}_1 = \mathbf{M}$.

Now

$$\mathbf{M} = \mathbf{R}(\mathbf{N}_n;\ n = 1, 2, \ldots) = \mathbf{R}(\mathbf{S})$$

and $\mathbf{S} \subseteq \mathbf{S}_1 \subseteq \mathbf{M}$. Hence $\mathbf{S}_1 = \mathbf{M}$ is established if we can show that $\mathbf{S}_1$ is a ring.

$\mathbf{S}$ is an algebra and αA, A^* (as one-variable functions), $A + B$ (as a two-variable function), and AB (as a one-variable function with respect to either variable) are continuous for metric convergence. (This follows from the evaluations of [6, p. 242, property II°].) Hence $\mathbf{S}_1$ is also an algebra. Therefore its closure with respect to metric convergence implies its closure in the weak topology by Theorem I. Consequently, $\mathbf{S}_1$ is a ring.

This completes the proof.

We can now prove

Theorem XI. *The (algebraical) type $\overline{\mathbf{M}}$ of an $\mathbf{M}$ which is approximately finite of type $[p_1, p_2, \ldots]$ depends on the sequence $[p_1, p_2, \ldots]$ alone and not on the particular choice of $\mathbf{M}$.*

(Provided such an $\mathbf{M}$ exists at all. Cf. above.)

Proof. Consider two such $\mathbf{M}^{(h)}$, $h = 1, 2$, and form their $\mathbf{N}_1^{(h)}, \mathbf{N}_2^{(h)}, \ldots$ by Definition 4.1.1 and their $\mathbf{S}^{(h)}$ by Lemma 4.1.4. We wish to prove that $\mathbf{M}^{(1)}$ and $\mathbf{M}^{(2)}$ are algebraically isomorphic.

We shall choose by induction for every $n = 1, 2, \ldots$ an (algebraical) isomorphism $\mathfrak{K}_n^{(h)}$ between $\mathbf{N}_n^{(h)}$ and the system of all (complex) p_n-order matrices in the sense of (α) in Lemma 4.1.1. For $n = 1$ we choose a $\mathfrak{K}_1^{(h)}$ which meets this requirement (cf. Remark 2 after Lemma 4.1.1). If for any $n = 2, 3, \ldots$, $\mathfrak{K}_{n-1}^{(h)}$ has already been chosen, then we choose $\mathfrak{K}_n^{(h)}$ in the following way: Apply Lemma 4.1.2 to $\mathbf{N}_{n-1}^{(h)}, \mathbf{N}_n^{(h)}, \mathfrak{K}_{n-1}^{(h)}$, in place of its $\mathbf{N}, \mathbf{O}, \mathfrak{K}$ (with $\mathbf{M}^{(h)}$ in place of $\mathbf{M}$) and put $\mathfrak{K}_n^{(h)} = \mathfrak{L}$. Thus $\mathfrak{K}_{n-1}^{(h)}$ and $\mathfrak{K}_n^{(h)}$ are related to each other as $\mathfrak{K}$ and $\mathfrak{L}$ in (iii) in Lemma 4.1.2.

Now form the (algebraical) isomorphism $\mathfrak{J}_n = (\mathfrak{K}_n^{(2)})^{-1}\mathfrak{K}_n^{(1)}$, of $\mathbf{N}_n^{(1)}$ and $\mathbf{N}_n^{(2)}$. In view of the relationship given in the last sentence of the preceding paragraph, $\mathfrak{J}_n$ is a part of $\mathfrak{J}_{n+1}$. Thus the isomorphisms $\mathfrak{J}_n$ of $\mathbf{N}_n^{(1)}$ and $\mathbf{N}_n^{(2)}$, for $n = 1, 2, \ldots$, are compatible with each other and they merge to an (algebraical) isomorphism $\mathfrak{J}$ of $\mathbf{S}^{(1)}$ and $\mathbf{S}^{(2)}$. In $\mathbf{N}_n^{(1)}$, our $\mathfrak{J}$ agrees with $\mathfrak{J}_n$. Hence $\mathfrak{J}$ maps $\mathbf{N}_n^{(1)}$ on $\mathbf{N}_n^{(2)}$.

In $\mathbf{N}_n^{(h)}$, $\mathrm{Tr}_{\mathbf{N}_n^{(h)}}(A)$ is purely algebraical (cf. §1.2). Clearly $\mathrm{Tr}_{\mathbf{M}^{(h)}}(A)$ will do as $\mathrm{Tr}_{\mathbf{N}_n^{(h)}}(A)$ for the $A \in \mathbf{N}_n^{(h)}$ in the sense of [6, p. 219, property IV]. And as the trace is uniquely determined by that property, so $\mathrm{Tr}_{\mathbf{M}^{(h)}}(A) = \mathrm{Tr}_{\mathbf{N}_n^{(h)}}(A)$ for $A \in \mathbf{N}_n^{(h)}$. Hence

$$[\![A]\!] = \left(\mathrm{Tr}_{\mathbf{M}^{(h)}}(A^*A)\right)^{1/2} = \left(\mathrm{Tr}_{\mathbf{N}_n^{(h)}}(A^*A)\right)^{1/2} \quad \text{for } A \in \mathbf{N}_n^{(h)},$$

and therefore $[\![A]\!]$, and with it $[\![A-B]\!]$, is purely algebraical in $\mathbf{N}_n^{(h)}$.

Thus $\mathfrak{J}$ leaves $[\![A-B]\!]$ invariant in $\mathbf{N}_n^{(h)}$ for all $n = 1, 2, \ldots$. Hence the same is true in all $\mathbf{S}^{(h)}$.

Summing up: $\mathfrak{J}$ is an isomorphism of $\mathbf{S}^{(1)}$ and $\mathbf{S}^{(2)}$ with respect to

$$1, \quad \alpha A, \quad A^*, \quad A+B, \quad AB, \tag{4.1.γ}$$

and

$$[\![A-B]\!]\,. \tag{4.1.δ}$$

By (4.1.δ), $\mathfrak{J}$ is isometric. Now extend $\mathfrak{J}$ as far as possible in $\mathbf{B}$ by metric convergence, thus obtaining a new isometric (and hence one-to-one) correspondence $\mathfrak{F}$. By (ii) in Lemma 4.1.4, $\mathfrak{F}$ maps all of $\mathbf{M}^{(1)}$ on $\mathbf{M}^{(2)}$.

The continuity of the operations (4.1.γ) (cf. the proof of (ii) in Lemma 4.1.4) guarantees that $\mathfrak{F}$ too leaves (4.1.γ) invariant. Hence $\mathfrak{F}$ is an (algebraical) isomorphism of $\mathbf{M}^{(1)}$ and $\mathbf{M}^{(2)}$. This completes the proof.

§4.2 As a preliminary before introducing other definitions of approximate finiteness, we develop further the ideas which appear in §1.5.

Lemma 4.2.1. *For any $\epsilon > 0$ there exists an $w_3 = w_3(\epsilon) > 0$ with the following property:*

Consider a ring $\mathbf{N} \subseteq \mathbf{M}$ and a projection $E \in \mathbf{M}$. If there exists an $A \in \mathbf{N}$ with $[\![A-E]\!] < w_3$, then there exists also a projection $F \in \mathbf{N}$ with $[\![F-E]\!] < \epsilon$.

Proof. Consider first the hypothetical A. We have $[\![A^*-E]\!] = [\![(A-E)^*]\!] = [\![A-E]\!] < w_3$. Since

$$\tfrac{1}{2}(A+A^*) - E = \tfrac{1}{2}\big((A-E)+(A^*-E)\big),$$

these inequalities imply $\left[\!\left[\tfrac{1}{2}(A+A^*)-E\right]\!\right] < w_3$ by the triangle inequality. Thus we may replace A by $\tfrac{1}{2}(A+A^*)$, i.e. we may assume that A is Hermitian.

Now form the following functions

$$\psi_1(\lambda) = \begin{cases} 1, & |\lambda| \geq 1, \\ 2|\lambda|-1, & \tfrac{1}{2} \leq |\lambda| < 1, \\ 0, & |\lambda| \leq \tfrac{1}{2}, \end{cases} \qquad \psi_2(\lambda) = \begin{cases} 1, & |\lambda| \geq \tfrac{1}{2}, \\ 2|\lambda|, & |\lambda| < \tfrac{1}{2}. \end{cases}$$

Clearly Lemma 1.5.3 applies to both of them. We form the $w_2(\epsilon)$ of that lemma for both functions and denote their minimum by $w_2'(\epsilon)$. We put

$$w_3(\epsilon) = w_2'\left(\frac{\epsilon}{2}\right). \tag{4.2.α}$$

Form the further function

$$\varphi(\lambda) = \begin{cases} 1, & |\lambda| \geq \frac{1}{2}, \\ 0, & |\lambda| < \frac{1}{2}. \end{cases}$$ [31]

The following facts are clear: $\overline{\varphi(\lambda)} = \varphi(\lambda) = \varphi(\lambda)^2$. Hence $F = \varphi(A)$ is a projection which belongs to $\mathbf{N}$ along with A. $\psi_1(\lambda) = \psi_2(\lambda) = \varphi(\lambda) = \lambda$ for $\lambda = 0, 1$, the entire spectrum of E. Hence

$$\psi_1(E) = \psi_2(E) = \varphi(E) = E. \tag{4.2.β}$$

Finally $(1 - \varphi(\lambda))\psi_1(\lambda) + \varphi(\lambda)\psi_2(\lambda) = \varphi(\lambda)$. Hence

$$(1 - F)\psi_1(A) + F\psi_2(A) = F. \tag{4.2.γ}$$

Now $[\![A - E]\!] < w_3(\epsilon)$ implies by Lemma 1.5.3 and (4.2.α) that

$$\left[\!\left[\psi_1(A) - \psi_1(E)\right]\!\right] < \frac{\epsilon}{2}, \qquad \left[\!\left[\psi_2(A) - \psi_2(E)\right]\!\right] < \frac{\epsilon}{2}.$$

Hence (4.2.β) now yields that $\left[\!\left[\psi_1(A) - E\right]\!\right] < \epsilon/2$ and $\left[\!\left[\psi_2(A) - E\right]\!\right] < \epsilon/2$. Since $|||F|||$ and $|||1 - F|||$ are ≤ 1[32] we can therefore conclude that

$$\left[\!\left[(1 - F)(\psi_1(A) - E) + F(\psi_2(A) - E)\right]\!\right] < \epsilon.$$

This may be rewritten

$$\left[\!\left[(1 - F)\psi_1(A) + F\psi_2(A) - E\right]\!\right] < \epsilon.$$

With (4.2.γ) this gives $[\![F - E]\!] < \epsilon$, as desired.

Lemma 4.2.2. *For any $\epsilon > 0$, there exists an $w_4 = w_4(\epsilon) > 0$ with the following property.*

Consider two projections $E, F \in \mathbf{M}$ with the closed linear sets $\mathfrak{M}, \mathfrak{N}$. If $[\![E - F]\!] < w_4$, then there exists a partially isometric $W' \in \mathbf{M}$ with an initial set $\mathfrak{M}' \subseteq \mathfrak{M}$, a final set $\mathfrak{N}' \subseteq \mathfrak{N}$, and with $[\![W' - E]\!] < \epsilon$.

[31]Observe that $\psi_1(\lambda)$, $\psi_2(\lambda)$ are continuous, while $\varphi(\lambda)$ is not.

[32]Actually we have an equality in the case of each of these operators except when it is zero.

Proof. Form the canonical decomposition of $A = FE$ in the sense of [5, p. 143, Definition 4.4.1]:

$$FE = A = W'B, \qquad EF = A^* = BW'^*. \tag{4.2.δ}$$

Here B is Hermitian and definite, W' is partially isometric with initial set

$$\mathfrak{M}' = [\text{Range}\, A^*] = [\text{Range}\, EF] \subseteq [\text{Range}\, E] = \mathfrak{M}$$

and final set

$$\mathfrak{N}' = [\text{Range}\, A] = [\text{Range}\, FE] \subseteq [\text{Range}\, F] = \mathfrak{N}.$$

The range of W'^* is $\mathfrak{M}' \subseteq \mathfrak{M} = [\text{Range}\, E]$. Thus $EW'^* = W'^*$, and taking adjoints, we obtain

$$W'E = W'. \tag{4.2.ϵ}$$

For any canonical decomposition $B^2 = A^*A$. In our case $A^*A = EF \cdot FE = EFE$, so

$$B^2 = EFE. \tag{4.2.ζ}$$

Observe also that due to the character of these operators

$$|||E||| \leq 1, \qquad |||F||| \leq 1, \qquad |||W'||| \leq 1. \tag{4.2.η}$$

Thus (4.2.ζ) implies $\left|\left|\left|B^2\right|\right|\right| \leq 1$, and so

$$|||B||| \leq 1. \tag{4.2.θ}$$

Now form the function

$$\psi(\lambda) = \begin{cases} 1, & |\lambda| \geq 1, \\ \sqrt{|\lambda|}, & |\lambda| \leq 1. \end{cases}$$

Clearly Lemma 1.5.3 applies to it. We form the $w_2(\epsilon)$ of that lemma for this function and put

$$w_4(\epsilon) = \text{Min.}\left(w_2\left(\frac{\epsilon}{2}\right), \frac{\epsilon}{2}\right). \tag{4.2.ι}$$

It is clear that $\psi(\lambda^2) = \lambda$ for $0 \leq \lambda \leq 1$, which includes the entire spectrum of B, since B is definite and (4.2.θ) holds. Thus $\psi(B^2) = B$, or using (4.2.ζ) we obtain

$$\psi(EFE) = B. \tag{4.2.κ}$$

On the other hand, $\psi(\lambda) = \lambda$ for $\lambda = 0, 1$, the spectrum of E, and hence

$$\psi(E) = E. \tag{4.2.λ}$$

Now our assumption, $[\![E-F]\!] < w_4$, implies by (4.2.η) that $[\![(E-F)E]\!] < w_4$ and $[\![E(E-F)E]\!] < w_4$, or

$$[\![E-FE]\!] < w_4, \qquad [\![E-EFE]\!] < w_4. \tag{4.2.μ}$$

(4.2.μ), Lemma 1.5.3, and (4.2.ι) together imply

$$\left[\!\left[\psi(E)-\psi(EFE)\right]\!\right] < \frac{\epsilon}{2}.$$

This and (4.2.κ) and (4.2.λ) yield $[\![E-B]\!] < \epsilon/2$. This and (4.2.η) imply $[\![W'(E-B)]\!] < \epsilon/2$. Hence (4.2.$\epsilon$) and (4.2.$\delta$) give

$$[\![W'-FE]\!] < \frac{\epsilon}{2}. \tag{4.2.ν}$$

(4.2.μ) and (4.2.ι) imply $[\![E-FE]\!] < \epsilon/2$. Hence (4.2.$\nu$) yields $[\![W'-E]\!] < \epsilon$, as desired.

Lemma 4.2.3. *If $U \in \mathbf{M}$ is unitary and $W \in \mathbf{M}$ is partially isometric, and if U agrees with W on its initial set, then*

$$[\![U-W]\!] \leq \left(2\,[\![W-1]\!]\right)^{1/2}.$$

Proof. The initial set of W is the range or final set of W^*. Therefore $UW^* = WW^*$. This implies for the adjoints $WU^* = WW^*$. Hence

$$\begin{aligned}(U-W)(U-W)^* &= UU^* - UW^* - WU^* + WW^* \\ &= 1 - WW^* - WW^* + WW^* = 1 - WW^*.\end{aligned}$$

Therefore $[\![U-W]\!]^2 = \mathrm{Tr}_{\mathbf{M}}((U-W)(U-W)^*) = \mathrm{Tr}_{\mathbf{M}}(1-WW^*)$, or

$$[\![U-W]\!] = \left(\mathrm{Tr}_{\mathbf{M}}(1-WW^*)\right)^{1/2}. \tag{4.2.o}$$

We now make use of Schwartz's inequality in the sense of [6, p. 241, Theorem VIII], remembering that $|||W||| \leq 1$, and so $[\![W]\!] \leq 1$. Thus

$$\begin{aligned}\mathrm{Tr}_{\mathbf{M}}(1-WW^*) &= \mathrm{Tr}_{\mathbf{M}}\left(-W(W-1)^* - (W-1)1^*\right) \\ &\leq 2\,[\![W-1]\!]\,.\end{aligned}$$

This and (4.2.o) yield the desired inequality of the lemma.

Lemma 4.2.4. *For any $\epsilon > 0$, there exists an $w_5 = w_5(\epsilon) > 0$ with the following property:*

Consider two projections $E, F \in \mathbf{M}$ with $D_{\mathbf{M}}(E) = D_{\mathbf{M}}(F)$. If $[\![E-F]\!] < w_5$, then there exists a unitary $U \in \mathbf{M}$ with $F = UEU^{-1}$ and with $[\![U-1]\!] < \epsilon$.

Proof. Put $w_5(\epsilon) = w_4(\epsilon')$, where $\epsilon' = \epsilon'(\epsilon) > 0$ will be chosen later.

Apply Lemma 4.2.2 to E, F and their closed linear sets $\mathfrak{M}, \mathfrak{N}$. It gives a partially isometric $W' \in \mathbf{M}$ with an initial set $\mathfrak{M}' \subseteq \mathfrak{M}$, a final set $\mathfrak{N}' \subseteq \mathfrak{N}$, and with

$$[\![W' - E]\!] < \epsilon'. \tag{4.2.ξ}$$

In view of the relation of W' to $\mathfrak{M}'$ and $\mathfrak{N}'$, $D_{\mathbf{M}}(\mathfrak{M}') = D_{\mathbf{M}}(\mathfrak{N}')$. By hypothesis, $D_{\mathbf{M}}(\mathfrak{M}) = D_{\mathbf{M}}(\mathfrak{N})$. Hence

$$D_{\mathbf{M}}(\mathfrak{M} \ominus \mathfrak{M}') = D_{\mathbf{M}}(\mathfrak{N} \ominus \mathfrak{N}').$$

Choose a partially isometric $V' \in \mathbf{M}$ with initial set $\mathfrak{M} \ominus \mathfrak{M}'$ and final set $\mathfrak{N} \ominus \mathfrak{N}'$. Then $U' = W' + V' \in \mathbf{M}$ is also partially isometric, with initial set $\mathfrak{M}$, final set $\mathfrak{N}$, and it agrees with W' on the initial set $\mathfrak{M}'$ of W'.

Apply all this to $1 - E, 1 - F$ and $\mathfrak{M}^{\perp}(= \mathfrak{H} \ominus \mathfrak{M}), \mathfrak{N}^{\perp}$ in place of E, F and $\mathfrak{M}, \mathfrak{N}$. (Clearly $[\![(1 - E) - (1 - F)]\!] = [\![E - F]\!]$.) Thus we obtain a partially isometric $W'' \in \mathbf{M}$ with an initial set $\mathfrak{M}'' \subseteq \mathfrak{M}^{\perp}$, a final set $\mathfrak{N}'' \subseteq \mathfrak{N}^{\perp}$, and with

$$[\![W'' - (1 - E)]\!] < \epsilon'. \tag{4.2.π}$$

Furthermore, there exists a partially isometric $U'' \in \mathbf{M}$ with the initial set $\mathfrak{M}^{\perp}$, final set $\mathfrak{N}^{\perp}$, and agreeing with W'' on the initial set $\mathfrak{M}''$ of W''.

Now put

$$W = W' + W'', \qquad U = U' + U''.$$

Then $W, U \in \mathbf{M}$ are partially isometric, with initial sets $[\mathfrak{M}', \mathfrak{M}'']$, $[\mathfrak{M}, \mathfrak{M}^{\perp}] = \mathfrak{H}$, respectively, and with the respective final sets $[\mathfrak{N}', \mathfrak{N}'']$ and $[\mathfrak{N}, \mathfrak{N}^{\perp}] = \mathfrak{H}$. So U is unitary.

U agrees with U' on the initial set $\mathfrak{M}$ of U'. Hence it maps $\mathfrak{M}$ on $\mathfrak{N}$. Therefore

$$F = UEU^{-1}. \tag{4.2.ρ}$$

U agrees with W on the initial set $[\mathfrak{M}', \mathfrak{M}'']$ of W. Therefore by Lemma 4.2.3,

$$[\![U - W]\!] = \left(2\,[\![W - 1]\!]\right)^{1/2}. \tag{4.2.σ}$$

(4.2.ξ) and (4.2.π) imply

$$[\![W - 1]\!] < 2\epsilon'. \tag{4.2.τ}$$

(4.2.σ) and (4.2.τ) give together $[\![U - W]\!] < (4\epsilon')^{1/2}$. This and (4.2.$\tau$) yield

$$[\![U - 1]\!] < (4\epsilon')^{1/2} + 2\epsilon'. \tag{4.2.υ}$$

Hence if we choose ϵ' so that $(4\epsilon')^{1/2} + 2\epsilon' \leq \epsilon$, (4.2.$\upsilon$) yields

$$[\![U - 1]\!] < \epsilon. \tag{4.2.ϕ}$$

In view, therefore, of (4.2.ρ) and (4.2.ϕ), the w_5 which we have specified satisfies the condition of our lemma.

§4.3 We shall recast the definition of approximate finiteness in a series of steps, i.e. Definitions 4.3.1, 4.5.2 and 4.6.1, below:

Definition 4.3.1. **M** *is approximately finite (A) if this is true:*

Given any $A_1, \ldots, A_m \in \mathbf{M}$ *and any* $\epsilon > 0$, *there exists an* $n = n_1(A_1, \ldots, A_m, \epsilon)$ *such that for every* $q \geq n$ *there exists a factor* $\mathbf{N} = \mathbf{N}_1(q, A_1, \ldots, A_m, \epsilon)$ *with the following properties:*

(*i*) **N** *is in case* (I_q).
(*ii*) $\mathbf{N} \subseteq \mathbf{M}$.
(*iii*) *There exist* $B_1, \ldots, B_m \in \mathbf{N}$ *with*

$$[\![B_h - A_h]\!] < \epsilon \qquad \textit{for } h = 1, \ldots, m.$$

In the present section we are interested in certain consequences of this definition, in particular what other properties of the approximating factor **N** may be assumed. It will be assumed therefore in the five lemmas of this section that **M** is approximately finite (A).

Lemma 4.3.1. *Given any* $A_1, \ldots, A_m \in \mathbf{M}$, *any projection* $E \in \mathbf{M}$ *with* $D_{\mathbf{M}}(E) = 1/p$ *for* $p = 1, 2, \ldots$, *and any* $\epsilon > 0$, *there exists an* $n_2 = n_2(A_1, \ldots, A_m, E, \epsilon)$ *such that for every* $q \geq n_2$ *which is divisible by* p *there exists a factor* $\mathbf{N} = \mathbf{N}_2(q, A_1, \ldots, A_m, E, \epsilon)$ *with the following properties:*

(*i*) **N** *is in case* (I_q).
(*ii*) $\mathbf{N} \subseteq \mathbf{M}$.
(*iii*) *There exist* $B_1, \ldots, B_m \in \mathbf{N}$ *with*

$$[\![B_h - A_h]\!] < \epsilon \qquad \textit{for } h = 1, \ldots, m.$$

(*iv*) *There exists a projection* $F \in \mathbf{N}$ *with* $D_{\mathbf{M}}(F) = D_{\mathbf{M}}(E) = 1/p$, *and* $[\![F - E]\!] < \epsilon$.[33]

Proof. Apply Definition 4.3.1 with $A_1, \ldots, A_m, E$ in place of its $A_1, \ldots, A_m$, and with ϵ' in place of ϵ, where $\epsilon' = \epsilon'(\epsilon)$ will be chosen later.

[33] Observe that this is a strengthened form of Definition 4.3.1.

Put

$$n_2(A_1, \ldots, A_m, E, \epsilon) = n_1(A_1, \ldots, A_m, E, \epsilon'),$$
$$\mathbf{N}_2(q, A_1, \ldots, A_m, E, \epsilon) = \mathbf{N}_1(q, A_1, \ldots, A_m, E, \epsilon').$$

Now consider a $q \geq n_2$ which is divisible by p, and its $\mathbf{N} = \mathbf{N}_2$.

In view of the above definition of $\mathbf{N}_2$, (i) and (ii) of the present lemma follow from (i) and (ii) of Definition 4.3.1. Furthermore, (iii) of Definition 4.3.1 states that

$$[\![B_h - A_h]\!] < \epsilon' \qquad \text{for } h = 1, \ldots, m, \tag{4.3.α}$$

and

$$[\![B_{m+1} - E]\!] < \epsilon'. \tag{4.3.β}$$

Put

$$\epsilon' = \text{Min.}\big(w_3(\epsilon''), \epsilon\big), \tag{4.3.γ}$$

where $\epsilon'' = \epsilon''(\epsilon)$ will be chosen later. Now (4.3.α) and (4.3.γ) imply (iii) in the lemma, so we must only derive (iv) from (4.3.β).

By Lemma 4.2.1, (4.3.β) and (4.3.γ) imply the existence of a projection $F_1 \in \mathbf{N}$ with

$$[\![F_1 - E]\!] < \epsilon''. \tag{4.3.δ}$$

This implies

$$\big|D_{\mathbf{M}}(F_1) - D_{\mathbf{M}}(E)\big| < \epsilon''.\text{[34]} \tag{4.3.ϵ}$$

As $\mathbf{N}$ is in case (I_q), and q is divisible by p, there exists a projection $F \in \mathbf{N}$ with

$$D_{\mathbf{M}}(F) = D_{\mathbf{M}}(E) = 1/p, \tag{4.3.ζ}$$

and we can even choose $F_1 \gtreqless F$. This latter implies

$$[\![F_1 - F]\!] = \big(|D_{\mathbf{M}}(F_1) - D_{\mathbf{M}}(F)|\big)^{1/2}.\text{[35]}$$

[34]For any two projections $E, F \in \mathbf{M}$,

$$\big|D_{\mathbf{M}}(F) - D_{\mathbf{M}}(E)\big| \leq [\![F - E]\!].$$

Proof. We have

$$\big|D_{\mathbf{M}}(F) - D_{\mathbf{M}}(E)\big| = \big|\mathrm{Tr}_{\mathbf{M}}(F) - \mathrm{Tr}_{\mathbf{M}}(E)\big| = \big|\mathrm{Tr}_{\mathbf{M}}(F - E)\big|.$$

This is $\leq [\![1]\!] \cdot [\![F - E]\!]$ (cf. [6, p. 241, Theorem VIII]); hence $\leq [\![F - E]\!]$, as desired.

[35] For any two projections $E, F \in \mathbf{M}$ with $E \gtreqless F$,

$$[\![F - E]\!] = \big(|D_{\mathbf{M}}(F) - D_{\mathbf{M}}(E)|\big)^{1/2}.$$

This equation, (4.3.ϵ), and (4.3.ζ) together imply $[\![F_1 - F]\!] < (\epsilon'')^{1/2}$. Hence (4.3.$\delta$) now implies

$$[\![F - E]\!] < (\epsilon'')^{1/2} + \epsilon''. \tag{4.3.η}$$

Thus if we choose $\epsilon'' = \epsilon''(\epsilon) > 0$, so that $(\epsilon'')^{1/2} + \epsilon'' \leq \epsilon$, we can conclude from (4.3.η) that $[\![F - E]\!] < \epsilon$. In view of (4.3.ζ) and $F \in \mathbf{N}$, our last inequality shows that $\mathbf{N}$ has property (iv), as desired.

Lemma 4.3.2. *Given any $A_1, \ldots, A_m \in \mathbf{M}$, any projection $E \in \mathbf{M}$ with $D_{\mathbf{M}}(E) = 1/p$ for $p = 1, 2, \ldots$, and any $\epsilon > 0$. Then there exists an $n_3 = n_3(A_1, \ldots, A_m, E, \epsilon)$ such that for every $q \geq n_3$ which is divisible by p there exists a factor $\mathbf{N} = \mathbf{N}_3(q, A_1, \ldots, A_m, E, \epsilon)$ with the properties (i), (ii), (iii) of Lemma 4.3.1, and in addition*
(iv′) $E \in \mathbf{N}$.[33]

Proof. Our hypotheses are identical with those of Lemma 4.3.1, and we apply the latter to $A_1, \ldots, A_m, E$, but with ϵ' in place of ϵ, where

$$\epsilon' = \mathrm{Min.}\big(w_5(\epsilon''), \epsilon''\big), \tag{4.3.θ}$$

and $\epsilon'' = \epsilon''(\epsilon) > 0$ will be chosen later.

Denote the $n_2, \mathbf{N}_2$, and $B_1, \ldots, B_m, F$ which we obtain in this way by $n_2', \mathbf{N}_2^+$, and $B_1', \ldots, B_m', F'$. We put $n_3 = n_2'$ and $\mathbf{N}^+ = \mathbf{N}_2^+$. In view of (4.3.$\theta$), Lemma 4.3.1(iii) gives

$$\big[\!\big[B_h' - A_h\big]\!\big] < \epsilon'' \qquad \text{for } h = 1, \ldots, m, \tag{4.3.ι}$$

while (iv) of that lemma yields

$$[\![F' - E]\!] < w_5(\epsilon'') \tag{4.3.κ}$$

and

$$D_{\mathbf{M}}(F') = D_{\mathbf{M}}(E). \tag{4.3.λ}$$

(4.3.κ) and (4.3.λ) allow us to apply Lemma 4.2.4 with F', E in place of its E, F. Consequently there exists a unitary $U \in \mathbf{M}$ with

$$[\![U - 1]\!] < \epsilon'' \tag{4.3.μ}$$

Proof. By symmetry we may assume $E \leq F$. Then $F - E$ is a projection, $D_{\mathbf{M}}(F) - D_{\mathbf{M}}(E) = D_{\mathbf{M}}(F - E)$. Put $G = F - E$. We must prove $[\![G]\!] = (D_{\mathbf{M}}(G))^{1/2}$. Now

$$[\![G]\!]^2 = \mathrm{Tr}_{\mathbf{M}}(G^*G) = \mathrm{Tr}_{\mathbf{M}}(G) = D_{\mathbf{M}}(G).$$

and

$$E = UF'U^{-1}. \tag{4.3.ν}$$

(4.3.μ) implies

$$\begin{aligned}\left[\!\left[A_h - U^{-1}A_hU\right]\!\right] &= \left[\!\left[A_h - A_hU + A_hU - U^{-1}A_hU\right]\!\right]\\ &\leq |||A_h|||\,[\![1-U]\!] + \left[\!\left[(1-U^{-1})A_hU\right]\!\right]\\ &\leq 2\,|||A_h|||\,\epsilon''.\end{aligned}$$

Hence by (4.3.ι), $\left[\!\left[B'_h - U^{-1}A_hU\right]\!\right] < (2\,|||A_h||| + 1)\epsilon''$, and so

$$\left[\!\left[UB'_hU^{-1} - A_h\right]\!\right] < (2\,|||A_h||| + 1)\epsilon''. \tag{4.3.o}$$

Now we choose $\epsilon'' = \epsilon''(\epsilon) > 0$ as

$$\epsilon'' = \frac{\epsilon}{2\,\mathrm{Max}_{h=1,\dots,m}\,|||A_h||| + 1}. \tag{4.3.ξ}$$

Put $\mathbf{N} = \mathbf{N}_3 = U\mathbf{N}^+U^{-1}$. Then $\mathbf{N}$ is a factor in the case (I_q) along with $\mathbf{N}^+$. Thus (i) holds; (ii) holds since $\mathbf{N}^+ \subseteq \mathbf{M}$, $U \in \mathbf{M}$ implies $\mathbf{N} \subseteq \mathbf{M}$. $B'_h \in \mathbf{N}^+$ gives $B_h = UB'_hU^{-1} \in \mathbf{N}$, while (4.3.$o$) and (4.3.$\xi$) imply $[\![B_h - A_h]\!] < \epsilon$, so (iii) holds. And finally $F' \in \mathbf{N}^+$ and (4.3.ν) imply $E = UF'U^{-1} \in \mathbf{N}$, so that (iv′) holds.

Lemma 4.3.3. *If we add to the assumptions of Lemma 4.3.2 that*

$$EA_h = A_hE = A_h, \tag{α}$$

then we can add to its assertions that

$$EB_h = B_hE = B_h.\text{[36]} \tag{β}$$

Proof. Apply Lemma 4.3.2, but write B'_h for its B_h. Now put $B_h = EB'_hE$. Then (β) is satisfied and $B'_h, E \in \mathbf{N}$ give $B_h \in \mathbf{N}$. By (α), $EA_hE = A_h$. By (iii) of Lemma 4.3.2, $\left[\!\left[B'_h - A_h\right]\!\right] < \epsilon$. Hence $\left[\!\left[E(B'_h - A_h)E\right]\!\right] < \epsilon$, or $[\![B_h - A_h]\!] < \epsilon$. Thus (iii) holds for B_h as well as for B'_h.

The other assertions ((i), (ii), (iv′)) are not affected by our replacing B'_h by B_h. Thus the proof is completed.

Lemma 4.3.4. *If we add to the assumption of Lemma 4.3.3 that $E \in \mathbf{N}^0$, where $\mathbf{N}^0$ is a factor in case (I_p), then we can add to its assertions*

[36] With the same $n_3, \mathbf{N}_3$ as in that lemma.

(v) $\mathbf{N}^0 \subseteq \mathbf{N}$.[33, 36]

Proof. Apply Lemma 4.3.3, but write $\mathbf{N}^+$ for its $\mathbf{N}$.

The considerations at the beginning of the proof of (i) in Lemma 4.1.2 show again that $D_{\mathbf{M}}(G) = D_{\mathbf{N}^0}(G)$ for all projections $G \in \mathbf{N}^0 (\subseteq \mathbf{M})$.

Hence in particular $D_{\mathbf{N}^0}(E) = D_{\mathbf{M}}(E) = 1/p$. Then the argument used in the proof of Lemma 2.6.2 establishes the existence of a system of p-order matrix units $W^0_{u,v}$, $u, v = 1, \ldots, p$, in $\mathbf{N}^0$, with $\sum_{u=1}^{p} W^0_{u,u} = 1$ and $W^0_{1,1} = E$. Thus in terms of (β) of Lemma 4.1.1,

$$\begin{gathered}\text{There exists a matrix base } W^0_{u,v}, \quad u, v = 1, \ldots, p, \text{ of } \mathbf{N}^0, \\ \text{with } W^0_{1,1} = E.\end{gathered} \tag{4.3.π}$$

On the other hand, the argument of Lemma 2.6.3 applied to $\mathbf{N}^+$ establishes the existence of a system of p-order matrix units $W^+_{u,v}$, $u, v = 1, \ldots, p$, in $\mathbf{N}^+$, with $\sum_{u=1}^{p} W^+_{u,u} = 1$ and $W^+_{1,1} = E$. (Observe that this is not a matrix base of $\mathbf{N}^+$, since $\mathbf{N}^+$ is in case (I_q) and not (I_p)!)

Now put

$$U = \sum_{u=1}^{p} W^0_{u,1} W^+_{1,u}.$$

As $W^0_{u,1} \in \mathbf{N}^0 \subseteq \mathbf{M}$, $W^+_{1,u} \in \mathbf{N}^+ \subseteq \mathbf{M}$, so $U \in \mathbf{M}$. One verifies by direct computation that $U^* U = UU^* = 1$, i.e. U is unitary. Also

$$UW^+_{u,v} = W^0_{u,v} U \quad \text{or} \quad W^0_{u,v} = UW^+_{u,v} U^{-1}, \tag{4.3.ρ}$$

and, since $W^0_{1,1} = W^+_{1,1} = E$,

$$EU = UE = E. \tag{4.3.σ}$$

Apply the mapping

$$X \longmapsto UXU^{-1} \tag{4.3.τ}$$

to $\mathbf{N}^+$. This carries $\mathbf{N}^+$ into $\mathbf{N} = U\mathbf{N}^+ U^{-1}$. Now we see: $\mathbf{N}$ is a factor in case (I_q) along with $\mathbf{N}^+$; thus (i) holds. $\mathbf{N}^+ \subseteq \mathbf{M}$, $U \in \mathbf{M}$ imply $\mathbf{N} \subseteq \mathbf{M}$; so (ii) holds. By (β) in Lemma 4.3.3 and (4.3.σ), the mapping (4.3.τ) leaves all B_h fixed; hence they belong to $\mathbf{N}$ as well as to $\mathbf{N}^+$. Now (iii) and (iv′) in Lemma 4.3.2 are unaffected and hence hold in the new case. By (4.3.ρ) the mapping (4.3.τ) carries $W^+_{u,v}$ into $W^0_{u,v}$. Hence $W^+_{u,v} \in \mathbf{N}^+$ gives $W^0_{u,v} \in \mathbf{N}$. Hence by (4.3.π) and (ii) in Lemma 4.1.1, every $A \in \mathbf{N}^0$ has the form

$$A = \sum_{u=1}^{p} \sum_{v=1}^{p} a_{v,u} W^0_{u,v},$$

so $A \in \mathbf{N}$. Thus $\mathbf{N}^0 \subseteq \mathbf{N}$, which is the desired relation (v).

Thus the proof is completed.

Lemma 4.3.5. *Assume that* $\mathbf{M}$ *is approximately finite (A). Given any* $A_1, \ldots, A_m \in \mathbf{M}$*, any* $p = 1, 2, \ldots$*, and any* $\epsilon > 0$*, then there exists an*

$$n_4 = n_4(A_1, \ldots, A_m, p, \epsilon)$$

such that for every $q \geq n_4$ *which is divisible by* p *and every factor* $\mathbf{N}^0 \subseteq \mathbf{M}$ *in case* (I_p)*, there exists a factor*

$$\mathbf{N} = \mathbf{N}_4(q, A_1, \ldots, A_m, \mathbf{N}^0, p, \epsilon)$$

with the following properties:

(i) $\mathbf{N}$ *is in case* (I_q).

(ii) $\mathbf{N} \subseteq \mathbf{M}$.

(iii) There exist $B_1, \ldots, B_m \in \mathbf{N}$ *with* $[\![B_h - A_h]\!] < \epsilon$ *for* $h = 1, \ldots, m$.

(v) $\mathbf{N}^0 \subseteq \mathbf{N}$.

(Note. This lemma is quite similar to Lemma 4.3.4. However, (α) has been dropped from the assumptions and (β) from the conclusion. Thus E is no longer mentioned.)

Proof. Choose a matrix base $W_{u,v}$, $u, v = 1, \ldots, p$, of $\mathbf{N}^0$, and let $E = W_{1,1}$. Put

$$A_{u,v;h} = W_{1,u} A_h W_{v,1}.$$

We have $D_{\mathbf{M}}(E) = D_{\mathbf{M}}(W_{1,1}) = 1/p$ (cf. the argument of (2.7.α)). Clearly

$$E A_{u,v;h} = A_{u,v;h} E = A_{u,v;h}.$$

Now apply Lemma 4.3.4 with the mp^2 operators $A_{u,v;h}$, $h = 1, \ldots, m$, $u, v = 1, \ldots, p$, in place of $A_1, \ldots, A_m$, and ϵ/p^2 in place of ϵ.

Denote the $n_3, \mathbf{N}_3$ which obtain in this way by $n_4, \mathbf{N}_4$. Write $\mathbf{N} = \mathbf{N}_4$. Denote the operators which correspond to the $A_{u,v;h}$ (as the B_h do to the A_h in Lemma 4.3.4) by $B_{u,v;h}$. Clearly:

$$A_h = \sum_{u=1}^{p} \sum_{v=1}^{p} W_{u,1} A_{u,v;h} W_{1,v}. \tag{4.3.ν}$$

Let

$$B_h = \sum_{u=1}^{p} \sum_{v=1}^{p} W_{u,1} B_{u,v;h} W_{1,v}. \tag{4.3.ϕ}$$

Now we argue as follows. (i), (ii), and (v) in Lemma 4.3.4 are unaffected. $B_{u,v;h} \in \mathbf{N}$ and $W_{u,v} \in \mathbf{N}^0 \subseteq \mathbf{N}$ imply $B_h \in \mathbf{N}$. We have $\left[\!\left[B_{u,v;h} - A_{u,v;h}\right]\!\right] < \epsilon/p^2$. Hence

$$\left[\!\left[W_{u,1}(B_{u,v;h} - A_{u,v;h})W_{1,v}\right]\!\right] < \epsilon/p^2$$

and

$$\left[\!\!\left[\sum_{u=1}^{p}\sum_{v=1}^{p} W_{u,1}(B_{u,v;h} - A_{u,v;h})W_{1,v}\right]\!\!\right] < \epsilon.$$

Thus (4.3.ν), (4.3.ϕ), and this inequality yield $[\![B_h - A_h]\!] < \epsilon$. So (iii) holds and the proof is complete.

§4.4 We now use the results of the preceding two sections to compare the two notions of approximate finiteness which we have introduced.

Lemma 4.4.1. *Assume that* **M** *is approximately finite (A) and that the sequence* $[p_1, p_2, \ldots]$ *fulfills (i), (ii) in Lemma 4.1.3. Then* $[p_1, p_2, \ldots]$ *possesses a subsequence* $[p'_1, p'_2, \ldots]$ *such that* **M** *is approximately finite of type* $[p'_1, p'_2, \ldots]$.

Proof. By [2], pp. 387–388, there exists a sequence $A_1, A_2, \ldots \in \mathbf{M}$ which is everywhere dense in **M** in the sense of strong convergence. Put $p'_0 = 1$ and $\mathbf{N}_0 = (\alpha \cdot 1)$. This is a factor $\subseteq \mathbf{M}$ in case (I_1). We shall choose by induction, for every $n = 1, 2, \ldots$, a p'_n and a factor $\mathbf{N}_n \subseteq \mathbf{M}$ in the case ($\mathrm{I}_{p'_n}$). If, for any $n = 1, 2, \ldots$, p'_{n-1} and $\mathbf{N}_{n-1}$ have already been chosen, then we choose p'_n and $\mathbf{N}_n$ in the following way. Apply Lemma 4.3.5 to

$$A_1, \ldots, A_n, \quad p'_{n-1}, \quad 1/n, \quad \mathbf{N}_{n-1}$$

in place of its $A_1, \ldots, A_m, p, \epsilon, \mathbf{N}^0$. Choose p'_n from the sequence $[p_1, p_2, \ldots]$ with $p'_n > \mathrm{Max}(n_4, p'_{n-1})$. Thus p'_n is divisible by p'_{n-1}. Put $q = p'_n$, and then put $\mathbf{N}_n = \mathbf{N}$ (from the above application of Lemma 4.3.5).

Thus $[p'_1, p'_2, \ldots]$ is a subsequence of $[p_1, p_2, \ldots]$. We have $\mathbf{N}_{n-1} \subseteq \mathbf{N}_n$, hence

$$\mathbf{N}_1 \subseteq \mathbf{N}_2 \subseteq \cdots . \tag{4.4.α}$$

$\mathbf{N}_n$ is a factor $\subseteq \mathbf{M}$ of class ($\mathrm{I}_{p'_n}$). Clearly

$$\mathbf{R}(\mathbf{N}_n; n = 1, 2, \ldots) \subseteq \mathbf{M}. \tag{4.4.β}$$

Now for $n \geq h$, there exists a $B_h^{(n)} \in \mathbf{N}_n$ with $\left[\!\!\left[B_h^{(n)} - A_h\right]\!\!\right] < 1/n$. So A_h is a metric limit of $\mathbf{R}(\mathbf{N}_n; n = 1, 2, \ldots)$ and hence, by Theorem I, $A_h \in \mathbf{R}(\mathbf{N}_n; n = 1, 2, \ldots)$. Thus $A_1, A_2, \ldots \in \mathbf{R}(\mathbf{N}_n; n = 1, 2, \ldots)$. Since the $A_1, A_2, \ldots$ are everywhere dense in **M** in the sense of strong convergence, this and (4.4.β) yield

$$\mathbf{R}(\mathbf{N}_n; n = 1, 2, \ldots) = \mathbf{M}. \tag{4.4.γ}$$

Equations (4.4.α), (4.4.γ), and our observations establish that **M** is approximately finite of type $[p_1', p_2', \ldots]$, as stated.

We need one more auxiliary lemma. In view of a later application, we shall prove it in a stronger form than that immediately needed.

Lemma 4.4.2. *Let* $\mathbf{N}, \mathbf{O}$ *be two factors in the cases* (I_p) *and* (I_q) *or* (II_1)*, respectively. Let* p *be a divisor of* r*, and (if* q *exists)* r *a divisor of* q*. Suppose* $\mathbf{N} \subseteq \mathbf{O}$*. Then there exists a factor* $\mathbf{R}$ *in case* (I_r) *such that* $\mathbf{N} \subseteq \mathbf{R} \subseteq \mathbf{O}$*.*

Proof. By Lemma 2.6.2 or Lemma 2.6.3, since **O** is in case (II_1), or in the case (I_q) with q divisible by r, there exists a factor $\mathbf{R}^+ \subseteq \mathbf{O}$ in the case (I_r) with a matrix base $W^+_{o,w}$, $o, w = 1, \ldots, r$. Choose also a matrix base $W^0_{u,v}$, $u, v = 1, \ldots, p$, of **N**. Thus $W^+_{o,w}$ is a system of r-order matrix units with $\sum_{o=1}^r W^+_{o,o} = 1$, and $W^0_{u,v}$ is a system of p-order matrix units with $\sum_{u=1}^p W^0_{u,u} = 1$.

Put

$$W^{++}_{u,v} = \sum_{i=1}^{r/p} W^+_{(u-1)r/p+i,(v-1)r/p+i}. \tag{4.4.δ}$$

Then one verifies by direct calculation that the $W^{++}_{u,v}$, $u, v = 1, \ldots, p$, form a system of p-order units with $\sum_{u=1}^p W^{++}_{u,u} = 1$. One can readily prove that $D_{\mathbf{O}}(W^{++}_{1,1}) = 1/p = D_{\mathbf{O}}(W^0_{1,1})$ in the standard normalization. Thus there exists a partially isometric $V \in \mathbf{O}$ such that $V^*V = W^{++}_{1,1}$, $VV^* = W^0_{1,1}$. Now put

$$U = \sum_{u=1}^p W^0_{u,1} V W^{++}_{1,u}.$$

Clearly $U \in \mathbf{O}$. One verifies by direct computation that $U^*U = UU^* = 1$, and hence that U is unitary. Also

$$UW^{++}_{u,v} = W^0_{u,v}U \quad \text{or} \quad W^0_{u,v} = UW^{++}_{u,v}U^{-1}. \tag{4.4.ϵ}$$

Hence the replacement of $\mathbf{R}^+$ by $U\mathbf{R}^+U^{-1}$, and of the system $W^+_{o,w}$ by the system $UW^+_{o,w}U^{-1}$, leaves all their properties unaffected, but it carries $W^{++}_{u,v}$ into $W^0_{u,v} = UW^{++}_{u,v}U^{-1}$. Consequently, we may assume that we had $W^{++}_{u,v} = W^0_{u,v}$, i.e.

$$W^0_{u,v} = \sum_{i=1}^{r/p} W^+_{(u-1)r/p+i,(v-1)r/p+i}, \tag{4.4.ζ}$$

to begin with.

Thus all $W^0_{u,v} \in \mathbf{R}$, hence $\mathbf{N} \subseteq \mathbf{R}$. We know already that $\mathbf{R}$ is a factor in the case (I_r) and that $\mathbf{R} \subseteq \mathbf{O}$. Thus the proof is completed.

We can now conclude our deductions from Definition 4.3.1.

Lemma 4.4.3. *Under the same assumptions as Lemma 4.4.1, $\mathbf{M}$ is approximately finite of type $[p_1, p_2, \ldots]$.*

Proof. Apply Lemma 4.4.1. Denote its $\mathbf{N}_1, \mathbf{N}_2, \ldots$ by $\mathbf{N}_{1'}, \mathbf{N}_{2'}, \ldots$. Write $0' = 0$, $\mathbf{N}_{0'} = \mathbf{N}_0 = (\alpha \cdot 1)$.

Thus $\mathbf{N}_n$ is now defined for $n = 0', 1', 2', \ldots$ only. Consider a fixed $m' = 1', 2', \ldots$. We have $\mathbf{N}_{(m-1)'} \subseteq \mathbf{N}_{m'}$, and $\mathbf{N}_{(m-1)'}, \mathbf{N}_{m'}$ are factors in the cases $(\mathrm{I}_{p_{(m-1)'}})$, $(\mathrm{I}_{p_{m'}})$, respectively. Each of

$$p_{(m-1)'}, p_{(m-1)'+1}, \ldots, p_{m'-1}, p_{m'}$$

is a divisor of its successor. Hence we obtain, by repeated applications of Lemma 4.4.2, a succession of factors $\mathbf{N}_{(m-1)'+1}, \mathbf{N}_{(m-1)'+2}, \ldots, \mathbf{N}_{m'-1}$, with these properties:

$$\mathbf{N}_{(m-1)'} \subseteq \mathbf{N}_{(m-1)'+1} \subseteq \cdots \subseteq \mathbf{N}_{m'-1} \subseteq \mathbf{N}_{m'},$$

and $\mathbf{N}_n$, $n = (m-1)' + 1, \ldots, m' - 1$, is in the case (I_{p_n}).

So we have defined $\mathbf{N}_n$ for all $n = 1, 2, \ldots$. We have

$$\mathbf{N}_1 \subseteq \mathbf{N}_2 \subseteq \cdots, \qquad (4.4.\eta)$$

and $\mathbf{N}_n$ is a factor in the case (I_{p_n}). Finally, by (4.4.η) and Lemma 4.4.1, we have

$$\mathbf{R}(\mathbf{N}_n; n = 1, 2, \ldots) = \mathbf{R}(\mathbf{N}_{m'}; m = 1, 2, \ldots) = \mathbf{M}. \qquad (4.4.\theta)$$

Equations (4.4.η), (4.4.θ), and our other observations establish that $\mathbf{M}$ is approximately finite of type $[p_1, p_2, \ldots]$, as stated.

§4.5 We proceed now to the second operation announced at the beginning of §4.3.

Definition 4.5.1. *A ring $\mathbf{N}$ is of finite order if it possesses a finite linear basis, i.e. a (fixed finite) system $A^0_1, \ldots, A^0_q$ such that $\mathbf{N}$ is the set of all*

$$A = \sum_{u=1}^{q} x_u A^0_u \qquad (x_1, \ldots, x_q \text{ any complex numbers}).$$

Definition 4.5.2. **M** *is approximately finite (B) if this is true: Given any* $A_1, \ldots, A_m \in \mathbf{M}$ *and any* $\epsilon > 0$*, there exists a ring* **N** *with the following properties:*

(i) **N** *is of finite order;*

(ii) $\mathbf{N} \subseteq \mathbf{M}$*;*

(iii) *There exist* $B_1, \ldots, B_m \in \mathbf{N}$ *with*

$$[\![B_h - A_h]\!] < \epsilon \qquad \textit{for } h = 1, \ldots, m.$$

Lemma 4.5.1. *If* **M** *is approximately finite of type* $[p_1, p_2, \ldots]$*, then it is also approximately finite (B).*

Proof. Suppose $A_1, \ldots, A_m \in \mathbf{M}$ and an $\epsilon > 0$ are given. Form the $\mathbf{N}_1, \mathbf{N}_2, \ldots$ of Definition 4.1.1. By (ii) in Lemma 4.1.4, there exists for each $h = 1, \ldots, m$ an n_h such that, for some $B_h \in \mathbf{N}_{n_h}$,

$$[\![B_h - A_h]\!] < \epsilon. \tag{4.5.α}$$

Put $n = \mathrm{Max}(n_1, \ldots, n_m)$; then $B_h \in \mathbf{N}_{n_h} \subseteq \mathbf{N}_n$, i.e.

$$B_1, \ldots, B_m \in \mathbf{N}_n. \tag{4.5.β}$$

Now $\mathbf{N}_n$ is a factor in the case (I_{p_n}). By (ii) in Lemma 4.1.1 it is a ring of finite order. This, together with (4.5.α) and (4.5.β), completes the proof.

Comparison of Lemma 4.4.3 and Lemma 4.5.1 shows that if we can derive approximate finiteness (A) from (B), then the equivalence of all kinds of approximate finiteness is established. This we accomplish in the next section. In the present section we obtain certain preliminary lemmas on rings of finite order; indeed we shall determine the structure of these.[37] Thus for the present section **N** is to be a ring of finite order.

Lemma 4.5.2. (i) *There is a maximum* r *such that there is a system* $E_1, \ldots, E_r$ *of mutually orthogonal projections* $\neq 0$ *and in* **N**.

(ii) *For such a system (with* r *maximum),* $E = \sum_{s=1}^{r} E_s$ *is the unit of* **N**.

(iii) *Every* E_s *in such a system (with* r *maximum) is minimal (with respect to* **N**) *in the sense of [5], p. 143, Definition 5.1.2.*

[37] The four lemmas which follow are merely an *ad hoc* derivation of Wedderburn's structure theorems for semi-simple algebras—specialized to the present situation. They can also be interpreted as a summary of certain results of unitary representation theory including the theorem of Burnside–Frobenius–Schur for that case.

Proof. Ad (i): Any such system is necessarily linearly independent. Hence the number of its elements must be $\leq$ the q of Definition 4.5.1. Hence the numbers for which there is such a system have a maximum.

Ad (ii): Let E be the unit of $\mathbf{N}$. $\sum_{s=1}^{r} E_s$ is a projection in $\mathbf{N}$; hence $\sum_{s=1}^{r} E_s \leq E$. So $E_{r+1} = E - \sum_{s=1}^{r} E_s$ is a projection in $\mathbf{N}$ which is orthogonal to the original E_s. Hence the assumed maximum property of r necessitates $E_{r+1} = 0$, i.e. $E = \sum_{s=1}^{r} E_s$.

Ad (iii): Consider a projection $F \leq E_s$ in $\mathbf{N}$. Then $E'_s = F$ and $E''_s = E_s - F$ are mutually orthogonal projections in $\mathbf{N}$ which are also orthogonal to all $E_{s'}$, $s' \neq s$. Hence the assumed maximum property of r necessitates $E'_s = 0$ or $E''_s = 0$, i.e. $F = 0$ or E_s. This, however, is the defining property of minimality.

Remark. The center $\mathbf{N} \cdot \mathbf{N}'$ of $\mathbf{N}$ is of finite order along with $\mathbf{N}$. Hence we can apply the above lemma to it, thus obtaining a system $\bar{E}_1, \ldots, \bar{E}_{\bar{r}}$.

By this lemma, $\bar{E} = \sum_{s=1}^{\bar{r}} \bar{E}_s$ is the unit of $\mathbf{N} \cdot \mathbf{N}'$; hence by [2], p. 393, Theorems 3 and 4, it is also the unit of $\mathbf{N}$.

Lemma 4.5.3. *(i) $\bar{E}_s$ (cf. above) is such that its range $\bar{\mathfrak{M}}_s$ reduces $\mathbf{N}, \mathbf{N}', \mathbf{N} \cdot \mathbf{N}'$.*
(ii)

$$(\mathbf{N} \cdot \mathbf{N}')_{(\bar{\mathfrak{M}}_s)} = (\bar{E}_s)_{(\bar{\mathfrak{M}}_s)}.$$

Proof. Ad (i): This is implied by $\bar{E}_s \in \mathbf{N} \cdot \mathbf{N}'$.

Ad (ii): This is equivalent to the statement $\bar{E}_s A = A \bar{E}_s = \alpha \bar{E}_s$ for all $A \in \mathbf{N} \cdot \mathbf{N}'$. Put $\bar{E}_s A = A \bar{E}_s = A'$. Clearly $A' \in \mathbf{N} \cdot \mathbf{N}'$, $\bar{E}_s A' = A' \bar{E}_s = A'$. Since $\bar{E}_s$ is a minimal projection in $\mathbf{N} \cdot \mathbf{N}'$, this implies $A' = \alpha \bar{E}_s$ (cf. the last part of the proof of Lemma 5.1.3 in [5], p. 144).

Lemma 4.5.4. $\mathbf{N}_{(\bar{\mathfrak{M}}_s)}$ *is a factor (in $\bar{\mathfrak{M}}_s$) in a case* (I_{q_s})*, $q_s = 1, 2, \ldots$.*

Proof. If A' is in the center of $\mathbf{N}_{(\bar{\mathfrak{M}}_s)}$, $A' \bar{E}_s$ defines an operator A in $\mathfrak{H}$ which is readily seen to be in the center of $\mathbf{N}$, and for which $A \bar{E}_s = \bar{E}_s A = A$. Thus the center of $\mathbf{N}_{(\bar{\mathfrak{M}}_s)}$ is included in $(\mathbf{N} \cdot \mathbf{N}')_{(\bar{\mathfrak{M}}_s)}$. The converse inclusion is obvious.

It follows that the center of $\mathbf{N}_{(\bar{\mathfrak{M}}_s)}$ is $(\mathbf{N} \cdot \mathbf{N}')_{(\bar{\mathfrak{M}}_s)}$, and hence $\mathbf{N}_{(\bar{\mathfrak{M}}_s)}$ is a factor by (ii) in Lemma 4.5.3.

$\mathbf{N}_{(\bar{\mathfrak{M}}_s)}$ is of finite order along with $\mathbf{N}$. Therefore Lemma 4.5.2 implies that $\mathbf{N}_{(\bar{\mathfrak{M}}_s)}$ contains a minimal projection. Thus $\mathbf{N}_{(\bar{\mathfrak{M}}_s)}$ is a direct factor (cf. [5], pp. 150, 173, Theorem IV and Lemma 8.6.1). Obviously, since it is of finite order, $\mathbf{N}_{(\bar{\mathfrak{M}}_s)}$ is not in a case (I_∞), and hence it is in a case (I_{q_s}), $q_s = 1, 2, \ldots$. This completes the proof.

Lemma 4.5.5. $\mathbf{N}$ *possesses a finite linear base which consists of the operators $W^s_{u,v}$, $s = 1, \ldots, \bar{r}$, $u, v = 1, \ldots, q_s$, where for any fixed s the $W^s_{u,v}$, $u, v = 1, \ldots, q_s$, are a* *system of q_s-order matrix units with $\bar{E}_s = \sum_{u=1}^{q_s} W^s_{u,u}$. Thus for the q of Definition 4.5.1,*

$$q = \sum_{s=1}^{\bar{r}} (q_s)^2.$$

Proof. In virtue of Lemma 4.5.4, we may apply Lemma 4.1.1 and Remark 2 after it to $\mathbf{N}_{(\bar{\mathfrak{M}}_s)}$. So $\mathbf{N}_{(\bar{\mathfrak{M}}_s)}$ possesses a finite linear base of operators $\widetilde{W}^s_{u,v}$ (in $\bar{\mathfrak{M}}_s$), $u, v = 1, \ldots, q_s$, which form a system of q_s-order matrix units in $\bar{\mathfrak{M}}_s$, with $(\bar{E}_s)_{(\bar{\mathfrak{M}}_s)} = \sum_{u=1}^{q_s} \widetilde{W}^s_{u,u}$.

Since $\widetilde{W}^s_{u,v} \in \mathbf{N}_{(\bar{\mathfrak{M}}_s)}$, there is a $W^{s0}_{u,v}$ in $\mathbf{N}$ for which the part in $\bar{\mathfrak{M}}_s$ is $\widetilde{W}^s_{u,v}$. Let $W^s_{u,v} = W^{s0}_{u,v}\bar{E}_s$. Since $\bar{E}_s \in \mathbf{N} \cdot \mathbf{N}'$, we have $W^s_{u,v} \in \mathbf{N}$, and also $W^s_{u,v} = W^{s0}_{u,v}\bar{E}_s = \bar{E}_s W^{s0}_{u,v}$. Clearly $(W^s_{u,v})_{(\bar{\mathfrak{M}}_s)} = \widetilde{W}^s_{u,v}$, and one can readily show that the $W^s_{u,v}$ form a system of q_s-order matrix units in $\mathfrak{H}$ with $\bar{E}_s = \sum_{u=1}^{q_s} W^s_{u,u}$.

Consider an $A_s \in \mathbf{N}$ with $\bar{E}_s A_s = A_s \bar{E}_s = A_s$. Then A_s has the part $(A_s)_{(\bar{\mathfrak{M}}_s)}$ in $\bar{\mathfrak{M}}_s$ and 0 in $\bar{\mathfrak{M}}_s^{\perp}$. In the sense, then, that $\bar{E}_s$ is a transformation from $\mathfrak{H}$ to $\bar{\mathfrak{M}}_s$, $A_s = (A_s)_{(\bar{\mathfrak{M}}_s)}\bar{E}_s$, and also $W^s_{u,v} = \widetilde{W}^s_{u,v}\bar{E}_s$. As $(A_s)_{(\bar{\mathfrak{M}}_s)} \in \mathbf{N}_{(\bar{\mathfrak{M}}_s)}$, we have, from the first paragraph of the proof,

$$(A_s)_{(\bar{\mathfrak{M}}_s)} = \sum_{u=1}^{q_s} \sum_{v=1}^{q_s} a^s_{u,v} \widetilde{W}^s_{u,v},$$

and hence

$$A_s = (A_s)_{(\bar{\mathfrak{M}}_s)}\bar{E}_s = \sum_{u=1}^{q_s} \sum_{v=1}^{q_s} a^s_{u,v} \widetilde{W}^s_{u,v}\bar{E}_s = \sum_{u=1}^{q_s} \sum_{v=1}^{q_s} a^s_{u,v} W^s_{u,v}.$$

Consider now any $A \in \mathbf{N}$. Put $A_s = \bar{E}_s A$, which also equals $A\bar{E}_s$, since $\bar{E}_s \in \mathbf{N} \cdot \mathbf{N}'$. By the above, $A_s = \sum_{u=1}^{q_s} \sum_{v=1}^{q_s} a^s_{u,v} W^s_{u,v}$. By the remark after Lemma 4.5.2, we have

$$A = \bar{E}A = \sum_{s=1}^{\bar{r}} \bar{E}_s A = \sum_{s=1}^{\bar{r}} \sum_{u=1}^{q_s} \sum_{v=1}^{q_s} a^s_{u,v} W^s_{u,v}.$$

Thus the $W^s_{u,v}$, $s = 1, \ldots, \bar{r}$, $u, v = 1, \ldots, q_s$, form indeed a finite linear basis of $\mathbf{N}$.

We now prove a lemma which possesses a certain interest of its own. It applies to any ring of finite order $\mathbf{N} \subseteq \mathbf{M}$. We assume that a representation of $\mathbf{N}$ in the sense of Lemma 4.5.5 is given, with the $W^s_{u,v}$ as described there.

Lemma 4.5.6. *Given a (fixed) $q = 1, 2, \ldots$, there exists a factor $\mathbf{O}$ in case (I_q), with $\mathbf{N} \subseteq \mathbf{O} \subseteq \mathbf{M}$, if and only if this is true for $\mathbf{N}$:*

$$\textit{All } qD_{\mathbf{M}}(W^s_{1,1}) \textit{ are integers.} \tag{4.5.γ}$$

Proof. Assume that an $\mathbf{O}$ with the specified properties exists. The considerations at the beginning of the proof of (i) in Lemma 4.1.2 show again that $D_{\mathbf{M}}(G) = D_{\mathbf{O}}(G)$ for all projections $G \in \mathbf{O}$ ($\subseteq \mathbf{M}$). Hence, in particular, $D_{\mathbf{M}}(W^s_{1,1}) = D_{\mathbf{O}}(W^s_{1,1})$, since $W^s_{1,1}$ is a projection in $\mathbf{N} \subseteq \mathbf{O}$. But as $\mathbf{O}$ is in case (I_q), $D_{\mathbf{M}}(W^s_{1,1}) = D_{\mathbf{O}}(W^s_{1,1})$ must have a value $0, 1/q, \ldots, 1$; i.e. $qD_{\mathbf{M}}(W^s_{1,1})$ must be an integer. Thus (4.5.γ) holds.

Sufficiency: Assume that $\mathbf{N}$ fulfills (4.5.γ). Considering Lemma 4.5.5, we must prove this:

$$\begin{cases} \text{Let a system of mutually orthogonal projections } \neq 0 \text{ from } \mathbf{M} \text{ be} \\ \text{given: } \bar{E}_1, \ldots, \bar{E}_{\bar{r}}. \text{ For each } s, \text{ let a system of } q_s\text{-order matrix units} \\ W^s_{u,v} \text{ from } \mathbf{M} \text{ be given, with } \bar{E}_s = \sum_{u=1}^{q_s} W^s_{u,u} \text{ and } q_s = 1, 2, \ldots. \\ \text{Let, for a given } q, \text{ every } qD_{\mathbf{M}}(W^s_{1,1}) \text{ be an integer. Then there} \\ \text{exists a factor } \mathbf{O} \subseteq \mathbf{M} \text{ in a case } (\mathrm{I}_q) \text{ such that all } W^s_{u,v} \in \mathbf{O}. \end{cases} \tag{4.5.δ}$$

We therefore disregard $\mathbf{N}$ completely and deal with (4.5.δ) alone.

Proof of (4.5.δ). We have $D_{\mathbf{M}}(W^s_{1,1}) = p_s/q$, $p_s = 1, 2, \ldots$. The projections $W^s_{1,1}$ and $W^s_{u,u}$ are equivalent in virtue of the partially isometric $W^s_{u,1}$ (cf. Lemma 2.6.1). Hence

$$D_{\mathbf{M}}(W^s_{u,u}) = p_s/q, \qquad D_{\mathbf{M}}(\bar{E}_s) = p_s q_s/q. \tag{4.5.ϵ}$$

We may assume that

$$\sum_{s=1}^{\bar{r}} \bar{E}_s = 1. \tag{4.5.ζ}$$

Otherwise we could put $\bar{E}_{\bar{r}+1} = 1 - \sum_{s=1}^{\bar{r}} \bar{E}_s$. Then $\bar{E}_{\bar{r}+1}$ is a projection $\neq 0$. Also $qD_{\mathbf{M}}(\bar{E}_{\bar{r}+1}) = q - \sum_{s=1}^{\bar{r}} qD_{\mathbf{M}}(\bar{E}_s)$ is an integer. Thus we could add $\bar{E}_{\bar{r}+1}$ to the system $\bar{E}_1, \ldots, \bar{E}_{\bar{r}}$, with $q_{\bar{r}+1} = 1$ and $W^{\bar{r}+1}_{1,1} = \bar{E}_{\bar{r}+1}$.

By (4.5.ϵ) and (4.5.ζ), $\sum_{s=1}^{\bar{r}} p_s q_s = q$. We put $l_s = \sum_{t=1}^{s} p_t q_t$, for $s = 0, 1, \ldots, \bar{r}$. Then

$$\begin{cases} l_s = l_{s-1} + p_s q_s, & s = 1, \ldots, \bar{r}, \\ l_0 = 0, & l_{\bar{r}} = q. \end{cases} \tag{4.5.η}$$

As $\mathbf{M}$ is in case (II_1), by Lemma 2.6.2 there exists a factor $\mathbf{O}^+ \subseteq \mathbf{M}$ in the case (I_q), with its matrix base $W^+_{o,w}$, $o, w = 1, \ldots, q$. Thus $W^+_{o,w}$ is a system of q-order matrix units, with $\sum_{u=1}^{q} W^+_{u,u} = 1$.

Put

$$E^+_s = \sum_{u=l_{s-1}+1}^{l_s} W^+_{u,u} \qquad (s = 1, \ldots, \bar{r}), \tag{4.5.θ}$$

$$\begin{cases} W^{+s}_{u,v} = \displaystyle\sum_{i=1}^{p_s} W^{+}_{l_{s-1}+(u-1)p_s+i,\, l_{s-1}+(v-1)p_s+i} \\ \qquad\qquad \text{for } s = 1,\dots,\bar{r}, \quad u,v = 1,\dots,q_s. \end{cases} \tag{4.5.ι}$$

Then one verifies by a direct computation that the E^+_s are mutually orthogonal projections $\neq 0$, with $\sum_{s=1}^{\bar{r}} E^+_s = 1$, and (using Definition 2.6.1) that the $W^{+s}_{u,v}$ form, for any fixed s, a system of q_s-order matrix units with $\sum_{u=1}^{q_s} W^{+s}_{u,u} = E^+_s$. All these operators are in $\mathbf{M}$.

The $W^+_{u,u}$ are equivalent (due to the $W^+_{u,v}$), and since $\sum_{u=1}^{q} W^+_{u,u} = 1$, $D_{\mathbf{M}}(W^+_{u,u}) = 1/q$. Hence by (4.5.$\iota$), $D_{\mathbf{M}}(W^{+s}_{1,1}) = p_s/q$, and by (4.5.$\epsilon$), $D_{\mathbf{M}}(W^{+s}_{1,1}) = D_{\mathbf{M}}(W^{s}_{1,1})$. Consequently there exists a partially isometric operator $V_s \in \mathbf{M}$, with initial projection $W^{+s}_{1,1}$ and final projection $W^{s}_{1,1}$.

Now put

$$U = \sum_{s=1}^{\bar{r}} \sum_{u=1}^{q_s} W^{s}_{u,1} V_s W^{+s}_{1,u}.$$

Clearly $U \in \mathbf{M}$. One verifies, by direct computation, $U^*U = UU^* = 1$, which implies that U is unitary, and similarly that

$$UW^{+s}_{u,v} = W^{s}_{u,v}U \quad \text{or} \quad W^{s}_{u,v} = UW^{+s}_{u,v}U^{-1}. \tag{4.5.κ}$$

$\mathbf{O} = U\mathbf{O}^+U^{-1}$ is a factor $\subseteq \mathbf{M}$ and in case (I_q), along with $\mathbf{O}^+$. By (4.5.ι), $W^{+s}_{u,v} \in \mathbf{O}^+$, and hence by (4.5.$\kappa$), $W^{s}_{u,v} = UW^{+s}_{u,v}U^{-1} \in U\mathbf{O}^+U^{-1} = \mathbf{O}$. Thus all $W^{s}_{u,v} \in \mathbf{O}$.

§4.6 We are now in a position to complete our proof of the relationships between the various kinds of approximate finiteness.

Lemma 4.6.1. *Suppose $\mathbf{N}, \mathbf{M}, \bar{E}_1, \dots, \bar{E}_{\bar{r}}$, and the $W^{s}_{u,v}$ are as in Lemma 4.5.5, with $\bar{E} = \sum_{s=1}^{\bar{r}} \bar{E}_s = 1$. For any $\epsilon > 0$ there exists an $n_5 = n_5(\mathbf{N}, \epsilon)$ such that for every $q \geq n_5$ there exists a projection $E = E(q, \mathbf{N}, \epsilon)$ which is in $\mathbf{M} \cdot \mathbf{N}'$ and has the following properties:*
(i) $D_{\mathbf{M}}(1 - E) < \epsilon$;
(ii) all $qD_{\mathbf{M}}(EW^{s}_{1,1})$ are $\neq 0$ and integers.

Proof. Given a $q = 1, 2, \dots$, we wish to choose for each $s = 1, \dots, \bar{r}$ a $k_s = 1, 2, \dots$ with

$$q\left(D_{\mathbf{M}}(W^{s}_{1,1}) - \frac{\epsilon}{\sum_{s=1}^{\bar{r}} q_s}\right) < k_s \leq qD_{\mathbf{M}}(W^{s}_{1,1}). \tag{4.6.α}$$

This is certainly feasible if $qD_{\mathbf{M}}(W^{s}_{1,1}) \geq 1$ and $q\epsilon/\sum_{s=1}^{\bar{r}} q_s \geq 1$, for $s = 1, \dots, \bar{r}$. We assume therefore that $q \geq n_5$, with

$$n_5 = n_5(\mathbf{N}, \epsilon) = \mathrm{Max}\left(\frac{\sum_{s=1}^{\bar{r}} q_s}{\epsilon}, \frac{1}{D_{\mathbf{M}}(W_{1,1}^s)}, \ s = 1, \ldots, \bar{r} \right). \tag{4.6.β}$$

Now (4.6.α) and $k_s \neq 0$ give

$$D_{\mathbf{M}}(W_{1,1}^s) - \frac{\epsilon}{\sum_{s=1}^{\bar{r}} q_s} < \frac{k_s}{q} \leq D_{\mathbf{M}}(W_{1,1}^s), \qquad \frac{k_s}{q} \neq 0.$$

As $\mathbf{M}$ is in case (II_1), this implies that we can find a projection $G^s \in \mathbf{M}$, with

$$D_{\mathbf{M}}(G^s) = k_s/q \neq 0, \tag{4.6.γ}$$

and

$$G^s \leq W_{1,1}^s, \tag{4.6.δ}$$

and that then

$$D_{\mathbf{M}}(W_{1,1}^s - G^s) < \frac{\epsilon}{\sum_{s=1}^{\bar{r}} q_s}. \tag{4.6.ϵ}$$

$W_{u,1}^s$ is a partially isometric operator with initial projection $W_{1,1}^s$ and final projection $W_{u,u}^s$. Its mapping carries therefore the $G^s \leq W_{1,1}^s$ into a projection $G_u^s \in \mathbf{M}$, with

$$G_u^s \leq W_{u,u}^s. \tag{4.6.ζ}$$

The same mapping carries (4.6.ϵ) into

$$D_{\mathbf{M}}(W_{u,u}^s - G_u^s) < \frac{\epsilon}{\sum_{s=1}^{\bar{r}} q_s}. \tag{4.6.η}$$

Finally, $W_{u,v}^s W_{v,1}^s = W_{u,1}^s$ implies that the mapping of $W_{u,v}^s$ carries G_v^s into G_u^s. This may be written

$$W_{u,v}^s G_v^s = G_u^s W_{u,v}^s. \tag{4.6.θ}$$

For $s \neq t$ or $v \neq w$, the initial projection $W_{v,v}^s$ of $W_{u,v}^s$ is orthogonal to $W_{w,w}^t$, hence, a fortiori, to $G_w^t \leq W_{w,w}^t$. So we have

$$W_{u,v}^s G_w^t = 0 \qquad \text{for } s \neq t \text{ or } v \neq w, \tag{4.6.ι}$$

$$G_w^t W_{u,v}^s = 0 \qquad \text{for } s \neq t \text{ or } u \neq w. \tag{4.6.κ}$$

Now put $E = \sum_{s=1}^{\bar{r}} \sum_{u=1}^{q_s} G_u^s$. Clearly $E \in \mathbf{M}$, and since $\sum_{s=1}^{\bar{r}} \sum_{u=1}^{q_s} W_{u,u}^s = \sum_{s=1}^{\bar{r}} \bar{E}_s = 1$, we have $1 - E = \sum_{s=1}^{\bar{r}} \sum_{u=1}^{q_s} (W_{u,u}^s - G_u^s)$. Hence (4.6.$\eta$) yields

$$D_{\mathbf{M}}(1 - E) < \epsilon. \tag{4.6.λ}$$

Equations (4.6.θ), (4.6.ι), and (4.6.κ) give

$$W^s_{u,v}E = EW^s_{u,v} \; (= W^s_{u,v}G^s_v = G^s_u W^s_{u,v}).$$

So E commutes with $W^s_{u,v}$, and hence, by Lemma 4.5.5, with all of $\mathbf{N}$. Since $E \in \mathbf{M}$, this yields

$$E \in \mathbf{M} \cdot \mathbf{N}'. \tag{4.6.μ}$$

Finally, direct computation shows that

$$EW^s_{1,1} = G^s_1 = G^s. \tag{4.6.ν}$$

Our assertions now follow immediately. Equation (4.6.μ) gives the preliminary one, (4.6.λ) implies (i), and (4.6.ν) and (4.6.γ) yield (ii). Thus the lemma is proven.

It is now possible to take the final step.

Lemma 4.6.2. *If $\mathbf{M}$ is approximately finite (B), then it is also approximately finite (A).*

Proof. We must verify the criteria of Definition 4.3.1 for $\mathbf{M}$.

Let therefore a system $A_1, \dots, A_m \in \mathbf{M}$ and an $\epsilon > 0$ be given. Apply Definition 4.5.2 to these $A_1, \dots, A_m$ and to $\epsilon/2$, thus obtaining the ring $\mathbf{N}$ and $B_1, \dots, B_m \in \mathbf{N}$ described there. Thus

$$[\![B_s - A_s]\!] < \epsilon/2. \tag{4.6.o}$$

$\mathbf{N}$ is a ring of finite order $\subseteq \mathbf{M}$. We form $\bar{E}_1, \dots, \bar{E}_{\bar{r}}$ and the $W^s_{u,v}$ in the sense of Lemma 4.5.5 for this $\mathbf{N}$. To these and to

$$\epsilon' = \left(\frac{\epsilon}{2\,\mathrm{Max}(|\!|\!|B_s|\!|\!|\,;\; s = 1, \dots, m)}\right)^2 \tag{4.6.π}$$

we apply Lemma 4.6.1, thus obtaining $n_5 = n_5(\mathbf{N}, \epsilon')$. For any $q \geq n_5$, we form the projection $E = E(q, \mathbf{N}, \epsilon')$ of Lemma 4.6.1.

Accordingly we have $E \in \mathbf{M} \cdot \mathbf{N}'$, and $\mathbf{N}_E = E \cdot \mathbf{N} = \mathbf{N}E$ is a finite-order ring $\subseteq \mathbf{M}$ along with $\mathbf{N}$. Thus $E\bar{E}_1, \dots, E\bar{E}_{\bar{r}}$ and the $EW^s_{u,v}$ perform for $\mathbf{N}_E$ the functions of $\bar{E}_1, \dots, \bar{E}_{\bar{r}}$ and the $W^s_{u,v}$ for $\mathbf{N}$ (cf. Lemma 4.5.5). By (ii) in Lemma 4.6.1, all $qD_{\mathbf{M}}(EW^s_{1,1})$ are integers. Thus Lemma 4.5.6 is applicable to $\mathbf{N}_E$, and we obtain a factor $\mathbf{O}$ in case (I_q), with

$$\mathbf{N}_E \subseteq \mathbf{O} \subseteq \mathbf{M}. \tag{4.6.ρ}$$

$B_s \in \mathbf{N}$ implies

$$C_s = EB_s \in \mathbf{N}_E \subseteq \mathbf{O}. \tag{4.6.σ}$$

Now (i) in Lemma 4.6.1 gives $D_{\mathbf{M}}(1-E) < \epsilon'$. Hence $[\![1-E]\!] < \epsilon'^{1/2}$[35], and therefore

$$[\![B_s - C_s]\!] = [\![B_s(1-E)]\!] < |||B_s|||\,\epsilon'^{1/2}.$$

Hence by (4.6.π),

$$[\![B_s - C_s]\!] < \epsilon/2. \tag{4.6.τ}$$

Combining (4.6.o) and (4.6.τ) gives

$$[\![C_s - A_s]\!] < \epsilon. \tag{4.6.υ}$$

Thus we can satisfy Definition 4.3.1 by putting

$$n_1 = n_1(A_1, \ldots, A_m; \epsilon) = n_5(\mathbf{N}, \epsilon')^{38},$$

and then, for $q \geq n_1 = n_5$, $\mathbf{N}_1 = \mathbf{N}_1(q, A_1, \ldots, A_m, \epsilon) = \mathbf{O}$.[38] Then our $C_1, \ldots, C_m$ replace the $B_1, \ldots, B_m$ of Definition 4.3.1, and (i)–(iii) there obtain as follows: (i) holds by virtue of the construction of $\mathbf{O}$, (ii) follows from (4.6.ρ), and (iii) coincides with (4.6.υ). Thus the proof is completed.

We could now make the final steps as indicated after Lemma 4.5.1 above, but we introduce one more variant of the concept of approximate finiteness.

Definition 4.6.1. $\mathbf{M}$ *is approximately finite (C) if there exists a sequence of rings* $\mathbf{N}_1, \mathbf{N}_2, \ldots$ *with the following properties:*

(i) $\mathbf{N}_n$ *is of finite order;*

(ii) $\mathbf{N}_1 \subseteq \mathbf{N}_2 \subseteq \cdots$;

(iii) the ring $\mathbf{M}$ *is generated by the* $\mathbf{N}_1, \mathbf{N}_2, \ldots$:

$$\mathbf{M} = \mathbf{R}(\mathbf{N}_n; n = 1, 2, \ldots).$$

Lemma 4.6.3. *If* $\mathbf{M}$ *is approximately finite of type* $[p_1, p_2, \ldots]$, *then it is also approximately finite (C).*

Proof. Immediate by comparison of Definitions 4.1.1 and 4.6.1.

Lemma 4.6.4. *If* $\mathbf{M}$ *is approximately finite (C), then it is also approximately finite (B).*

Proof. (ii) and (iii) in Definition 4.6.1 coincide with those in Definition 4.1.1. Hence Lemma 4.1.4 applies to our $\mathbf{N}_1, \mathbf{N}_2, \ldots$.

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[38]Observe that the (indirectly defined) expression at the extreme right has the proper dependence—i.e. that one indicated by the expression in the middle.

Let then the $A_1, \ldots, A_m \in \mathbf{M}$ and the $\epsilon > 0$ of Definition 4.5.2 be given. Owing to Lemma 4.1.4, A_k is the limit of a metrically convergent sequence from the set-theoretical sum of the $\mathbf{N}_1, \mathbf{N}_2, \ldots$. Hence there exists a $B_k \in \mathbf{N}_{n_k}$ with $[\![B_k - A_k]\!] < \epsilon$.

Now put $n = \mathrm{Max}(n_1, \ldots, n_m)$. Then every $B_k \in \mathbf{N}_{n_k} \subseteq \mathbf{N}_n$. Thus $\mathbf{N} = \mathbf{N}_n$ meets all the requirements (i)–(iii) of Definition 4.5.2.

We can now prove:

Theorem XII. *All kinds of approximate finiteness are equivalent to each other: (A), (B), (C), and every type $[p_1, p_2, \ldots]$ (for every sequence $[p_1, p_2, \ldots]$ which fulfills (i), (ii) of Lemma 4.1.3).*

We shall therefore use the expression approximately finite from now on, without any further qualification.

Proof. The implications

$$(A) \longrightarrow \text{type } [p_1, p_2, \ldots] \longrightarrow (B) \longrightarrow (A)$$

were established in Lemmas 4.4.3, 4.5.1, and 4.6.2, respectively. Consequently

$$\text{type } [p_1, p_2, \ldots] \Longleftrightarrow (A) \Longleftrightarrow (B).$$

Further,

$$\text{type } [p_1, p_2, \ldots] \longrightarrow (C) \longrightarrow (B)$$

by Lemmas 4.6.3, 4.6.4, respectively; hence

$$\text{type } [p_1, p_2, \ldots] \Longleftrightarrow (C)$$

too.

§4.7 We have not shown so far that approximately finite factors exist at all. In §§5.2, 5.3 and 5.6 we shall give various examples of such factors. Actually it is more difficult to show that there are factors in case (II_1) which are not approximately finite.

Our immediate goal is to prove a general theorem which yields the existence of approximately finite factors as a byproduct.

A preparatory lemma is necessary:

Lemma 4.7.1. *Let a factor $\mathbf{N} \subseteq \mathbf{M}$ in a case (I_p), $p = 1, 2, \ldots$, be given. If an $A \in \mathbf{N}$ possesses these two properties* *(i)* $[\![AX - XA]\!] \leq \epsilon \, |||X||| $ *for every* $X \in \mathbf{N}$,

(ii) $\mathrm{Tr}_{\mathbf{M}}(A) = 0$,

then $[\![A]\!] \leq \epsilon$.

Proof. For any unitary $U \in \mathbf{N}$ we have $|||U||| = \left|\left|\left|U^{-1}\right|\right|\right| = 1$. Hence by (i),

$$\left[\!\left[UAU^{-1} - A\right]\!\right] = \left[\!\left[U(AU^{-1} - U^{-1}A)\right]\!\right] \leq \left[\!\left[AU^{-1} - U^{-1}A\right]\!\right] \leq \epsilon.$$

Also by (ii)

$$\mathrm{Tr}_{\mathbf{M}}(UAU^{-1}) = \mathrm{Tr}_{\mathbf{M}}(A) = 0.$$

If $U_1, \ldots, U_m$ are unitary and $\in \mathbf{N}$, then we have by the above $\left[\!\left[U_iAU_i^{-1} - A\right]\!\right] \leq \epsilon$, $\mathrm{Tr}_{\mathbf{M}}(U_iAU_i^{-1}) = 0$. So for

$$B = \frac{1}{m}\left(\sum_{i=1}^{m} U_iAU_i^{-1}\right) \tag{4.7.α}$$

the evaluations

$$[\![B - A]\!] \leq \epsilon, \tag{4.7.β}$$

$$\mathrm{Tr}_{\mathbf{M}}(B) = 0 \tag{4.7.γ}$$

obtain.

Now choose an isomorphism $\mathcal{K}$ between $\mathbf{N}$ and the system of all (complex) p-order matrices, in the sense of (α) in Lemma 4.1.1. Put

$$\mathcal{K}A = a = \{a_{t,s}\}, \qquad \mathcal{K}B = b = \{b_{t,s}\}.$$

Form next the operators $V, X_1, \ldots, X_p \in \mathbf{N}$ with

$$\mathcal{K}V = v = \{v_{s,t}\}, \qquad v_{s,t} = \begin{cases} 1, & s - t \equiv 1 \pmod{p}, \\ 0, & \text{otherwise}, \end{cases}$$

$$\mathcal{K}X_r = x_r = \{x^r_{s,t}\}, \qquad x^r_{s,t} = \begin{cases} 1, & t = s \neq r, \\ -1, & t = s = r, \\ 0, & \text{otherwise}. \end{cases}$$

The matrices v, x_r are unitary. Hence the operators V, X_r are also unitary.

Now put $m = p2^p$ and let $U_1, \ldots, U_m$ be the operators $V^h X_1^{k_1} \cdots X_p^{k_p}$, $h = 0, 1, \ldots, p-1$; $k_1, \ldots, k_p = 0, 1$. Then a combination of these formulas gives, due to (4.7.α) by direct computation,[39]

$$b_{t,s} = \begin{cases} \dfrac{1}{p}\displaystyle\sum_{u=1}^{p} a_{u,u} = a_0, & t = s, \\ 0, & \text{otherwise.} \end{cases}$$

Consequently

$$B = a_0 \cdot 1. \tag{4.7.δ}$$

(4.7.δ) and (4.7.γ) give together $0 = \mathrm{Tr}_{\mathbf{M}}(B) = a_0$, i.e. $B = 0$. Hence (4.7.β) becomes $[\![A]\!] \le \epsilon$ as desired.

We are able to prove:

---

[39]This computation may be outlined as follows. Let

$$\epsilon_{s,t}^{(r)} = (-1)^{\delta_{s,r}+\delta_{r,t}},$$

i.e.

$$\epsilon_{s,t}^{(r)} = \begin{cases} 1, & r \neq t \text{ and } s \neq r \text{ or if } r = s = t, \\ -1, & \text{if either } r = s \text{ and } r \neq t \text{ or } r \neq s \text{ and } r = t. \end{cases}$$

Then if $\mathcal{K} X_r A X_r^{-1} = c^{(r)} = (c_{s,t}^{(r)})$,

$$c_{s,t}^{(r)} = \epsilon_{s,t}^{(r)} a_{s,t}.$$

Let

$$B_0 = \sum_{k_1,\ldots,k_p=0,1} (X_1^{k_1} \cdots X_p^{k_p}) A (X_1^{k_1} \cdots X_p^{k_p})^{-1}$$

and $\mathcal{K} B_0 = d = (d_{t,s})$. Then

$$\begin{aligned} d_{t,s} &= \sum_{k_1,\ldots,k_p=0,1} (\epsilon_{s,t}^{(1)})^{k_1} \cdots (\epsilon_{s,t}^{(p)})^{k_p} a_{s,t} \\ &= \big((\epsilon_{s,t}^{(1)})^0 + (\epsilon_{s,t}^{(1)})^1\big) \cdots \big((\epsilon_{s,t}^{(p)})^0 + (\epsilon_{s,t}^{(p)})^1\big) a_{s,t} \\ &= (1 + \epsilon_{s,t}^{(1)}) \cdots (1 + \epsilon_{s,t}^{(p)}) a_{s,t}. \end{aligned}$$

Unless $s = t$, $\epsilon_{s,t}^{(s)} = -1$ and $d_{s,t} = 0$. If $s = t$, $d_{s,s} = 2^p a_{s,s}$. Thus B_0 is a diagonal matrix.

The effect of V on a diagonal matrix is just to permute the diagonal terms cyclicly. Hence if B and $b_{s,t}$ are as in the text,

$$B = \frac{1}{p 2^p} \sum_{h=0}^{p-1} V^h B_0 V^{-h}$$

and

$$b_{s,t} = 0 \quad \text{if } s \neq t, \qquad b_{s,s} = \frac{1}{p} \sum_{u=1}^{p} a_{u,u}.$$

Theorem XIII. *If* $\mathbf{M}$ *is a factor in case* (II_1), $\mathbf{M}$ *contains a subring* $\mathbf{O}$ *which is a factor in case* (II_1), *and approximately finite.*

Proof. Choose a sequence $[p_1, p_2, \ldots]$ which fulfills (i), (ii) in Lemma 4.1.3, e.g. $p_n = 2^n$.

Put $p_0 = 1$ and $\mathbf{N}_0 = (\alpha 1)$. This is a factor $\subseteq \mathbf{M}$ and in case (I_1). We shall choose by induction for every $n = 1, 2, \ldots$ a factor $\mathbf{N}_n \subseteq \mathbf{M}$ in the case (I_{p_n}). If for any $n = 1, 2, \ldots$, $\mathbf{N}_{n-1}$ has already been chosen, we choose $\mathbf{N}_n$ in the following way: $\mathbf{N}_{n-1}, \mathbf{M}$ are factors in the cases ($I_{p_{n-1}}$), (II_1) respectively; $\mathbf{N}_{n-1} \subseteq \mathbf{M}$, p_{n-1} is a divisor of p_n. Apply therefore Lemma 4.4.2 to $\mathbf{N}_{n-1}, \mathbf{M}, p_{n-1}, p_n$ in place of its $\mathbf{N}, \mathbf{O}, p, r$. Put $\mathbf{N}_n = \mathbf{R}$. Then $\mathbf{N}_n$ is a factor $\subseteq \mathbf{M}$ in the case (I_{p_n}).

Thus

$$\mathbf{N}_1 \subseteq \mathbf{N}_2 \subseteq \cdots \subseteq \mathbf{M}. \tag{4.7.ϵ}$$

Denote by S the set-theoretical sum of the sequence $\mathbf{N}_1, \mathbf{N}_2, \ldots$. Owing to (4.7.$\epsilon$), S is an algebra along with $\mathbf{N}_1, \mathbf{N}_2, \ldots$.

Denote the set of the limits of all metrically convergent sequences from S by S_1. The same argument as in the proof of Lemma 4.1.4 shows that S_1 is an algebra along with S and then that it is a ring. Hence $S \subseteq S_1$ implies $\mathbf{R}(S) \subseteq S_1$. On the other hand $\mathbf{R}(S)$ is closed in the sense of metric convergence by Theorem I. Hence $S \subseteq \mathbf{R}(S)$ implies $S_1 \subseteq \mathbf{R}(S)$. Consequently $\mathbf{R}(S) = S_1$ and so

$$\mathbf{M} \supseteq \mathbf{R}(\mathbf{N}_n; n = 1, 2, \ldots) = \mathbf{R}(S) = S_1. \tag{4.7.ζ}$$

Next we prove that the ring S_1 is a factor, i.e. that every $A \in S_1 \cdot S_1'$ belongs to $(\alpha \cdot 1)$.

Consider an $A \in S_1 \cdot S_1'$. Choose $\epsilon > 0$. As $A \in S_1$, there exists a $B_\epsilon \in \mathbf{N}_{p_{n_\epsilon}}$ with

$$[\![B_\epsilon - A]\!] < \epsilon. \tag{4.7.η}$$

Put $C_\epsilon = B_\epsilon - \mathrm{Tr}_{\mathbf{M}}(B_\epsilon) \cdot 1$. Then $C_\epsilon \in \mathbf{N}_{p_{n_\epsilon}}$ too, and

$$\mathrm{Tr}_{\mathbf{M}}(C_\epsilon) = 0. \tag{4.7.θ}$$

Consider an $X \in \mathbf{N}_{p_{n_\epsilon}}$. As $A \in S_1'$, $X \in S_1$, so $AX - XA = 0$. Also, clearly $B_\epsilon X - XB_\epsilon = C_\epsilon X - XC_\epsilon$. Hence $C_\epsilon X - XC_\epsilon = (B_\epsilon - A)X - X(B_\epsilon - A)$ and so $[\![C_\epsilon X - XC_\epsilon]\!] \leq 2\,|||X|||\,[\![B_\epsilon - A]\!]$. This and (4.7.$\eta$) yield

$$[\![C_\epsilon X - XC_\epsilon]\!] \leq 2\,|||X|||\,\epsilon. \tag{4.7.ι}$$

(4.7.θ) and (4.7.ι) permit us to apply Lemma 4.7.1 to $C_\epsilon, 2\epsilon$ in place of its A, ϵ. Thus

$$[\![C_\epsilon]\!] \leq 2\epsilon.$$

Put $\mathrm{Tr}_{\mathbf{M}}(B_\epsilon) = a_\epsilon$. Then the above equation means

$$[\![B_\epsilon - a_\epsilon \cdot 1]\!] \leq 2\epsilon.$$

Hence by (4.7.η)

$$[\![A - a_\epsilon \cdot 1]\!] < 3\epsilon. \tag{4.7.κ}$$

As (4.7.κ) holds for all $\epsilon > 0$, it proves that A is the limit of a metrically convergent sequence from $(\alpha \cdot 1)$. As $(\alpha \cdot 1)$ is obviously closed in this sense, we have $A \in (\alpha \cdot 1)$ as desired.

Thus S_1 is a factor. The considerations at the beginning of the proof of (i) in Lemma 4.1.2 show again that $D_{S_1}(E) = D_{\mathbf{M}}(E)$ for all projections $E \in S_1$. Hence S_1 is in a finite case, along with $\mathbf{M}$. If S_1 were in a case (I_q), $q = 1, 2, \ldots$, we could argue as follows: $\mathbf{N}_{p_n} \subseteq S_1$ and $\mathbf{N}_{p_n}$ is a factor in case (I_{p_n}). Hence p_n is a divisor of q by (i) in Lemma 4.1.2. But this is impossible, since $p_n \to \infty$ as $n \to \infty$. Therefore S_1 must be in the case (II_1).

Now (4.7.ϵ), (4.7.ζ) prove that S_1 is approximately finite of type $[p_1, p_2, \ldots]$—i.e. approximately finite by Theorem XII.

Thus $\mathbf{O} = S_1$ meets all our requirements.

And to conclude:

Theorem XIV. *Approximately finite factors* $\mathbf{M}$ *(in the case II$_1$) exist and they have all the same algebraical type* $\overline{\mathbf{M}}$.

We denote this algebraical type by $\overline{\mathbf{A}}$.

Proof. The existence follows from Theorem XIII.

Consider two approximately finite $\mathbf{M}_1, \mathbf{M}_2$. Choose a sequence $[p_1, p_2, \ldots]$ which fulfills (i), (ii) in Lemma 4.1.3, e.g. $p_n = 2^n$. Then $\mathbf{M}_1, \mathbf{M}_2$ are both approximately finite of type $[p_1, p_2, \ldots]$ by Theorem XII. Hence they are algebraically isomorphic by Theorem XI, i.e. $\overline{\mathbf{M}}_1 = \overline{\mathbf{M}}_2$.

Thus $\overline{\mathbf{M}}$ ($\mathbf{M}$ approximately finite) is uniquely determined and the definition of $\overline{\mathbf{A}}$ is justified.

§4.8 The algebraical isomorphism invariants described at the end of §2.10 will now be studied for the algebraical type $\overline{\mathbf{A}}$. These invariants are

(1) Validity of $\overline{\mathbf{A}}^c = \overline{\mathbf{A}}$,

(2) The fundamental group: $\mathfrak{G} = \mathfrak{G}(\overline{\mathbf{A}})$.

Lemma 4.8.1. $\overline{\mathbf{A}}^c = \overline{\mathbf{A}}$.

Proof. By Theorem XIV on one hand and Theorem III on the other, it suffices to show that a definition of approximate finiteness is unaffected by a conjugate or by a dual isomorphism. This is obviously true, even for all the definitions given in Theorem XII and for both types of general isomorphisms (cf. B^*) in §2.2) mentioned above.

Lemma 4.8.2. $\mathfrak{G} = \mathfrak{G}(\overline{\mathbf{A}})$ *contains all rational numbers* θ *in* $0 < \theta < \infty$.

Proof. Since $\mathfrak{G}$ is a group, it suffices to prove that it contains all $1/p$, $p = 2, 3, \ldots$.

Consider such a $1/p$. Choose a sequence $[p_1, p_2, \ldots]$ which fulfills (i), (ii) in Lemma 4.1.3 and with all p_m divisible by p, e.g. $p_n = p^n$.

Choose an approximately finite $\mathbf{M}$ and the $\mathbf{N}_1, \mathbf{N}_2, \ldots$, for this $\mathbf{M}$ and the sequence $[p_1, p_2, \ldots]$ in the sense of Definition 4.1.1.

The considerations at the beginning of the proof of (i) in Lemma 4.1.2 show again that $D_{\mathbf{M}}(G) = D_{\mathbf{N}_n}(G)$ for all projections $G \in \mathbf{N}_n$ $(\subseteq \mathbf{M})$.

As $\mathbf{N}_1$ is in the case (I_{p_1}), p_1 divisible by p, there exists a projection $E \in \mathbf{N}_1$ with $D_{\mathbf{N}_1}(E) = 1/p$. Thus

$$E \in \mathbf{N}_1 \subseteq \mathbf{N}_2 \subseteq \cdots \subseteq \mathbf{M} \tag{4.8.α}$$

while $D_{\mathbf{N}_1}(E) = 1/p$ yields

$$\frac{D_{\mathbf{O}}(E)}{D_{\mathbf{O}}(1)} = D_{\mathbf{O}}(E) = \frac{1}{p} \quad \text{for } \mathbf{O} = \mathbf{N}_1, \mathbf{N}_2, \ldots, \mathbf{M}. \tag{4.8.β}$$

Denote the range of E by $\mathfrak{M}$.

By (4.8.α) we can form the factors $\mathbf{N}_{1(\mathfrak{M})}, \mathbf{N}_{2(\mathfrak{M})}, \ldots, \mathbf{M}_{(\mathfrak{M})}$ in $\mathfrak{M}$, and we have

$$\mathbf{N}_{1(\mathfrak{M})} \subseteq \mathbf{N}_{2(\mathfrak{M})} \subseteq \cdots \subseteq \mathbf{M}_{(\mathfrak{M})}. \tag{4.8.γ}$$

By (4.8.β), $\overline{\mathbf{N}_{n(\mathfrak{M})}} = \overline{\mathbf{N}}_n^{1/p}$, $\overline{\mathbf{M}_{(\mathfrak{M})}} = \overline{\mathbf{M}}^{1/p}$ (cf. the discussion before Theorem VI). The former implies by Lemma 2.10.3 that

$$\mathbf{N}_{n(\mathfrak{M})} \text{ is in the case } (\mathrm{I}_{p_n/p}). \tag{4.8.δ}$$

We restate the latter:

$$\overline{\mathbf{M}_{(\mathfrak{M})}} = \overline{\mathbf{M}}^{1/p}. \tag{4.8.ϵ}$$

We know that $\mathbf{M}$ is the closure of the set-theoretical sum of $\mathbf{N}_1, \mathbf{N}_2, \ldots$ in the strong topology.[40] Hence it follows from (4.8.α) that $\mathbf{M}_{(\mathfrak{M})}$ is the closure of the

[40] That closure is obviously $= \mathbf{R}(\mathbf{N}_n, n = 1, 2, \ldots)$. Hence it is $\mathbf{M}$ by (iii) in Definition 4.1.1.

set-theoretical sum of $\mathbf{N}_{1(\mathfrak{M})}, \mathbf{N}_{2(\mathfrak{M})}, \ldots$ in the strong topology. This and (4.8.γ) give together

$$\mathbf{M}_{(\mathfrak{M})} = \mathbf{R}(\mathbf{N}_{n(\mathfrak{M})}; n = 1, 2, \ldots). \tag{4.8.ζ}$$

Now (4.8.δ), (4.8.γ) and (4.8.ζ) coincide with (i), (ii), (iii) in Definition 4.1.1 as applied to $\mathbf{M}_{(\mathfrak{M})}, \mathbf{N}_{1(\mathfrak{M})}, \mathbf{N}_{2(\mathfrak{M})}, \ldots, p_1/p, p_2/p, \ldots$ in place of its $\mathbf{M}, \mathbf{N}_1, \mathbf{N}_2, \ldots, p_1, p_2, \ldots$. Hence $\mathbf{M}_{(\mathfrak{M})}$ is approximately finite.

As $\mathbf{M}$ is approximately finite too, this shows that $\overline{\mathbf{M}_{(\mathfrak{M})}} = \overline{\mathbf{M}}$ by Theorem XIV. Owing to (4.8.ϵ), this gives $\overline{\mathbf{M}}^{1/p} = \overline{\mathbf{M}}$. Hence $1/p \in \mathfrak{G}$ as desired.

Lemma 4.8.3. *If there exists for every $\epsilon > 0$ an α in $1 - \epsilon < \alpha < 1$ such that $\overline{\mathbf{M}}^{\alpha}$ is approximately finite, then $\mathbf{M}$ itself is approximately finite.*

Proof. Assume that $\mathbf{M}$ possesses the property described above. We shall establish its approximate finiteness in the sense of Definition 4.5.2.

Let accordingly $A_1, \ldots, A_m \in \mathbf{M}$ and an $\epsilon > 0$ be given. We shall construct an $\mathbf{N}$ which fulfills (i)–(iii) in that definition.

Apply first our hypothesis concerning $\mathbf{M}$ to

$$\epsilon' = \left(\frac{\epsilon}{4\,\mathrm{Max}(|||A_s|||\,,\ s = 1, \ldots, m)}\right)^2 \tag{4.8.η}$$

in place of its ϵ. Choose the α mentioned there with $1 - \epsilon' < \alpha < 1$. Then choose a projection $E \in \mathbf{M}$ with $D_{\mathbf{M}}(E) = \alpha$. Let $\mathfrak{M}$ be the range of E. Form $\mathbf{M}_{(\mathfrak{M})}$. Then we have $\overline{\mathbf{M}_{(\mathfrak{M})}} = \overline{\mathbf{M}}^{\alpha}$ (cf. the discussion before Theorem VI). It follows from our assumptions concerning α that $\mathbf{M}_{(\mathfrak{M})}$ is approximately finite.

By [5], p. 187, (i), in Lemma 11.3.3,

$$A \longleftrightarrow A_{(\mathfrak{M})}$$

is a one-to-one correspondence between all $A \in \mathbf{M}$ with $EA = AE = A$ and all $A_{(\mathfrak{M})} \in \mathbf{M}_{(\mathfrak{M})}$. Now we can repeat, with a slight variation an argument that was used in the proof of Theorem XI: Clearly $c\,\mathrm{Tr}_{\mathbf{M}}(A)$ will do as $\mathrm{Tr}_{\mathbf{M}_{(\mathfrak{M})}}(A_{(\mathfrak{M})})$ in the sense of [6], p. 219, Property IV, for the $A, A_{(\mathfrak{M})}$ described above, if we choose the normalizing factor $c > 0$ so that $c\,\mathrm{Tr}_{\mathbf{M}}(A) = 1$ when $A_{(\mathfrak{M})}$ is the unit of $\mathbf{M}_{(\mathfrak{M})}$. This means $A = E$, so we want $c\,\mathrm{Tr}_{\mathbf{M}}(E) = 1$. Since $\mathrm{Tr}_{\mathbf{M}}(E) = D_{\mathbf{M}}(E) = \alpha$ this means $c = 1/\alpha$. And as the trace is uniquely characterized by the above property,

$$\mathrm{Tr}_{\mathbf{M}_{(\mathfrak{M})}}(A_{(\mathfrak{M})}) = \frac{1}{\alpha}\,\mathrm{Tr}_{\mathbf{M}}(A).$$

From this

$$\frac{1}{\alpha}\,\mathrm{Tr}_{\mathbf{M}}(A^*A) = \mathrm{Tr}_{\mathbf{M}_{(\mathfrak{M})}}((A^*A)_{(\mathfrak{M})}) = \mathrm{Tr}_{\mathbf{M}_{(\mathfrak{M})}}((A^*)_{(\mathfrak{M})}A_{(\mathfrak{M})})$$

for the same $A, A_{(\mathfrak{M})}$, hence

$$\left[\!\left[A_{(\mathfrak{M})}\right]\!\right]_{\mathbf{M}_{(\mathfrak{M})}} = \left(\frac{1}{\alpha}\right)^{1/2} [\![A]\!]_{\mathbf{M}}.^{41} \tag{4.8.θ}$$

Now put $B_s = EA_sE$, $s = 1, \dots, m$. Then we have $D_{\mathbf{M}}(1-E) < \epsilon'$. Hence $[\![1-E]\!]_{\mathbf{M}} < \epsilon'^{1/2}$ and therefore

$$[\![A_s - B_s]\!]_{\mathbf{M}} = [\![A_s(1-E) + (1-E)A_sE]\!]_{\mathbf{M}} < |\!|\!|A_s|\!|\!|\,\epsilon'^{1/2}.$$

Hence by (4.8.η)

$$[\![A_s - B_s]\!]_{\mathbf{M}} < \epsilon/2. \tag{4.8.ι}$$

Now $B_s \in \mathbf{M}$ and $EB_s = B_sE = B_s$ imply $B_{s(\mathfrak{M})} \in \mathbf{M}_{(\mathfrak{M})}$. We now apply Definition 4.5.2 to the approximately finite $\mathbf{M}_{(\mathfrak{M})}$ with $B_{1(\mathfrak{M})}, \dots, B_{m(\mathfrak{M})}, \epsilon/(2\alpha^{1/2})$ in place of its $A_1, \dots, A_m$ and ϵ. This yields a ring $\mathbf{N}^+ \subseteq \mathbf{M}_{(\mathfrak{M})}$ of finite order with $C_1^+, \dots, C_m^+ \in \mathbf{N}^+$ such that

$$\left[\!\left[C_s^+ - B_{s(\mathfrak{M})}\right]\!\right]_{\mathbf{M}_{(\mathfrak{M})}} < \frac{\epsilon}{2\alpha^{1/2}}. \tag{4.8.κ}$$

As $\mathbf{N}^+ \subseteq \mathbf{M}_{(\mathfrak{M})}$ we extend every $C^+ \in \mathbf{N}^+$ to a $C \in \mathbf{M}$ with $EC = CE = C$ and $C^+ = C_{(\mathfrak{M})}$.[42] These C form obviously a ring $\mathbf{N} \subseteq \mathbf{M}$ of finite order with $\mathbf{N}^+ = \mathbf{N}_{(\mathfrak{M})}$. By this process the $C_1^+, \dots, C_m^+ \in \mathbf{N}^+$ give rise to the $C_1, \dots, C_m \in \mathbf{N}$. Now (4.8.$\theta$) gives

$$\left[\!\left[C_s^+ - B_{s(\mathfrak{M})}\right]\!\right]_{\mathbf{M}_{(\mathfrak{M})}} = \left[\!\left[(C_s - B_s)_{(\mathfrak{M})}\right]\!\right]_{\mathbf{M}_{(\mathfrak{M})}} = \frac{1}{\alpha^{1/2}}\,[\![C_s - B_s]\!]_{\mathbf{M}}$$

and so (4.8.κ) becomes

$$[\![C_s - B_s]\!]_{\mathbf{M}} < \epsilon/2. \tag{4.8.λ}$$

(4.8.ι) and (4.8.λ) together give

$$[\![C_s - A_s]\!]_{\mathbf{M}} < \epsilon. \tag{4.8.μ}$$

Thus the ring $\mathbf{N} \subseteq \mathbf{M}$ of finite order meets the requirements (i)–(iii) in Definition 4.5.2. Hence $\mathbf{M}$ is approximately finite as asserted.

[41] As a rule we did not indicate the factor (in a finite case) with respect to which $[\![A]\!]$ is formed. However, here it is desirable.

[42] C is reduced by $\mathfrak{M}$, its part in $\mathfrak{M}$ is C^+ and its part in $\mathfrak{M}^\perp$ is 0.

Lemma 4.8.4. *$\mathfrak{G} = \mathfrak{G}(\overline{\mathbf{A}})$ is the set P of all (real) numbers θ in $0 < \theta < \infty$.*

Proof. Consider a θ in $0 < \theta < \infty$. By Lemma 4.8.2 we have for every rational ξ in $0 < \xi < \infty$, $\overline{\mathbf{A}}^{\xi} = \overline{\mathbf{A}}$. Hence $\overline{\mathbf{A}}^{\xi}$ is approximately finite. Therefore $(\overline{\mathbf{A}}^{\theta})^{\alpha} = \overline{\mathbf{A}}^{\theta\alpha}$ is approximately finite if $\theta\alpha$ is rational.

Now it is obviously possible to find for any given $\epsilon > 0$ an α in $1 - \epsilon < \alpha < 1$ for which $\theta\alpha$ is rational. This makes $(\overline{\mathbf{A}}^{\theta})^{\alpha}$ approximately finite. Hence $\overline{\mathbf{A}}^{\theta}$ itself is approximately finite by Lemma 4.8.3 with $\overline{\mathbf{A}}^{\theta}$ in place of its $\mathbf{M}$. Consequently Theorem XIV gives $\overline{\mathbf{A}}^{\theta} = \overline{\mathbf{A}}$. Hence $\theta \in \mathfrak{G}$ as desired.

We restate the results of Lemmas 4.8.1 and 4.8.4.

Theorem XV. *The algebraical isomorphism invariants in (1), (2) at the beginning of this section, behave for the approximate finite type $\overline{\mathbf{A}}$ as follows:*

(1) $\overline{\mathbf{A}}^{c} = \overline{\mathbf{A}}$,

(2) $\mathfrak{G}(\overline{\mathbf{A}}) = P$.

It is worth pointing out that the two invariants (1), (2) are not sufficient to characterize an algebraical type $\overline{\mathbf{M}}$ in the case (II_1). A proof of this is only possible if we show first that not all $\overline{\mathbf{M}}$ are approximately finite: If all were, then Theorem XIV would give $\overline{\mathbf{M}} = \overline{\mathbf{A}}$, i.e. only one $\overline{\mathbf{M}}$ in case (II_1) would exist and thus no invariants for these $\overline{\mathbf{M}}$ would be needed at all.

Now we shall prove in Theorems XVI, XVI′ that not approximately finite $\overline{\mathbf{M}}$ in the case (II_1) exist. By making advance use of this result we prove:

Lemma 4.8.5. *The algebraical isomorphism invariants in (1), (2) at the beginning of this paragraph are not sufficient to characterize the algebraical type $\overline{\mathbf{M}}$ in the case (II_1).*

Proof. Assume the opposite.

Consider a type $\overline{\mathbf{M}}$ in the case (II_1).

Owing to the formulae (2.3.α) and (2.8.β) the validity of $\overline{\mathbf{M}}^{c} = \overline{\mathbf{M}}$ is unaffected if we replace $\overline{\mathbf{M}}$ by $\overline{\mathbf{M}}^{c}$ or by an $\overline{\mathbf{M}}^{\theta}$ ($0 < \theta < \infty$). By Lemma 2.10.2 $\mathfrak{G}(\overline{\mathbf{M}})$ is unchanged if we replace $\overline{\mathbf{M}}$ by $\overline{\mathbf{M}}^{c}$ or by an $\overline{\mathbf{M}}^{\theta}$ ($0 < \theta < \infty$). In other words: The invariants (1) and (2) are unaffected by a replacement of $\overline{\mathbf{M}}$ by $\overline{\mathbf{M}}^{c}$ or by an $\overline{\mathbf{M}}^{\theta}$.

Therefore our assumptions concerning these invariants permit us to conclude $\overline{\mathbf{M}} = \overline{\mathbf{M}}^{c}$ and $\overline{\mathbf{M}} = \overline{\mathbf{M}}^{\theta}$. The latter equations imply $\theta \in \mathfrak{G}(\overline{\mathbf{M}})$ and therefore $\mathfrak{G}(\overline{\mathbf{M}}) = P$.

Thus the two invariants (1) and (2) are uniquely determined, independently of the choice of $\overline{\mathbf{M}}$ in the case (II_1).

Hence a second application of our assumption concerning these invariants yields that there exists only one type $\overline{\mathbf{M}}$ in the case (II_1). $\overline{\mathbf{A}}$ is such a type; hence all $\overline{\mathbf{M}}$ in the case (II_1) must equal $\overline{\mathbf{A}}$, i.e. be approximately finite.

However, as stated above, the opposite will be proven in Theorems XVI, XVI′, and we propose to use this now. Thus we have a contradiction and the proof is completed.

We conclude with the observation that the existence of not approximately finite factors $\mathbf{M}$ in the case (II_1) will be established with the help of algebraic invariants other than (1), (2) above (cf. the beginning of §6.1).

CHAPTER V. ON VARIOUS EXAMPLES

§5.1 We established the existence of approximately finite factors (i.e. of their type $\overline{\mathbf{A}}$) by the indirect procedure of Theorems XIII and XIV.[43] It is therefore desirable to furnish examples of this by explicit and direct construction and also to investigate those examples of factors in the case (II_1) which we possess already as to their approximate finiteness. The latter investigation is also necessary in order to find not approximately finite factors in the case (II_1).

In connection with this work we shall develop a new technique to construct examples of factors in the case (II_1), which is a simplification of the original one of [5], pp. 192–209, Part IV.

We begin by establishing the approximate finiteness of some of our old examples: certain ones in [5] and that one in [7], pp. 72–77, §7.5.

§5.2

Lemma 5.2.1. *The ring $\mathbf{C}^{*1}$ of [7], p. 72, Lemma 7.5.1 is approximately finite.*

Proof. Using the notations loc. cit., particularly pp. 68, 72, we have

$$\mathbf{C}^{*1} = \mathbf{R}(\overline{\mathbf{B}}_{(n,1)}; n = 1, 2, \ldots).$$

Hence

$$\mathbf{C}^{*1} = \mathbf{R}(\mathbf{N}_m; m = 1, 2, \ldots) \tag{5.2.α}$$

with

$$\mathbf{N}_m = \mathbf{R}(\overline{\mathbf{B}}_{(n,1)}; n = 1, \ldots, m). \tag{5.2.β}$$

[43]The decisive passage from Theorem XIII to Theorem XIV makes use of the existence of factors in the case (II_1), for instance the examples of [5].

Thus $\mathbf{N}_1 \subseteq \mathbf{N}_2 \subseteq \cdots$ and therefore $\mathbf{C}^{*1}$ is approximately finite by Definition 4.6.1 if the $\mathbf{N}_m$ are shown to be of finite order.[44]

These considerations belong in $\prod_{n=1,2,\dots}(\mathfrak{H}_{(n,1)} \otimes \mathfrak{H}_{(n,2)})$ where $\mathbf{C}^{*1}$ is a factor in the case (II_1), but in establishing the finite order of $\mathbf{N}_m$, we may as well consider the entire space $\prod_{n=1,2,\dots}(\mathfrak{H}_{(n,1)} \otimes \mathfrak{H}_{(n,2)})$.

We proceed as in the lower part of p. 68, loc. cit. $B_{(n,1)}$ is in $\mathfrak{H}_{(n,1)}$. It is first extended to $\overline{B}_{(n,1)}$ in $\mathfrak{H}_{(n,1)} \otimes \mathfrak{H}_{(n,2)}$ and then to $\overline{\mathbf{B}}_{(n,1)}$ in $\prod_{n=1,2,\dots}(\mathfrak{H}_{(n,1)} \otimes \mathfrak{H}_{(n,2)})$. Hence the $\overline{\mathbf{B}}_{(n,1)}$ commute with each other. Therefore the $\mathbf{N}_m$ of (5.2.β) is of finite order if the $\overline{\mathbf{B}}_{(n,1)}$ are. Now $B_{(n,1)}$ is of finite order because it is in the finite 2-dimensional space $\mathfrak{H}_{(n,1)}$ and so $\overline{\mathbf{B}}_{(n,1)}$ is of finite order too.

Lemma 5.2.2. *The ring* $\mathbf{M}$ *of [5], pp. 203–204, Theorem XI, is approximately finite if the group* $\mathfrak{G}$ *possesses this property:*

(i) $\mathfrak{G}$ *is the set-theoretical sum of a sequence* $\mathfrak{G}_1 \subseteq \mathfrak{G}_2 \subseteq \cdots$ *of finite groups.*

(We assume, of course, that the requirements of loc. cit. [5], p. 206, Lemma 13.1.2, are fulfilled, placing $\mathbf{M}$ *in the case* (II_1)*.)*

Proof. Using the notations loc. cit., particularly those on pp. 198–200, we have

$$\mathbf{M} = \mathbf{R}(\overline{U}_{a_0}, \overline{L}_{\varphi(x)}; a_0 \in \mathfrak{G},\ \varphi(x) \text{ bounded and measurable}). \tag{5.2.γ}$$

A familiar argument shows that the $\varphi(x)$ in (5.2.γ) can be restricted considerably without changing the ring $\mathbf{M}$. Indeed:

First. It suffices for reasons of continuity to consider the $\varphi(x)$ with finite ranges only.

Second. These $\varphi(x)$ are finite linear combinations of the characteristic functions

$$\chi_S(x) = \begin{cases} 1, & x \in S, \\ 0, & \text{otherwise.} \end{cases}$$

Third. Let $T_1, T_2, \dots$ be a countable basis for the T ($\subseteq S$ and measurable).[45] Then

[44] It is not difficult to show that $\mathbf{N}_n$ is a factor in the case (I_{2^n}). Hence Definition 4.1.1 would also be applicable, yielding approximate finiteness of type $[2, 4, 8, \dots]$.

[45] The space of all measurable sets T (disregarding sets of measure zero) can be metrized by the distance

$$\overline{T'T''} = \mu(T' + T'') - \mu(T'T'').$$

This is obviously equal to

$$|\chi_{T'} - \chi_{T''}|^2 = \int_S |\chi_{T'}(x) - \chi_{T''}(x)|^2\, dx.$$

it suffices for reasons of continuity to consider these $\varphi(x) = \chi_{T_i}(x)$, $i = 1, 2, \ldots$, only.

Finally, these properties of the system $T_1, T_2, \ldots$ are not lost if we replace it by another countable system which contains it. We do this in such a way that the increased system $\overline{T}_1, \overline{T}_2, \ldots$ possesses these further properties:

(1) It contains $T' \cdot T''$ along with T', T''.

(2) It contains $a_0 T'$ along with T' for any $a_0 \in \mathfrak{G}$.

These being understood, we have

$$\mathbf{M} = \mathbf{R}(\overline{U}_{a_0}, \overline{L}_{\chi_{\overline{T}_i}(x)}; a_0 \in \mathfrak{G}_1,\ i = 1, 2, \ldots). \tag{5.2.δ}$$

Consider now the finite groups $\mathfrak{G}_1, \mathfrak{G}_2, \ldots$ of (i). $\overline{T}_1, \ldots, \overline{T}_n$ generates, with the help of the operations (1) and (2) when a_0 in the latter is restricted to $\mathfrak{G}_n$, a finite subsystem of $\overline{T}_1, \overline{T}_2, \ldots$,—say the set $(\overline{T}_i; i \in I_n)$. So I_n is a finite set. Clearly $I_1 \subseteq I_2 \subseteq \cdots$. I_n contains $1, 2, \ldots, n$. Hence $(1, 2, \ldots)$ is the set-theoretical sum of $I_1, I_2, \ldots$.

Consequently (5.2.δ) gives

$$\mathbf{M} = \mathbf{R}(\mathbf{N}_n; n = 1, 2, \ldots) \tag{5.2.ϵ}$$

with

$$\mathbf{N}_n = \mathbf{R}(\overline{U}_{a_0}, \overline{L}_{\chi_{\overline{T}_i}(x)}; a_0 \in \mathfrak{G}_n,\ i \in I_n). \tag{5.2.ζ}$$

Thus $\mathbf{N}_1 \subseteq \mathbf{N}_2 \subseteq \cdots$ and therefore $\mathbf{M}$ is approximately finite by Definition 4.6.1 if the $\mathbf{N}_n$ are shown to be of finite order.[46]

Now the group character of $\mathfrak{G}_n$ and the construction of I_n show that the operators

$$\overline{U}_{a_0}\overline{L}_{\chi_{\overline{T}_i}(x)}, \qquad a_0 \in \mathfrak{G}_n, \quad i \in I_n \tag{5.2.η}$$

are reproduced by multiplication. As $\mathfrak{G}_n, I_n$ are finite, these operators (5.2.η) form a finite set too. Hence they are a finite basis of $\mathbf{N}_n$, completing the proof.

This last result establishes the approximate finiteness of the examples (β), (γ) in [5], p. 208. Thus our Theorem XIV establishes their algebraic isomorphism and [6], p. 244, Theorem XI, their spatial isomorphism as announced by (v) in [5], p. 209. In order to include (α), p. 208, eod., we need this:

Since $\mathfrak{H}_S$ is separable, there exists a sequence $\chi_{T_1}, \chi_{T_2}, \ldots$ dense in the set of all χ_T, and correspondingly the sequence $T_1, T_2, \ldots$ is dense in the set of all T. This is what we mean by a basis.

[46]This time we really must use Definition 4.6.1 and it would not be possible to replace it by Definition 4.1.1—in contrast with footnote 44.

Lemma 5.2.3. *The condition (i) in Lemma 5.2.2 can be replaced by this condition:*
(ii) $\mathfrak{G}$ is abelian.

The proof of this lemma is somewhat complicated. It requires some rather deep results on the decompositions of mappings of measurable sets, which will be published elsewhere. We shall not pursue this matter further on this occasion.

§5.3 We proceed to expound the new (simplified) method for the construction of examples of factors in the case (II_1), announced in §5.1.

Let a group $\mathfrak{G}$ be given. For the moment $\mathfrak{G}$ could be finite or countably infinite, but we shall be forced to assume the latter at a subsequent stage (cf. Lemma 5.3.4 and the Remark after it).

Form a unitary vector space $\mathfrak{H}$ by using the elements of $\mathfrak{G}$ as indices for the vector components, i.e. $\mathfrak{H}$ is the set of all complex vectors $f = [x_a; a \in \mathfrak{G}]$ with a finite $\sum_{a\in\mathfrak{G}} |x_a|^2$ and for $f = [x_a; a \in \mathfrak{G}]$ and $g = [y_a; a \in \mathfrak{G}]$, $(f, g) = \sum_{a\in\mathfrak{G}} x_a\overline{y_a}$.[47]

Now we introduce and discuss some operators in $\mathfrak{H}$.

Lemma 5.3.1. *Define the operators*

$$\begin{aligned}
&\text{(i)} \quad U_{a_0}[x_a; a \in \mathfrak{G}] = [x_{aa_0}; a \in \mathfrak{G}],\\
&\text{(ii)} \quad V_{a_0}[x_a; a \in \mathfrak{G}] = [x_{a_0^{-1}a}; a \in \mathfrak{G}],\\
&\text{(iii)} \quad W[x_a; a \in \mathfrak{G}] = [x_{a^{-1}}; a \in \mathfrak{G}] \qquad \textit{for any } a_0 \in \mathfrak{G}.
\end{aligned}$$

These operators are unitary and

$$W = W^{-1}, \qquad WU_{a_0}W = V_{a_0}.^{48}$$

In other words, $f \mapsto Wf$ is an involutory spatial automorphism of $\mathfrak{H}$ which interchanges U_{a_0} with V_{a_0}.

Proof. All assertions are immediately verified.

We introduce, as usual, the complete normalized orthogonal system

$$\varphi_a, \qquad a \in \mathfrak{G}, \tag{5.3.α}$$

[47] This $\mathfrak{H}$ coincides with the $\mathfrak{H}_{\mathfrak{G}}$ of [5], p. 194, Definition 12.1.4. The only difference is that we now write x_a in place of the $f(a)$, loc. cit.

[48] One also verifies immediately

$$U_{a_1}U_{a_0} = U_{a_1a_0}, \qquad V_{a_1}V_{a_0} = V_{a_1a_0},$$

i.e. $a \mapsto U_a$ and $a \mapsto V_a$ are both representations of $\mathfrak{G}$. They correspond in fact to the regular representation of $\mathfrak{G}$.

in $\mathfrak{H}$, where $\varphi_a = [\delta_{a,b}; b \in \mathfrak{G}]$. Thus $f = [x_a; a \in \mathfrak{G}]$ is equivalent to $f = \sum_{a\in\mathfrak{G}} x_a\varphi_a$ and

$$U_{a_0}\varphi_a = \varphi_{aa_0^{-1}}, \qquad V_{a_0}\varphi_a = \varphi_{a_0a}, \qquad W\varphi_a = \varphi_{a^{-1}}. \tag{5.3.β}$$

Lemma 5.3.2. *Let I be the set of all U_{a_0} and J the set of all V_{a_0}. Form for each bounded operator A in $\mathfrak{H}$, its matrix $\{\alpha_{a,b}\}$ $(a, b \in \mathfrak{G})$ in the usual way.*[49]

Then $A \in I'$ if and only if $\alpha_{a,b}$ has the form $\alpha_{a,b} = \eta_{ab^{-1}}$ and $A \in J'$ if and only if $\alpha_{a,b}$ has the form $\alpha_{a,b} = \eta_{a^{-1}b}$, i.e. $\alpha_{a,b}$ depends on the value of ab^{-1} only or on the value of $a^{-1}b$ only, respectively. (We do not determine for which systems η_c, $c \in \mathfrak{G}$ a bounded A to which the given η_c corresponds, actually does exist.)

Proof. If $A \sim \{\alpha_{a,b}\}$, then (5.3.β) gives $U_{a_0}^{-1}AU_{a_0} \sim \{\alpha_{aa_0^{-1},ba_0^{-1}}\}$. Now $A \in I'$ means that A commutes with all U_{a_0}, i.e. that

$$\alpha_{a,b} = \alpha_{aa_0^{-1},ba_0^{-1}} \quad \text{for all } a_0 \in \mathfrak{G}. \tag{5.3.γ}$$

Write $\eta_c = \alpha_{c,1}$. Then (5.3.γ) with $a_0 = b$ gives

$$\alpha_{a,b} = \eta_{ab^{-1}}. \tag{5.3.δ}$$

Conversely (5.3.δ) implies (5.3.γ). Thus (5.3.δ) is characteristic for $A \in I'$. This proves our statement concerning I'.

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[49]The definition is

$$\alpha_{a,b} = (A\varphi_a, \varphi_b) = (\varphi_a, A^*\varphi_b). \tag{49,α}$$

Hence

$$|||A\varphi_a|||^2 = \sum_{b\in\mathfrak{G}} |(A\varphi_a, \varphi_b)|^2 = \sum_{b\in\mathfrak{G}} |\alpha_{a,b}|^2,$$
$$|||A^*\varphi_b|||^2 = \sum_{a\in\mathfrak{G}} |(\varphi_a, A^*\varphi_b)|^2 = \sum_{a\in\mathfrak{G}} |\alpha_{a,b}|^2,$$

i.e.

$$\text{all } \sum_{b\in\mathfrak{G}} |\alpha_{a,b}|^2, \quad \sum_{a\in\mathfrak{G}} |\alpha_{a,b}|^2 \text{ are finite.} \tag{49,β}$$

Finally

$$(Af, \varphi_b) = (f, A^*\varphi_b) = \sum_{a\in\mathfrak{G}} (f, \varphi_a)(\varphi_a, A^*\varphi_b) = \sum_{a\in\mathfrak{G}} \alpha_{a,b}(f, \varphi_a),$$

i.e.

$$\left.\begin{aligned} f &= \sum_{a\in\mathfrak{G}} x_a\varphi_a, \\ Af &= \sum_{a\in\mathfrak{G}} y_a\varphi_a \end{aligned}\right\} \implies y_b = \sum_{a\in\mathfrak{G}} \alpha_{a,b}x_a. \tag{49,γ}$$

The matrix operations for these (complex numerical) matrices are defined in the same way as for the (operator) matrices of B_p in §2.4.

If $A \sim \{\alpha_{a,b}\}$, then $WAW \sim \{\alpha_{a^{-1},b^{-1}}\}$.[50] Hence the automorphism $f \mapsto Wf$ of $\mathfrak{H}$ carries (5.3.δ) into

$$\alpha_{a,b} = \eta_{a^{-1}b}. \tag{5.3.ϵ}$$

Since it carries I into J it takes I' into J'. Therefore (5.3.ϵ) is characteristic for $A \in J'$. This proves our statement concerning J'.

Lemma 5.3.3. $\mathbf{R}(I) = J'$, $\mathbf{R}(J) = I'$.

Proof. Clearly V_{a_0} commutes with all U_{a_0}. So $V_{a_0} \in I'$ and hence $J \subseteq I'$. As I' is a ring, this implies $\mathbf{R}(J) \subseteq I'$.

If $A \in I'$, $B \in J'$, then $A \sim \{\eta_{ab^{-1}}\}$, $B \sim \{\theta_{a^{-1}b}\}$. Direct computation shows that $AB = BA$ (use the matrix computation rules as mentioned at the end of footnote 49). Thus $A \in J''$ and hence $I' \subseteq J''$. Since $1 \in J$, $J'' = \mathbf{R}(J)$ (cf. [2], p. 397). Therefore $I' \subseteq \mathbf{R}(J)$.

The results of the two preceding paragraphs imply $\mathbf{R}(J) = I'$. The spatial automorphism $f \mapsto Wf$ of $\mathfrak{H}$ interchanges I with J and thus we obtain $\mathbf{R}(I) = J'$ too. The proof is now complete.

Lemma 5.3.4. *Put* $\mathbf{M} = \mathbf{R}(I) = J'$. *Then* $\mathbf{M}' = \mathbf{R}(J)$. I' *is a factor if and only if* $\mathfrak{G}$ *fulfills this condition:*

(i) For every $a \in \mathfrak{G}$, $a \neq 1$, *the class of* a,

$$\mathfrak{L}_a = (c^{-1}ac; c \in \mathfrak{G}),$$

is infinite.

Proof. Clearly $\mathbf{M}' = (\mathbf{R}(I))' = I'$.

Now consider an $A \in \mathbf{M} \cdot \mathbf{M}'$. Put $A \sim \{\alpha_{a,b}\}$. $A \in \mathbf{M} \cdot \mathbf{M}'$ implies that both

$$\alpha_{a,b} = \eta_{a^{-1}b} \qquad \text{and} \qquad \alpha_{a,b} = \theta_{ab^{-1}} \tag{5.3.ζ}$$

hold. Put $b = 1$. This gives $\eta_{a^{-1}} = \theta_a$. Thus (5.3.ζ) is equivalent to

$$\alpha_{a,b} = \eta_{a^{-1}b} = \eta_{ba^{-1}}. \tag{5.3.η}$$

[50] Use (49,α) and (5.3.β). Then

$$\begin{aligned}(U_{a_0}^{-1}AU_{a_0}\varphi_a, \varphi_b) &= (AU_{a_0}\varphi_a, U_{a_0}\varphi_b)\\ &= (A\varphi_{aa_0^{-1}}, \varphi_{ba_0^{-1}}),\\ (WAW\varphi_a, \varphi_b) &= (AW\varphi_a, W\varphi_b) = (A\varphi_{a^{-1}}, \varphi_{b^{-1}}).\end{aligned}$$

Hence $A \sim \{\alpha_{a,b}\}$ implies $U_{a_0}^{-1}AU_{a_0} \sim \{\alpha_{aa_0^{-1},ba_0^{-1}}\}$ and $WAW \sim \{\alpha_{a^{-1},b^{-1}}\}$.

The second equation in (5.3.η) may be rewritten by replacing a, b by b, ab. Then it becomes $\eta_{b^{-1}ab} = \eta_a$, i.e. it states that η_d is a constant for all $d \in \mathfrak{L}_a$. So (5.3.η) is equivalent to this:

$$\alpha_{a,b} = \eta_{a^{-1}b}, \qquad \eta_d \text{ is constant for all } d \in \mathfrak{L}_a. \tag{5.3.θ}$$

We now prove that (i) is characteristic.

Sufficiency: Assume that (i) is true. If $a \neq 1$ then $\mathfrak{L}_a$ is infinite. Hence the finiteness of $\sum_{d\in\mathfrak{G}} |\eta_d|^2$[51] together with (5.3.$\theta$) imply $\eta_a = 0$ if $a \neq 1$. Thus

$$\alpha_{a,b} = \eta_{a^{-1}b} = \begin{cases} \eta_1, & a = b, \\ 0, & \text{otherwise}, \end{cases}$$

or $A = \eta_1 \cdot 1$, and so $\mathbf{M}$ is a factor.

Necessity: Assume that (i) is not true. Choose an $a_0 \neq 1$ with $\mathfrak{L}_{a_0}$ finite. Define

$$\alpha_{a,b} = \eta_{a^{-1}b}, \qquad \eta_d = \begin{cases} 1, & d \in \mathfrak{L}_{a_0}, \\ 0, & \text{otherwise}. \end{cases} \tag{5.3.ι}$$

One verifies that the $A \sim \{\alpha_{a,b}\}$ may be expressed as $A = \sum_{d\in\mathfrak{L}_{a_0}} U_d$.[52] Thus $A \in \mathbf{M} \cdot \mathbf{M}'$ by (5.3.θ) and (5.3.ι).

If $\mathbf{M}$ were a factor, this would necessitate $A = \alpha \cdot 1$ or $A \sim \{\alpha\delta_{a,b}\}$, where $\delta_{a,b} = 1$ for $a = b$ and $\delta_{a,b} = 0$ otherwise. Consequently

$$\alpha_{a,b} = \alpha\delta_{a,b}. \tag{5.3.κ}$$

Now (5.3.ι) and (5.3.κ) give for $a = 1$ and $b = a_0$ (remember $a_0 \neq 1$) $\alpha_{1,a_0} = 1$ and $\alpha_{1,a_0} = 0$ respectively. This contradiction shows that $\mathbf{M}$ is not a factor.

This last result forces us to assume that $\mathfrak{G}$ is infinite, since even its classes $\mathfrak{L}_a$, $a \neq 1$, are to be infinite. Groups $\mathfrak{G}$ with this property are easy to construct. We shall give some examples at the end of §5.5 and in §5.6. Clearly no such group can be abelian.

There is, however, another circumstance worth pointing out before we go further. Our construction is obviously nothing but an extension of Frobenius's

[51] The finiteness of $\sum_{d\in\mathfrak{G}} |\eta_d|^2$ follows immediately from the first formula of (49,β) with $a = 1$:

$$\sum_{b\in\mathfrak{G}} |\alpha_{1,b}|^2 = \sum_{b\in\mathfrak{G}} |\eta_b|^2.$$

[52] Observe that the sum can be formed because $\mathfrak{L}_{a_0}$ is finite.

group numbers to infinite groups.[53] The unitary space $\mathfrak{H}$ and its operators are only technical devices to take care of convergence difficulties. Both the U_a and V_a furnish representations of $\mathfrak{G}$ (cf. footnote 48), and so both $\mathbf{M} = \mathbf{R}(I)$ and $\mathbf{M}' = \mathbf{R}(J)$ are the equivalents of Frobenius's group numbers. But in the case of finite $\mathfrak{G}$ this procedure does not give a factor. The group numbers notoriously possess other central elements than the $\alpha \cdot 1$. There are as many linearly independent ones as there are different classes $\mathfrak{L}_a$ in $\mathfrak{G}$. The real meaning of our Lemma 5.3.4 appears to be this: The role of all classes in $\mathfrak{G}$, when $\mathfrak{G}$ is finite, is now taken over by the finite classes.[54] For finite $\mathfrak{G}$ this distinction is vacuous, but for the infinite $\mathfrak{G}$ it is now seen to be decisive.

In other words: We have essentially merely extended the construction of Frobenius group numbers to infinite $\mathfrak{G}$. For finite $\mathfrak{G}$ this can never give a factor; indeed, the usefulness of the group numbers lies in that case in the opposite direction. For infinite $\mathfrak{G}$ however, our treatment of the convergence questions makes this construction perfectly adequate to produce factors.

We now determine the case to which these factors $\mathbf{M}, \mathbf{M}'$ belong, and a few other characteristics.

The condition (i) in Lemma 5.3.4 is assumed to be fulfilled.

Lemma 5.3.5. *(i)* $\mathbf{M}, \mathbf{M}'$ *are coupled factors both in case (II_1).*

(ii) In their standard normalization, the C of [5], p. 182, Theorem X, is 1.

(iii) If $A \in \mathbf{M}$, $(A \in \mathbf{M}')$ *then* $\mathrm{Tr}_{\mathbf{M}}(A)$, $(\mathrm{Tr}_{\mathbf{M}'}(A))$ *has the value* η_1 *for the* η_c *of Lemma 5.3.2.*

Proof. We proceed in a somewhat changed order.

Define for $A \in \mathbf{M}$, a quantity $T'(A) = \eta_1$ for the η_c of Lemma 5.3.2. Let us check for this $T'(A)$ the properties (i)–(vi′) in [6], p. 218.

(i)–(iii), (v) loc. cit., are obvious.

If $A \sim \{\alpha_{a,b}\} = \{\eta_{a^{-1}b}\}$, $B \sim \{\beta_{a,b}\} = \{\theta_{a^{-1}b}\}$, then clearly $AB \sim \{\gamma_{a,b}\} = \{\zeta_{a^{-1}b}\}$, where $\zeta_a = \sum_{c\in\mathfrak{G}} \eta_c\theta_{ac^{-1}}$. (Use the matrix computation rules indicated at the end of footnote 49.) Hence $T'(AB) = \zeta_1 = \sum_{c\in\mathfrak{G}} \eta_c\theta_{c^{-1}}$. This expression is symmetric

[53] We treat these groups as discrete groups and use everywhere sums not integrals. The familiar generalization of Peter–Weyl to continuous groups and of von Neumann to almost-periodic groups where integrals and means are used, do not show the peculiarities of our present discussion. In those cases the infinite group behaves almost entirely like the finite group.

[54] Concerning the connection between factors and representation theory, cf. also [5], pp. 118–120, Introduction, §3.

in η_a and θ_a. Hence

$$T'(AB) = T'(BA) = \sum_{c \in \mathfrak{G}} \eta_c \theta_{c^{-1}}. \tag{5.3.μ}$$

This proves (vi), loc. cit., and (vi′) eod., by replacing A, B by $U^{-1}A, U$.

As $A^* = \{\overline{\alpha}_{b,a}\} = \{\xi_{a^{-1}b}\}$ with $\xi_a = \overline{\eta}_{a^{-1}}$, (5.3.$\mu$) yields for $B = A^*$

$$T'(A^*A) = \sum_{c \in \mathfrak{G}} |\eta_c|^2. \tag{5.3.ν}$$

This proves (iv), (iv′) loc. cit.

Thus all desired relations are established.

If we consider $T'(E)$ only for projections $E \in \mathbf{M}$ then the above relations allow us to assert the validity of (ii), (iii) in [5], p. 165, Definition 8.2.1. Also $T'(1) = 1 \neq 0, \infty$. Hence, by [5], p. 170, Lemma 8.3.5, this $T'(E)$ is a relative dimension function of $\mathbf{M}$ and $\mathbf{M}$ is in a finite case.

The cases (I_n), $n = 1, 2, \ldots$, are excluded, since $\mathbf{M}$ is not a finite ring. Hence $\mathbf{M}$ is in the case (II_1).

Now we conclude by [6], p. 218, Property IV, that $\mathrm{Tr}_{\mathbf{M}}(A) = T'(A)$, i.e. that $\mathrm{Tr}_{\mathbf{M}}(A) = \eta_1$.

Thus our (i), (iii) hold for $\mathbf{M}$. The spatial automorphism $f \mapsto Wf$ in $\mathfrak{H}$ shows that they hold for $\mathbf{M}'$ in view of Lemma 5.3.1.

Let us now consider our (ii). Observe that by (5.3.β) $\mathfrak{M}^{\mathbf{M}}_{\varphi_1}$ as well as $\mathfrak{M}^{\mathbf{M}'}_{\varphi_1}$ (cf. [5], p. 143, Definition 5.1.1) contain all φ_a, $a \in \mathfrak{G}$. Hence both are $\mathfrak{H}$. It follows that in the standard normalization,

$$D_{\mathbf{M}}(\mathfrak{M}^{\mathbf{M}'}_{\varphi_1}) = D_{\mathbf{M}'}(\mathfrak{M}^{\mathbf{M}}_{\varphi_1}) = 1.$$

Therefore [5], p. 182, Theorem X, gives $C = 1$ as asserted.

Lemma 5.3.6. *Consider an $A \in \mathbf{M}$ ($A \in \mathbf{M}'$) and its η_c by Lemma 5.3.2. Then we have*

(i) $[\![A]\!] = \left(\sum_{c \in \mathfrak{G}} |\eta_c|^2\right)^{1/2}$,

(ii) $A = \sum_{c \in \mathfrak{G}} \eta_c U_c$, ($A = \sum_{c \in \mathfrak{G}} \eta_c V_c$) *in the sense of metric convergence in* $\mathbf{M}$, ($\mathbf{M}'$) *irrespective of the order in which the $c \in \mathfrak{G}$ are gone through.*

Proof. The spatial automorphism $f \mapsto Wf$ of $\mathfrak{H}$ interchanges $\mathbf{M}$ and $\mathbf{M}'$. Therefore it suffices to consider the $A \in \mathbf{M}$.

Ad (i): By Equation (5.3.ν) in the proof of Lemma 5.3.5,

$$[\![A]\!]^2 = \mathrm{Tr}_{\mathbf{M}}(A^*A) = T'(A^*A) = \sum_{c \in \mathfrak{G}} |\eta_c|^2.$$

Hence $[\![A]\!] = (\sum_{c\in\mathfrak{G}} |\eta_c|^2)^{1/2}$.

Ad (ii): Choose any enumeration $a^{(1)}, a^{(2)}, \ldots$ of $\mathfrak{G}$. One verifies immediately that for $A^{(n)} = \sum_{k=1}^{n} \eta_{a^{(k)}} U_{a^{(k)}}$,

$$A^{(n)} \sim \{\eta^{(n)}_{a^{-1}b}\}, \qquad \eta^{(n)}_c = \begin{cases} \eta_c, & c = a^{(1)}, \ldots, a^{(n)}, \\ 0, & \text{otherwise.} \end{cases}$$

Hence

$$A - A^{(n)} \sim \{\eta^{(n)1}_{a^{-1}b}\}, \qquad \eta^{(n)1}_c = \begin{cases} 0, & c = a^{(1)}, \ldots, a^{(n)}, \\ \eta_c, & \text{otherwise.} \end{cases}$$

Consequently (i) above gives

$$[\![A - A^{(n)}]\!] = \left(\sum_{c\in\mathfrak{G}} |\eta^{(n)1}_c|^2\right)^{1/2} = \left(\sum_{c\in\mathfrak{G},\ c\neq a^{(1)},\ldots,a^{(n)}} |\eta_c|^2\right)^{1/2} = \left(\sum_{i=n+1}^{\infty} |\eta_{a^{(i)}}|^2\right)^{1/2}.$$

This is equivalent to

$$\left[\!\!\left[A - \sum_{k=1}^{n} \eta_{a^{(k)}} U_{a^{(k)}}\right]\!\!\right] = \left(\sum_{k=n+1}^{\infty} |\eta_{a^{(k)}}|^2\right)^{1/2}. \tag{5.3.o}$$

Now $\sum_{k=1}^{\infty} |\eta_{a^{(k)}}|^2 = \sum_{c\in\mathfrak{G}} |\eta_c|^2$ is finite by (i) above. Hence the right hand side of (5.3.o) tends to 0 as $n \to \infty$. This means that $\sum_{k=1}^{\infty} \eta_{a^{(k)}} U_{a^{(k)}}$ converges metrically to A.

Since this holds for any enumeration $a^{(1)}, a^{(2)}, \ldots$ of $\mathfrak{G}$, our assertion is established.

§5.4 As we pointed out, the construction of §5.3 is closely related to that of [5], pp. 192–209, Part IV. The group $\mathfrak{G}$ plays the same role in both cases, and the difference consists in the disappearance of the space S.

This space played an important part in the construction referred to above: Every $a \in \mathfrak{G}$ had to induce a mapping $x \mapsto xa$ in S (cf. loc. cit., p. 195, Definition 12.1.5). The absence of S from our present construction may also be interpreted by saying that S is now a one-element set: $S = (x_0)$ and that for every $a \in \mathfrak{G}$, $x_0 a = x_0$.

Now the difference between our two constructions can be expressed as follows: In [5], $\mathfrak{G}$ was entirely unrestricted, but S had to obey (among other things) (iii) in Definition 12.1.5, loc. cit., p. 195: for $a \neq 1$ and $x \in S$ always $xa \neq x$.[55] In §5.3,

[55] Except for an x set of measure zero.

$\mathfrak{G}$ had to fulfill (i) of Lemma 5.3.4, but the (iii) referred to was most flagrantly violated.[56,57] Thus our two procedures correspond to two different ways of securing the factor character of $\mathbf{M}$ in what is fundamentally the same construction. In one case the whole burden of the necessary restrictions was thrown on S, in the other case, on $\mathfrak{G}$.

It would be possible to use a more general arrangement of which both these procedures are special cases. The necessary restriction would then be of a mixed type, affecting the structure of S and $\mathfrak{G}$.

We shall not consider this question in more detail at this time.

§5.5 A second connection between our two constructions which deserves some attention can be obtained as follows.

Let an arbitrary countably infinite group $\mathfrak{G}$ be given. We shall try to use it for the construction of [5] referred to above, that is to find an S which fulfills all preliminary conditions of that construction in conjunction with the given $\mathfrak{G}$.

This construction runs as follows:[58]

(I) Let S be the set of all systems $x = [\alpha_a; a \in \mathfrak{G}]$ where each $\alpha_a = 0, 1$. Let $\mathfrak{S}$ be the set of those $x = [\alpha_a; a \in \mathfrak{G}]$ for which $\alpha_a = 0$ except for a finite number of a's.

Define in S: If $x = [\alpha_a; a \in \mathfrak{G}]$, $y = [\beta_a; a \in \mathfrak{G}]$, then $x \oplus y = [\gamma_a; a \in \mathfrak{G}]$, where $\gamma_a = \alpha_a + \beta_a \pmod 2$, $\gamma_a = 0$ or 1. S is clearly a (commutative) group with the "unit" $0 = [0; a \in \mathfrak{G}]$, and $\mathfrak{S}$ is an enumerably infinite subgroup of S.

Define further in S: If $x = [\alpha_a; a \in \mathfrak{G}]$ and $a_0 \in \mathfrak{G}$, then $xa_0 = [\alpha_{a_0 a}; a \in \mathfrak{G}]$. Then the mappings $x \mapsto xa_0$ represent the group $\mathfrak{G}$ by permutations of S, all of which carry $\mathfrak{S}$ into itself.

Now choose an arbitrary but fixed enumeration $a^{(1)}, a^{(2)}, \dots$ of $\mathfrak{G}$. For S we use the mapping

$$\Xi : \quad x = [\alpha_a; a \in \mathfrak{G}] \longmapsto \xi(x) = \sum_{m=1}^{\infty} \frac{\alpha_{a^{(m)}}}{2^m}$$

of S on the numerical interval $0 \le \xi \le 1$. Except for the Ξ image of $\mathfrak{S}$, the set of all dyadically rational numbers, which is a set of Lebesgue measure 0, this mapping is one-to-one. So the common (exterior) Lebesgue measure in $0 \le \xi \le 1$ is mapped

[56] We have $x_0 a = x_0$ and $\mu((x_0)) = \mu(S) \neq 0$.

[57] There is no difference between the two constructions as regards the other conditions of Definition 12.1.5, loc. cit., p. 195. In particular, $\mathfrak{G}$ is ergodic in our present construction, since $S = (x_0)$ is a one-element set.

[58] The construction (I)–(V) which follows is in many respects analogous to the construction (I)–(VII) in [7], pp. 72–77. The main differences are these: We use $\mathfrak{G}$ and not $\mathfrak{S}$ as the group of mappings in S, and the latter parts of the two constructions pursue different aims.

by the inverse of Ξ on a Lebesgue measure in S with all its usual properties.

We shall now consider $\mathfrak{G}$ as the "group" and S (with the "mappings" $x \mapsto xa_0$ for $x \in S, a_0 \in \mathfrak{G}$) as the "space" described in [5], pp. 192–195. In the sense of [5], p. 195, Definition 12.1.5, $\mathfrak{G}$ is an m-group and ergodic in S.

First we show the m-group character. Ad (i), loc. cit. This is concerned with the preservation of outer measure of the measure determining sets $\{T_i\}$. In view of the definition of measure in S given in the previous paragraph, we may take for a set of T_i's the set of images, under the inverse of Ξ, of ξ-intervals $k/2^m \leq \xi < (k+1)/2^m$. An image of a ξ-interval $k/2^m \leq \xi < (k+1)/2^m$ is given by specifying the values of $\alpha_{a^{(1)}}, \dots, \alpha_{a^{(m)}}$. Hence it is mapped by $x \mapsto xa_0$ onto a set, in which the values of $\alpha_{a_0^{-1}a^{(1)}}, \dots, \alpha_{a_0^{-1}a^{(m)}}$ are specified. Choose n so that $a_0^{-1}a^{(1)}, \dots, a_0^{-1}a^{(m)}$ occur among the $a^{(1)}, \dots, a^{(n)}$. Then the above set is the sum of 2^{n-m} sets, each of which is defined by specifying the values of $\alpha_{a^{(1)}}, \dots, \alpha_{a^{(n)}}$, i.e. each of which corresponds to an interval $l/2^n \leq \xi < (l+1)/2^n$. Thus the set corresponding to the ξ-interval $k/2^m \leq \xi < (k+1)/2^m$, having measure $1/2^m$, is mapped on a set of measure $2^{n-m}/2^n = 2^{-m}$. Thus the measure of each such is conserved. Ad (ii), which states that $(xa_0)b_0 = x(a_0b_0)$, can be verified by a direct computation.[59] Ad (iii) states that if $a_0 \neq 1$, $xa_0 = x$ holds only for a set of measure zero. For $x = [\alpha_a; a \in \mathfrak{G}]$, $xa_0 = x$ is equivalent to $\alpha_{a_0a} = \alpha_a$ for all a. But since $a_0 \in \mathfrak{G}$, $a_0 \neq 1$, we can construct an infinite sequence $a_1, a_2, \dots$ such that the $a_1, a_2, \dots, a_0a_1, a_0a_2, \dots$ are all different.[60] For any m, let $n = n(m)$ be such that $a^{(1)}, \dots, a^{(n)}$ includes $a_1, \dots, a_m, a_0a_1, \dots, a_0a_m$. Then the x-set $S^{(m)}$ for which $\alpha_{a_0a_i} = \alpha_{a_i}$, $i = 1, \dots, m$, consists of 2^{n-m} sets $I^{(n)}$ in each of which the $\alpha_{a^{(1)}}, \dots, \alpha_{a^{(n)}}$ have specified values. Such a set $I^{(n)}$ is the image of a ξ-interval of measure 2^{-n}, and hence $S^{(m)}$ has measure $2^{n-m} \cdot 2^{-n} = 2^{-m}$. Now the x-set for which $\alpha_{a_0a} = \alpha_a$ for all $a \in \mathfrak{G}$ is in $S^{(m)}$ for every m. Thus it has measure zero. Since this is also the set for which $xa_0 = x$, we have shown Ad (iii).

The ergodicity will be established in (III) below.

(II) Form for these $S, \mathfrak{G}$ the spaces $\mathfrak{H}_S$ and $\mathfrak{H}_{\mathfrak{G}S}$ of all complex-valued functions

[59] Let $x = [\alpha_a; a \in \mathfrak{G}]$, $xa_0 = [\beta_a; a \in \mathfrak{G}]$, $(xa_0)b_0 = [\gamma_a; a \in \mathfrak{G}]$. Then $\beta_a = \alpha_{a_0a}$ and $\gamma_a = \beta_{b_0a} = \alpha_{a_0(b_0a)} = \alpha_{(a_0b_0)a}$. Hence $(xa_0)b_0 = [\alpha_{(a_0b_0)a}; a \in \mathfrak{G}] = x(a_0b_0)$.

[60] If the $a_0, a_0^2, \dots$ are all distinct, let $a_i = a_0^{2i-1}$. The result readily follows in this case. Otherwise, a_0 is of finite order, $p = 2, 3, \dots$ ($p \neq 1$ since $a_0 \neq 1$). For each a we can define S_a as the set of $b \in \mathfrak{G}$ for which $b = a_0^k a$ for some $k = 0, 1, \dots, p-1$. It is easily seen how we can choose a sequence $a_1, a_2, \dots$ such that the S_{a_i} are mutually exclusive. For this sequence we have that the $a_1, a_2, \dots, a_0a_1, a_0a_2, \dots$ are all different. This readily follows from the facts that the S_{a_i} are mutually exclusive and $a_0 \neq 1$.

$f(x)$, respectively $F(x,a)$, $x \in S$, $a \in \mathfrak{G}$, which are measurable in x and with a finite

$$\int_S |f(x)|^2\,dx, \qquad \text{respectively} \qquad \sum_{a\in\mathfrak{G}} \int_S |F(x,a)|^2\,dx.$$

Thus

$$(f,g) = \int_S f(x)\overline{g(x)}\,dx, \qquad (F,G) = \sum_{a\in\mathfrak{G}} \int_S F(x,a)\overline{G(x,a)}\,dx$$

(cf. [5], p. 194). Form the bounded operators

$$U_{a_0} f(x) = f(xa_0), \qquad \overline{U}_{a_0} F(x,a) = F(xa_0, aa_0) \qquad (a_0 \in \mathfrak{G}),$$

$$\overline{L}_{\varphi(x)} F(x,a) = \varphi(x)F(x,a) \qquad (\varphi(x) \text{ bounded and measurable})$$

(loc. cit., pp. 196, 198–199) and the ring $\mathbf{M}$ in $\mathfrak{H}_{\mathfrak{G}S}$ which the $\overline{U}_{a_0}$, $\overline{L}_{\varphi(x)}$ generate (loc. cit., p. 200):

$$\mathbf{M} = \mathbf{R}(\overline{U}_{a_0}, \overline{L}_{\varphi(x)}; a_0 \in \mathfrak{G},\ \varphi(x) \text{ bounded and measurable}). \tag{5.5.α}$$

Before we go on we form the following system of functions in $\mathfrak{H}_S$:

$$\omega_{\bar{x}}(x) = (-1)^{\sum_{a\in\mathfrak{G}} \bar{\alpha}_a \alpha_a} \quad (\bar{x} = [\bar{\alpha}_a; a \in \mathfrak{G}] \in \mathfrak{S},\ x = [\alpha_a; a \in \mathfrak{G}] \in S).^{62}$$

It is easy to verify that these functions (for all $\bar{x} \in \mathfrak{S}$) form a complete normalized orthogonal system in $\mathfrak{H}_S$.[63] Consequently the functions

$$F_{\bar{x},\bar{a}}(x,a) = \omega_{\bar{x}}(x)\delta_{a,\bar{a}} \qquad (x \in S,\ \bar{a} \in \mathfrak{G})$$

form a complete normalized orthogonal system in $\mathfrak{H}_{\mathfrak{G}S}$.

A familiar argument shows that the $\varphi(x)$ in (5.5.α) can be restricted considerably without changing the ring $\mathbf{M}$. Indeed:

First it suffices, for reasons of continuity, to consider those $\varphi(x)$ only which are finite linear aggregates of the $\omega_{\bar{x}}(x)$.[64] Second, we may now restrict the $\varphi(x)$ to the $\omega_{\bar{x}}(x)$ themselves. So we have

$$\mathbf{M} = \mathbf{R}(\overline{U}_{a_0}, \overline{L}_{\omega_{\bar{x}}(x)}; a_0 \in \mathfrak{G},\ \bar{x} \in \mathfrak{S}). \tag{5.5.β}$$

[62]Owing to $\bar{x} = [\bar{\alpha}_a; a \in \mathfrak{G}] \in \mathfrak{S}$ we have $\bar{\alpha}_a \neq 0$ only for a finite set of a's. Hence the sum $\sum_{a\in\mathfrak{G}} \bar{\alpha}_a \alpha_a$ is really finite.

[63]In this particular case the group nature of $\mathfrak{G}$ is irrelevant. Hence we can replace $\mathfrak{G}$ by any other countably infinite set. We replace the $a \in \mathfrak{G}$ by $a = 1, 2, \ldots$. After this substitution our $\omega_{\bar{x}}(x)$ become the well-known complete normalized orthogonal set of Walsh–Rademacher functions. It was also considered in [7], p. 74, under the name of $\omega_a(x)$.

[64]Cf. the argument of [7], pp. 73–74 (II).

And as 1 occurs among the $\overline{U}_{a_0}$ (for $a_0 = 1$) and among the $\overline{L}_{\omega_{\bar{x}}(x)}$ (for $\bar{x} = 0$), we can equally write:

$$\mathbf{M} = \mathbf{R}(\overline{U}_{a_0}\overline{L}_{\omega_{\bar{x}}(x)};\, a_0 \in \mathfrak{G},\ \bar{x} \in \mathfrak{S}). \tag{5.5.γ}$$

(III) Clearly

$$U_{a_0}\omega_{\bar{x}}(x) = \omega_{\bar{x}}(xa_0) = \omega_{\bar{x}a_0^{-1}}(x).$$

Now consider any $f \in \mathfrak{H}_S$ with $U_{a_0}f = f$ for all $a_0 \in \mathfrak{G}$. Use the orthogonal expansion $f(x) = \sum_{\bar{x}\in\mathfrak{S}} \lambda_{\bar{x}}\omega_{\bar{x}}(x)$. Then the above formula gives

$$f = \sum_{\bar{x}\in\mathfrak{S}} \lambda_{\bar{x}}\omega_{\bar{x}}, \qquad U_{a_0}f = \sum_{\bar{x}\in\mathfrak{S}} \lambda_{\bar{x}}\omega_{\bar{x}a_0^{-1}} = \sum_{\bar{x}\in\mathfrak{S}} \lambda_{\bar{x}a_0}\omega_{\bar{x}}.$$

Hence $\lambda_{\bar{x}} = \lambda_{\bar{x}a_0}$. As $|||f|||^2 = \sum_{\bar{x}\in\mathfrak{S}} |\lambda_{\bar{x}}|^2$ is finite, this implies that $\lambda_{\bar{x}} = 0$ if there are infinitely many distinct $\bar{x}a_0$, $a_0 \in \mathfrak{G}$. Now this is the case for all $\bar{x} \in \mathfrak{S}$, $\bar{x} \neq 0$.[65] So $\bar{x} \neq 0$ implies $\lambda_{\bar{x}} = 0$. Thus $f(x) = \lambda_0\omega_0(x) = \lambda_0$.

So $U_{a_0}f = f$ (for all $a_0 \in \mathfrak{G}$) implies that f is a constant. Hence $\mathfrak{G}$ is ergodic.

(IV) Clearly

$$\omega_{\bar{x}}(x)\omega_{\bar{y}}(x) = \omega_{\bar{x}\oplus\bar{y}}(x),$$

$$\overline{L}_{\omega_{\bar{x}}(x)}F_{\bar{y},\bar{b}}(x,a) = \omega_{\bar{x}}(x)F_{\bar{y},\bar{b}}(x,a) = F_{\bar{x}\oplus\bar{y},\bar{b}}(x,a).$$

Next

$$\overline{U}_{\bar{a}}F_{\bar{y},\bar{b}}(x,a) = F_{\bar{y},\bar{b}}(x\bar{a}, a\bar{a}) = F_{\bar{y}\bar{a}^{-1},\bar{b}\bar{a}^{-1}}(x,a).$$

Combining these gives

$$\overline{U}_{\bar{a}}\overline{L}_{\omega_{\bar{x}}(x)}F_{\bar{y},\bar{b}}(x,a) = F_{(\bar{x}\oplus\bar{y})\bar{a}^{-1},\bar{b}\bar{a}^{-1}}(x,a). \tag{5.5.ϵ}$$

Now treat the index $\bar{y},\bar{b}$ of $F_{\bar{y},\bar{b}}$ as one pair. Make these pairs into a group $\{\mathfrak{S},\mathfrak{G}\}$ by defining

$$\{\bar{y},\bar{b}\}\{\bar{z},\bar{c}\} = \{\bar{y}\bar{c} \oplus \bar{z}, \bar{b}\bar{c}\}. \tag{5.5.ζ}$$

As $\bar{x} \oplus \bar{x} = 0$, (5.5.$\zeta$) implies that

$$\{\bar{y},\bar{b}\}\{\bar{x},\bar{a}\}^{-1} = \{(\bar{x}\oplus\bar{y})\bar{a}^{-1}, \bar{b}\bar{a}^{-1}\}.$$

[65]We prove this as follows: Put $\bar{x} = [\bar{\alpha}_a; a \in \mathfrak{G}]$. Since $\bar{x} \neq 0$ there is an a^* such that $\bar{\alpha}_{a^*} \neq 0$. Now as in footnote 60, we can construct an infinite sequence of the a's $\in \mathfrak{G}$ such that $a_1a^*, a_2a^*, \ldots$ are all different. Then $\bar{x}a_i^{-1} = [\bar{\alpha}_{a_i^{-1}a}; a \in \mathfrak{G}]$ has for its a_ia^* component $\bar{\alpha}_{a^*} = 1$. Thus if we consider the $\bar{x}a_1^{-1}, \bar{x}a_2^{-1}, \ldots$, we find that there is an infinite number of a's $\in \mathfrak{G}$ such that the a component is not zero for at least one of the $\bar{x}a_i^{-1}$. Since each $\bar{x}a_i^{-1}$ can have at most a finite number of components different from zero, there must be an infinite number of distinct $\bar{x}a_i^{-1}$.

Hence (5.5.ϵ) becomes

$$\overline{U}_{\bar{a}}\overline{L}_{\omega_{\bar{x}}(x)}F_{\bar{y},\bar{b}}(x,a) = F_{\{\bar{y},\bar{b}\}\{\bar{x},\bar{a}\}^{-1}}(x,a). \tag{5.5.η}$$

(V) Apply the construction of §5.3 to the group $\{\mathfrak{S},\mathfrak{G}\}$ of (5.5.ζ) above, instead of its $\mathfrak{G}$. This is possible since $\{\mathfrak{S},\mathfrak{G}\}$ fulfills (i) in Lemma 5.3.4; i.e. if $\{\bar{x},\bar{a}\} \neq \{0,1\}$ then there exist infinitely many different $\{\bar{y},\bar{b}\}^{-1}\{\bar{x},\bar{a}\}\{\bar{y},\bar{b}\}$. Indeed, if $\bar{x} \neq 0$, then form

$$\{0,\bar{b}\}^{-1}\{\bar{x},\bar{a}\}\{0,\bar{b}\} = \{\bar{x}\bar{b},\bar{b}^{-1}\bar{a}\bar{b}\}.$$

There are infinitely many distinct $\bar{x}\bar{b}$ for $\bar{b} \in \mathfrak{G}$. Cf. footnote 65. Thus our statement is true for $\bar{x} \neq 0$. If however $\bar{x} = 0$, then necessarily $\bar{a} \neq 1$. Form

$$\{\bar{y},1\}^{-1}\{0,\bar{a}\}\{\bar{y},1\} = \{\bar{y}\bar{a} \oplus \bar{y},\bar{a}\}.$$

There exist infinitely many different $\bar{y}\bar{a} \oplus \bar{y}$.[66]

Now write $\mathfrak{H}_I, \mathbf{M}_I$ for its $\mathfrak{H}, \mathbf{M}$. We have

$$\mathbf{M}_I = \mathbf{R}(U_{\{\bar{x},\bar{a}\}}; \bar{x} \in \mathfrak{S},\ \bar{a} \in \mathfrak{G}) \tag{5.5.θ}$$

and by (5.3.β)

$$U_{\{\bar{x},\bar{a}\}}\varphi_{\{\bar{y},\bar{b}\}} = \varphi_{\{\bar{y},\bar{b}\}\{\bar{x},\bar{a}\}^{-1}}. \tag{5.5.ι}$$

We can map $\mathfrak{H}_{\mathfrak{G}S}$ isomorphically on $\mathfrak{H}_I$ by carrying the complete orthonormalized system $F_{\{\bar{y},\bar{b}\}}$ of the former into the set of $\varphi_{\{\bar{y},\bar{b}\}}$ of the latter. Then comparison of (5.5.ι) and (5.5.η) shows that this isomorphism carries $\overline{U}_{\bar{a}}\overline{L}_{\omega_{\bar{x}}(x)}$ into $U_{\{\bar{x},\bar{a}\}}$. Hence (5.5.γ) and (5.5.θ) show that it carries $\mathbf{M}$ into $\mathbf{M}_I$. Thus $\mathbf{M}$ and $\mathbf{M}_I$ are spatially isomorphic.

Summing up:

Lemma 5.5.1. *Given any countably infinite group $\mathfrak{G}$, define $S, \mathfrak{S}$ as described in (I) above, and the group $\mathfrak{G}_I = \{\mathfrak{S},\mathfrak{G}\}$ as described by (5.5.ζ) (IV) above. Then the construction of [5], pp. 192–209, Part IV, can be applied to $\mathfrak{G}$ and S, and our construction of §5.3 to $\{\mathfrak{S},\mathfrak{G}\}$ (in place of its $\mathfrak{G}$).*

These two constructions produce spatially isomorphic factors $\mathbf{M}$.

The mappings

$$x\{\bar{y},\bar{b}\} = x\bar{b} + \bar{y} \tag{5.5.κ}$$

[66]Put $\bar{y}_b = [\delta_{a,b}; a \in \mathfrak{G}] \in \mathfrak{S}$. As $\bar{a} \neq 1$, $\bar{y}_b\bar{a} \oplus \bar{y}_b = \bar{y}_c\bar{a} \oplus \bar{y}_c$ means either $b = c$ or $\bar{a}b = c, \bar{a}c = b$. In other words: If b runs over all $\mathfrak{G}$ there can never coincide more than two $\bar{y}_b\bar{a} \oplus \bar{y}_b$. Now $\mathfrak{G}$ is infinite. Hence there are infinitely many different $\bar{y}_b\bar{a} \oplus \bar{y}_b$.

provide a convenient representation of the group $\{\mathfrak{S}, \mathfrak{G}\}$ of (5.5.ζ) in (V). If we specify $x \in S$, then (5.5.κ) appears as a permutation of S. But as it maps $\mathfrak{S}$ on itself, we can also prescribe $x \in \mathfrak{S}$ and view (5.5.κ) as a permutation of $\mathfrak{S}$.

(5.5.κ) makes it clear that $\{\mathfrak{S}, \mathfrak{G}\}$ is the simplest combination of $\mathfrak{S}$ and $\mathfrak{G}$ apart from the "direct group product".

Our above result can also be interpreted in this way:

We start with an arbitrary (countably infinite) group $\mathfrak{G}$ and should like to apply the construction of §5.3. Since $\mathfrak{G}$ may not fulfill the condition (i) of Lemma 5.3.4, this cannot always be done directly. We therefore propose to expand $\mathfrak{G}$ to a group $\mathfrak{G}_I$ which fulfills that condition. (5.5.ζ) provides a very simple generally valid mechanism to do just that. This has the further interesting consequence that within $\mathbf{M}$, the set of group numbers for any $\mathfrak{G}$, the metric closure of a subalgebra is equivalent to the topological closure for any of the ring topologies. For we extend $\mathbf{M}$ to $\mathbf{M}_I$ as above, and Theorem I (of §1.6 above) shows that our statement is true in the extended space $\mathfrak{H}_I$. Since any subalgebra of $\mathbf{M}$ will be reduced by the original space $\mathfrak{H}$, topological closure in $\mathfrak{H}_I$ will yield topological closure. Similarly the discussion of Chapter I may be carried through on the assumption that $\mathbf{M}$ is a group algebra.

§5.6 The analogue of Lemma 5.2.2 is true for the construction of §5.3 too, and this time it is even easier to prove.

Lemma 5.6.1. *The ring* $\mathbf{M}$ *of §5.3 is approximately finite if* $\mathfrak{G}$ *possesses the property (i) of Lemma 5.2.2 (besides the property (i) of Lemma 5.3.4).*

Proof. Form the $\mathfrak{G}_1, \mathfrak{G}_2, \ldots$ of (i) in Lemma 5.2.2. Now

$$\mathbf{M} = \mathbf{R}(U_a; a \in \mathfrak{G}).$$

Hence

$$\mathbf{M} = \mathbf{R}(\mathbf{N}_n; n = 1, 2, \ldots)$$

with

$$\mathbf{N}_n = \mathbf{R}(U_a; a \in \mathfrak{G}_n).$$

Thus $\mathbf{N}_1 \subseteq \mathbf{N}_2 \subseteq \cdots$ and $\mathbf{N}_n$ is of finite order because $\mathfrak{G}_n$ is finite. Therefore $\mathbf{M}$ is approximately finite by Definition 4.6.1 (cf. the author's note following (5.2.ζ)).

Observe that if $\mathfrak{G}$ fulfills (i) in Lemma 5.2.2 (but not necessarily (i) in Lemma 5.3.4), then the group $\mathfrak{G}_I = \{\mathfrak{S}, \mathfrak{G}\}$ of Lemma 5.5.1 possesses that property too. Put $\mathfrak{G} = \mathfrak{G}_1 + \mathfrak{G}_2 + \cdots$, $\mathfrak{G}_1 \subseteq \mathfrak{G}_2 \subseteq \cdots$, all $\mathfrak{G}_n$ finite. Let $\mathfrak{S}_n$ be the set of all

$\bar{x} = [\bar{\alpha}_a; a \in \mathfrak{G}]$ with $\bar{\alpha}_a = 0$, when a is not in $\mathfrak{G}_n$. Then $\mathfrak{S}_1 \subseteq \mathfrak{S}_2 \subseteq \cdots$ and all the $\mathfrak{S}_n$ are finite. So

$$\{\mathfrak{S}, \mathfrak{G}\} = \{\mathfrak{S}_1, \mathfrak{G}_1\} + \{\mathfrak{S}_2, \mathfrak{G}_2\} + \cdots, \qquad \{\mathfrak{S}_1, \mathfrak{G}_1\} \subseteq \{\mathfrak{S}_2, \mathfrak{G}_2\} \subseteq \cdots,$$

and all the $\{\mathfrak{S}_n, \mathfrak{G}_n\}$ are finite.

Thus it is easy to form groups $\mathfrak{G}$ which fulfill both (i) in Lemma 5.2.2 and (i) in Lemma 5.3.4. The following example is more direct.

Lemma 5.6.2. *The group $\mathfrak{G}$ of all those permutations of $(1, 2, \ldots)$ which move only a finite number of elements, fulfills both (i) in Lemma 5.2.2 and (i) in Lemma 5.3.4.*

Proof. Ad (i) in Lemma 5.3.4: Given an $a \in \mathfrak{G}$, $a \neq 1$, choose n so that a moves no number other than $1, \ldots, n$. Assume that a does move i (which is therefore $= 1, \ldots, n$). Let c_j be the transposition of i and j, where $j = n+1, n+2, \ldots$. Then the only $k = n+1, n+2, \ldots$ moved by $c_j^{-1} a c_j$ is $k = j$. Thus the $c_j^{-1} a c_j$, $j = n+1, n+2, \ldots$, are pairwise different and so $\mathfrak{L}_a$ is infinite.

Ad (i) in Lemma 5.2.2: Let $\mathfrak{G}_n$ be the set of those permutations which move no number other than $1, \ldots, n$. Then $\mathfrak{G} = \mathfrak{G}_1 + \mathfrak{G}_2 + \cdots$, $\mathfrak{G}_1 \subseteq \mathfrak{G}_2 \subseteq \cdots$, and each $\mathfrak{G}_n$ is a finite subset of $\mathfrak{G}$.

The analysis of the invariants (1), (2) of §4.8, gives no new information. As to (1), all examples of [5] as well as those of §5.3 have $\overline{\mathbf{M}}^c = \overline{\mathbf{M}}$. Indeed: Neither of these definitions ever mentions any specific non-real (complex) number. Hence $\mathfrak{H}_c, \mathbf{M}_c$ coincide with $\mathfrak{H}, \mathbf{M}$, so that $\overline{\mathbf{M}}^c = \overline{\mathbf{M}_c} = \overline{\mathbf{M}}$ (cf. §2.3).

Thus we possess no examples of a factor $\mathbf{M}$ in the case (II_1) with $\overline{\mathbf{M}}^c \neq \overline{\mathbf{M}}$. Possibly a generalization of the construction of §5.3, with the introduction of complex numerical factors in (i), (ii) of Lemma 5.3.1 (and (5.3.β) after that lemma) might achieve this.[67] But we have not obtained so far any conclusive results in this respect.

As to (2), $\mathfrak{G}(\overline{\mathbf{M}})$, we know nothing beyond Theorem XV.

CHAPTER VI. NON-APPROXIMATELY FINITE FACTORS

§6.1 As pointed out at the end of §4.8 and again at the end of §5.6, we have not succeeded so far to establish the not approximately finite character of any factor with the help of the invariants (1), (2) of §4.8. We shall now prove the existence of

[67] Corresponding to the technique of "multipliers" in forming skew fields over a given center in abstract algebra.

not approximately finite factors, but it is necessary to introduce a new invariant in order to achieve this aim.

Throughout what follows, **M** is again a factor in case (II_1).

Definition 6.1.1. $\mathbf{M}$ *has the property* Γ *if this is true:*

Given any system $A_1, \dots, A_m \in \mathbf{M}$ *and any* $\epsilon > 0$ *there exists a unitary*

$$U = U(A_1, \dots, A_m, \epsilon) \in \mathbf{M}$$

with

$$\mathrm{Tr}_{\mathbf{M}}(U) = 0$$

and

$$[[U^{-1}A_h U - A_h]] < \epsilon \qquad \textit{for } h = 1, \dots, m.$$[68,69]

This property Γ is clearly a purely algebraic property—i.e. one concerning the algebraic type $\overline{\mathbf{M}}$ only.

Observe first this

Lemma 6.1.1. *Let a group* $\mathfrak{G}$ *fulfilling (i) in Lemma 5.3.4 be given, which possesses this property:*

(i) Given any system $a^{(1)}, \dots, a^{(n)} \in \mathfrak{G}$ *there exists a* $c_0 \in \mathfrak{G}$*,* $c_0 \neq 1$ *which commutes with* $a^{(1)}, \dots, a^{(n)}$*.*

Then the $\mathbf{M}$ *of §5.3 possesses the property* Γ*.*

Proof. Let $A_1, \dots, A_m \in \mathbf{M}$ and $\epsilon > 0$ be given as indicated in Definition 6.1.1. Following (ii) in Lemma 5.3.6, we can choose $a^{(1)}, \dots, a^{(n)} \in \mathfrak{G}$ and $\zeta_{h,1}, \dots, \zeta_{h,n}$ (these are the $\eta_{a^{(1)}}, \dots, \eta_{a^{(n)}}$ mentioned there for $A = A_h$, for $h = 1, \dots, m$) so that

$$\left[\left[A_h - \sum_{i=1}^{n} \zeta_{h,i} U_{a^{(i)}}\right]\right] < \epsilon/2 \qquad \text{for } h = 1, \dots, m. \tag{6.1.α}$$

Now apply our assumption (i) to these $a^{(1)}, \dots, a^{(n)}$, obtaining a $c_0 \in \mathfrak{G}$, $c_0 \neq 1$, which commutes with them.

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[68]The requirement $\mathrm{Tr}_{\mathbf{M}}(U) = 0$ is necessary, since otherwise we could choose $U = 1$. Outright commutativity of U with $A_1, \dots, A_m$ (instead of the last inequality) would not do for the following reason: If $\mathbf{M} = \mathbf{R}(A_1, \dots, A_m)$ then this commutativity would mean $U \in \mathbf{M}'$. Hence $U \in \mathbf{M} \cap \mathbf{M}'$, $U = \alpha \cdot 1$. Since $\mathrm{Tr}_{\mathbf{M}}(U) = 0$ this would mean $U = 0$, contradicting the unitarity of U. Hence no $\mathbf{M} = \mathbf{R}(A_1, \dots, A_m)$ could then have this property. Now it can be shown that an approximately finite $\mathbf{M} = \mathbf{R}(A_1, A_2)$ for two suitable A_1, A_2. Hence an approximately finite $\mathbf{M}$ would not have this property—but we need Lemma 6.1.2.

[69]This property could be further subdivided, according to the values of m and of expressions like $\epsilon/\mathrm{Max}(|||A_1|||, \dots, |||A_m|||)$. We could also require $A_1, \dots, A_m$ to be unitary, etc. For our present purpose, however, these refinements are not needed.

Then U_{c_0} and $U_{a^{(i)}}$ commute (cf. footnote 48) and so

$$U_{c_0}^{-1}\left(A_h - \sum_{i=1}^{n} \zeta_{h,i} U_{a^{(i)}}\right) U_{c_0} = U_{c_0}^{-1} A_h U_{c_0} - \sum_{i=1}^{n} \zeta_{h,i} U_{a^{(i)}}.$$

Hence (6.1.α) yields

$$\left[\left[U_{c_0}^{-1} A_h U_{c_0} - \sum_{i=1}^{n} \zeta_{h,i} U_{a^{(i)}} \right]\right] < \epsilon/2 \tag{6.1.β}$$

and (6.1.α) and (6.1.β) yield together

$$[[U_{c_0}^{-1} A_h U_{c_0} - A_h]] < \epsilon. \tag{6.1.γ}$$

Clearly

$$U_{c_0} \sim \{\eta'_{ab^{-1}}\}, \qquad \eta'_c = \begin{cases} 1, & c = c_0, \\ 0, & \text{otherwise.} \end{cases}$$

As $c_0 \neq 1$ this implies $\eta'_1 = 0$, i.e.

$$\mathrm{Tr}_{\mathbf{M}}(U_{c_0}) = 0. \tag{6.1.δ}$$

Thus (6.1.γ), (6.1.δ) show that $U = U_{c_0}$ meets all our requirements.

We can now prove

Lemma 6.1.2. *Approximate finiteness implies the Property* Γ.

Proof. [70] It suffices to find a group $\mathfrak{G}$ which fulfills the condition (i) in Lemma 5.3.4 (so that the $\mathbf{M}$ of §5.3 is a factor, of course, in case (II_1)), (i) in Lemma 5.2.2 (so that $\mathbf{M}$ is approximately finite by Lemma 5.6.1), and (i) in Lemma 6.1.1 (so that $\mathbf{M}$ possesses the property Γ). Then Theorem XIV and our remark after Definition 6.1.1 take care of everything.

Now the group $\mathfrak{G} = \overline{\mathfrak{G}}$ of Lemma 5.6.2 possesses the first two properties. As to (i) in Lemma 6.1.1: Let $a^{(1)}, \ldots, a^{(n)}$ be the given elements of $\mathfrak{G}$. Choose p so that none of the $a^{(1)}, \ldots, a^{(n)}$ move any element other than $1, \ldots, p$. Let c_0 be the transposition of $p+1$ with $p+2$. Then $c_0 \neq 1$ and it commutes with $a^{(1)}, \ldots, a^{(n)}$.

§6.2 We now proceed to establish the decisive negative result.

[70] There exists also a relatively simple direct proof, based on Def. 4.1.1.

Lemma 6.2.1. *Let a group $\mathfrak{G}$ fulfilling (i) in Lemma 5.3.4 be given, which possesses this property:* [71]

(i) There exists a set $\mathfrak{F} \subseteq \mathfrak{G}$ with these properties:

(i_1) There exists a $c_1 \in \mathfrak{G}$ such that

$$\mathfrak{F} + c_1 \mathfrak{F} c_1^{-1} = \mathfrak{G} - (1).$$

(i_2) There exists a $c_2 \in \mathfrak{G}$, such that the three sets $c_2^l \mathfrak{F} c_2^{-l}$, $l = 0, \pm 1$, are disjoint.

Then the **M** *of §5.3 does not possess the property* Γ.

Proof. Assume the opposite, i.e. that **M** possesses the property Γ. Apply Definition 6.1.1 with $n = 2$. Put $A_1 = U_{c_1}$, $A_2 = U_{c_2}$, while $\epsilon > 0$ will be chosen subsequently. Form the $U = U(A_1, A_2; \epsilon)$ described there.

Then we have:

$$\mathrm{Tr}_{\mathbf{M}}(U) = 0, \tag{6.2.α}$$

$$[[U^{-1} U_{c_h} U - U_{c_h}]] < \epsilon \qquad \text{for } h = 1, 2. \tag{6.2.β}$$

As U_{c_h}, U are both unitary and

$$U_{c_h}^{-1} U (U^{-1} U_{c_h} U - U_{c_h}) = U - U_{c_h}^{-1} U U_{c_h},$$

(6.2.β) is equivalent to

$$[[U - U_{c_h}^{-1} U U_{c_h}]] < \epsilon \qquad \text{for } h = 1, 2. \tag{6.2.γ}$$

Now determine the η_c in the sense of Lemma 5.3.2 for $U, U_{c_h}^{-1} U U_{c_h}, U - U_{c_h}^{-1} U U_{c_h}$ in succession. If the first is θ_c it is easy to verify that the second is $\theta_{c_h c c_h^{-1}}$ [50] and hence the third is $\theta_c - \theta_{c_h c c_h^{-1}}$. Therefore the application of (i) in Lemma 5.3.6 to U and to $U - U_{c_h}^{-1} U U_{c_h}$ gives

$$[[U]]^2 = \sum_{c \in \mathfrak{G}} |\theta_c|^2,$$

$$[[U - U_{c_h}^{-1} U U_{c_h}]]^2 = \sum_{c \in \mathfrak{G}} |\theta_c - \theta_{c_h c c_h^{-1}}|^2.$$

As U is unitary, $[[U]]^2 = 1$. Considering this and (6.2.γ) these equations yield

$$\sum_{c \in \mathfrak{G}} |\theta_c|^2 = 1, \tag{6.2.δ}$$

[71] This proof copies to a certain extent Hausdorff's famous $1/2 - 1/3$ division of the sphere. In this connection the use of the free group in Lemma 6.2.2 should be noted.

$$\left(\sum_{c\in\mathfrak{G}} |\theta_c - \theta_{c_h c c_h^{-1}}|^2\right)^{1/2} < \epsilon \qquad \text{for } h = 1, 2. \tag{6.2.ϵ}$$

After these preparations we introduce a measure in $\mathfrak{G}$ by defining

$$\nu(\mathfrak{A}) = \sum_{c\in\mathfrak{A}} |\theta_c|^2 \qquad \text{for } \mathfrak{A} \subseteq \mathfrak{G}.$$

Then (6.2.δ) becomes

$$\nu(\mathfrak{G}) = 1, \tag{6.2.ζ}$$

and (6.2.α) means $\theta_1 = 0$, i.e.

$$\nu((1)) = 0. \tag{6.2.η}$$

The triangle inequality in infinitely many dimensions gives

$$\left|\left(\sum_{c\in\mathfrak{A}} |\theta_c|^2\right)^{1/2} - \left(\sum_{c\in\mathfrak{A}} |\theta_{c_h c c_h^{-1}}|^2\right)^{1/2}\right| \leq \left(\sum_{c\in\mathfrak{G}} |\theta_c - \theta_{c_h c c_h^{-1}}|^2\right)^{1/2}.$$

The left-hand side is clearly $|\nu(\mathfrak{A})^{1/2} - \nu(c_h\mathfrak{A}c_h^{-1})^{1/2}|$. The right-hand side is $\leq (\sum_{c\in\mathfrak{G}} |\theta_c - \theta_{c_h c c_h^{-1}}|^2)^{1/2}$, which is $< \epsilon$ by (6.2.ϵ). So we have

$$|\nu(\mathfrak{A})^{1/2} - \nu(c_h\mathfrak{A}c_h^{-1})^{1/2}| < \epsilon. \tag{6.2.θ}$$

Now by (6.2.ζ), $\nu(\mathfrak{A})$ and $\nu(c_h\mathfrak{A}c_h^{-1})$ are $\leq \nu(\mathfrak{G}) = 1$. Hence

$$\begin{aligned} |\nu(\mathfrak{A}) - \nu(c_h\mathfrak{A}c_h^{-1})| &= |\nu(\mathfrak{A})^{1/2} - \nu(c_h\mathfrak{A}c_h^{-1})^{1/2}|(\nu(\mathfrak{A})^{1/2} + \nu(c_h\mathfrak{A}c_h^{-1})^{1/2}) \\ &\leq 2|\nu(\mathfrak{A})^{1/2} - \nu(c_h\mathfrak{A}c_h^{-1})^{1/2}|. \end{aligned}$$

Therefore (6.2.θ) becomes

$$|\nu(\mathfrak{A}) - \nu(c_h\mathfrak{A}c_h^{-1})| < 2\epsilon. \tag{6.2.ι}$$

Let us apply (6.2.ι) to $\mathfrak{F}, c_1$; $\mathfrak{F}, c_2$; $c_2^{-1}\mathfrak{F}c_2, c_2$ in place of its $\mathfrak{A}, c_h$. Then

$$|\nu(\mathfrak{F}) - \nu(c_1\mathfrak{F}c_1^{-1})| < 2\epsilon, \tag{6.2.κ}$$

$$|\nu(\mathfrak{F}) - \nu(c_2\mathfrak{F}c_2^{-1})| < 2\epsilon, \tag{6.2.λ}$$

$$|\nu(c_2^{-1}\mathfrak{F}c_2) - \nu(\mathfrak{F})| < 2\epsilon \tag{6.2.μ}$$

obtain. Now (i_1) and (6.2.ζ), (6.2.η) and (6.2.κ) give

$$\nu(\mathfrak{F}) + (\nu(\mathfrak{F}) + 2\epsilon) > 1,$$

i.e.

$$\nu(\mathfrak{F}) > \frac{1}{2} - \epsilon. \tag{6.2.ν}$$

On the other hand (i_2) and (6.2.ζ), (6.2.λ) and (6.2.μ) give

$$\nu(\mathfrak{F}) + (\nu(\mathfrak{F}) - 2\epsilon) + (\nu(\mathfrak{F}) - 2\epsilon) < 1,$$

i.e.

$$\nu(\mathfrak{F}) < \frac{1}{3} + \frac{4}{3}\epsilon. \tag{6.2.o}$$

(6.2.ν) and (6.2.o) imply

$$\frac{1}{2} - \epsilon < \frac{1}{3} + \frac{4}{3}\epsilon \quad \text{or} \quad \frac{1}{14} < \epsilon.$$

Hence it suffices to choose $\epsilon = 1/14$ in order to have a contradiction. Thus we have shown that **M** cannot possess the property Γ.

Corollary 1. *Under the above assumptions,* **M** *is not approximately finite.*

Proof. Combine the above lemma with Lemma 6.1.2.

Corollary 2. *We may replace condition (i_2) on $\mathfrak{G}$ and $\mathfrak{F}$ by*
(i_2'). There exist two elements c_2 and c_3 in $\mathfrak{G}$ such that $\mathfrak{F}, c_2\mathfrak{F}c_2^{-1}, c_3\mathfrak{F}c_3^{-1}$ are mutually exclusive.

Proof. In the first paragraph of the above proof we let $A_3 = U_{c_3}$ and then take $U = U(A_1, A_2, A_3, \epsilon)$ instead of $U = U(A_1, A_2, \epsilon)$. We have $h = 1, 2, 3$ in (6.2.β), (6.2.γ), (6.2.ϵ), (6.2.θ) and (6.2.ι), instead of $h = 1, 2$. Finally we obtain an equation

$$|\nu(\mathfrak{F}) - \nu(c_3\mathfrak{F}c_3^{-1})| < 2\epsilon \tag{6.2.μ'}$$

instead of (6.2.μ), by applying (6.2.ι) to $\mathfrak{F}, c_3$ instead of $c_2^{-1}\mathfrak{F}c_2, c_2$. (6.2.$\mu'$) is used instead of (6.2.μ) in order to obtain (6.2.o), and the rest of the proof is the same.

We conclude by giving a specific instance of an **M** which does not possess the property Γ and is not approximately finite. We do this by giving an example of a group $\mathfrak{G}$ which fulfills the requirements of Lemma 6.2.1.

Lemma 6.2.2. *Let $\mathfrak{G}$ be the free group of two generators c_1, c_2. Then $\mathfrak{G}$ fulfills the requirements of Lemma 6.2.1.*

Proof. Ad (i) in Lemma 5.3.4: Consider an $a \in \mathfrak{G}$, $a \neq 1$. Write a as a power product of c_1, c_2 of minimum length. Then it is easy to verify that $c_1^{-n}ac_1^n$,

$n = 1, 2, \ldots$, are pairwise different unless $a = c_1^m$. In the latter case we have that the $c_2^{-n} a c_2^n = c_2^{-n} c_1^m c_2^n$ are distinct unless $m = 0$, i.e. $a = 1$. Thus unless $a = 1$, $\mathfrak{L}_a$ is infinite.

Ad (i) in Lemma 6.2.1: Let $\mathfrak{F}$ be the set of $a \in \mathfrak{G}$ which when written as a power product of c_1, c_2 of minimum length end with a c_1^n, $n = \pm 1, \pm 2, \ldots$. Then (i_1) and (i_2) in Lemma 6.2.1, i.e. (i) eod., are immediately verified.

Combining Lemma 6.2.2 with Corollary 1 to Lemma 6.2.1 gives

Theorem XVI. *There exist not approximately finite factors* $\mathbf{M}$ *in the case* (II_1).

Or, considering Theorem XV,

Theorem XVI′. *There exist in case* (II_1) *more than one algebraical type* $\overline{\mathbf{M}}$.

§6.3 We wish in this section to present a number of examples of groups $\mathfrak{G}$ for which the corresponding $\mathbf{M}$ of §5.3 is not approximately finite.

Lemma 6.3.1. *Let $\mathfrak{G}$ be a group which contains two multiplicative subsets $\mathfrak{F}_1$ and $\mathfrak{F}_2$. Suppose that $\mathfrak{G}$ is the free product of $\mathfrak{F}_1$ and $\mathfrak{F}_2$ and that $\mathfrak{F}_1$ contains at least two elements and $\mathfrak{F}_2$ contains at least three. Then $\mathfrak{G}$ satisfies condition (i) of Lemma 5.3.4, condition (i_1) of Lemma 6.2.1 and condition (i'_2) of Corollary 2 of that lemma (for the same $\mathfrak{F}$).*

Proof. Let c_1 be an element of $\mathfrak{F}_1$ which is not 1, and c_2, c_3 elements of $\mathfrak{F}_2$ which are each not 1.

We prove first that (i) of Lemma 5.3.4 holds, i.e. that for $a \in \mathfrak{G}$, $a \neq 1$, the $\mathfrak{L}_a$ is infinite.

Suppose an $a \in \mathfrak{G}$, $a \neq 1$, is given. Since $\mathfrak{G}$ is a free product of $\mathfrak{F}_1$ and $\mathfrak{F}_2$, each a has a unique factorization $a_1 \cdot a_2 \cdots a_p$ into a product in which the factors are alternately in $\mathfrak{F}_1$ and $\mathfrak{F}_2$ and no factor is 1.

We note for later use that if $c \in \mathfrak{F}_i$, then $c^{-1} \in \mathfrak{F}_i$. Suppose $i = 1$. Then c^{-1} will have a factorization $b_1 \cdot b_2 \cdots b_r$. If $b_1 \in \mathfrak{F}_2$ we have $1 = cc^{-1} = c \cdot b_1 \cdot b_2 \cdots b_r \neq 1$. Thus $b_1 \in \mathfrak{F}_1$. But $1 = cc^{-1} = (cb_1) \cdot b_2 \cdots b_r$ implies $r = 1$. Consequently $c^{-1} = b_1 \in \mathfrak{F}_1$.

Now suppose firstly that a_1 (the first factor) and a_p (the last factor) are both from $\mathfrak{F}_1$. Let $b_n = (c_2 c_1)^n$. Then the set of $b_n^{-1} a b_n$ are for $n = 1, 2, \ldots$ all distinct. (The factorization is immediately apparent.) Hence $\mathfrak{L}_a$ is infinite in this case.

If a_1 and a_p are both from $\mathfrak{F}_2$, we let $b_n = (c_1 c_2)^n$ and the result is again apparent.

If a_1 is from $\mathfrak{F}_1$ and a_p is from $\mathfrak{F}_2$, we consider c_2 and c_3 and denote by c' that one of c_2, c_3 which is not equal to a_p^{-1}. Let $b_n = (c' c_1)^n$. Then for $n = 1, 2, \ldots$ the factorization of $b_n^{-1} a b_n$ is obtained by simply writing out the expressions for these elements and coalescing a_p and the first c' in b_n. a_pc' is in $\mathfrak{F}_2$ but is not 1 since $c' \neq a_p^{-1}$. Hence the factorization has been obtained. These factorizations for $n = 1, 2, \ldots$ show that the $b_n^{-1}ab_n$ are distinct, and hence $\mathfrak{L}_a$ is infinite.

If a_1 is in $\mathfrak{F}_2$ and a_p is in $\mathfrak{F}_1$ we define c' as that one of c_2, c_3 which is not equal to a_1. An argument similar to that of the preceding paragraph will show that in this case also $\mathfrak{L}_a$ is infinite.

Thus we have established (i) of Lemma 5.3.4.

To establish (i_1) of Lemma 6.2.1 and (i_2') of Corollary 2 of that lemma for the same $\mathfrak{F}$, we define $\mathfrak{F}$ as the set of a's, $a \neq 1$, having a factorization $a_1 \cdots a_p$ in which a_p is in $\mathfrak{F}_1$.

We show firstly (i_1) in Lemma 6.2.1, i.e.

$$\mathfrak{F} + c_1^{-1}\mathfrak{F}c_1 = \mathfrak{G} - (1).$$

If $a \in \mathfrak{G}$, $a \neq 1$, then a has a factorization $a_1 \cdots a_p$. Now if a is not $\in \mathfrak{F}$, a_p is not $\in \mathfrak{F}_1$ but $a_p \in \mathfrak{F}_2$. In this case we consider $c_1ac_1^{-1} = c_1a_1 \cdots a_pc_1^{-1}$. If p is even, i.e. $p = 2, 4, \ldots$, then $a_1 \in \mathfrak{F}_1$ (since $a_p \in \mathfrak{F}_2$) and the factorization of $c_1ac_1^{-1}$ is either $(c_1a_1) \cdot a_2 \cdots a_p \cdot c_1^{-1}$ or (if $c_1a_1 = 1$) $a_2 \cdots a_pc_1^{-1}$. If p is odd, i.e. $p = 1, 3, \ldots$, $a_1 \in \mathfrak{F}_2$ and $c_1ac_1^{-1}$ has the factorization $c_1 \cdot a_1 \cdots a_p \cdot c_1^{-1}$. These factorizations all show that $c_1ac_1^{-1}$ is $\in \mathfrak{F}$. Thus if $a \neq 1$ and a is not $\in \mathfrak{F}$, $c_1ac_1^{-1} \in \mathfrak{F}$ or $a \in c_1^{-1}\mathfrak{F}c_1$. This is equivalent to $\mathfrak{G} - (1) \subseteq \mathfrak{F} + c_1^{-1}\mathfrak{F}c_1$. However this inclusion is not proper. For 1 is not in $\mathfrak{F}$. Consequently 1 is not in $c_1^{-1}\mathfrak{F}c_1$ and $\mathfrak{F} + c_1^{-1}\mathfrak{F}c_1$. We can now conclude

$$\mathfrak{F} + c_1^{-1}\mathfrak{F}c_1 = \mathfrak{G} - (1).$$

One can establish (i_2') of Corollary 2 of Lemma 6.2.1 in a similar (indeed easier) way.

Thus our Lemma has been proven.

This lemma permits us to offer many other examples similar to the $\mathfrak{G}$ of Lemma 6.2.2. For we may obtain $\mathfrak{F}_1$ and $\mathfrak{F}_2$ in a number of ways, and then $\mathfrak{G} = \mathfrak{G}(\mathfrak{F}_1, \mathfrak{F}_2)$ is constructed by adding to 1 the "free" products $a_1 \cdot a_2 \cdots a_p$ where the a_i's are never 1 and are alternately from $\mathfrak{F}_1$ and $\mathfrak{F}_2$. As an example we may take $\mathfrak{F}_1$ as the group generated by a single generator a of order $p = 2, 3, \ldots, \infty$ and $\mathfrak{F}_2$ as the group generated by an element b of order $q = 3, 4, \ldots, \infty$. $\mathfrak{G}(a, b) = \mathfrak{G}(\mathfrak{F}_1, \mathfrak{F}_2)$ is then an example. Of course we may take for $\mathfrak{F}_1$ any group of order ≥ 2 and $\mathfrak{F}_2$ of order ≥ 3.

The restriction that $\mathfrak{F}_2$ contains at least 3 members rules out the case in which $\mathfrak{F}_1$ and $\mathfrak{F}_2$ are both groups of order 2, i.e. $\mathfrak{F}_1$ consists of $(1, a)$ and $\mathfrak{F}_2$ consists of

$(1, b)$. Here of course we do not have the (i_2) of Lemma 6.2.1 or the (i_2') of its second corollary. However, condition (i) of Lemma 5.3.4 also fails. For the element ab has only ba and itself as conjugates, i.e. $\mathfrak{L}_{ab} = (ab, ba)$. Thus $\mathfrak{L}_{ab}$ is finite. Hence by Lemma 5.3.4, the $\mathbf{M}$ for this $\mathfrak{G}$ is not even a factor.

Another lemma, which we shall use in an appendix, is readily shown now.

Lemma 6.3.2. *If $\mathfrak{G}$ is such that to every $a \in \mathfrak{G}$, $a \neq 1$, there is a $b \in \mathfrak{G}$, b of order ≥ 3, such that $\mathfrak{G}$ contains the free group $\mathfrak{G}(a, b)$, then $\mathfrak{G}$ fulfills (i) of Lemma 5.3.4. (If a is not of order 2, b need only differ from the identity.)*

Proof. Let $\overline{\mathfrak{L}}_a$ denote the class of a relative to the subgroup $\mathfrak{G}(a, b)$. Our discussion following Lemma 6.3.1 shows that $\mathfrak{G}(a, b)$ satisfies the hypotheses of Lemma 6.3.1. Consequently that Lemma and Lemma 5.3.4 imply that $\overline{\mathfrak{L}}_a$ is infinite. Since $\overline{\mathfrak{L}}_a \subseteq \mathfrak{L}_a$ we have $\mathfrak{L}_a$ infinite for $\mathfrak{G}$ itself, and thus (i) of Lemma 5.3.4 holds.

APPENDIX[72]

In this appendix we give examples of two Class (II_1) factors which are not isomorphic and yet each of which is isomorphic to a subset of the other. The factors will be the factors associated with certain groups $\mathfrak{G}_1$ and $\mathfrak{H}_1$ as in §5.3. These groups are best presented as subgroups of a group $\mathfrak{G}$ which we shall now define.

$\mathfrak{G}$ is determined by generators x_r, one for each rational number $r = p/q$. The commutation rules for these generators are determined as follows. Let A denote the set of dyadic rationals $q/2^n$ in the most reduced form where $n = 0, 1, 2, \ldots$ and q is odd except possibly when $n = 0$. Let B denote the set of all other rationals.

The commutation rules for x_r are the following:

(i) If $r \in A$, then x_r combines freely with any product $x_{s_1} \cdot x_{s_2} \cdots x_{s_p}$ if $s_1, s_2, \ldots, s_p$ are all $< r$.

(ii) If $r \in B$, x_r commutes with all products $x_{s_1} \cdot x_{s_2} \cdots x_{s_p}$ if $s_1, s_2, \ldots, s_p$ are all $< r$.

(iii) All x_r are of order ∞.

One can show that the set of products $x_{r_1}^{n_1} \cdots x_{r_p}^{n_p}$, $n_i = \pm 1, \pm 2, \ldots$ (with a finite number of factors) constitutes a denumerably infinite group $\mathfrak{G}$.[73] This is also

[72]This example was originally part of a later investigation by the second-named author, but it can be most easily presented here, using our present methods.

[73]This can be most readily shown by obtaining a "normal form" for each such product $x_{r_1}^{n_1} \cdots x_{r_p}^{n_p}$. This normal form may be obtained as follows. Let r_i denote the largest subscript which appears

true for the set of products in which the x_r's are restricted, so that r is in a fixed non-empty subset S of the rational numbers.

Let $\mathfrak{H}_r$ be the subgroup of $\mathfrak{G}$ determined by the generators $x_{r'}$, with $r' \leq r$. Let $\mathfrak{G}_r$ be the subgroup of $\mathfrak{G}$ determined by the generators $x_{r'}$, with $r' < r$. We can show

Lemma A.1.

(i) $\mathfrak{G}_r \subseteq \mathfrak{H}_r$.

(ii) If $r' < r$, $\mathfrak{H}_{r'} \subseteq \mathfrak{G}_r$.

(iii) If $n = \pm 1, \pm 2, \ldots$, *then* $\mathfrak{H}_{r+n}$ *is isomorphic to* $\mathfrak{H}_r$.

(iv) $\mathfrak{G}_r$ *has property (i) of Lemma 5.3.4.*

(v) $\mathfrak{G}_r$ *has property (i) of Lemma 6.1.1.*

(vi) Assume $r \in A$. *Then* $\mathfrak{H}_r$ *satisfies the hypotheses of Lemma 6.3.1, and consequently it satisfies* (i_1) *and* (i'_2) *of Lemma 6.2.1 and its Corollary 2 and (i) in Lemma 5.3.4.*

Proof. Ad (i) and (ii) are clear.

Ad (iii). The correspondence $x_{r+n} \sim x_r$ between the generators of $\mathfrak{H}_{n+r}$ and $\mathfrak{H}_r$ is one-to-one and preserves the properties $r \in A$ and $r \in B$. Consequently it determines an isomorphism between $\mathfrak{H}_{n+r}$ and $\mathfrak{H}_r$.

Ad (iv). Suppose $a \in \mathfrak{G}_r$. Then $a = x_{r_1}^{n_1} \cdots x_{r_p}^{n_p}$ with each $r_i < r$. Let r' be the $\mathrm{Max}_{i=1,\ldots,p}\, r_i$. Then of course $r' < r$ and there is an $r'' \in A$ such that $r' < r'' < r$. By the first commutation rule above $x_{r''}$ combines freely with a. By the definition of $\mathfrak{G}_r$, $\mathfrak{G}(a, x_{r''}) \subseteq \mathfrak{G}_r$. The third rule above tells us that $x_{r''}$ is of infinite order. These last two results show that the hypotheses of Lemma 6.3.2 are satisfied. It follows from that lemma that the condition (i) of Lemma 5.3.4 is satisfied.

Ad (v). $\mathfrak{G}_r$ has property (i) of Lemma 6.1.1: For let $a_1, \ldots, a_k$ be any k elements of $\mathfrak{G}_r$ and let $x_{r_1}, x_{r_2}, \ldots, x_{r_l}$ denote the generators which appear explicitly in $a_1, \ldots, a_k$. Let $r' = \mathrm{Max}_{i=1,\ldots,l}\, r_i$. Then $r' < r$ by the definition of $\mathfrak{G}_r$. Hence there is an $r'' \in B$ such that $r' < r'' < r$. Let $c = x_{r''}$. Then $c \neq 1$, $c \in \mathfrak{G}_r$ and c commutes with $a_1, \ldots, a_k$ by the second commutation rule above. This result shows that (i) in Lemma 6.1.1 is satisfied.

Ad (vi). For $\mathfrak{H}_r$ we let $\mathfrak{F}_1 = \mathfrak{G}_r$ and $\mathfrak{F}_2 = (x_r^n)$. Now x_r is of infinite order and

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and is in B. We move $x_{r_i}^{n_i}$ as far to the left as possible. Let $r_{i'}$ be the next largest subscript, which is also in B. We then move $x_{r_{i'}}^{n_{i'}}$ as far to the left as possible. We continue on until we have considered all the $r_i \in B$. The resulting product is then in a form in which the $x_{r_j}^{n_j}$, $r_j \in A$, have the same order as before, but each such $x_{r_j}^{n_j}$, $r_j \in A$, is followed by a product $x_{r_{i_1}}^{n_{i_1}} \cdots x_{r_{i_k}}^{n_{i_k}}$ with the r_{i_t} each in B, each $< r_j$, and with $r_{i_1} < r_{i_2} < \cdots < r_{i_k}$. We may use this form to establish the associativity of the composition rule. The existence of an inverse is clear.

the first commutation rule shows that $\mathfrak{H}_r$ is the free product of $\mathfrak{G}_r$ and $\mathfrak{F}_2$. These show that the hypotheses of Lemma 6.3.1 are fulfilled. That lemma itself now shows that (i$_1$) and (i$'_2$) of Lemma 6.2.1 and its Corollary 2 are satisfied, and (i) in Lemma 5.3.4.

We next state a quite general lemma.

Lemma A.2. *Let* $\mathbf{M}$ *be a ring which contains* $(\alpha 1)$ *and suppose* $f \in \mathfrak{H}$ *is such that* $Af = 0$ *and* $A \in \mathbf{M}$ *imply* $A = 0$. *Let* $\mathfrak{M} = \mathfrak{M}_f^{\mathbf{M}}$. *Then if* A *is closed, has domain dense, and* $A \,\eta\, \mathbf{M}$, A *is bounded if and only if it is bounded on* $\mathfrak{M}$.

Proof. It is clear that we need only show that if A is unbounded, it is unbounded on $\mathfrak{M}$. Let $A = WB$ where W is partially isometric and B is definite (cf. [5], §4.4). Since $A \,\eta\, \mathbf{M}$, we have $B \,\eta\, \mathbf{M}$ and also for all g, $|Ag| = |Bg|$. (This may be ∞ of course.) Hence we may substitute B for A, or what is the same thing assume that A is definite. But A definite implies that there is a resolution $\{E(\lambda)\} \subseteq \mathbf{M}$ such that

$$A = \int_0^\infty \lambda \, dE(\lambda).$$

Since A is unbounded, $1 - E(\lambda) \neq 0$ for $\lambda < \infty$. Let $g = (1 - E(\lambda))f$ for any fixed λ. By our hypothesis on f, $g \neq 0$ since $1 - E(\lambda) \neq 0$. From the formula for A we see $|Ag| > \lambda |g|$. Furthermore $g \in \mathfrak{M}_f^{\mathbf{M}} = \mathfrak{M}$. Thus if A is unbounded, it is unbounded on $\mathfrak{M}$.

Corollary. *If* $A \in \mathbf{M}$ *the bound of* A *on* $\mathfrak{M}$ *is the same as the bound of* A ($|||A|||$).

Proof. It is clear that the bound of A on $\mathfrak{M}$ is $\leq |||A|||$. To show that it can't be less one uses an argument similar to the above.

We wish to apply Lemma A.2 to the case in which $\mathbf{M}$ is a subring of the set of group numbers in the sense of §5.3. We may do this by taking $f = \varphi_1$ in the sense of (5.3.α). For if $A \in \mathbf{M}$, we have that $A \sim \{\eta_{a^{-1}b}\}$ by Lemma 5.3.2 and

$$A\varphi_1 = \sum_{c \in \mathfrak{G}} \eta_c \varphi_c$$

by (49,γ). Thus $A\varphi_1 = 0$ implies $\eta_c = 0$ for all $c \in \mathfrak{G}$ and hence $A = 0$.

It is also useful to notice in applying this to group numbers that if we have any sequence $\{\eta_c\}$ with $\sum_{c \in \mathfrak{G}} |\eta_c|^2 < \infty$ then the matrix $\{\eta_{a^{-1}b}\}$ determines an A with a closure and domain dense. For it is clear from (49,γ) that A is defined for all φ_a, $a \in \mathfrak{G}$ of (5.3.α) and hence for the linear combinations of these. Since the latter are dense, A has domain dense. Furthermore the matrix $\{\overline{\eta}_{b^{-1}a}\}$ is adjoint to it. Since this latter also has domain dense, A must have a closure. We shall consider A to be the least closed transformation, with matrix $\{\eta_{a^{-1}b}\}$. On its domain, A is the limit of finite linear combinations of the U_c and hence is $\eta\mathbf{M}$. Cf. Lemmas 5.3.1, 5.3.2 and 5.3.6.

Lemma A.3. *Suppose $\mathfrak{G}^0$ and $\mathfrak{G}^{00}$ are two groups and $\mathbf{M}^0$ and $\mathbf{M}^{00}$ their respective group numbers in the sense of §5.3. Then if $\mathfrak{G}^0 \subseteq \mathfrak{G}^{00}$, $\mathbf{M}^0$ is algebraically isomorphic with a subring of $\mathbf{M}^{00}$.*

Proof. For an $A \in \mathbf{M}^{00}$ we have a sequence $\{\eta_c; c \in \mathfrak{G}^{00}\}$ as in Lemma 5.3.2. Let $\mathbf{M}^{0+}$ denote the set of A's $\in \mathbf{M}^{00}$ for which $\eta_c = 0$ if c not $\in \mathfrak{G}^0$. Since $\mathfrak{G}^0$ is a subgroup, this set of A's is closed under the operations $\alpha A, A^*, A + B, AB$. Furthermore this set of A's is metrically closed, and as we mentioned at the end of §5.5, for a subalgebra of group numbers, this implies ring closure. Thus $\mathbf{M}^{0+}$ is a ring, and indeed is the ring determined by the U_c, $c \in \mathfrak{G}^0$.

Let $\mathfrak{M}^0$ be the set $\subseteq \mathfrak{H}^{0,0}$ determined by the φ_a with $a \in \mathfrak{G}^0$ (cf. (5.3.α)). It is clear that we can identify this with $\mathfrak{H}^0$, the space on which we have $\mathbf{M}^0$ defined. Now for every $A \in \mathbf{M}^0$ we have a sequence $\{\zeta_c; c \in \mathfrak{G}^0\}$ and corresponding matrix $\{\zeta_{a^{-1}b}\}$. Correspondingly, we have a sequence $\{\eta_c; c \in \mathfrak{G}^{00}\}$ with $\eta_c = \zeta_c$ for $c \in \mathfrak{G}^0$ and $\eta_c = 0$ otherwise. As we pointed out above, the matrix $\{\eta_{a^{-1}b}\}$ determines a closed operator $\widetilde{A}$ with domain dense. This is clearly $\eta\mathbf{M}^{0+}$ and hence by the application of Lemma A.2 described above, it is bounded if and only if it is bounded on $\mathfrak{M}_f^{\mathbf{M}^{0+}}$. Now one can show by our above definition of $\mathbf{M}^{0+}$ that $\mathfrak{M}_f^{\mathbf{M}^{0+}} = \mathfrak{M}^0$. Now $\widetilde{A} = A$ on $\mathfrak{M}^0$ and thus is bounded there. Hence, by Lemma A.2, $\widetilde{A}$ is bounded and $\in \mathbf{M}^{0+}$. We now see that to every $A \in \mathbf{M}^0$ there corresponds an $\widetilde{A} \in \mathbf{M}^{0+}$ such that $\widetilde{A}$ agrees with A on $\mathfrak{H}^0 = \mathfrak{M}^0$. On the other hand, it is clear that to every $\widetilde{A} \in \mathbf{M}^{0+}$ we have an $A \in \mathbf{M}^0$ with this property.

One can readily verify that this is an algebraic isomorphism.

We are now ready for the final step. Let $\mathbf{M}_1$ be the set of group numbers for $\mathfrak{G}_1$, $\mathbf{M}_2$ the set of group numbers for $\mathfrak{H}_1$ (cf. §5.3).

Lemma A.4.

(i) $\mathbf{M}_1$ *and* $\mathbf{M}_2$ *are factors in Case* (II$_1$).
(ii) $\mathbf{M}_1$ *is algebraically isomorphic with a subring of* $\mathbf{M}_2$.
(iii) $\mathbf{M}_2$ *is algebraically isomorphic with a subring of* $\mathbf{M}_1$.
(iv) $\mathbf{M}_1$ *has property* Γ *of Definition 6.1.1.*
(v) $\mathbf{M}_2$ *does not have property* Γ.

Proof. Ad (i). $\mathbf{M}_1$ is a factor by Lemma A.1 (iv) and Lemma 5.3.4. $\mathbf{M}_2$ is a factor by Lemma A.1 (vi) and Lemma 5.3.4. $\mathbf{M}_1$ and $\mathbf{M}_2$ are both in Case (II_1) by Lemma 5.3.5 (i).

Ad (ii). This follows from Lemma A.1 (i) and Lemma A.3.

Ad (iii). Let $\mathbf{M}_0$ denote the set of group numbers for $\mathfrak{H}_0$. Lemma A.1 (iii) for $n = -1$ shows that $\mathfrak{H}_0$ and $\mathfrak{H}_1$ are isomorphic, and this of course implies that $\mathbf{M}_2$ is algebraically isomorphic to $\mathbf{M}_0$. Lemma A.1 (ii) and Lemma A.3 imply that $\mathbf{M}_0$ is isomorphic to a subring of $\mathbf{M}_1$. Hence $\mathbf{M}_2$ is algebraically isomorphic to a subring of $\mathbf{M}_1$.

Ad (iv). This is implied by Lemma A.1 (v) and Lemma 6.1.1.

Ad (v). This is implied by Lemma A.1 (vi) and Corollary 2 of Lemma 6.2.1.

Since property Γ is an isomorphism invariant, (iv) and (v) imply that $\mathbf{M}_1$ and $\mathbf{M}_2$ are not algebraically isomorphic. Combining this with (ii), (i) and (iii), we have

Theorem A. *There exist two rings $\mathbf{M}_1, \mathbf{M}_2$, each a factor in Case* (II_1)*, which are not algebraically isomorphic to each other but are such that each is algebraically isomorphic to a subring of the other.*

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